

EXHALE: Update Log (stage 2, from 2026-09-05)

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This log starts fresh on 2026-09-05. The two earlier logs are kept unchanged under their new names:

- docs/Update_EXHALE_stage0.{tex,pdf}: the early phase (the ATES fork up to 2026-08-15).
- docs/Update_EXHALE_stage1.{md,tex,pdf}: sections 1-171, 2026-06 to 2026-09-05. Every "Update_EXHALE section N" reference in the source comments and in the older memos points into that file; the references were rewritten to the new name on 2026-09-05.

Each entry below is one dated section: what changed, why, which files, the gate that was run and its result, and the movement it produced against the scratch baseline or the goldens. Section numbers restart at 1. The plan the entries execute is docs/development_plan_20260905_rev3.md (sections 1-10) with the order of docs/development_plan_20260905_execution.md; item labels (A1, A8, 10.1-1, D0, C1 ...) are the plan's.

1. Plan, audit and decisions that open stage 2 (2026-09-05)

State of the tree at the start: HEAD 35d9dd5 (clean apart from docs/), binary rebuilt from it with gfortran 16.2 (md5 16cde610347c10c0a48db5c2a7fa6cbb). All twelve regression goldens differ from the case outputs (they are the pre-section-170 snapshots); by decision they are not refreshed until the end of Phase 1 (below).

Documents written today, all under docs/:

- `physics_numerics_audit_20260905.md` (external audit of 35d9dd5), its diagnostics in `audit_20260905/`.
- `development_plan_20260905.md`, `_rev1.md`, `_rev2.md`, `_rev3.md` and the three reviews `_review.md`, `_review2.md`, `_review3.md`: the plan and its revisions after each review. Rev 3 is the plan of record; its section 10 holds the independent five-group source audit (three new P0, twelve new P1, about fifty P2), with the reports and the verification probe in `audit_20260905/source_audit/`.
- `development_plan_20260905_execution.md`: Steps 0-7, who acts, gates.

Decisions taken by the user today (recorded in rev 3 sections 8 and 10.5 and in the execution document, Step 0):

1. GoldenS are refreshed once, at the end of Phase 1; until then a scratch baseline (`backup/regression/baseline_post170_<date>/`, read through a `REGRESSION_GOLDEN_DIR` variable to be added to `run_check.sh`) is the control for byte-identity. `golden/` is not touched before that.
2. ROE interface: `speed_estimate_ROE` returns a status and, in the vacuum case, the two gas-edge speeds; `Num_flux` gains an HLLE vacuum branch. Signature shown before coding.
3. One spectrum type builds every band. Power-law input: power law on the whole grid, XUV and below 13.6 eV alike (the present construction is the stated meaning of the key). A new Spectrum type: Planck gives the Planck photosphere from Stellar Teff and Stellar radius on the whole grid. Load: the table on the whole grid. The Balmer continuum follows the run's type. The setup report states the sub-13.6 eV source and the integrated grid flux.
4. $m_{\text{He}}/m_{\text{H}} = 3.9715$ in the species table; $R_J = 7.1492e9$ cm (IAU nominal equatorial) in the code and the Wind-AE bridge alike (R_0 of every run +2.26%); MJ and MSUN to the IAU nominal values.
5. The du stop uses the signed spread of $\rho v r^2$ over the mean, as `flux_spread` does.
6. Every planet folder, benchmark and planet example runs on the SED its reference literature used or assumed, one file per planet under `inputdata/sed/` with provenance; the input switch is a configuration change, the reruns stay instruction-gated.
7. A loaded SED that ends above the band an active absorber needs ($\text{He } 2^3\text{S}$ at 2583 Å, a low-IP metal at its threshold) stops the run; the message offers the two remedies (turn the absorber off in the input, or supply a covering SED). No continue key.

Open: decision 4 (A3 retry design: carrier substep retry or outer step controller); 2, 5, 6 inside D0; 7 before B2; 10 in Phase 5.

Documentation restructuring done today (this section): `Update_EXHALE.*` renamed to `Update_EXHALE_stage1.*`, `Update_EXHALE_early_phase.*` to `Update_EXHALE_stage0.*`; every reference in `src/`, `docs/`, `python/`, `README*.md`, the notebooks and the workspace `CLAUDE.md` rewritten to the new names. Check: `git diff --numstat` shows additions equal to deletions in every one of the 112 touched files (no line moved; the three exceptions are the renamed logs themselves, which git sees as deleted paths). The rebuilt binary's md5 moved from 16cde610347c10c0a48db5c2a7fa6cbb to e779298d0b482bfb40ae2f9aa80d60a because four of the rewritten references are inside printed message strings (`EXHALE_main.f90` 2602, `ionization_equilibrium.f90` 2948,

input_read.f90 1004, diffusion_tests.f90 1292), which is text the binary carries; no executable statement changed. The pointers in README.md, README_HOWTO.md and the workspace CLAUDE.md name this file as the current log and the stage 1 file as the history.

Scratch baseline created: backup/regression/baseline_post170_20260905/, the Hydro_ioniz.txt, Ion_species.txt, *_adv.txt and run.log of the twelve cases as they sat in the case directories (48 files, 26 MB; provenance in the headers: git=6d07d48afd41 tree=dirty, the working tree that became 35d9dd5, runs of 2026-09-05). It is the byte-identity control for Phase 1 and is deleted after the golden refresh at the end of Phase 1.

2. Phase 0: the tests exist and are red on the known defects (2026-09-05)

Plan reference: development_plan_20260905_execution.md Step 1. Four workers, disjoint files; every result below was re-run by the advisor after the worker's report (times are the advisor's runs).

2.1 src/tests/physics_probe/ (W1): 41 assertions, 15 PASS, 26 FAIL, 3.9 s

Derived from the frozen diagnostics in docs/audit_20260905/ (which stay); links the production modules through a computed use closure (source_closure.py), so no module can be left out of the link by hand. Drivers named for the physics: riemann_wave_speeds.f90 (LLF signal speed 1 against the required 3, mass flux -3.5 against -4.5, mirror symmetry; ROE normalization invariance broken by 0.661 and 3.162 in the two branches; NaN star speeds on the equal-pressure expansion; the production ROE flux leaves thermal energy -2.93 after one Euler update on the vacuum-producing expansion), h2_channel_detector_ratio.f90 ((S+D)/M 0.2545 against the measured 0.2827, D/(S+D) 1/9 against 0.20; channel sum and stoichiometry GREEN), h3p_cooling_limits.f90 (cooling independent of $n(\text{H}_2)$ at 0, 1, 100 cm^{-3} ; joins +0.430, -0.00113, -0.0235 at 300, 800, 1800 K; factor jump $1.5\text{e-}3$ at $1\text{e}14$), h2_rovibrational_identity.f90 (chemistry $T^2 \text{dln}Z/\text{d}T$ against the EOS u_{rv} : -0.63% to -8.29% over 300-8000 K), photoevent_energy_ledger.f90 (H event at 20 eV closes exactly through the production photoelectron_share; the H2 double event at 80 eV accounts for 60.28 of 80 eV, 19.72 eV unassigned), radiation_exchange_ledger.py (two-cell ownership rule, GREEN by construction).

2.2 src/tests/grid_and_gates/ (W2): 6 programs, all RED, 13.4 s

grid_width_identity.f90 (production define_grid): $\text{dr}_j(j)$ against $r_{\text{edg}}(j) - r_{\text{edg}}(j-1)$ off by 1.0145 on the Mixed grid at $r_{\text{max}} = 10$, 1.0100 at $r_{\text{max}} = 2$, 1.0046 on Stretched, exact on Uniform (where every cell has the same width, which is why the defect was invisible). photon_grid_quadrature.f90 (production set_energy_vectors, J_{inc} , cross sections; wasp_full spectrum): code/exact P_HI 1.0951 (floor 4.8 eV), 1.0890 (5.139 eV), 1.0003 (13.6 eV); P_HeI 1.0157 and P_HeII 1.0299 in every configuration; heating 0.99984. New from this test: P_HeI carries the same threshold-bin defect (24.6 eV is a grid point whatever the floor), and the X-ray block ends at 1184.19 eV, not 1240 (share of P_HI $1.9\text{e-}10$). mass_flux_spread_functional.py: the coded du and the signed flux_spread on the baseline states (lower_profile 823.4 vs 3.61 with two sign changes in the window; mol_base_handoff 1.29 vs 0.696 with none; wasp_full $9.85\text{e-}4$ vs $9.85\text{e-}4$), and $du = 0$ on a synthetic +F/-F window. output_state_consistency.sh (200-step mol_base_handoff): header sec_ion=T, heat_total/heat median 0.2752,

largest difference 0.822. `base_level_single_statement.sh` (lower_profile, 2 steps): setup $2.0035e-5$ bar against the profile's $1.00e-6$ bar, factor 20.035. `restart_grid_guard.sh` (roundtrip on Mixed, restarted on Uniform with the same N): exit 0, the restart's r equals the file's Mixed radii exactly and sits 0.60 relative from the run's own grid.

2.3 inputdata/sed/ (W4): the SED inventory of decision 16

Four files built from data already in the workspace, each with a provenance header, a build script under `inputdata/sed/build/`, and the coverage test `sed_coverage_and_order_check.py` (31 assertions, all PASS): WASP-52 b (eps Eri, MUSCLES v22, reproduces the existing planet-folder file to round-off), hot Uranus (the benchmark's Ribas band table continued to 2995 Å from the Gueymard composite it came from), HD 209458 b (WHI 2008 solar minimum at 0.04707 au, the p-winds file reproduced exactly), LHS 1140 b (GJ 1132 proxy). Not built, data missing: WASP-121 b (Huang 2023's LLmodels F5V spectrum above 1700 Å) and HD 189733 b (Salz 2016's composite; its solar table is not here). `inputdata/scaled_solar_hd189.sed` judged an orphan (no provenance, hardness inconsistent with its own normalization, ends at 1899.5 Å). Verbatim paper quotes and the open items are in `inputdata/sed/README.md`; rev 3 decision 16 records the status. Also found: WASP-52b/`input.inp` points at a directory that no longer exists (EXHALE/), the top-level Koskinen benchmark runs at 0.048 au against the paper's 0.05, HD209458b/`input.inp` at 0.048 au against the literature 0.047.

2.4 Regression harness and new cases (W3)

`run_check.sh`: REGRESSION_GOLDEN_DIR (default golden/; golden mode still writes only to golden/); the rebuild is fatal with its log kept in `backup/regression/.build.log`; the summary counts only the cases of the run; the binary copy is status-tested; FILES gains the two `_adv.txt` profiles (the post-processing pass was outside the regression); a file without a reference is MISS, not FAIL, and a run that compared nothing prints NO VERDICT; a restart case keeps its IC pair in `<case>/IC/` and the script restores it after clearing output/; phase0-cases prints the new case list. Makefile: `make check CASES=...`; `make test` runs `element_census_tests`, `diffusion_tests`, `residual_determinism` and every executable `src/tests/*/run.sh` found at run time, every suite even after a failure, nonzero exit if any failed. New cases, not in DEFAULT_CASES (PHASE0_CASES): `hydrostatic_column` (hot-Uranus gravity, irradiation 18 orders below normal, no key exists to switch cooling off so the gas is held where it is negligible; 300 steps, isothermal to 2%, but draining through the outflow boundary at 0.8 Mach, so it is a configuration for the Phase 4 D2 measurement, not a hydrostatic state), `oxygen_chemistry` (`examples/18_oxygen_chemistry` verbatim, 1000 steps, 3 min 52 s single-threaded under load), `hp_front`, `hp_zero_seed`, `hp_trace_seed` (restarts of the `mol_base_handoff` state with `Ionization transport: True`; the H II column set to 0 and to $1e-12$ of the H nuclei by moving the protons into H I, since `load_IC` refuses a changed He/H; the three 100-step states are mutually distinct, so the seed reaches the solver). No dump hook asserts the seed at the entry of `solve_carriers`; the README names the two lines where one belongs. Measured (advisor re-run of two cases: 6.2 s, byte-identical on all four files): `check wasp_full mol_base_handoff` against the scratch baseline is byte-identical on all four files per case (W3, 13 min); the five new cases reproduce their own baseline copies; `make check CASES=mol_base_handoff` against the stale golden/ reports the known $8.3e-4$ and $3.6e-4$ movements and MISS on the two `_adv` files; `make test` (31 s) runs five suites: `diffusion_tests` and `residual_determinism` PASS, `grid_and_gates` and `physics_probe` RED on the known defects as intended, and

element_census_tests does not compile at HEAD (pre-existing: carrier_state gained nrho and wfac in section 158 and the test's two calls still pass ten arguments; carrier_write_back likewise takes nrho where the test passes ntot). Fixed in section 3 below.

Also found by W3: with the flux made negligible, the He recombination coupling warning (ionization_equilibrium.f90 947, 952) fires in every cell with a ratio of 1e80 because both sides sit at their guards; diagnostic only.

3. element_census_tests compiles again (2026-09-05)

src/tests/element_census_tests.f90: the two carrier_state calls pass the nrho and wfac outputs the routine has had since stage 1 section 158, and the two carrier_write_back calls pass nrho (the density the carrier fraction is a fraction of) in place of ntot. Test only; no production source. Result: builds, 20 assertions PASS (E1 stoichiometry and charge, E2 elemental closure through the write-back including the CO collapse, E3 the H2 photoevent nuclei ledger, E4 the base H2 fraction). Until now the suite had been unbuildable, so its assertions had not been checked since section 158.

4. Phase 1 batch 2a: bounded corrections with no golden movement (2026-09-05)

Seven items, seven workers on disjoint files in one tree (private OBJDIR/EXE builds, no shared regression runs); one rebuild and one gate by the advisor afterwards. Plan references: rev 3 sections 4.1 (A2), 10.2 items 7, 8, 10, 12, 10.3, 10.5 decisions 3, 13, 17.

- **A2, ROE estimator and caller** (speed_estimate_ROE.f90 rewritten; Num_Fluxes.f90 ROE branch; input_read.f90; EXHALE_main.f90): docs/a2_roe_interface.md implemented as approved: roe_star_state, status OK/VACUUM/INADMISSIBLE, the vacuum test first, PVRS kept only when admissible, two-rarefaction exponent $1/z$, two-shock densities times the upstream densities, HLLE flux on faces without an admissible star state with a printed counter, refusal of Numerical flux: ROE with molecular chemistry and the ladder EOS. Four deviations accepted: a positivity guard on the input states at the top of the estimator; the refusal placed in the consistency block after the optional keys (where thereis_mol is known); the old p_min floor removed (the guard supersedes it); the counter printed for ROE runs only. Test: 30 ROE assertions GREEN (scale invariance, equal-pressure expansion $c_* 0.9576611$, stationary contact, asymmetric shock and rarefaction against exact Riemann solutions at 5%, vacuum edges $-6.127017/+6.127017$, positive thermal energy after one Euler update).
- **2a-SED, decision 17** (sed_read.f90): a table that stops above the band an active absorber needs stops the run with the file, the missing band in eV and A, the absorbers (He 2^3S ; each active neutral metal below the file minimum, K I at 4.341 eV included) and the two remedies; no continue key. Test sed_coverage_stop.sh (six assertions) GREEN; no configuration of record trips it (the Koskinen runs have the metastable off).
- **2a-OPAC** (util_ion_eq.f90 PH_heat_H/PH_heat_HHe, opacity_models.f90): model P's factor multiplies the shared flux weight int_f of every absorption integrand (rates, heating, secondary ionization, acc_q), so the

beam's loss equals the local absorption; test `opacity_model_p_ledger.f90`: a cell with $f = 10$ closes energy and H I ionizations to $3e-8$ (was a factor 10 off).

- **2a-LYA** (`lya_rt.f90`, `util_ion_eq.f90` `fuv_lw_photon_field`, `water_photolysis.f90` header, plus the two call sites in `ionization_equilibrium.f90` and `diffusive_photochemistry.f90` that the new `nHI` argument required): band B2 is attenuated by `lya_stellar_beam_transmission(T, tau) = erfc(x1/(sqrt(2) X_s))`, the expression `jlya_escape_prob` carried inline and now calls (bitwise equal at 4941 test points). Test `lya_band_transmission.f90` (ten assertions) GREEN. Movement, `oxygen_chemistry` 1000-step snapshot: `du` 903 -> 596, `log10 Mdot` 9.62 -> 9.61, B2 transmission $1e-291$ at the base, $1e-3$ at $1.07 R_p$, 0.98 at $1.5 R_p$; H₂O moves up to 6.2x and OH 7.3x in the layer. Scratch baseline of that case re-snapshotted (previous copy kept).
- **2a-WAE** (`wae_exhale_input.f90`): mass fractions from the same `m_He/m_H` the solver is handed; test: reconstructed He/H equals the input to $1e-12$ (was 1.0071696x). RJ/MJ/MSUN untouched (decision 14 is 2c).
- **2a-HPFIX** (`constrained_chemical_equilibrium.f90`): `hp_is_fixed` pins `species_fixed(is_HII)` like H₂ and the oxygen carriers; new suite `src/tests/constrained_network_layout/` (eight assertions) GREEN. The three `hp_*` cases never enter the continuation (every cell is a class-1 root), so no run exercises the fix yet.
- **2a-PLANCK** (`J_inc.f90`, `set_energy_vectors.f90`, `input_read.f90`, `write_setup_report.f90`, `excited_hydrogen.f90`, `docs/input_schema.md`): Spectrum type: Planck (requires Stellar Teff and Stellar radius) builds the whole grid from $\pi B_{\nu}(T_{\text{eff}}) (R_{\text{star}}/a)^2$; the Balmer routines call the same shared function (unchanged result); the setup report states the spectrum type, the source of the band below 13.6 eV and the integrated grid flux against the nominal `J_XUV`. New suite `src/tests/spectrum_type/` (12 assertions) GREEN. Measured on the `wasp_full` star: Planck/power-law 1642 at 4.8 eV, 358 at 6.2 eV, 2.9 at 10 eV, 0.014 at 13.6 eV; He 2³S rate 45.2 s^{-1} (Planck) vs 0.100 (power law); integrated grid flux / nominal 1.42 (power law, floor 4.8 eV), 0.995 (floor 13.6), 112.8 (Planck).
- **2a-GUARDS** (`define_grid.f90`, `set_IC.f90`, `EXHALE_main.f90`, `viscous_conduction.f90`, `steady_newton.f90`, `input_read.f90`, `diffusive_photochemistry.f90`, `water_photolysis.f90`, `oxygen_rates.f90`, `lyman_werner.f90`, `parameters.f90`, `molecular_infrared_data.f90`): `cell_nearest_radius` (the bare `minloc` read `r(503) = 0` out of bounds on the default grid), `j_min` ≥ 1 , `dead_report_unmet_steady_gates` removed, `Mdot` guarded for small N and non-outflow, conduction floor counted and reported, `okres` read in the ray test, correct sweep count, `get_word` initialized, `dead_close(unit=1)` removed, `adv_corr` zeroed on allocation, `dead Tcode` argument removed from four carrier routines and their callers, `carrier_mass_amu` through the named species indices, `wp_ready` used as the guard it names, `dead` constants and stale headers fixed, `molecular_infrared_data's i` private. Test `grid_window_indices` (seven assertions) GREEN.

Gate, advisor's single rebuild (`md5 b52c79ebb15ee9014858425bfea4cb1c`): make test runs seven suites; every assertion added in this batch is GREEN and only the known defects of later items remain RED (two test drivers had linked the program object `element_census_tests.o` from `build/`; their link lists now exclude program objects). Regression, sixteen cases against the scratch baseline, single-threaded: fifteen byte-identical on all four files (the eleven of `DEFAULT_CASES`, `hydrostatic_column`, the three `hp_*`); `oxygen_chemistry` moved as stated under 2a-LYA.

Follow-ups recorded for batch 2b and the docs pass: `carrier_mass_amu` returns 0 for a species missing from the table (needs a stop); `write_output.f90` 806-816's FUV ledger closed form carries no line transmission for B2 (its `rel_diff` reads 1.0 now; the LW row was already 0.84); `inputdata/sed/README.md` describes the old warning in six places; `docs/input_schema.md` row 11a should state the coverage requirement; `docs/a2_oxygen_option_design.md` 402 describes the pre-2a B2 field; the `physics_probe/README.md` summary counts are stale; `J_XUV` (used by the Wind-AE bridge), the FUV band fluxes and the monochromatic flux do not yet follow the spectrum type (decision 13 is enforced for the photon grid only); `backup/regression/parse_golden/` was already stale at 35d9dd5; the three audit probe scripts under `docs/audit_20260905/` no longer compile against the tree and are marked as frozen records.

5. Phase 1 batch 2b: bounded corrections that move some outputs (2026-09-05)

Six workers (five code items and one documentation pass), disjoint files, private builds; one rebuild (`md5 d2173f0210abdc84f41cb6fb5076e18`) and one gate by the advisor. Plan references: rev 3 sections 4.1 (A1, A3 increment (a), A8) and 10.3 (`k_lw` cell mean, He I singlet, He II recombination cooling).

- **A1, LLF signal speed** (`Num_Fluxes.f90`, `RK_rhs.f90`, `EXHALE_main.f90`): $a1 = \max(\text{abs}(vL)+aL, \text{abs}(vR)+aR)$, the spectral radius of the Euler flux Jacobian, replacing $\max(\text{abs}(vL+aL), \text{abs}(vR+aR))$ (1 instead of 3 on the inflow test pair); the header's positivity claim now describes the coefficient the code carries. Reporting: calls of the positivity repair, repaired interfaces, repaired interfaces in accepted steps, and LLF faces, all printed even when zero. Test: the three LLF assertions GREEN. Movement: none (no case selects LLF and the repair path is never entered in the matrix: `flux correction: 0 positivity repair call(s)` in every run).
- **A3 increment (a), carrier reference scales and status** (`diffusive_photochemistry.f90`, `write_setup_report.f90`, `EXHALE_main.f90`): one `carrier_element_reference_density(ic, nH, nO, nC)` (H_2 and H^+ hydrogen, OH and H_2O oxygen, CO min(O , C)) read by both the Jacobian step (`carrier_source_derivative_step`) and the residual floor; the reported residual is that of the state after `limit_to_element_budget`; `solve_carriers` returns converged / iteration cap / line search failed and the caller stops on a failed status in a carrier the limiter did not move (rejection and retry remain increment (b), decision 4 open); a cumulative CO-ceiling record (cells, applications, molecules removed) in the run summary and `EXHALE_resolved.out`; `carrier_mass_amu` stops on a species missing from the table. New suite `src/tests/carrier_reference_scales/` (14 assertions; before the change the proton row's diagonal derivative was exactly zero at $\text{H}^+ = 0$ and $1\text{e-}12$ `n_H` because the step had collapsed to $1\text{e-}300$). Movement of this item alone: `hp_zero_seed` up to $7.8\text{e-}4$ at the ionization front, `hp_trace_seed` $2.8\text{e-}5$, `hp_front` round-off, `mol_carrier` and `mol_base_handoff` byte-identical. Cost of the extra residual assembly +1.5% wall time on `hp_zero_seed`.
- **A8, Chung detector inversion** (`h2_photo_channels.f90`): Table II is the detector event ratio $(N_S+N_D)/N_M$ (Chung et al. 1993 p. 886, one pulse per event, quoted at the code site), $q_D = N_D/(N_S+N_D) = 0.20$ at 80 eV; $\sigma_M = (1-f_{di}) \sigma_i$, $\sigma_S = (1-q_D) f_{di} \sigma_i$, $\sigma_D = q_D f_{di} \sigma_i$; `yan_rho` converted with $q_D = \rho_D/(1+\rho_D)/f_{di}$; an inadmissible `q_D` stops the run instead of being

clipped; frac_H2_double_of_protons renamed frac_H2_double_of_proton_events. At 80 eV: D 0.02254 -> 0.04408, M 0.7972 -> 0.7796, S 0.1803 -> 0.1763. Test: 18 assertions GREEN including the stored-channel and yield integration checks; E1 test 4 prints the same 6.951e-3. Movement of this item alone: mol_base_handoff n_e up to 1.8e-2 at r = 1.018, x(H+) 1.9e-2 at the base, T 4.6e-4; mol_sec_ion at the 1e-3 to 5e-3 level, with a 0.078 cm/s velocity shift moving the v = 0 crossing by one cell at r = 1.024-1.026 and flipping the advection-correction branch of the _adv files there. Mdot unchanged.

- **k_LW cell mean** (lyman_werner.f90, util_ion_eq.f90 fuv_lw_photon_field): new lyman_werner_dissociation_rate_cell_mean(F, N_out, N_in, T, n_H, tau_out, tau_in), a composite three-point Gauss rule on segments of at most 0.05 dex of column and 0.5 of continuum depth (the level-resolved cross section is a piecewise power law with slope up to 3.3, so the water bands' exponential mean does not apply; measured accuracy 2.5e-4 worst over 1500 random cells); the star-ward face carries NH2col(j+1). Test: seven assertions GREEN (the old inner-face rate was 3.18x too small on a cell spanning 1.12 dex). Movement: mol_lyman_werner k_LW +1.4% to +28.1% (median +2.7%), n(H2) -20% and n(H3+) -20% at the front r = 1.071, T up to 8e-3, n_e 1.7e-2; mol_base_handoff and wasp_full identical (no LW flux). Cost under 0.5%.
- **He I singlet and He II recombination cooling** (util_ion_eq.f90 eval_cool/eval_cool_cells, Cool_coeff.f90, ionization_equilibrium.f90 comment): the 24.6 eV collisional ionization and the Cen (1992) excitation of He I are charged to the ground singlet he_ground_singlet_density(nhei, nheiTR) (the metastable had paid both levels); lambda_rec_HeII = kT alpha_rec_HeII_total(T) with alpha_rec_HeII_total = alpha_11S + alpha_23S when the metastable is tracked (the coefficient the balance removes He+ at) and alpha_B otherwise; the coefficient is built inside eval_cool from T and the run's flags so that energy_semi_implicit's finite-difference dC/dT sees one function (a coefficient-passing form would have left an offset the derivative multiplies by 1e5). At 1e4 K the He II coefficient rises 2.807e-13 -> 3.640e-13 (+30%); the He recombination coupling's own weight (y_net alpha_1) is still not charged (about 10% of this channel, stated at the code site). Test: 15 assertions GREEN. Movement of this item alone: wasp_full median 2e-5 / max 1.7e-4 in rho, T max 5.9e-4, Ca I and Na I up to 4.7e-3; mol_base_handoff cool column median 1% / max 2.6% (the recombination channel is 64% of the cooling there), state 1e-4.
- **Documentation pass:** inputdata/sed/README.md (seven places, the stop), docs/input_schema.md (row 11a coverage rule; rows 11, 11b, 11d, K3, K4 line references corrected), docs/a2_oxygen_option_design.md (B2 row), src/tests/physics_probe/README.md (two summaries: 41/15/26 at 35d9dd5; 83 assertions, 69 PASS, 14 FAIL after 2a with the LLF fix already in the tree) and grid_and_gates/README.md, rev 3 section 10.2 and 10.5 "landed" notes, the execution document Steps 1 and 2a marked done, README.md and README_HOWTO.md (Spectrum type: Planck, the coverage rule, make test, the regression section's case list corrected from eight to eleven).

Gate (advisor): make test runs eight suites; every 2b assertion GREEN, RED only on the 2c and Phase 3 targets. Regression, sixteen cases against the post-2a scratch baseline, single-threaded: wasp_he23off and hydrostatic_column byte-identical (metastable off, no carriers); the other fourteen moved, Mdot unchanged to two decimals in all sixteen. Largest relative differences over all cells and columns (single cells; the large values are v near a zero crossing or trace

species in one cell, see the items):

case	Hydro_ioniz	Ion_species	Hydro_ioniz_adv	Ion_species_adv
wasp_full	1.06e-3	4.63e-3	1.37e-3	4.63e-3
wasp_full_newton	1.48e-3	4.69e-3	1.91e-3	4.69e-3
mol_base_handoff	2.54e-2	1.99e-2	5.58e-2	4.21e-2
mol_metals	2.16e-2	2.20e-2	2.70e-2	2.20e-2
mol_lyman_werner	1.25e-1	2.06e-1	5.63e-2	2.06e-1
mol_diffusion	2.54e-2	1.99e-2	5.86e-1	1.95e-2
mol_ir_bands	2.16e-2	2.20e-2	9.65e-2	2.20e-2
mol_sec_ion	1.70 (v at a zero crossing)	6.31e-3	1.70	1.0 (trace ion, one cell)
mol_carrier	2.52e-2	1.98e-2	3.87e-2	1.95e-2
lower_profile	5.38e-1	5.19e-2	5.38e-1	4.96e-2
oxygen_chemistry	1.94	1.0	1.94	1.0
hp_zero_seed	8.13e-2	9.54e-2	4.20e-2	9.54e-2
hp_trace_seed	4.25e-1	1.0	4.24e-2	1.0
hp_front	2.59e-2	6.30e-3	4.42e-2	6.29e-3

The scratch baseline was re-snapshotted from these outputs; the post-2a state of every case is kept in `backup/regression/baseline_post2a_20260905/`.

Follow-ups recorded: `T_equation.f90` (the temperature root of the advection post-process) carries no He 2³S cooling channel and charges the He I ground-state coefficients to the summed neutral column, so the `_adv` temperature and the hydro temperature are not solved with one cooling function (Phase 3 A5/A6 or the `_adv` consistency item of Phase 6); `post_process_adv.f90` 392 calls `eval_cool` without `nheiTR`; `cross_sec.f90` 519-528, `parameters.f90` 475-501 and `input_read.f90` 856-858 still describe the H⁺ column as a proton count and quote a "factor 1.6" between the two double-ionization models that is now reversed and much smaller; `write_output.f90` 200-203 says the `f_shield` column is the factor the rate carries (the rate is now the cell mean); `pct_limited` counts only overshoots above 1e-10 while the clamp acts at any size (same class as the CO counter, now recorded cumulatively); the give-up path of `retry_step` leaves the loop through the same exit as an accepted step (decision 4); `set_oxygen_coeffs` is called with strided sections that create array temporaries in the carrier Jacobian; `docs/EXHALE_user_manual.tex` is stale on the spectrum types and the SED stop.

6. Phase 1 batch 2c: corrections that move every output (2026-09-05)

Fourteen worker items on disjoint files plus advisor edits, every one measured alone against a frozen snapshot of the tree as it stood when the item started (private builds; the items therefore chain, and the "before" of one item is the "after" of those that landed earlier). One rebuild by the advisor with the conda gfortran 16.2 and its OpenBLAS (md5 5845df587aa7948153f1bed3becb28ac), `make test`, and one regression run against the post-2b scratch baseline. Plan references: rev 3 sections 10.1 (items 1-3), 10.2 (items 1-6, 9, 11), 10.5 (decisions 13-15, 17, 2c-CONST). Movement

is quoted as the largest relative change over the 500 physical cells of wasp_full (WASP-121b, He 2³S + metals, du stop, no Newton finish, so its Mdot carries the path-dependent spread of that stop) and mol_base_handoff (hot Uranus, 12000-step relaxation snapshot, not steady, so its outer-tail and base-cell numbers compare two points of one transient).

- **2c-DRJ, the cell width is the cell's own width** (define_grid.f90, set_IC.f90): $dr_j(j) = r_edg(j) - r_edg(j-1)$, an exact identity of the finite-volume grid; the stored width had been the width of cell $j+1$, 0.7 to 1.25 percent too large over the stretched region. Every consumer confirmed to use $dr_j(j)$ for cell j ; none changed. grid_width_identity GREEN. wasp_full: T 2.0e-3, v 5.4e-3 (base cell), n_e 6.3e-3, x(H+) 1.1e-2, mass flux +0.65 percent, printed log10 Mdot 13.24 both. mol_base_handoff: bulk profiles 1e-2 class with the transient tail at 1e-1, mass flux -0.96 percent, printed Mdot 10.52 -> 10.51.
- **2c-QUAD, thresholds on bin edges** (set_energy_vectors.f90): the photon grid is built from bin edges, e_v the geometric center of each bin and de_v its exact width; H I, He I, He II, the metastable edge and e_mid, e_top are edges, and the X-ray block reaches e_top. photon_grid_quadrature GREEN except P_HeI at 1.2e-3 (closed by 2c-THR). wasp_full: T 7.5e-2 at 1.12 Rp, n_e 7.8e-2, x(H+) 0.23, base v 0.47, mass flux +0.82 percent, log10 Mdot 13.24 -> 13.25. mol_base_handoff: T 4.7e-2 (tail), n_e 2.5e-2, mass flux +0.08 percent, Mdot 10.51 both. This is the single largest movement of the batch after the Jupiter radius, and it is the removal of the 9.5 percent overcount of P_HI found by the audit.
- **2c-BALMER, the Balmer continuum follows the run's spectrum type** (excited_hydrogen.f90, J_inc.f90; decision 13): the $n = 2$ photoionization rate and heating integrate the field the photon grid uses, the power law for a power-law run, the Planck field for Planck, the loaded table for Load (stellar_flux_eV, spectrum_covers_eV); a table that does not reach 3647 Å stops the run with the decision-17 message naming H(n=2). balmer_continuum_field GREEN (11 assertions). wasp_full (n = 2 coupling armed): gamma_2 falls by a factor 2939, n_e at 1.02 Rp -12.5 percent, x(H+) at the base -27 percent, T 9.6e-3, wind above 1.2 Rp and Mdot unchanged (13.25). mol_base_handoff: byte-identical (no stellar lines).
- **2c-KICK, the momentum kick removed** (EXHALE_main.f90): the $u(2, :) + 1e-16$ diagnostic aid is gone; the relative step change skips cells whose old momentum is exactly zero. momentum_row_from_fluxes_only GREEN. Movement round-off on wasp_full (1e-9), 4e-6 on mol_base_handoff, 4.4e-3 in v on lower_profile (cold-start snapshot with a sign-changing base flux amplifies it); Mdot unchanged everywhere.
- **2c-STATE, one state per output file** (EXHALE_main.f90): when the secondary-ionization coupling never switched on during the march, the final write no longer forces it on; the header says sec_ion=F, and the _adv files and the heating breakdown are built from the relaxed state. Hydro_ioniz.txt / Ion_species.txt byte-identical in every case; the _adv files and Heating_breakdown.txt of the fixed-step snapshots whose flip never fired (mol_base_handoff, lower_profile, and the other molecular gates) change by up to 0.97 in n(H II)_adv at the base. output_state_consistency stays RED at 1.2e-5: the heat column is assembled from the pre-sweep rates and the breakdown re-evaluates on the post-sweep composition (Phase 3, together with the excited_H_update call between the loop and the final write).
- **2c-DU, the signed mass-flux spread** (EXHALE_main.f90; decision 15): one functional mass_flux_spread, $(\max - \min) / \max(|\text{mean}|, 1e-30)$ of the signed $\rho v r^2$ over $j_{\min} \dots N$, read by the stop, the Newton arm-

ing and the stage flip, and written to `run.log` at the final step (`mass_flux_spread_functional.py` GREEN). `wasp_full` byte-identical (every threshold crossing on the same step); `mol_base_handoff` byte-identical (fixed steps; the reported `du` falls from 1.29 to 0.70); `wasp_full_newton` flips and hands off one step earlier, solution moves by $7e-8$.

- **2c-RESTART, a restart keeps the run's grid** (`load_IC.f90`): the file's cell centers are compared with the run's over the physical cells and a relative difference above $1e-10$ refuses the restart, naming both grids. `restart_grid_guard.sh` GREEN; `roundtrip` and the three `hp_*` cases byte-identical on the same grid. Consequence: after the grid-moving items of this batch (RJ +2.26 percent of the domain in R_p units, `dr_j`) the shipped `hp_*` IC pairs are refused, and are regenerated with `backup/regression/make_hp_initial_conditions.py` from a fresh `mol_base_handoff` output (below).
- **2c-BASE, one base level with a lower-atmosphere profile** (`input_read.f90`, `write_setup_report.f90`): with a profile, `n0` follows from `p_match_bar` exactly as from `p_base` of `base.inp`; a density key beside a profile is refused unless it agrees within 1 percent (`base_level_single_statement.sh` GREEN). `lower_profile` starts at the profile's level: base density $1.0e14 \rightarrow 5.0e12 \text{ cm}^{-3}$, its 12000-step snapshot changes entirely (printed $\log_{10} \text{Mdot}$ 8.44 $\rightarrow$ 7.92, ionization front inward, base `n_e` $\times 8$). The case README and `examples/17_lower_profile/input.inp` drop the density key. Nothing else moves.
- **2c-CI, the C I Voronov row** (`Cool_coeff.f90`): $P = 0$, $X = 0.193$, $K = 0.25$, $A = 6.85e-8$, $dE = 11.26$ (Voronov 1997 Table 1; the transcription had X and K in the wrong slots, $2.56x$ at 2000 K, $1.69x$ at 1e4 K). `voronov_carbon_ionization` GREEN. `wasp_full` $1e-5$ class, `x(C I)` $6e-4$, `Mdot` $-1e-5$ dex; `mol_metals` bitwise identical (collisional ionization of C I is negligible at its temperatures).
- **2c-MASS and 2c-MASS2, helium mass and Jupiter radius** (`decision 14`; `species_table.f90`, `parameters.f90`, `wae_exhale_input.f90`, `wae_ic_writer.f90`, `binary_element_diffusion.f90`, `diffusive_photochemistry.f90`, `lower_column.f90`, `utilities.f90`, `examples/exhale_io.py` and the Python tools): He I / He II / He III / He 2^3S carry 3.9715 m_H (HeH+ 4.9715), $RJ = 7.1492e9 \text{ cm}$, $MJ = 1.8982e30 \text{ g}$, $M_{\text{sun}} = 1.98842e33 \text{ g}$ (IAU 2015 nominal), one definition each, the Python tools importing `RJ_CM`, `MJ_G`, `MSUN_G` from `exhale_io`. Tests `atomic_mass_and_radius_constants`, `jupiter_radius_python_tools.py`, suite `species_masses` GREEN. Measured as two sub-steps: the helium mass alone moves `wasp_full` by $2.3e-3$ in T, $1.1e-2$ in `x(H+)`, `Mdot` +0.095 percent, and `mol_base_handoff` by $1.2e-2$ (tail), `Mdot` -0.004 percent; the Jupiter radius then rescales every physical length (R_0 , the domain top `r_max` in R_p units, the gravity) and moves `wasp_full` by $7.3e-2$ in T, 0.16 in `n_e`, 0.34 in `x(H+)`, raw `Mdot` +11.0 percent (printed 13.24 $\rightarrow$ 13.29), and `mol_base_handoff` raw `Mdot` +15.8 percent (printed 10.52 $\rightarrow$ 10.58). MASS2 alone (the remaining copies): `mol_diffusion` $2.8e-5$, `mol_carrier` $6.5e-8$, `wasp_full` byte-identical.
- **2c-THR, one definition of each threshold** (`parameters.f90`, `cross_sec.f90` and the files that restated a threshold): `e_th_HI` = 13.598434599, `e_th_H2` = 15.425927, `e_th_HeI` = 24.587389, `e_th_HeII` = 54.417765, `e_th_HeTR` = 4.767775 eV (NIST ASD), every cross section turning on exactly there; `sigma(E, Z)` takes the ion's measured potential as an optional argument so He II is not restated as 13.6 Z^2 . `ionization_threshold_turn_on` and `threshold_literal_uniqueness.py` GREEN; the `P_HeI` residual of

- QUAD closes (0.99878 -> 0.99995). wasp_full 7e-4 at most, mass flux +0.044 percent; mol_base_handoff 1.3e-3 (tail); printed Mdot unchanged.
- **2c-N2FLOOR, H(n=2) is an absorber of the grid-floor rule** (sed_read.f90 photon_grid_floor_eV, J_inc.f90, set_energy_vectors.f90, excited_hydrogen.f90): with the excited-hydrogen coupling armed the grid floor is $e_{th_HI}/4 = 3647$ Å, a loaded SED is read down to it and the decision-17 stop is about the file, not the type (WASP-52b with the eps Eri table to 3700 Å now runs). Tests sed_coverage_stop.sh and photon_grid_n2_floor GREEN: the 20 bins added below the lowest bulk-gas absorber change no rate the grid computes, exactly. wasp_full 2e-4 at most (the edge written as $e_{th_HI}/4$ instead of 3.40 eV), Mdot 0.0000 dex; wall clock +2.3 percent for the 20 bins.
 - **2c-CONST, the particle masses and the astronomical constants in one place** (parameters.f90, species_table.f90, lower_column.f90, Cool_coeff.f90, binary_element_diffusion.f90, diffusive_photochemistry.f90, lyman_werner.f90, oxygen_rates.f90): $m_{He_atom} = 6.6464790722e-24$ g (4.002603254 u x CODATA 2018 u; the brief's 6.6464731e-24 was 9.2e-7 low and is not the helium-4 atom), amu, m_e, m_p in parameters.f90, m_He_over_m_H and amu_over_m_H derived, bsp_mass built from them, $melem_A = melem_A_u * amu/mu$. This closed a UNIT ERROR: the standard atomic weights in u had been added to mass sums carried in hydrogen atoms, every metal nucleus 0.78 percent too heavy in rho, the mean molecular weight, the base composition, the census closure and the metal element diffusion. The advisor closed the remaining local copies (Cool_coeff line-center opacity now takes the atomic weight in u and uses the global amu; $amu_C/N/O = melem_A_u(iel_*)$; the local m_amu_g and kb_erg/m_h2 copies of the diffusion, Lyman-Werner and oxygen modules replaced by the globals). wasp_full 1.0e-3 at most (base v), mass flux +3.2e-5; mol_base_handoff 2.8e-5; printed Mdot unchanged.
 - **2c-METALEDGE, every active threshold a bin edge** (set_energy_vectors.f90): each of the six bands keeps its bin budget and is cut at every ionization threshold strictly inside it (H2 when the molecular chemistry is on, every photo-ionizable stage of an element with nonzero abundance), the budget shared over the sub-intervals in proportion to their logarithmic width with at least one bin each; NL is derived from the partition and equals the previous count in every configuration (240 wasp_full, 220 mol_metals, 220 mol_base_handoff). A band with no interior threshold keeps exactly the partition it had. photon_grid_threshold_edges GREEN: seventeen thresholds at relative distance 0 from an edge (eighteen with H2), P_m of C I 1.045 -> 1.000, Ca I 0.933 -> 0.997, Na I 1.030 -> 0.998, Fe I 0.965 -> 0.994 of a 400001-point reference, total H2 photoabsorption 0.985 -> 1.000. wasp_full: T 9.6e-3, n_e 5.2e-3, x(C I) 0.14 at the base, mass flux -0.17 percent, log10 Mdot 13.2860 -> 13.2852. mol_metals: bulk 2e-3, n_e 4.5e-2 in the neutral layer at 1.1 Rp (ionization degree 1e-6), Mdot unchanged. mol_base_handoff: 1e-4 class, by the H2 edge alone. Open (decision needed, below): Mg I, Ca I, Na I, Fe I still miss 1e-3 by 0.25 to 0.58 percent, the rectangle-rule error of the 20-bin sub-Lyman band.
 - **Advisor edits after the worker:** the H2 channel thresholds $e_{th_H2_di}$ (18.076 eV) and $e_{th_H2_dd}$ (51.4 eV) join the edge set when the molecular chemistry is on, because the channel shares of one cross section switch on there (the 32-41.5 eV neutral-dissociation window is continuous and is not cut; the comment at the code site says so); the setup report prints the grid floor as the lower edge of the first bin (photon_grid_floor_eV, or the table floor for a loaded SED) instead of the first bin center, which the new partition had moved to 4.95 eV against a floor

of 4.768 eV. The documents that restated the changed quantities were brought in line: docs/input_schema.md (profile level, R_J), docs/a2_oxygen_option_design.md (molecular masses in m_H units), rev 3 and the execution document (landed marks).

Decision handed to the user: the sub-Lyman band budget

The band [4.768, 13.598] eV has 20 bins (edge ratio 1.054), 4.4 times coarser in $\ln E$ than the pure-H I band (50 bins, 1.012), and it is where every low-IP metal is photoionized; Fe I's cross section falls by a factor 2.5 across the single bin above its threshold. Measured ladder (code/exact):

num_TR	20	30	40	50	80
Mg I	0.99753	0.99871	0.99915	0.99955	0.99982
Ca I	0.99737	0.99877	0.99937	0.99961	0.99984
Na I	0.99752	0.99912	0.99958	0.99967	0.99987
Fe I	0.99417	0.99685	0.99805	0.99883	0.99953
Nl (wasp_full)	240	250	260	270	300

Decided by the user 2026-09-05: num_TR = 89, the budget that gives the band the bin ratio of the pure-H I band (0.0118 in $\ln E$). Measured with the same driver: Mg I 0.999853, Ca I 0.999870, Na I 0.999902, Fe I 0.999615, so the four rates are asserted at the 1e-3 of every other ion. Nl becomes 309 for wasp_full (was 240) and 289 for the hot-Uranus cases (was 220); the H(n=2) band keeps its 20 bins. Every He 2³S and metals-on case moves again by this, and the gate below was re-run on the 89-bin binary (md5 b0a4ada9f84ee8410c84fda6780ba097) before the baseline re-snapshot. The user also decided the LHS1140b/examples/photochem_profile_base/input.inp density key: deleted (the profile fixes the level).

Gate (advisor)

make test on the single rebuild: every assertion of this batch GREEN (threshold_edges, n2_floor, balmer_continuum_field, atomic_mass_and_radius_constants, species_masses, ionization_threshold_turn_on, threshold_literal_uniqueness, voronov_carbon_ionization, grid_width_identity, photon_grid_quadrature, restart_grid_guard, base_level_single_statement, mass_flux_spread_functional, momentum_row_from_fluxes_only, sed_coverage_stop); diffusion_tests had 13 of 35 assertions fail because its own references carried the helium mass as 4 m_H (7.2e-4 = 1 - 3.9715/4), and now derives every helium mass from m_He_over_m_H (35 of 35, no tolerance changed, the T9 ambipolar targets written as functions of the mass ratio). RED, as intended, only on the Phase 3 targets: output_state_consistency (1.2e-5), h3p_cooling_limits, h2_rovibrational_identity, h2_double_photoevent_energy_sum, helium_collisional_ionization_by_level.

Regression, sixteen cases against the post-2b scratch baseline, single-threaded: the thirteen cases that ran all moved beyond the 1e-3 tolerance, as every item of this batch predicts; the three hp_* restart cases were refused by the new grid guard (their IC pairs sat on the old grid: domain top 4.7253 -> 4.6215 R_p from R_J, base spacing from dr_j)

and were regenerated with `make_hp_initial_conditions.py` from the fresh `mol_base_handoff/output` (nuclei conserved row by row; H II as written / 0 / 1e-12 of the nuclei) and re-run. Movement of the thirteen (the cell-wise maxima compare the same cell index, whose radius moved by up to 2.2 percent with the domain, so they overstate the change at fixed radius; the `Mdot` column is the run's printed value):

case	log10 Mdot before -> after	mass flux at cell N-20	max rel T	max rel v	max rel rho	domain top [R_p]
wasp_full	13.24 -> 13.29	+6.2 %	6.7e-02	7.6e-01	4.1e-01	1.6014 -> 1.5666
wasp_he23off	13.24 -> 13.29	+6.2 %	6.7e-02	7.7e-01	4.1e-01	1.6014 -> 1.5666
wasp_full_newton	13.21 -> 13.26	+6.7 %	1.3e-01	1.9e-01	3.4e-01	1.6014 -> 1.5666
mol_base_handoff	10.52 -> 10.57	+8.3 %	2.5e-01	1.1e-01	4.5e-01	4.7253 -> 4.6215
mol_metals	10.52 -> 10.58	+8.3 %	2.4e-01	1.1e-01	4.4e-01	4.7253 -> 4.6215
mol_lyman_werner	10.50 -> 10.55	+8.2 %	1.0e-01	1.2e-01	2.5e-01	4.7253 -> 4.6215
mol_diffusion	10.52 -> 10.57	+8.3 %	2.5e-01	1.1e-01	4.5e-01	4.7253 -> 4.6215
mol_ir_bands	10.52 -> 10.58	+8.3 %	2.4e-01	1.1e-01	4.4e-01	4.7253 -> 4.6215
mol_sec_ion	10.47 -> 10.52	+8.2 %	2.1e-01	3.7e+02 (sign change of a stagnant base cell)	3.8e-01	4.7253 -> 4.6215
mol_carrier	10.52 -> 10.58	+8.4 %	2.5e-01	1.1e-01	4.6e-01	4.7253 -> 4.6215
lower_profile	8.44 -> 7.95	-68.9 %	2.0e+00	1.5e+03 (sign change)	9.7e-01	4.1487 -> 4.0577
hydrostatic_column	8.25 -> 8.27	-0.1 %	5.7e-02	4.3e-01	4.9e-01	2.9600 -> 2.9595
oxygen_chemistry	9.61 -> 9.60	-6.9 %	7.3e-01	3.1e+01 (sign change)	8.9e-01	4.1487 -> 4.0577

The +0.05 dex of every WASP-121b and hot-Uranus case is the Jupiter radius (item MASS, +11 and +16 percent in the raw flux at fixed cell, partly compensated by the shrunken domain at cell N-20); `lower_profile` is item BASE (the run now starts at the profile's 1 microbar level); the `oxygen_chemistry` and `hydrostatic_column` moves are the grid and the threshold edges. The baseline was re-snapshotted after the `hp_*` re-run (the post-2b copy kept as

baseline_post2b_20260905/).

Golden refresh: prepared, awaiting the user's command

golden/ was archived to backup/regression/golden_pre170_20260905/ with a README (provenance from the headers, unknowns marked), roundtrip was re-run on the post-2c binary, and DEFAULT_CASES of run_check.sh now lists the five Phase 0 cases as well. The refresh itself, run_check.sh golden over the seventeen cases followed by make check and one run_fcheck.sh, is the user's command to give; until it is given the reference of record is the pre-2c state and make check reports every case as moved.

Gate on the 89-bin band (2026-09-06)

Fourteen cases re-run against the post-2c (20-bin) baseline with md5 b0a4ada9f84ee8410c84fda6780ba097: hydrostatic_column byte-identical (no sub-Lyman absorber), every other case within 1.3e-3 in any cell (mol_metals rho 1.3e-3, wasp_full v 1.3e-3 at the base) and printed Mdot unchanged, the size the 20 -> 89 ladder predicts; the hp_* IC pairs were regenerated on the 89-bin mol_base_handoff output and the three cases run (Mdot 10.56).

oxygen_chemistry stopped on the 89-bin grid, diagnosed. At step 13, cell 7, the carrier Newton reached its iteration cap (30) with a relative OH row residual of 0.63 and the status stop of item 2b-A3a ended the run. Diagnosis (worker, measured on frozen snapshots): the Newton is correct (a root exists; the same solver with a larger budget reaches it in 141 iterations, the last three 5.4e-2, 3.1e-9, 3.9e-12) and its residual scale and the element limiter are not involved. The stiffness comes from upstream: at step 12 the ionization sweep accepted a non-root at cell 7 under the relaxation amnesty (info 4, normalized residual 5.2e4), which put $n_e = 3.5e10$ and $n(H^+) = 2.6e8$ into the frozen background where the previous step had 1.1e3 and 0, so the $H^+ + H_2$ channel asked for two thirds of the cell's H_2 in one 2.85 s step. On the 20-bin grid cell 7 never took that path. The "7.8e-2 carrier row" of the 20-bin summary is the STEADY residual of a relaxation snapshot (dt -> infinity), not the Newton's; the backward-Euler Newton converged to below 6e-12 on all 999 steps there. Two things landed: the iteration budget is 200 (the comment at the code site states the measurement), and the attribution test of the status stop, which asked "did an element constraint act" by carrier over the whole grid, now asks it by cell (the CO ceiling, acting in 160 cells of this case, had made any CO failure anywhere excusable, and had left OH rows it moved by 49 percent through the oxygen closure reported as the solver's); new suite src/tests/carrier_constraint_attribution/, 7 assertions, RED before. Open for the user: decision 4 (reject and retry the step on a shorter dt, the structure that also covers a step without a root) and the ionization sweep's relaxation amnesty, which handed the carrier system a non-root (acceptance policy, ionization_equilibrium.f90 class 4).

Golden refresh executed (2026-09-06, on the user's instruction)

run_check.sh golden over sixteen cases: the eleven of the previous DEFAULT_CASES, hydrostatic_column, the three hp_* restart cases and roundtrip, all from the 89-bin binary b0a4ada9f84ee8410c84fda6780ba097 (conda gfortran 16.2, OpenBLAS); the previous golden is golden_pre170_20260905/. golden/ now carries the *_adv.txt files as well. oxygen_chemistry has no golden entry yet (it stops, above); it stays in DEFAULT_CASES, so make check reports it as the one failing case until the carrier solver question is settled. make check and one run_fcheck.sh follow in this

section.

After the refresh, same day (2026-09-06): the loaded-SED grid, the carrier budget, hygiene, the SED files

- **2c-SEDGRID, the photon grid of a loaded SED** (sed_read.f90, J_inc.f90, parameters.f90, set_energy_vectors.f90; item C5 of the D0 draft): the Load path kept the table rows as bin points with central-difference widths, so no threshold was a bin edge there, on the path every planet calculation of record uses (decision 16). Now the bins are built from edges (geometric means of neighbouring rows, the table span as the two end edges), every active ionization threshold inside the span is inserted as an edge with the split bin's two halves carrying the row's flux (exact for the integrated flux), e_v the geometric center, de_v the exact width; read_sed receives the threshold list from ionization_thresholds_active, one definition. The table rows themselves are kept in e_sed_node/F_sed_node (undiluted), which stellar_flux_eV, loaded_table_floor_eV and spectrum_covers_eV now read, so the loaded field no longer depends on the binning. Test sed_edges (src/tests/spectrum_type/, 15 assertions) on the eps Eri table of WASP-52 b: thresholds at distance 0, flux identity to 2e-16, rate deviations from the exact table integral P_HI 1.8e-3 -> 4.6e-7, P_HeI 1.1e-3 -> 1.1e-6, P_HeII 8.4e-3 -> 8.5e-6, worst metal 3.2e-3 -> 5.5e-4 (Mg II, the table's own 1 A resolution). Movement: benchmarks/wasp52 2000-step marching stop, mass flux at cell N-20 2.3e-6, log10 Mdot 9.68 both, profiles up to 2e-2 at a relaxation point; wasp_full (power law) byte-identical on seven output files. No regression case loads a table, so no golden moves. Noticed, not fixed: e_mid is not an edge of the loaded grid (the LX/LEUV split it feeds moved 0.02 percent); stop_if_balmer_band_unstated prints bin centers as the band.
- **Carrier Newton budget** (diffusive_photochemistry.f90): newton_maxit 30 -> 200 after the diagnosis above; oxygen_chemistry runs its 1000 steps on the 89-bin grid (log10 Mdot 9.60) and is added to golden/ below. The constraint attribution of the status stop, by cell, and its test suite carrier_constraint_attribution (7 assertions) landed with it.
- **Hygiene, D0 items C16 and C4:** the dummy alpha, threaded through reconstruction_continuation_rhs, RK_rhs and Num_flux and never read, is removed from the three signatures and ten call sites (hydrostatic_column byte-identical against a same-compiler control); output_state_consistency.sh and its README describe the rate lag (measured RED 1.19e-5 with the header now reading sec_ion=F) instead of the removed post-loop block. docs/audit_20260905/roe_flux_vacuum_probe.f90 is an archived probe that no longer compiles against speed_estimate_ROE; left as the frozen record it is.
- **Document sync** (docs/EXHALE_user_manual.tex + PDF rebuilt, docs/p44_base_sawtooth.md, write_output.f90 and util_ion_eq.f90 comments, write_setup_report.f90): the manual carries the Planck type, the decision-17 stop, the six-band edge grid, the IAU constants and NIST thresholds, the signed du, the profile base level and the restart grid guard; NLTR is now counted on the Load path (the "no grid point below 13.6 eV" sentence had been printed for every loaded SED); src/utis/glob.py, which shadowed the standard library module, is src/utis/EXHALE_interface_state.py with its two importers updated. Reported, not changed: the GUI offers no Planck type; global_parameters exports an integer named count that shadows the intrinsic.

- **Stellar SED files** (inputdata/sed/, decision 16, source rule of 2026-09-06: MUSCLES data normalized as the planet's reference paper did): wasp121b_solar_huang2023.txt (recommended; WHI 2008 solar shape below 1700 Å, $F(1-912 \text{ Å}) = 1.6e6$ and $L_{\text{ya}} = 1.0e5$ as Huang et al. 2023 section 2.2 prescribe, 6459 K photosphere above; the $1.6e6$ confirmed as the $\lambda < 912 \text{ Å}$ band by Salz et al. 2019 section 7), wasp121b_wasp17_huang2023.txt (MUSCLES WASP-17 alternative; its 1700-2583 Å photosphere is 1.8x fainter than the blackbody, the size of the He 2^3S photoionization bias of the solar candidate), hd189733b_epseri_salz2016.txt (eps Eri, MUSCLES v22, normalized to Salz et al. 2016 Table 3; the proxy pairing is the one Salz et al. 2018 section 4.1.1 make), and hd189733b_bourrier2020.txt (the released MOVES III tables, mean of Visits B-E, measured to be the flux at 1 au, scaled to 0.031 AU, eps Eri photosphere above 1600 Å identical to the Salz file's; 30 percent fainter in the Lyman-Werner band, 1.4x brighter in the EUV). Each file was read by EXHALE.x for two steps on the final binary without error. User decision: HD 189733 b runs at $a = 0.031 \text{ AU}$ (Salz 2016 Table 2, the modelling paper of the comparison); the planet folder, benchmarks/hd189, examples 01-11 and 13, and params_table.txt carry it (the regression inputs jfnk_hd189*, tpm_hd189 and the archived copies keep 0.0330). No input.inp has been switched to a loaded SED yet (2d).

Phase 1 closing gate (2026-09-06)

make test on the final binary b2c15e53c888d09fa90ffdad80443c3a: ten suites, eight PASS; RED only on the Phase 3 targets (output_state_consistency 1.19e-5, h3p_cooling_limits, h2_rovibrational_identity, h2_double_photoevent_energy_sum, helium_collisional_ionization_by_level). make check: sixteen DEFAULT_CASES against the refreshed golden/, all four files byte-identical in every case (REGRESSION PASS); the sixteen goldens taken from the earlier 89-bin binary reproduce on the final one, which is the measured statement that the loaded-SED grid, the carrier budget, the alpha removal and the report edits leave the analytic-spectrum matrix untouched. run_fcheck.sh result recorded below. run_fcheck.sh: the -fcheck=bounds, do, mem build ran 1500 bounded HD 209458 b steps with no runtime trap (fcheck regression: CLEAN, $\log_{10} \dot{M}$ 9.59), and the production build was restored to b2c15e53c888d09fa90ffdad80443c3a. **Phase 1 is closed.** Open for the user before Phase 3: decision 4 (carrier step rejection/retry), the relaxation amnesty of the ionization sweep that handed the carrier system a non-root, the D0 draft (docs/d0_governing_system_20260906.md, 33 consistency items, 15 questions) for acceptance, the switch of the planet inputs to their loaded SEDs (2d), the WASP-121 b SED choice (solar shape, WASP-17 shape, or the two joined at 1700 Å), and the sub-Lyman band budget of the Load grid (the table's own resolution is kept; Mg II stays at $5.5e-4$).

7. PLAN_20260906_rev2, Step A: acceptance contracts and run modes (2026-09-06)

Plan of record for this stage: docs/PLAN_20260906_rev2.md (rev 0 and rev 1 and their two reviews are kept beside it; the execution document of 2026-09-05 points at rev 2). The decisions of To_be_determined_by_user_20260906.md were judged against To_be_determined_by_user_recommend_20260906.md and adopted as rev 2 section 2 states. Documents written (advisor): docs/a0_run_mode_contract_20260906.md (the three run states; Time stepping:

Local and PTC are initialization only; physical time advances only by accepted global steps; headers gain mode= and t_phys=; the proposed Run mode: init | phys key awaits the user's confirmation before A0-impl); docs/a2_certification_contract_20260906.md (one evaluator, four contexts: probe, stationary trial, physical step, certification; statuses not_applicable / evaluated / unavailable; one condition per active balance; what replaces what). Documents drafted by workers and verified for structure and wording: docs/b1a_active_equation_inventory_20260906.md (flags, equations per flag, constraints, five validity states with producers, the prerequisite table for eight configurations; finding: only the hydro rows have a state-consistent evaluator; the composition projection, the rho rewrite and thirteen closures with stated domains are recorded nowhere) and docs/b1_target_system_20260906_draft.md (energy ledger with the shared H2 zero at $v=0$, $J=0$; chemistry mass-preserving; Lyman-alpha loss and return from one 2p balance, an escape-probability multiplier on the existing cooling explicitly not the target; the zero-net-current closure derived from Koskinen 2022 B9/B10 and shown to give exactly the ambipolar field binary_element_diffusion already computes; the trace-metal counter-flux; the rollback table of fourteen state-changing operations showing u_{old} is not a checkpoint; CO alternative A blocked on rate literature, alternative B specified and recommended; ten open questions).

- **A1 (carrier half)** (diffusive_photochemistry.f90; suite carrier_returned_state_acceptance, 36 assertions, plus one in carrier_constraint_attribution): the carrier update is accepted on the RETURNED state, every unconstrained row of every cell, $|res|/\max(row_terms, 1e-300) \leq 1e-8$ with the absolute floor inside the scale, finiteness of every row, independent of the termination branch; the relative-drop exit is carrier_solve_stagnated, a diagnostic; the decision is carrier_returned_state_verdict, a side-effect-free function returning verdict and reason for A3 to consume. Defect found and fixed: limit_to_element_budget marked a cell constrained only when the clamp exceeded limit_report = $1e-10$, so smaller clamps left cells presented as Newton-measured. Impact: mol_carrier byte-identical (no changed branch reachable: all 499 transport steps of the first 500 ended at the floor); **oxygen_chemistry refuses at step 4**: the CO thermal ceiling, a clamp without a rate, leaves a neighbouring cell's CO row at $1.0e-7$ of its terms (accepted steps before it: $1.1e-9$, $4.2e-9$), and the code no longer certifies it. That is the plan's position (B1 alternative B: a ceiling-active atmosphere is outside the validated set); the case can run only in A0's initialization mode once A0-impl exists, so it fails make check until then.
- **A1 (class half)** (steady_residual.f90, steady_newton.f90, ionization_equilibrium.f90 comment block; suite acceptance_classes, 12 assertions): n_cells_without_chemical_root(acc_n) = acc_n(4), one definition for the trial handling (a local-closure validity check, no stationary condition) and for the three points that declare a steady state solved; a class-4 cell in the state handed back is written but reported "NOT certified". Impact: wasp_full_newton and mol_base_handoff identical (no class-4 or class-5 acceptance in any ledger). Open: the marching stop cannot pass the count because the loop never receives the sweep ledger; A2 closes it.
- **C1** (in progress) and **A2** (steps 1-3 of the contract, in progress).
- **C1** (inputdata/sed/README.md, build/sed_coverage_and_order_check.py, suite spectrum_type driver loaded_sed_table_semantics, 78 assertions): a table value is a bin average on every shipped file (MUSCLES

const-res, WHI, Gueymard: 1 A or 1 nm bins; Bourrier 0.016 A samples with no stated convention) except the blackbody part of the solar WASP-121 b candidate (point sample); the reader treats a row as a histogram in energy, exact for the band energy of a bin-averaged table, second order in the bin width otherwise (MEASURED order 2.0 to 2.9 through the production consumers). Coverage is now checked per configuration from the active threshold inventory (2600.5 A metastable, 3647 A when both stellar keys arm H(n=2), the lowest active metal), 12 configurations. Findings closed the same day (advisor): sed_read.f90 single-precision 1e-8 at two sites (6e-9 relative shift of every loaded energy); the HD 209458 b WHI file rebuilt to 5000 A (its 3000 A cut stopped the run in read_sed naming H(n=2)); WASP-52b/input.inp SED path from before the 2026-07-14 move; the two solar build scripts' headers (2583 A / 4.80 eV). Remaining: the WASP-17 file begins at 10.0 A because the MUSCLES v24 table does (the grid top becomes 1239.84 eV; recorded).

- **BALMERQ** (excited_hydrogen.f90; suite spectrum_type driver balmer_band_quadrature, 21 assertions; balmer_continuum_field Planck references refined): the Balmer-continuum rate and heating integrals follow the field's own nodes in closed form (power-law segments of a loaded table and of the power-law type; 5-point Gauss on a geometric subdivision for Planck), replacing a fixed 400-point uniform rule that stepped over emission lines. On the eps Eri table of benchmarks/wasp52 the old rule was 2.9 percent low in the rate and 2.7 percent high in the heating (the Lyman-alpha line at 10.20 eV lies in the band); on the power law it carried 9.5e-5 / 6.0e-5 of truncation. Movement: benchmarks/wasp52 gamma_2 +2.99 percent, n_e 5.3e-4 at 1.09 Rp; wasp_full 1e-8 class (sub-0.1 percent, no golden refresh). One worker reported two scratch runs killed by a pkill matching a shared basename: the common brief now says kill by PID after checking /proc/<pid>/cwd, never by name.
- **A2, steps 1-3** (src/modules/time_step/certification.f90 new, steady_residual.f90, steady_newton.f90, EXHALE_main.f90 call sites, utilities.f90 header, diffusive_photochemistry.f90 save/restore pair; suite certification, 18 assertions): one evaluator builds a 19-entry inventory from the configuration flags and returns per entry a status (not_applicable / evaluated / unavailable), the row measure (max over physical cells of |res|/max(scale, floor)), the volume average, the worst cell, finiteness and the scale text; hydro rows and carrier balances are evaluated (carriers inside an isolated workspace that saves and restores every module array carrier_steady_residual writes, asserted after every call), elemental transport, level equations and the seven System_HeH_* closures are unavailable until step 4. The certification context runs at the marching stop (which now receives the sweep ledger, closing the A1 open item), the JFNK final and the PTC declaration; probe and trial decisions read one record, the merit rule unchanged. Exit status (user decision): a run that DECLARED a stationary state and failed certification exits 2 (stop 2); a run ending on a cap makes no claim, writes certified=F cert_reason=no_stationary_claim and exits 0; run_check.sh never tests the status. Five shell gates of grid_and_gates that asserted exit 0 from a full run now accept 2 with the reason stated at the code site (advisor). Impact: numeric content of wasp_full, wasp_full_newton, mol_carrier identical; headers gain certified=. **No configuration certifies today**, by construction (the closures are unavailable), and two measured findings stand on their own: the JFNK info = 0 is set at the loop top and never revised when the returned state is re-measured at its own composition (wasp_full_newton energy row 4.1e-4 against the 1e-5 target, 40x), and the du stop accepts hydro rows of

order one on their own terms (wasp_full at $du = 1.0e-3$: mass $1.3e-2$, momentum 1.8, energy 0.72; at $du = 2e-2$ the same rows read 1.7, 2.0, 0.96, so the spread and the rows measure different things). Not implemented in this item (it would change the solve): the probe rule that a class-4 chemistry makes a Newton direction unusable. Tolerances inherited; the convergence study is owed.

- **B1 accepted (2026-09-06).** The user decided the fourteen open questions of docs/b1_target_system_20260906_draft.md as recommended (metal reservoir zero; electron mass convention; ioniz_eq density intent(in); the residual H I excitation cooling under the Lyman-alpha ledger; trace metals in the diffusion matrix; H+ in the B9 set; CO exclusion adopted with the destruction model held for rate literature; H3+ low-collider policy; one H2 partition function; rollback of diagnostic extrema; identity row placement; the CO ceiling's validity category; O(1D) reservoir; the refused _adv product rule). The accepted specification is docs/b1_target_system_20260906.md (the draft kept and marked superseded). This is the Phase 2 gate of PLAN rev 2; B2 (the residual-controlled temperature solve) started the same day.
- **A3**(diffusive_photochemistry.f90; suite carrier_retry, 32 assertions): the carrier subsystem's checkpoint (carrier_checkpoint_take / restore / matches: carrier fractions, the module arrays of the A2 save/restore enumeration, every constraint and solve diagnostic, the verdict, the cumulative CO record), and the controller: a refused returned state restores the substep's entry state, halves the carrier substep with the background explicitly frozen (listed at the routine), covers the interval with accepted substeps, and on exhaustion (nine refusals, floor $dt/2^8$) restores the ENTRY state and stops; f_sp is written only for a covered interval. Two CO records (accepted, restored on refusal; attempted, never restored) and controller attempt statistics with the A0 ledger-family hook. Retried result equals the explicitly subdivided integration bitwise; step doubling gives first order (ratio 2.001). Impact: mol_carrier (0 retried intervals in 11999), mol_base_handoff, oxygen_chemistry (3 steps) byte-identical. **Finding:** the retry recovered the oxygen_chemistry refusal ($1.04e-7 \rightarrow 7.9e-9$ after ten halvings) because the row scale carries the $1/dt$ time term, so any fixed clamp imbalance passes a fixed relative floor at a short enough substep; that is a defect of the acceptance measure shared by A1 and A2, fixed next (A1scale: the returned-state and stationary measures are judged against the physical terms of the row, transport and reaction, with the absolute floor; the time term stays only in the Newton's own iteration test).
- **B2** (energy_semi_implicit.f90; suite energy_update, 20 assertions): the implicit temperature step of a cell, $R(T) = u(T) - u(T_{old}) - a [\text{heat} - \text{cool}(T)] = 0$ with $u(T)$ from the run's equation of state (caloric H2 included), is solved by a safeguarded Newton-bisection hybrid on a physical bracket ($0.01 T_0$ to $5e4 K$ with H2, $1e7 K$ otherwise) to $|R|/scale \leq 1e-9$, the cooling written back is cool($T_{returned}$), and every non-converged outcome is a named failure (ITER_CAP / NO_BRACKET / FLOOR / NONFINITE) with the cell, the residual and the missing volumetric source; no clamp remains, a floor is a failure, and the interim action is a stop that B3a replaces by rejecting the attempted step. The caller is untouched (optional status argument, energy_update_last_status). RED kept in the suite: the two-iteration update on a falling cooling branch left $|R|/scale = 2.9e-2$ and a 6.9 percent temperature error. Impact: wasp_full transient $5.4e-5$, converged (same stop step 16257, log10 Mdot 13.29) $1.6e-6$; mol_base_handoff $4.9e-5$; hydrostatic_column identical; no floor or failure fired anywhere; cost +9 percent end to end on wasp_full. Follow-ups assigned: the identical clamp in

the conduction stage (`viscous_conduction.f90` 572-589; B2b), the certification's counting of energy-floor hits as an accepted correction (A2 step 4), the stale end-of-run floor report (A0-impl).

- A1scale** (`diffusive_photochemistry.f90`; `carrier_retry` 42 assertions): the returned-state acceptance and the stationary carrier measure are judged against the row's PHYSICAL terms (`|transport| + |reaction| + floor`, the floor in rate units without the $1/dt$ form); the time term $|n - n_{old}|/dt$ stays only in the Newton's own iteration scale (`row_terms`, bitwise unchanged), beside which `row_terms_phys` is accumulated. RED: a residual of $1e-7$ of a row's physical terms read $3.3e-8$ at dt and $5.6e-9$ at $dt/16$ on the old scale (crossing the $1e-8$ floor with the defect unchanged); now $1.0e-7$ at every dt . Impact: `mol_carrier`, `mol_base_handoff` identical; `oxygen_chemistry` is refused at step 2 (CO row $5.5e-8$ beside a clamped cell; 37 retry attempts to the floor, all refused, entry state restored), and the old measure only postponed that refusal to step 88 on today's tree. Finding carried forward (A1scale2, needs its own impact measurement): with a dt -independent acceptance there is a shortest certifiable substep, since the Newton's stopping test is still relative to the full terms and the round-off it leaves grows as $1/dt$ against the physical terms; the same measure in both places closes it. Also: `cterm` in `steady_newton.f90` (534, 645) is written and never read. **Consequence for the matrix:** `oxygen_chemistry` cannot run under the physical acceptance while the rateless CO ceiling is active (B1 decision 7: exclusion); once A0-impl lands, the exhaustion action reads the run mode (initialization: record the refusal in the attempted ledger, write the state flagged, continue; physical: stop), which is the item A3b, and the case runs in init mode as the diagnostic it is.
- B3a and B4 designs** (`docs/b3a_attempted_step_controller_design_20260906.md`, `docs/b4_spatial_operator_design_20260906.md`, read-only workers; advisor decisions in their last sections). B3a: eight of the fourteen step operations have no returned-state verdict today (RK/BC, element diffusion, excited H, `ioniz_eq`, the projection, `Apply_BC`, conduction, Shapiro); checkpoint about 2.1 kB per cell. B4: species mass fluxes are not consistent with the mass flux today (equilibrium species are not transported and rewrite `rho`; carrier advection is a cell-centered product; element diffusion is in advective form; no species enthalpy transport); increments B4-1 to B4-6 with their expected movement. **Advisor correction to the B4 draft:** the formation-energy flux $F_{form} = \sum_s \epsilon_s F_s$ is the RESERVOIR's transport and enters no thermal equation (photoheating already carries $h_{nu} - I$; recombination emits I); the draft had written it as a thermal source, which would heat a cell by the ionization energy leaving it. B4-2 becomes the diagnostic reservoir balance that the ledger identity T1.5 tests, plus the face-composition enthalpy (T-B4.5). The B4-2 pre-measurement (`src/utls/formation_energy_flux_diagnostic.py`, section 7 of the design) stands as the size of that reservoir transport: on `wasp_full_newton` 28 percent of the absorbed heating leaves the column as advected ionization energy (median 14 percent of the local heating, 52 percent at the ionization front at 1.23 Rp); the present energy face flux is $v(E + p)$ with the caloric H₂ internal energy already inside E , addressed by the owning cell (T-B4.5). Findings: `lower_profile`'s golden mass flux is not flat ($1.1e-2$), so continuity-based diagnostics on it discriminate nothing until it is Newton-finished; `benchmarks/hd209/output` integrates to cooling 24 percent above heating on its saved state.
- B2b** (`viscous_conduction.f90`; `energy_update` suite second driver `conduction_floor_tests`, 24 asser-

tions): the Crank-Nicolson conduction stage no longer clamps; a solved temperature at or below the shared EOS floor (imported from `energy_semi_implicit`, one declaration) is a FAILURE with status, cell, the solved temperature and the energy the floor would have injected; the tridiagonal solve refuses a zero or non-finite pivot; no partial state is written. The operator is in flux form and conserves thermal energy exactly on a closed column (MEASURED 0.0); as boundary-conditioned in a run it exchanges heat through the base Dirichlet anchor by design ($3.4\text{e-}15$ up to that flux). Impact: byte-identical on `wasp_full`, `mol_base_handoff`, `hydrostatic_column` and on two conduction-on variants in which the stage runs; zero floor activations anywhere. B1 section 7.4 rows 9 and 11 updated (no clamp state left). Noticed: `benchmarks/koskinen2022_model_a/*` cannot restart (stored IC on the pre-2c grid, $4.5\text{e-}4$ off; the same class as the `hp_*` regeneration).

- **B4-5 measurement** (`src/tests/grid_and_gates/hydrostatic_residual.f90`, design section 8): on the analytic isothermal column, through the production `Reconstruct`, `Num_flux`, `RK_rhs` and `Source`, the pressure-gravity pair is at design order in the interior ($\text{dr}^{\wedge}1.96$ PLM, $\text{dr}^{\wedge}1.98$ WENO3 at $r = 2$; base cell $1\text{e-}4$ of its weight, first order in the base spacing), and the $O(1)$ hydrostatic imbalance of the mechanical column sits at the OUTER free-outflow cell: 0.75 of its own weight at every N under PLM (the zero-gradient ghost zeroes the MC forward difference, so cell N carries no pressure gradient), $6.4\text{e-}2$ falling at order 1.4 under WENO3. A well-balanced pair (B4-5b) would not remove the drain; the outer boundary fill is the item (B4-4a, started). Two latent defects found: `define_grid`'s `tol = 1.0` initializer implied SAVE, so a second call in one process skipped the stretch-factor Newton (production calls it once; fixed by the advisor: `tol` reset before the loop); the case README's single Grid cells ladder does not refine the base cell (README updated).
- **A2 step 4** (`certification.f90`, `binary_element_diffusion.f90`, `excited_hydrogen.f90`, `ionization_equilibrium.f90`; certification suite +7 driver and +15 shell assertions): the five entries that were unavailable are measured on the state without solving or writing anything (He/H partition and trace-element transport balances, He 2^3S and H($n=2$) level balances, the eliminated-species closure as the sweep's own normalized reaction residual at the given composition), each restoring every module array it installs, asserted on every call. Output content identical on seven cases. **What refuses certification now:** the three hydrodynamic rows on every marching case; the energy row alone on `wasp_full_newton` ($4.1\text{e-}4$ at cell 376, the JFNK `info = 0` finding); the H2 carrier row on `mol_carrier`; and, new, the elemental transport on the diffusive runs (`mol_diffusion` partition $6.4\text{e-}5$ at cell 203; `lower_profile` partition $4.8\text{e-}3$, trace metals $1.2\text{e-}2$, the first measurement of what the Picard alternation reaches). The closures and level balances are within tolerance everywhere ($1\text{e-}16$ to $1\text{e-}12$ of their turnover), so the equations that lacked an evaluator are the solved ones. Both floors are reported as attempt diagnostics (0 everywhere). `cert_tol_element` is inherited, not measured, and is now decisive for the diffusive runs: the convergence study of contract section 4 is owed.
- **A1scale2** (`diffusive_photochemistry.f90`; `carrier_retry` 51 assertions): the carrier Newton stops when its residual is within its own tolerance of the full terms AND within the acceptance tolerance of the physical terms, or when the arithmetic forbids the second (residual at 64 epsilon of the full terms), a named outcome; the interim reason `unreachable_at_substep` for a refusal whose rows are all at round-off was superseded the same day by A1scale3 (below); the stop message reports the halvings made. Option (a) (stopping on the physical

scale alone) was built, measured (one extra iteration every step, no rescue of the short substep) and declined. Correction of the A1scale report: the Newton reached one epsilon in two iterations on that column; the 7.7e-5 was the ninth attempt's state. Impact: mol_carrier, mol_base_handoff identical; oxygen_chemistry still refused after step 2 (now the unreachable reason, OH at cell 90, 15 attempts). Decision taken from its finding (A1scale3, started): a row at the arithmetic round-off of its own terms is not evidence of a defect and is accepted as "round-off limited" with a flag and a count, so a trace carrier at the element floor does not refuse an interval; the physical-terms acceptance stays for every other row.

- **A0-impl** (input_read.f90, parameters.f90, EXHALE_main.f90, utilities.f90, load_IC.f90, write_setup_report.f90, ionization_equilibrium.f90 ledgers; suite run_mode, 22 assertions): the run mode is in. Run mode: init | phys, default init in every configuration (advisor decision replacing the draft rule: the ordinary use is a stationary solution reached by marching and certified, which is initialization in the contract's terms; a physical integration is asked for); phys with Time stepping: Local refused at startup; in phys the handoff state is tested at step 0, the clock advances only after a complete accepted step (measured equal to the sum of accepted dt to the last bit, with and without forced rejections), two ledger families (ledger_family in global_parameters), headers gain mode= and, in phys, t_phys=; a restart of an init snapshot in phys mode is the handoff at t = 0 (contract section 3 clarified), and only a mode=phys header without a finite t_phys is refused; init never exits nonzero for certification. Impact: no input changed; wasp_full, hp_front, hydrostatic_column, mol_base_handoff bitwise identical. Both floor reports of EXHALE_main.f90 now state the B2/B2b behavior.
- **COLit** (docs/co_destruction_rates_literature_20260906.md; publisher PDFs of Westley 1980 NSRDS-NBS 67, Tsai 2017, Wakelam 2012 added to references/): B1's CO choice is settled on physics: every neutral CO destruction channel carries a barrier of 6.7 eV or more, so the one-sided destruction model's tau_dest << tau_res fails at the ceiling's onset at every density (factor 9 to 1.5e8) and the exclusion (decision 7) is the physics. Findings: molecular_reaction_heat.f90 (H2, H+, He+, H2+, H3+, HeH+) and the oxygen Shomate table of oxygen_rates.f90 carry different reference zeros (T1.9 one-table item, assigned as B3b-ENTH); CO2 is not in the species table (removes the one evaluated CO channel below 2500 K and an OH sink); the workspace VULCAN network's CO reactions all produce species EXHALE does not carry, VULCAN's stated validity is 500 to 2500 K, KIDA's 10 to 300 K. Requests for the user: the A&A papers Visser et al. 2009 (503, 323; the CO shielding functions), McElroy et al. 2013 (550, A36
 - RATE12 file), Venot et al. 2012 (546, A43) and 2020 (634, A78), plus Moses et al. 2011 (ApJ 737, 15) and Tsai et al. 2021 (ApJ 923, 264), which this host cannot download.
- **A1scale3** (diffusive_photochemistry.f90; carrier_retry 61 assertions): a row at the arithmetic round-off of its own full terms (64 epsilon) is counted "round-off limited", flagged per row, and left out of the acceptance decision; every other row is judged on the physical terms at 1e-8 as before; the controller halves only for genuine refusals; the stop prints both scales for every refused row. Impact: mol_carrier, mol_base_handoff bitwise identical (0 round-off limited rows); oxygen_chemistry still refused at the same step and row (CO at cell 281,

- |res|/phys = 3.2e-8, |res|/full = 2.2e-12, four orders above the round-off bound): the element limiter's imbalance one cell beyond a clamped cell, correctly refused. The count's accessor exists; its line in the run summary and the certification carrier entry are B3a's (owner of both files).
- **B3b-H2Q** (caloric_eos.f90, mol_rates.f90; h2_rovibrational_identity 19 assertions): one H2 partition function over the observed Roueff et al. 2019 ladder (302 levels, zero at v=0,J=0) gives u_rv(T) as its first moment and the chemistry's K_eq as its free energy; the Huber-Herzberg model constants are deleted. The RED (0.6 to 8 percent in u_rv) was the model ladder's level placement (no centrifugal distortion: J^4 signature, 100 percent of the gap to 2000 K), not a defect of the sum; independent check: K_eq against NIST-JANAF (Shomate) now 0.9997 to 1.0008 over 600 to 5000 K (was 1.010 to 1.101). Impact: wasp_full identical; mol_base_handoff T 1.3e-5, n(H2) 3.1e-4; mol_diffusion T 3.2e-5; mol_metals at 400 steps n(H2) 5.6e-3 at 1.17 Rp; Mdot unchanged. Findings: a third copy of the ladder sum in molecular_infrared_cooling.f90 212-215 (to fold into h2_partition_function); element_census_tests does not compile against the live tree (no_free argument, another worker's signature change; to fix at the next single build); the load average reached 38 with the parallel long runs, so future briefs pace the 12000-step impact runs.
 - **A3b** (diffusive_photochemistry.f90; carrier_retry 89 assertions): the carrier controller's exhaustion reads the run mode and returns a status (carrier_interval_covered / exhausted) the caller reads. phys: unchanged (restore, print, stop until B3a takes the status). init: the entry state is restored, the interval is recorded in the attempts ledger (count, first/last step, worst row and both scales), the carrier history is marked not certifiable (carrier_history_certifiable()), and the march continues from the entry carriers. The accepted CO record is indexed by A0-impl's ledger_family. Impact: mol_carrier, mol_base_handoff bitwise identical; oxygen_chemistry (init, its default) now runs its 1000 steps, exit 0, with 998 exhausted intervals (first at step 2, worst CO at cell 313), log10 Mdot 9.61 against the 20-bin 9.60, and a final certification refusing on four carrier rows at 1.0: its carriers are frozen at the step-2 values for the whole run, which is exactly what the mark and the ledger say. The refused row is the real imbalance beside a clamped cell (|res|/full 4e-12 against a 1.4e-14 round-off bound), open physics of the excluded CO model, not a controller defect. Its golden is re-taken at the end of this series with the mark noted in the case README.
 - **B5pre** (steady_newton.f90; suite_steady_completion_flag): the JFNK and PTC completion flag is decided AFTER the certification of the returned state, from the same record the certification prints; a solve that met the gate at the loop top and not at the returned state ends info = 2 naming the row. Measured on wasp_full_newton: loop-top ||R|| = 5.5e-6, returned state 4.2e-4; the existing own-composition consistency pass opens the gap (0, 1, 2, 3, 8 sweeps: 5.5e-6, 1.9e-4, 3.2e-4, 3.9e-4, 4.2e-4), so the hydro-only Newton converges to a state that is not a root of the coupled hydro + composition system: the loop-top gate measures a lagged residual. Consequence today: EXHALE_main treats info = 2 as a solver failure and falls back to marching, which ends on a du plateau with a state that certifies worse than the JFNK state; assigned to B3a (info = 2 = a failed stationarity claim, written uncertified, exit 2, no fallback) and to B5a (steady_newton.f90: the residual evaluated on the self-consistent composition of the current iterate, so gate, merit and returned state measure one thing). Also found: the JFNK stop message prints an unset resid_th (B3a); certification_row_measure and relnorm_over_cells

are one expression written twice.

- B3b-PE** (h2_photo_channels.f90; photoevent_energy_ledger 105 assertions): the 19.725 eV missing from the H2 double photoionization event is the kinetic energy release of the two protons: 51.4 eV is a vertical threshold (Yan et al. 1998 section 4), the asymptotic product energy is $D_0(\text{H}_2) + 2 I(\text{H}) = 31.675 \text{ eV}$, and energy conservation at the threshold fixes the fragment kinetic energy with no free parameter (Franck-Condon check $e^2/R_e = 19.42 \text{ eV}$, 1.5 percent). Second defect found: the 33 to 41 eV neutral window dissociates into $\text{H}(2p) + \text{H}(2s)$ (Chung et al. 1993 p. 887), so 20.398 eV per event is internal excitation leaving as prompt Lyman-alpha and two-photon continuum, which the integrand charged to heat. h2_channel_energy_recipients (reservoir, electron, fragment kinetic, radiated; sums to $h \nu$) with the per-channel table and sources. Channels M and S already closed; every atomic absorber and all 17 metal ions correct (wasp_full byte-identical). Impact: mol_sec_ion T 4.2e-4, heat_H2 4.5e-2; mol_base_handoff T 2.1e-5; Mdot unchanged. The four-line wiring of the two new terms into the heating integrand of util_ion_eq.f90 is handed to B4-6 (owner of that file). Open: suprathermal H2 dissociation credits heat but destroys no H2; the double channel has no absorber row of its own; channel S has no source for a fragment kinetic split.
- B3b-H3** (h3p_cooling.f90; h3p_cooling_limits 17 assertions): the Miller et al. 2013 fits as the paper defines them: Table 5 is the LTE emission of one molecule per steradian ($\Lambda = n(\text{H}3+) \int E(T) d\Omega$), Table 6's $s(T, n_{\text{H}2})$ is scaled LTE with H2 the only collider (Oka and Epp coefficient; the paper's upper-limit caveat at the code site); below the table's $1e6 \text{ cm}^{-3}$ the non-LTE factor follows the exact collisional limit (linear in $n_{\text{H}2}$, zero at zero), replacing the clamp at the lowest column; above $1e14$ the table edge is held; the four Table 5 segments do not join (measured with the paper's coefficients: +43 percent at 300 K, -0.11 percent at 800 K, -2.4 percent at 1800 K) and the code joins them by a stated linear blend in $\log_e E$ over the 5 percent below each join (Table 6's 300 K value supports the upper polynomial); B1 T6.3 reworded to say so. Published values reproduced per segment to the paper's precision. Impact: the H3+ cooling peak unmoved; T moves $1e-5$ to $2.7e-4$ in the three molecular cases; the clamp removal changes cells beyond 1.22 Rp where $n_{\text{H}2} < 1e6$ by more than 50 percent at $1e-10$ of the peak (emission from gas with no collision partners, gone); wasp_full identical. Four domain counters exposed (B6 wiring is the certification owner's).
- COLit, completed with the user's PDFs** (Visser 2009, McElroy 2013, Venot 2012, 2020; docs/co_destruction_rates_literature_20260906.md, 1419 lines): CO photodissociation as Visser et al. define it (37 bands inside EXHALE's 912 to 1110 Å beam, unshielded $2.6e-10 \text{ s}^{-1}$ in the Draine field, shielding tables in $N_{\text{CO}} \times N_{\text{H}2}$ for four parameter sets, T_{ex} raised only to 512 K because no data exist for dissociation out of $v'' = 1$); evaluated on the oxygen_chemistry run's own base columns ($\log N_{\text{CO}} 18.96$, $\log N_{\text{H}2} 21.80$) it gives $\tau_{\text{dest}} = 2.8e8 \text{ s}$, twelve decades faster than the barrier channels and still two to six decades short of $\tau_{\text{dest}} \ll \tau_{\text{res}}$ at the base; zero in mol_base_handoff, whose spectrum carries no flux longward of 911.8 Å. RATE12's paper prints no entries (the data file is still needed for the ion channels); Venot 2012 is valid 300 to 2500 K and 0.01 to 100 bar (EXHALE's base at $1e-6$ bar is four decades below); none of the six sources covers the 3000 to 4000 K band where the ceiling fires. The CO exclusion (B1 decision 7) stands, now quantitatively. If a CO photodissociation term is ever added it is a

fourth absorber on the existing Lyman-Werner beam with a second column (N_CO) and the cell-mean form of `lyman_werner_dissociation_rate`.

- **TESTINFRA** (`grid_and_gates/run.sh`, `physics_probe/run.sh`, their READMEs): both suites take EXHALE_OBJDIR, EXHALE_EXE, EXHALE_TEST_OUT with the semantics of `spectrum_type/run.sh`; default behavior byte for byte as before; against a private build both run without touching build/ objects (the six `grid_and_gates` shell tests still place their run copies under `build/tests/`). A dead fallback that wrote a build stamp the closure never compiled is fixed. Also (advisor): the two Python source scanners now skip `.ipynb_checkpoints/` directories, which hold editor auto-save copies of four modules (git-ignored) and had made `threshold_literals_written_once` red on a stale copy of `cross_sec.f90`; the Makefile source list is explicit and never reached them.
- **B3b-ENTH** (`molecular_reaction_heat.f90`, `oxygen_rates.f90`; 78 Fortran
 - 37 Python assertions): one table `species_formation_energy(isp)` for every species column (H/He stages, 27 metal stages cumulative from the neutral, H₂/H₂⁺/H₃⁺/HeH⁺, OH/H₂O/CO, plus free electron, H(n=2), O(1D)) from the single B1 reference; the 17 reactions of the H₂/He network are a stoichiometry table and each heat a difference of table entries; the Shomate table stays with its reference stated, every use of it being a balanced-reaction difference (the zero cancels), and the oxygen and carbon entries of the one table are differences of it; one D0(H₂) (`mol_rates::h2_dissociation_energy_eV`, reproduced by the fit to 1.5 meV). Physics defect fixed: the oxygen photolysis thresholds were 298.15 K reaction enthalpies; they are now 0 K dissociation energies (the lowest Shomate interval's F coefficient is the 0 K enthalpy), 0.9 to 1.0 percent lower, deposited heat +0.8 percent at the oxygen base cell. Impact: `mol_base_handoff`, `mol_sec_ion`, `mol_metals` byte-identical; oxygen_chemistry T +0.4 percent at 1.02 Rp, Mdot 9.61. Open for B3c: the undeposited collisional oxygen heats and the He 2³S associative branch (D0 C25; the table exists); the B4-2 Python diagnostic lacks four oxygen entries; `formation_energy_report` has no caller; a stale threshold comment in `water_photolysis.f90`: 281.
- **C3** (six `input.inp` and READMEs; `inputdata/sed/README.md`): the planet folders and benchmarks of HD 209458 b, HD 189733 b and WASP-121 b load their literature SED files (Spectrum type: Load from file.; Power-law index removed, LX/LEUV kept because mandatory and overwritten by the grid integral, which the setup report shows); every README records the configuration of record with the existing power-law products listed by mtime and the reruns left to a separate instruction; `read_sed` accepts every file at its configuration's floor (the H(n=2) edge in all six). `log F_XUV` at the orbit moves 3.193 -> 3.127 (HD 209458 b), 4.438 -> 4.311 (HD 189733 b; the recomputed LX/LEUV reproduce Salz Table 3), 6.205 -> 6.204 (WASP-121 b). Advisor follow-ups: HD 209458 b's orbital distance set to 0.04707 AU (Oklopčic and Hirata 2018, the reference the run is compared with; the input said 0.0480 while the file was scaled to 0.04707); the setup report's flux ratio format (F8.3 overflowed when the sub-Lyman band dominates); the three benchmark directories added to the coverage script's inventory (all pass). Known: Load IC? True in the two planet folders is broken by the grid guard (pre-rebaseline IC), the same class as the `hp_*` regeneration.

- **COLit, revised with RATE22, Moses 2011, Tsai 2021** (document now 1956 lines): RATE22 (Millar et al. 2024, A&A 682, A109) carries 'He+ + CO -> C+
 - O + Heat $1.6e-9 \text{ cm}^3 \text{ s}^{-1}$, measured, accuracy A, all products EXHALE species; it has noH+ + CO, CO + MorCO + H2destruction entry, and itsTu = 41000on 6562 entries is an undocumented "no upper bound" marker. On theoxygen_chemistryprofile the He+ channel with the Visser shielding givestau_dest/tau_resfrom 6.7e2 at the base to 9.6e-5 at 1.14 Rp: **the one-sided destruction model is constructible above the helium ionization front and not below it**; the boundary is compositional, not the ceiling's thermal switch. Decision 7 (exclusion) stays until T5.1 is implemented; B1 section 5.2 revised, B3b-CO scheduled after B3c. Moses 2011: no stated envelope, one fit interval 500 to 2500 K, no shielding. Tsai 2021: 0.1 nm continuum opacities, no CO self-shielding; EXHALE'slyman_werner.f90' is the working precedent. The literature requests are closed except Baulch 1976 vol. 3, which blocks nothing.
- **B4-6** (utilities.f90 cell_mean_attenuation, util_ion_eq.f90; xuv_cell_mean_attenuation 56 assertions): the photoionization and photoheating sums of a cell use the exact cell mean of the attenuation, $\exp(-\tau_{\text{out}}) (1 - \exp(-d\tau))/d\tau$ (composite Gauss when the 2D correction is on), instead of the inner-face value the top-down columns gave; one definition for H, He, He²S, H2's four channels, the metals, the single-absorber rates, q_abs and the secondary-ionization integrands. Inner-face error: 0.5 percent at $d\tau = 0.01$, 42 percent at
 1. Photon conservation on the written states: cell mean 1.000000000 of the incident flux, inner face 0.9725 (wasp_full), 0.9843, 0.9849 (the loss sat at the ionization front). Impact: wasp_full (24000 steps) x(H+) $8.3e-2$ at 1.32 Rp, Mdot +0.74 percent; mol_base_handoff Mdot +0.93; mol_sec_ion +1.01; hydrostatic_column unchanged. B3b-PE's two heating terms wired into the integrand (its own numbers reproduced cell for cell). Advisor: the single-precision $1.0e-18$ cross-section unit literal (33 sites, $4e-8$ relative on every rate) made double. Finding to resolve at the gate: wasp_full on the mid-series tree plateaus at $du = 4.2e-3$ from step 22000 and never reaches $1e-3$ (the golden stopped at 16230), identical in the item's before binary, so it belongs to an in-flight change (B3a edits EXHALE_main.f90) or to an earlier landed one; measured after B3a.
- **A2tol** (measurement; a2_certification_contract_20260906.md section 9): the du stop and the hydrodynamic row measures are not measures of one thing (over a decade of du on wasp_full the mass row falls at order 0.2, the momentum row rises, the energy row is not monotone; mol_carrier's carrier row falls at order 0.08 over 12000 to 48000 steps); the JFNK returns an energy row 41 to 47 times its target at every rung of a two-decade Resid tol ladder (structural; B5a); the elemental transport rows cannot be measured on the marching route (the operator advances $1e-6$ of the way per CFL step) and the diffusive cases never reach the steady route (du stalls at 5 to $6e-2$ against the $1e-2$ hand-off). Round-off floors on the converged JFNK state: mass $1.0e-13$, momentum $5.1e-12$, energy $1.6e-11$ (residual bitwise reproducible). Recommended and adopted

(applied by B3a in certification.f90, provisional until B5a): cert_tol_mass 3e-12, cert_tol_momentum 1e-8, cert_tol_energy 5e-5, cert_tol_element 1e-8 (with the homopause regime caveat), carrier, closure and level tolerances kept. At these values wasp_full_newton certifies with Resid tol: 1.0e-7 (marginally at 1e-6) and nothing as shipped. Noted for the user: the shipped default Resid tol 1e-5 is one to two decades looser than the solve reaches at no cost.

- **B5a** (steady_newton.f90; suite steady_selfconsistent_residual): eval_residual eliminates the composition AT the state it evaluates (the equilibrium sweep iterated at fixed conserved variables until the composition stops moving), so the loop-top gate, the line-search merit, the returned state and the certification measure one object; the gate equals the certification to 1e-12 by construction. Finding: the eliminated residual's Jacobian carries the summed Picard response $(I - M)^{-1}$, and a finite-difference probe with a convergence test stops after one pass (the seed already is the state's composition), so the Newton direction ascended; the probes now run a FIXED sweep count taken from an elimination seeded at a different state (about 15 terms), and the "frozen-composition Jacobian" the brief offered as a fallback is exactly the arrangement that fails. wasp_full_newton: info = 0 at $\|R\| = 4.4e-7$ in 9 iterations and CERTIFIED (was info = 2 at 4.8e-5 after 26), state moved 1.0e-4 at most, log10 Mdot 13.27, cost 1.49x. mol_base_handoff identical. Open: Resid tol values were calibrated on the lagged measure; the other Solver: Newton cases and the PTC route are measured at the series gate; n_selfconsistent_max comment stale (B3a); the duplicated row-measure expression (B3a).
- **HYG-absorbed** (water_photolysis.f90, utilities.f90 comment; driver absorbed_fraction_switch, 5 assertions): the FUV band rates take the XUV beam's absorbed_fraction_per_unit_depth by use association; the local copy was bitwise the same over 1501 points and is deleted; the series switch and its error bound (about 1e-8 on the closed-form side of $d = 1e-8$) stated once; mol_base_handoff, oxygen_chemistry byte-identical. Advisor: the B4-6 driver's reference carried the same single-precision 1.0e-18 the production code had; made double with the production change, the driver is green again (51 assertions had turned red on the 4e-8 mismatch).
- **B3a** (EXHALE_main.f90, new_attempted_step.f90, certification.f90, parameters.f90; suite attempted_step, 87 assertions): a marching step is one trial of the fourteen operations from a checkpoint covering everything rows 1 to 12 write, evaluated at one adoption boundary before update_map_end_step, retaken at half dt from the checkpoint when refused (cap 8, floor $dt/2^8$, a reduced dt bounding the next step); the clock and the accepted-step count move only there; exhaustion stops in phys and continues flagged in init. Impact: wasp_full (300 and its du stop at 14496 steps, log10 Mdot 13.33), hydrostatic_column, mol_base_handoff, mol_carrier bitwise identical against a control built from the same snapshot; cost within run-to-run spread. Measured and reported, not gating: the time-discrete hydro row (1.92 at a cold-start base where the momentum scale vanishes) and the energy identity T1.5 (-0.20 to -0.74 on wasp_full, the size of the row-8 projection, the RED of AT-1a); the step-doubling estimate gives order 0.89 for the whole step (the operator split is first order), cost 12 percent at n_err_every = 20. Two defects of the item caught by its own impact runs and fixed. The mid-item requests landed (A2tol tolerances, info = 2 written uncertified with no fallback, the ledgers in the summary, the H3+ domain counters, one row-measure definition, the n_selfconsistent_max

- comment). The B4-6 "du plateau" was the mid-edit tree; the final tree reaches its du stop. Advisor decision on the exit status: a run that DECLARES a stationary state and fails certification exits 2 in every mode (A0's init exemption covers physical-step claims only); the mode gate is removed (A2exit). Small items of its report assigned (HYG-B3a: operator status on every path, conduction accessor, step-level carrier verdict, optional volume measure) and the -00 floating-point exception on hydrostatic_column under diagnosis (O0FPE).
- **B4-6b** (lya_rt.f90, excited_hydrogen.f90 field evaluation; driver lya_beam_cell_mean, 37 assertions): the stellar Lyman-alpha depth was a trapezoid between cell centers (neither the cell's face depth nor its own, every depth short by half the outermost cell); it is now the rectangle rule of calc_column_dens on the cell's own opacity, with the cell's own depth and its star-ward face as outputs. The beam transmission is $\text{erfc}(c \sqrt{\tau})$, not exponential, so its cell mean is a 3-point Gauss rule in $u = \sqrt{\tau}$ (mean and mean of the square, since the trapping makes the field quadratic in the transmission); the photosphere factor's mean is closed form with $\log_{10}$. jlya_escape_prob keeps the cell center by Neufeld 1990 section II (the escape fraction is a point property of where the photon is created). Impact: hydrostatic_column, mol_base_handoff byte-identical; wasp_full reaches its du stop at step 14442 on both binaries, wind unmoved ($1e-7$), the field $1.6e-3$ and the Balmer proton source $6.5e-4$; the face rule over-counted the column's scatterings by 2.4 to 2.7 percent. Carried forward: the FUV band B2 line transmission in util_ion_eq.f90 still reads a face value (B4-6c).
 - **HYG-B3a** (diffusive_photochemistry.f90, certification.f90, steady_newton.f90, viscous_conduction.f90; suites carrier_retry 97, energy_update 48, certification 29): the carrier operator sets its interval status on every return path (already true since A3b; now stated and tested); carrier_last_verdict_of_run renamed for what it is, plus a step-level carrier_step_verdict() reset at every transport step; conduction_floor_cell_hits exported like the energy update's; the volume-weighted companion of certification_row_measure optional and no longer computed where discarded. All impact cases bitwise identical (wasp_full_newton finish: info = 2 at $||R|| = 7.3e-6$ against the new cert_tol_momentum = $1e-8$, re-anchored by A2tol2). Left for the gate (advisor): the now-unneeded pre-set in EXHALE_main.f90 and a stale field comment in attempted_step.f90.
 - **A2exit** (EXHALE_main.f90 exit block; run_mode suite 30 assertions, certification shell gate): the certification exit status follows the CLAIM, not the mode: a run that declared a stationary state (du stop, residual gate, JFNK finish) and was refused exits 2 in every mode with its outputs written; a capped or plateaued run exits 0. wasp_full_newton now exits 2 with info = 2; run_check.sh never reads the status. Advisor: run_newton.sh and run_baseline.sh (set -e loops) now report a status 2 instead of aborting. Open: the EXHALE_PTC=1 route stops before the certification and so claims a stationary state with no verdict (B5).
 - **B4-4a** (Apply_BC.f90; hydrostatic_residual +1 and new free_outflow_boundary 18 assertions): the outer free-outflow ghost is the isothermal hydrostatic continuation of the last cell ($\rho_g = \rho_N \exp[-(\phi_g - \phi_N)/(\rho_N/\rho_g)]$, p_g by the same factor, $v_g = v_N$), one rule for PLM and WENO3, replacing the zero-gradient copy (PLM) and the linear extrapolation (WENO3). The outer cell's momentum residual on the analytic column falls from 0.750 of its weight at every N to $4.4e-3$ to $1.0e-4$ over $N = 250$ to 2000 (order 1.8

PLM, 1.6 WENO3); interior and base unchanged; four ghost rules measured, this one 30 to 40x better. Velocity: the copy (a continuity-exact rule stopped fastest but divides by a ghost density the stratification makes small and was the one rule under which wasp_full_newton segfaulted at -O3). Two premises of the brief corrected: wasp_full's edge is subsonic (Mach 0.18 to 0.96; mol_base_handoff is the supersonic case), and a supersonic edge blocks the ghost only through the Riemann solution, not the stencil, so byte identity is not available. Impact: mol_base_handoff one cell above 1e-4; hydrostatic_column 3.5e-3 at cell N; wasp_full du stop 16208 -> 17135 steps with the base-breathing phase moved (11 percent in rho at 1.27 Rp, mean mass flux above r_flux 0.078 percent); wasp_full_newton Mdot 13.27 both. Removing the outer residual does not by itself stop the marched hydrostatic column's drain (still 0.79 c_s at 300 steps). Reported: a reproducible -O3-only segfault reachable from the rejected velocity rule (latent; O3SEGV to diagnose); EXHALE_main.f90 2538-2540 ghost-counter comment stale (B3c); B4 design section 8.3 (ii) historical.

- **O0FPE** (util_ion_eq.f90; shell test fpe_traps in grid_and_gates; run_fcheck.sh header): the -O0 failure was two defects. (1) An integer division by zero in EXHALE_main.f90 about 1234: mod(count, n_err_every) with n_err_every = 0 in init mode, reached because .and. does not short-circuit (-O1 and above happen to skip it); fix handed to B3c (owner of the file). (2) A real floating-point overflow in he_rec_coupling (util_ion_eq.f90:2813): 1100 exp(1.2/T4) sqrt(T4) at 17 K is 3.9e309, an Inf consumed as ne/Inf = 0 into the correct low-density limit; the product is now formed only above T4 = 1.71e-3 and the limit value taken below, byte-identical on hydrostatic_column, wasp_full, mol_base_handoff. A third site of the same class (the stall detector's huge sentinel) also handed to B3c. run_fcheck.sh states why it builds at -O1.
- **B4-6c** (util_ion_eq.f90 fuv_lw_photon_field; driver lya_band_transmission rewritten, 16 assertions): band B2's line factor is the cell mean of the stellar Lyman-alpha transmission over the cell's own line-center depth (product of the two means with the continuum mean formed inside water_photolysis_rate; the covariance shortfall 6.0e-6 against a 1.4e-4 bound, the face rule 1.2e-2 high on the budget); the top cell carries its own mean instead of a neutral 1. The face rule was 1.0018x on the oxygen_chemistry state, 1.095x on an exponential H I layer and 18.2x in the base cell of a molecular base whose line-center column reaches 1e7 in one scale height. Consumers: the H2O and OH photolysis in B2 (both ionization_equilibrium call sites and the carrier photolysis), the output columns, the FUV heating column; the H2 channel reads LW only. Impact: wasp_full, mol_base_handoff, mol_sec_ion identical; oxygen_chemistry moves by factors at 45 steps, shown by a control (a 4e-5 perturbation of the face value diverges the same way) to be the case's transient amplification, not the size of the change; the case is not usable as a controlled point until it converges. Finding: write_output.f90's FUV band ledger closed form N_ph (1 - exp(-tau)) carries no line factor for B2 and its other rows are 0.79 to 0.98 off (LEDGER-FUV assigned).
- **O3SEGV** (diagnosis, no change): the -O3-only segfault B4-4a reported was a null F0 argument of jv_product in a pgmres clone the compiler had specialized for a null pointer, i.e. a defect of the source text of one intermediate build (13:08) between two edits of steady_newton.f90; every later build carries a plain pgmres, HEAD plus the rejected ghost rule completes wasp_full_newton three times, AddressSanitizer finds nothing over a full run. Correction to B4-4a's premise: -g changes no generated code (.text bitwise identical at -O3 with and without

it). Noted: at the time of the diagnosis the working tree stopped wasp_full_newton at step 0 on the energy floor (B3c in progress).

- LEDGER-FUV** (write_output.f90 write_oxygen_chemistry; suite fuv_band_ledger, 15 assertions): the FUV band ledger is three labeled statements: (a) closure of the rated absorptions against the beam's loss between the cell faces (holds for any rate form, $1e-16$); (b) the beam budget with each band's own line factor and the un-rated share named per band (none for B1/B3/B4, which is therefore the cell-mean test; H I resonance scattering for B2, a share; for LW a sign-changing disagreement); (c) energy; (d) state drift. Two defects of the ledger fixed: it multiplied post-sweep densities by a pre-sweep field, and summed the two inner ghost cells. Data rows of every output identical; only the FUV_bands.txt trailer changes. **Finding (LW-NORM assigned):** the LW absorbers take 45 percent MORE photons than the beam loses ($1.27e13$ against $8.73e12$) and $249 \text{ erg cm}^{-2} \text{ s}^{-1}$ against an incident 171.5: two independent normalizations of one absorption, the level-resolved dissociation table over p_lw_absorbed against the DB96 eq. 39 band share (lyman_werner.f90, h2_self_shielding_table.f90). Also: docs/a2_oxygen_option_design.md 882 documents columns that no longer exist; the a2_m2 kinetics check asserts the beam identity for every band.
- A2tol2** (measurement and decision; a2_certification_contract_20260906.md section 10): the section-9 anchors re-measured on the self-consistent JFNK residual of B5a. The returned row is now 0.36 to 0.93 of the solve's target (section 9 measured 41 to 47 above it). wasp_full_newton at Resid tol: $1.0e-8$ certifies (info = 0, 14 iterations, 658 s, log10 Mdot 13.27; rows mass $2.6e-13$, momentum $1.3e-9$, energy $6.275e-10$); the PTC route stalls from a cold start; wasp_he23off_newton, newton_rsw01, newton_rsw05 at the default $1e-5$ refuse on the mass and momentum rows at cell 499; the four Load IC Newton cases (jfnk_hd189, jfnk_hd189_tight, ptc_warm, armD_D2_newton) are stopped by the grid guard and need their IC regenerated on the current grid before they can be measured. **Applied:** cert_tol_energy $5e-5 \rightarrow 1e-6$ in certification.f90 ($5e-5$ had been anchored on the lagged-gate energy row $4.700e-06$ and could refuse nothing; $1e-6$ is a decade above $1.212e-07$, the largest energy row of any state whose solve met its own target, $5.7e4x$ above the floor $1.765e-11$); mass $3e-12$ and momentum $1e-8$ kept; the report header now names sections 9 and 10. backup/regression/wasp_full_newton/input.inp gains Resid tol: $1.0e-8$ as its configuration of record, the only setting at which that case certifies (Mdot unchanged at this precision, cost +11 percent); its golden moves at the series gate with everything else. Verified: private build, certification suite green (29 assertions). Not changed: the other three Newton cases stay at the default so that the matrix keeps a refused claim (exit 2) under test.
- LW-NORM** (diagnosis; no executable statement changed): the 45 percent excess of the FUV band ledger is two WAVELENGTH BANDS, not two fits. The self-shielding table's line list runs to 1200 Å while its sigma_pump is normalized per photon of 912-1110 Å; the beam loss is the DB96 eq. (39) equivalent width of 912-1110 Å. MEASURED: the column integral of sigma_diss/p_eff reaches 1.51538 at the top of the column axis against 1.519288 for the photon content of 912-1200 Å over 912-1110 Å of a flat spectrum (0.26 percent), is exactly density-independent, and reproduces the shipped oxygen_chemistry ratio 1.4544 at its base column where eq. (39) clamps to 1. p_eff already separates dissociations from absorptions, so

the heating takes fragment energy only (not cause 2). Landed: `lyman_werner.f90` header section 3a with the invariant and the measurement, its section 3 "2 percent" statement corrected to the live table's thin-limit cross section $2.6396 \times 10^{-17} \text{ cm}^2$; `lyman_werner_cell_mean` probe gains two assertions and a photon-conservation diagnostic (9/9); `fuv_band_ledger/run.sh` comment. Advisor follow-up: the table's generator inputs (nine `abs_T####.npz`, 27 CLOUDY rate files, the line-by-line scripts, the Meudon level and line data, 29 MB) moved from the 2026-09-03 scratch directory into `src/utls/h2_shielding_lbl/` with a README, and `h2_shielding_table_line_by_line.py`'s template brought in line with the hand-renamed axis dummy of `h2_shield_locate`: the generator now reproduces the shipped module BYTE FOR BYTE (MEASURED). The repair, table on 912-1110 Å or beam widened to 912-1200 Å with the B1 flux joined, is item 6 of `To_be_determined_by_user_20260906.md` (recommendation: widen), because the second redefines the Stellar LW flux key. Still to correct after B3c releases `write_output.f90`: the ledger's LW label says "two fits" and should say "two bands" (with the `header_names_lw_fits` assertion of `fuv_band_ledger`).

- **HYG-HeCI** (test hygiene, advisor): `physics_probe` was red on three `helium_collisional_ionization_by_level` assertions (1×10^{-13} at 1×10^4 K to 1.0×10^{-4} at 5×10^4 K). Not a code defect: batch 2c replaced the single-precision literal $3.940 \times 10^{-11} \text{ erg}$ (24.6 eV) of `eval_cool` with the global `e_th_HeI_erg` (24.587389 eV), and the test's reference still carried the old literal with a comment calling it "a separate open item". The reference now uses `e_th_HeI_erg` and `e_th_HeTR_erg`; `physics_probe` PASSED on the current tree (private build). The LW-NORM worker's attribution of this to a concurrent `Cool_coeff.f90` edit was wrong: that file's uncommitted diff is the landed 2c-CI / MASS2 work.
- **LW-NORM-B** (user decision 2026-09-06: recommendation (B) adopted; landed): ONE Lyman-Werner band, 912-1201 Å, with one normalization.

- **The band.** `oxygen_rates.f90` carries four FUV bands, not five: the Lyman-Werner interval is 912-1201 Å and band B1 (1110-1201 Å) is gone into it. 1110 Å was an edge because Draine & Bertoldi end the Solomon process there for interstellar $v = 0$ gas; at the 700-3200 K of a planetary base the vibrationally excited levels pump longward of it, and the self-shielding table's line list always ran to 1200 Å. The two halves carried the same H₂O yield triplet, so no quantum yield moves; the band-averaged cross sections are re-formed over the whole interval and MEASURED again from the same Photochem files: $\sigma(\text{H}_2\text{O})$ 8.51337×10^{-18} , $\sigma(\text{OH})$ $4.74371 \times 10^{-18} \text{ cm}^2$, (H₂O) 12.0526 eV, (OH) 11.3829 eV, $= 2hc/(912+1201 \text{ Å}) = 11.7354 \text{ eV}$. They agree to seven digits with the merge of the two shipped halves, and they reproduce the values the band carried before it was split in 2026-08.
- **The key.** Stellar LW flux now means the 912-1201 Å band flux at the orbit; Stellar FUV B1 flux is retired and a file that states it stops the run with a message naming the merge and asking for the two numbers added. `Spectrum type: Load` integrates 912-1201 Å. The resolved dump loses `fuv_band_B1_flux`. The inputs that carried both keys were restated to their sum: $343.0 + 137.9 = 480.9$ for the HD 209458 b spectrum (`Update_EXHALE_stage1.md` records the 912-1201 Å integral of that spectrum as 481.0 and its halves as those two numbers), and the LW-only cases (`mol_lyman_werner`, `armA_LW`, `armD_D2_LW`, `armD_D2_LW_newton`) from 343.0 to 480.9 so that they keep the same star. The README_HOWTO HD

189733 b recipe goes from LW 600.1 + B1 648.4 to LW 1248.5.

- **The table.** `src/utls/h2_shielding_lbl/lbl_table.py` normalizes per photon of 912-1201 Å (WL_HI, wl_max, and E_LW_PHOTON_ERG the mean photon energy of that band); F_BAND_ERG is restated as $343.0 \times 289/198 = 500.6$ with the derivation in the file, and the file now records that it cancels out of sigma_pump exactly. The nine `abs_T####.npz` were recomputed (`run_abs.py`, 23 to 251 s each) and `h2_self_shielding_table.f90` regenerated. MEASURED: the generator reproduced the shipped module byte for byte before the change, so the only difference is the band. sigma_pump at the bottom of the column axis is $1.701\text{e-}17$ (700 K) to $1.887\text{e-}17$ cm², which is DB96's own $2.557\text{e-}17$ per 912-1110 Å photon restated per 912-1201 Å photon ($1.676\text{e-}17$) to 1-13 per cent, the same agreement as before.
- **The beam loses what the table rates.** The band share A is no longer the DB96 eq. (39) equivalent width; it is the column integral of the pump cross section $\sigma_{\text{diss}}/p_{\text{eff}}$ of the same table, in closed form (`h2_lw_band_photon_fraction_absorbed`, exact for the module's own interpolant knot interval by knot interval). MEASURED: A at the top of the column axis is 0.6903 to 0.9974 over the whole (T, n_H) grid, where it used to reach 1.5154; the closed form agrees with an independent 20000-step quadrature of the two public accessors to 2e-10. The band ledger's photon count is now the cell mean of the same pump cross section (`lyman_werner_band_absorption_rate_cell_mean`), on the quadrature the dissociation rate already used, and not the cell mean of sigma_diss divided by p_eff at one column: on cells spanning 2.5 H2 scale heights the second stood 8.0 per cent above the beam loss and the first closes on it to 1e-6 (`physics_probe/lyman_werner_cell_mean` sec. 6). `h2_band_equivalent_width` and eq. (39) are removed; the two published self-shielding fits stay, called by nothing, for the comparison of `docs/h2_self_shielding_cloudy.md`.
- **The ledger.** The LW row of `output/FUV_bands.txt` is a closure like B3 and B4, and its label says so. MEASURED on the `oxygen_chemistry` state, 5 steps: `unrated_frac` $-3.62\text{e-}1$ before, $-1.18\text{e-}2$ after, and the LW absorbed energy goes from 233.5 against an incident 171.5 (36 per cent over the band) to 202.3 against 200.0. The residual left is the part of the column above the top of the table's column axis: this case reaches $6.4\text{e}21$ cm⁻² against an axis top of $5\text{e}21$, where the clamped edge cross section drives A past 1 and the beam is capped while the rate is not. The run already warns about that column.
- **Impact.** `mol_lyman_werner` at its own 12000 steps, before at 343.0 against after at 480.9: $\log_{10} \text{Mdot}$ 10.53 -> 10.54, T and rho under $5\text{e-}3$, cooling $6.5\text{e-}2$, the molecular densities and the shielding $1.2\text{e-}1$ at the H2 front ($r = 1.075$), the base H2 column $3.3875\text{e}20$ -> $3.4067\text{e}20$ cm⁻² and the base x_{H2} 0.96288 -> 0.96365. The optically thin rate at the top of the column moves by $480.9/500.6 = 0.9606$ exactly: the band's photon content grew 46 per cent and the cross section fell by the same factor, so what is left is that the real spectrum gives 480.9 over 912-1201 Å where the flat continuation of the 912-1110 Å deck would give 500.6. Deeper cells move -4.7 to -6.0 per cent, the wider band's extra lines shielding differently; slightly less destruction leaves slightly more H2.
- **Not repaired here, and it blocks the fuv_band_ledger suite.** With its restated 480.9 the `oxygen_chemistry` case stops at step 0 on the energy floor. MEASURED, it is not the band merge: the same case stops with the OLD binary at the same restated flux, and both binaries stop or run at chance

along the flux axis (old: runs at 343.0 and 440.0, stops at 360.0, 400.0 and 480.9; new: runs at 343.0 and 400.0, stops at 360.0, 440.0 and 480.9). That is the step-0 acceptance knife edge of items B3a and 2 of To_be_determined_by_user_20260906.md. Every ledger assertion is GREEN on the same case at a flux where it runs.

- Also: p_lw_absorbed is now written by the sweep and checkpointed but read by nothing, since the ledger takes the pump rate directly. Removing it needs attempted_step.f90.

- **B3b-IR** (molecular_infrared_cooling.f90; h2_infrared_line_populations and h2_partition_sum_uniqueness probes): the third copy of the H2 ladder sum (212-215) is gone; the H2 infrared line populations are normalized by caloric_eos's h2_partition_function, the one Boltzmann sum over the Roueff et al. 2019 ladder whose energy the equation of state carries and whose free energy sets the $H + H \leftrightarrow H_2$ equilibrium. The probe that greps the tree for a second sum went RED (1) -> GREEN (0). Impact (advisor measured, 12000-step snapshots, OMP_NUM_THREADS=1): mol_ir_bands (9 output files) and mol_base_handoff (7 files) BYTE-IDENTICAL before and after, Mdot 7.99 both; the retired copy had an $\exp(-\min(E/T, 700))$ clamp whose only difference from the shared sum's $\exp(-E/T)$ is below double rounding. No golden moves. The worker's draft report stopped at "impact runs in progress"; this entry closes it.
- **LW-NORM-B, advisor verification** (2026-09-06): private build of the landed tree, physics_probe PASSED (554 assertions incl. the rewritten section 6: column integral of sigma_pump at the top of the axis 0.69 to 0.997 on the whole grid, was 1.515; photons spent / photons lost 0.999999), spectrum_type PASSED (154), fuv_band_ledger NOT RUNNABLE: its case is oxygen_chemistry, whose Stellar LW flux restated to 480.9 (the same spectrum over 912-1201 Å) stops at step 0 with energy update FAILURE at cells 85-88 ($T \ 2.1e4$ K, ionization sweep NON-ROOTs accepted by the relaxation amnesty with residual 1 to 4, then the balance asks for T below the 14.5 K floor). MEASURED by the worker along the flux axis with both binaries: 343 runs, 360 stops, 400 runs (after) / stops (before), 440 stops (after), 480.9 stops: a step-0 acceptance knife edge of the OLD code at cells with no H2, items 1 and 2 of To_be_determined_by_user_20260906.md (decision 4: reject-and-retry; amnesty cap), not of the band. The flux is NOT tuned to make the case run; the case and fuv_band_ledger stay red until those two decisions land. Code read: a_lines = min(A, 1) with transmission $1 - A$ (a band share, not an optical depth) from the closed-form power-law integral of the tabulated sigma_diss/p_eff; both the rate and the share are read at the local T and n_H of the cell for the whole column above it (homogeneous-column table), a comment in util_ion_eq.f90 now says so. Accepted. Goldens untouched; every LW case moves at the series gate (mol_lyman_werner 500-step pair: $T \ 3.3e-3$, $H_2 \ 0.125$ at the front, $k_{LW} \text{ thin-limit } x0.9606 = 480.9/500.6$).
- **ACCEPT-1/2** (user decisions 2026-09-06, items 1 and 2 of To_be_determined_by_user_20260906.md, both as recommended; brief plan_rev2_brief_ACCEPT12.md, to start when B3c releases EXHALE_main.f90, energy_semi_implicit.f90, ionization_equilibrium.f90): (1b) a failed energy update refuses the attempted step in both modes and the controller retakes it at half dt, exhaustion exits 2; (2a) the relaxation amnesty accepts a non-root only below a residual cap measured on the matrix, above it the cell keeps the composition it entered the sweep with (new acceptance class 6). Motivation: oxygen_chemistry at its restated LW flux dies at step 0 on exactly this pair (LW-NORM-B verification entry above).

- **B3b-IR impact runs, advisor correction** (2026-09-06): the worker's before/after pairs were run with binaries built from the tree as it stood at 12:30, when the B3a controller still refused on the time-discrete hydrodynamic row in init mode: MEASURED from their run logs, 12000 steps accepted of 104389 attempted (mol_ir_bands) and 104594 (mol_base_handoff), 103697 refusals all "the Runge-Kutta stages and the fluxes: the time-discrete hydrodynamic row was out of tolerance", dt collapsed and T stayed within 1 K of the 1140 K initial condition over the whole grid (golden mol_ir_bands reaches 2553 K at 2.18 Rp). The byte-identity of the pair therefore says little about the layer; the statement that the IR refactor changed no arithmetic rests on the emission assertions at five temperatures with identical digits and on retired/ $Q - 1 = 0$ at every probe temperature, which stand. On the current tree the same case accepts 12000 of 12000 attempts with no refusal (MEASURED, private build). The H2 IR tabulation grid extension (advisor, this entry's neighbor above): n_tab 401 -> 496 at the same log spacing to t_tab_hi = 30146 K, old nodes reproduced by the same expression, emission at 10000 K: clamped = direct sum = 1.57875E-18 (was 1.181e-18); molecular_infrared_bands.py template carries the private :: i of the generated data module and regenerates it byte for byte (MEASURED).
- **B3c** (EXHALE_main.f90, ionization_equilibrium.f90, energy_semi_implicit.f90, molecular_reaction_heat.f90; new suite coupled_source_step, 30 assertions in the advisor's run): rows 7 to 9 of the enumerated step are one local source step per cell at fixed volume, the temperature and the composition its unknowns, iterated to a fixed point ($|dT|/T \leq 1e-8$, max $|df_{sp}| \leq 1e-8$, cap 20 passes). The composition projection (comp_p_from_T + W_to_U, which rebuilt the energy row at the new particle count) is gone; the energy row is anchored on u_th_old, the thermal energy the cell had at its old composition; rho is intent(in) to the chemistry and the mass sum is a checked diagnostic (worst 6.4e-12, T2.2). The two collisional deposits D0 C25 named (oxygen O1/O1r/O2/O2r/O6, associative He 2^3S) are in as differences of the one formation-energy table. RED reference kept in the suite: the removed update changed u(3) by -0.600 with heat = cool = 0. **Advisor decision on the worker's deviation (accepted):** Delta u_form is NOT added to the energy row. In this code heat and cool are the net THERMAL sources (photoheating deposits $h\nu - E_{th}$, collisional ionization cooling removes E_{th} , every molecular, Penning, Lyman-Werner and oxygen reaction heat is a difference of the same table), so the formation reservoir is already accounted for and the term would double count; MEASURED with it added, wasp_full dies at step 0 with 368 of 504 cells at the floor. Consequence: T1.5/T1.6 as an identity on $u_{th} + u_{form}$ against $dt(heat - cool)$ cannot close; the B3a instrument (attempted_step.f90) is to be restated as $du_{th} = dt(heat - cool) +$ the transport contribution of the step, or evaluated on the source substep alone (item T15-INSTRUMENT, pending). Impact (control = same snapshot with the edits reversed): wasp_full to its du stop, log10 Mdot 13.34 -> 13.34, T 2.5e-3, rho 2.4e-3, metals 1.5e-2, one sign change of v at r = 1.062 (2.5e-4 of the wind); wasp_full 300 steps front one cell apart; molecular 12000-step snapshots up to 40 percent in rho beyond 4 Rp, Mdot 10.58->10.57, 10.53->10.51, 10.58->10.58; hydrostatic_column $\leq 6e-9$. **Cost: 5 to 12 passes per step, 8.1x on wasp_full 300, 10.3x to the du stop (18:59 -> 3:15:07).** Assigned as B3c-COST (re-sweep only the cells not yet at the fixed point; tolerance anchored on a measured convergence study, not chosen to pass). Also in and byte-identical: the carrier status pre-set removal (HYG-B3a) and the two OOFPE sites; fpe_traps passes at -O0 with traps. Left red on purpose: heat_column_matches_breakdown_total 6.886e-3, because write_heat_breakdown_eq

(util_ion_eq.f90 2280) is a second hand-maintained copy of the heating assembly that lacks the new deposit, and post_process_adv.f90 ~847 a third: assigned as HEATBRK (one assembly, three callers), not the ten-line patch. Advisor verification: private build of the live tree, coupled_source_step 30/0, energy_update 48/0, attempted_step 31/0, certification 29/0, run_mode 30/0, acceptance_classes 12/0, grid_and_gates 92/1 (the HEATBRK red).

- **HEATBRK, first pass BLOCKED** (2026-09-06): the heating assembly the sweep uses is 210 inline lines of ioniz_eq (ionization_equilibrium.f90 2119-2331), not a routine, so the one-assembly fix has to extract it there, a file ACCEPT-1/2 owned at the time; the worker stopped with no edit, as briefed. Re-measured RED: worst 6.886e-3 (cell 376, the associative He 2³S and collisional oxygen deposits missing from the breakdown) over a 1.24e-5 median floor (the breakdown RE-SOLVES the field and rates on the written state while the heat column carries the rates the sweep produced it with). Interface proposed: heating_of_composition(...) in utils_ion_eq returning heat and a non-optional heat_chan(:,17) (channel list and header owned by the module; LW dissociation and fluorescence as two columns), called by the sweep, by write_heat_breakdown_eq (which then forms no rate) and by post_process_adv (which today omits the Balmer and associative terms from its theat). **Advisor decision:** Heating_breakdown.txt writes the channel array the sweep computed for the written state, not a recomputation; this is the "one state per output file" rule (rev3 section 10.2 item 2, landed) applied to this file, and the only form in which the breakdown total can equal the heat column to round-off. Resumes when ACCEPT-1/2 releases ionization_equilibrium.f90.
- **ACCEPT-1/2, advisor verification** (2026-09-06): private build of the live tree, energy_update 53/0 (the driver was killed by error stop 1 before), acceptance_classes 22/0 (did not compile before: class 6 did not exist), attempted_step green (10 RED before: exit status, history lines, reason breakdown), coupled_source_step 30/0, certification 29/0, run_mode 30/0, carrier_retry 97/0. Code read: the kept state of class 6 is x_entry, the composition the cell entered the sweep with (built once, also the solve's starting point), heat and cool of the returned composition as section 170 requires. Accepted, including the two deviations: the two-line steady_residual.f90 change that makes n_cells_without_chemical_root count class 6 (without it the controller would not see a refused non-root), and as_op_projection -> as_op_composition (operation number 8 unchanged; B3c had removed the projection the name described). Findings taken over: cap 1e2 MEASURED on the whole default matrix at 400 steps (the amnesty population is a cold-start population: 111 events 1e-6 to 8.853 at step 0 of oxygen_chemistry, nothing until 8.6e4, then the four cells 85-88 whose energy update failed); with the cap oxygen_chemistry runs its 1000 steps and exits 0 (log10 Mdot 9.61), no step refused, fifteen of sixteen cases byte-identical; the comment's "res ~28 in the metals-on molecular gates" is no longer reproduced (they accept no non-root today) and was corrected in place. The energy-refusal path is exercised by the default-off hook EXHALE_ENERGY_FAIL_AT_STEP. Outside scope, assigned: "class 4" wording in certification.f90/steady_newton.f90 and steady_newton.f90:815 acc_resmax(4) missing class 6 (HYG-CLASS6, folded into HEATBRK's resume); an H-only run has no acceptance classes at all (pre-existing, B6 list); oxygen_chemistry cannot run in phys mode (56 handoff non-roots refused by A0), a statement about that configuration, recorded.

- HEATBRK landed, advisor verification** (2026-09-06; measured on mol_base_handoff 200 steps, wasp_full 300, oxygen_chemistry 1000, the mol_metals impact pair still running when this was written): heating_of_composition (util_ion_eq.f90 1503-1780) is the one heating assembly, 17 named channels, the total the running sum of the columns in the sweep's original order (sweep heat bit-identical); the sweep calls it once (181 inline lines of ioniz_eq gone), write_heat_breakdown_eq writes the module array heat_channel_state the sweep filled and forms no rate, post_process_adv calls the same routine. Private build of the live tree: grid_and_gates 93/0 with heat_column_matches_breakdown_total 1.678e-16 (was 6.886e-3), physics_probe 557/0 incl. heating_channel_closure and the scanner heating_sum_uniqueness (one exception recorded: the band ledger of write_output.f90 restates the Lyman-Werner and FUV photolysis event energies, a fourth copy, assigned HYG-LEDGER-EPS). Heating_breakdown.txt 18 -> 21 columns; _adv profiles move by up to 1.8e-2 (the Balmer and associative deposits the third copy lacked). heat_channel_state is class-1 state and enters the attempted-step checkpoint through B3c-COST (owner of attempted_step.f90), assigned. HYG-CLASS6 (class 4/6 wording, acc_resmax over both classes) in the same landing.
- B3b-CO landed: the CO row carries two published destruction rates** (2026-09-06). src(ic_CO) is no longer 0. It is $-(k_{D1} n(\text{He}^+) + k_{\text{CO}}) n_{\text{CO}}$ with $k_{D1} = 1.60\text{e-}9 \text{ cm}^3 \text{ s}^{-1}$ (UMIST RATE22 entry 4068, measured, accuracy better than 25 per cent, temperature independent, READ) and k_{CO} the cell mean of $(F_{\text{LW}}/\langle h\nu \rangle) \sigma_{\text{CO}} \Theta(N_{\text{CO}}, N_{\text{H}_2}) \exp(-\tau_{\text{cont}})$ on the 912-1201 Å beam. New: co_self_shielding_table.f90 (the 12CO block of Visser, van Dishoeck & Black 2009 Table 6, $T_{\text{ex}}(\text{CO}) = 50 \text{ K}$, with their Table 5 beside it for the spread) and co_photodissociation.f90 (the point rate, the two-column cell mean, the fragment heat). fuv_lw_photon_field gains the CO column and returns k_{CO} and Θ ; carrier_photolysis rebuilds k_{CO} from the current carrier columns, so the CO layer shields itself inside the relaxation. Constants, all MEASURED: $\sigma_{\text{CO_band}} = 1.0160\text{e-}17 \text{ cm}^2$ per beam photon (Heays et al. 2017 cross section over the 912-1118 Å window CO absorbs in), $\langle h\nu \rangle = 12.8674 \text{ eV}$ (oscillator-strength weighted over Visser's 37 bands), $D0(\text{CO}) = 11.1157 \text{ eV}$ and $q(D1) = +2.2117 \text{ eV}$, both differences of the one formation-energy table; the deposit is 1.7517 eV per dissociation. Both deposits are channels 18 and 19 of heating_of_composition (17 -> 19 channels) and are formed nowhere else; heat_per_co_dissociation is registered in the uniqueness scanner with the band ledger of write_output.f90 as its one recorded exception. output/FUV_bands.txt gains N_{CO} , k_{CO} and Θ_{CO} and a CO row in the beam ledger (CO is printed apart from rated_ph because beam_loss_ph carries no CO term: MEASURED, folding it in moved the LW budget residual from -1.8e-2 to -8.2e-2). output/Oxygen_chemistry.txt gains the domain record $\tau_{\text{dest}} \leq 0.1 \tau_{\text{res}}$ with the worst ratio and its radius, $\tau_{\text{res}}/\tau_{\text{form}}$ from RATE22 8597, the count of cells above the shielding table's 512 K excitation-temperature limit and the count in which the He^+ channel outruns He^+ recombination. Tests: physics_probe co_destruction (27 assertions, RED on a mutated table entry and a mutated photon energy), fuv_band_ledger extended. **Impact, MEASURED on oxygen_chemistry at its maxsteps of 1000, the only regression case that carries CO** (a control binary of the same tree with the CO source and the two deposits reverted): both run 1000 steps, exit 0 and give $\log_{10} \dot{M} = 9.61$; the after binary costs 628 s against 11 per cent less for the control at a matched 150 steps. The $\text{He}^+ + \text{CO}$ charge-transfer deposit is **65 per cent of the total heating at $r = 1.089$** , the helium ionization front where CO still survives, and under 1

per cent outside 1.05-1.17; the layer's temperature moves by up to 46 per cent at $r = 1.023$ and its CO by up to 52 per cent. `mol_carrier` at 300 steps is byte-identical in every output file, which is the gate that the new terms are on the same `thereis_oxychem` as the rest of the oxygen path; no other regression case sets Oxygen chemistry. The domain record of that run: 401374 out-of-domain cell visits, worst `tau_dest/tau_res` $1.34e5$ at $r = 1.0002$, worst `tau_res/tau_form` 0.219, 305125 cell visits above the 512 K excitation-temperature limit and 281325 in which the He+ channel outruns He+ recombination. Goldens not refreshed. **P0 recorded:** the He+ ledger is not closed. This reaction consumes a helium ion and neither the carrier row (ion stages frozen) nor the sweep's He+ balance returns it, so where `k_D1 n_CO` exceeds the He+ loss rate both the destruction and that heating peak are over-stated. The fix is a sink in the He+ row of `mol_heh_rows` (`System_HeH_mol.f90`), not this item's file; `dom_cells_HeP` is the observable. **Not done, carried over:** the thermodynamic CO ceiling is still in `limit_to_element_budget`; deleting it changes `EXHALE_main.f90`, `certification.f90` and `write_setup_report.f90`, none of them this item's files. The exact lines are in the item's report.

- **HYG-LEDGER-EPS closed in the same landing** (2026-09-06): the FUV band ledger of `output/FUV_bands.txt` is formed by `utils_ion_eq::fuv_band_absorption_ledger`, beside the heating assembly and reading the same imports, and `write_output.f90` writes what it returns; the `heat_LW` column of `output/Lyman_Werner.txt` is now the assembly's own channel (`heat_channel_state(:,ih_H2_LW_dissoc)`) instead of a second evaluation of `k_LW n_H2 e_lw_fragment_erg`. The heating_sum_uniqueness scanner's exception list is EMPTY: every per-event energy is named only in the module that defines it and in the one assembly. RED before (the pre-move import line, two deposits reported outside the assembly), GREEN after. `FUV_bands.txt` is byte-identical across the move (MEASURED, oxygen_chemistry at 5 steps); the `heat_LW` column moves by at most $5.57e-4$ relative, which is the one-sweep lag between the state the file wrote and the state the energy equation deposited, and the new number is the deposited one.
- **B3c-COST, the cost of the coupled step measured and the skip specified, not implemented** (2026-09-06; `EXHALE_main.f90`, `attempted_step.f90`, the `coupled_source_step` and `attempted_step` suites). WHERE THE COST IS, MEASURED on `wasp_full` 300 steps, one thread: the run timers give `ioniz_eq` 83.6 per cent and the energy solve 16.3 per cent of the marching wall time, and a `gprof` build puts **78 per cent of the whole marching time inside the MINPACK solve of one cell's chemistry** (one Jacobian and one QR per cell per pass; the warm start already reaches each root in one Newton step), against about 13 per cent in the whole-grid rate and field work a pass repeats. So the cost is the number of CELL solves and not the field, and reusing the field between passes is worth at most that 13 per cent. THE CONVERGENCE STUDY CONTRADICTS THE EXPECTATION THE ITEM WAS WRITTEN FROM: after pass 2, 308 of the 500 cells are still above the tolerances, after pass 5 still 130, and the cell that sets the pass count walks inward one cell per pass with the ionization front. Contraction 0.191 median over 2729 passes. A cell-local skip is therefore worth 2.41x on the cell solves with a one-pass rule -- but that rule is WRONG: a cell at rest moves again on 177 of the 3029 passes, 690 of the 809 wake-ups at pass 8, because the column keeps moving a cell after the cell has stopped. **Two consecutive passes of rest: zero wake-ups, 1.96x on the cell solves, 1.55x on the step.** The skip has to be a mask in the cell loop of `ioniz_eq`, which this item does not own (HEATBRK was writing to that file at 20:30

and 20:35), so it is SPECIFIED and not implemented: the optional `cell_at_rest` argument, the class-6 path (`sys_x = x_entry` straight to the extraction) as the body of the skip, and the one consequence that is not a pure skip -- the acceptance ledger describes the cells the sweep SOLVED, so a mask needs an acceptance class stored for each cell and inherited by masked cells, or `n_cells_without_chemical_root` reads a different population from the one it believes. TOLERANCE ANCHORED AND KEPT AT 1e-8: the anchor the comment gave (the 1e-6 heat-column gate) stopped depending on it when HEATBRK made the two assemblies one; the anchor that binds is `cert_tol_carrier = cert_tol_element = 1e-8` on the composition this step returns. Measured 1e-10 to 1e-5 on `wasp_full 300` and `mol_base_handoff 1500`, the adopted state moves by about the tolerance and a decade costs 1.3 passes and 15 per cent of the wall time, **so the tolerance is not the cost lever**; the step's own truncation error (step-doubling, phys, 5.3e-3 relative) is five decades above the loosest value and does not bind. Newton on T with the composition's response (item 4) stays live, and the CHEAP version of it is refuted by measurement: the contraction ratio spreads by 122 per cent globally and by 200 per cent cell by cell over the tail of a step, so the sequence is geometric on average over a run and not within one step at one cell -- which is why B3c's Aitken failed, and why the acceleration has to come from $dc/dT = -(dF/dc)^{-1} (dF/dT)$ against the factorization `hybrd` already forms, not from the iterates. ALSO FIXED, a defect of my own file found outside the task: the run's cost line divided passes accumulated over every ENTRY of the coupled loop by the number of adopted STEPS, two different populations, so it could report a mean above the worst single entry (MEASURED on the control, `wasp_full 300` in phys mode: "13.98 passes per step on average, worst 11"). It now divides by the entries and names them. ALSO: `heat_channel_state` is in the attempted-step checkpoint (save/restore/compare/perturb), the four lines HEATBRK's report asked for, with two assertions RED on the control (the moved array is not seen, and the restore leaves it 3.0 off). Tests: `coupled_source_step 36/0` (33 before), `attempted_step 97/0` (95 before). Impact, control = the same tree snapshot with this item's two production files reverted: `wasp_full 300`, `mol_base_handoff 1500` and `mol_metals 1500` all **byte-identical** in `Hydro_ioniz.txt` and `Ion_species.txt`.

- **B3c-COST** (measurement; EXHALE_main.f90 comments and debug counters, `attempted_step.f90` checkpoint; numerically inert, `wasp_full 300`, `mol_base_handoff 1500`, `mol_metals 1500` byte-identical): the coupled source step's cost is not where the brief assumed. MEASURED on `wasp_full 300` (3029 passes): after pass 2 still 308 of 500 cells above tolerance, after pass 5 still 130; the pass count is set by the ionization front walking inward one cell per pass, not by a fixed handful of cells, so a cell-at-rest skip is worth 1.96x on the cell solves and 1.55-1.62x on the step, and the naive rule (rest on the last pass) is wrong: 177 of 3029 passes wake a resting cell, two consecutive passes of rest wake none. gprof: 78 percent of marching time is the MIN-PACK solve of one cell's chemistry (Jacobian + QR per cell per pass), the whole-grid field and rate work 13 percent, the energy solve 16 percent (already deactivates converged cells). Tolerance anchored and KEPT at 1e-8 (the binding anchor is `cert_tol_carrier = cert_tol_element = 1e-8` on the returned composition): a decade costs 1.3 passes and 15 percent, not the lever. Aitken and any acceleration built from the iterates fail because the contraction ratio spreads 200 percent cell by cell within a step. **The lever is a per-cell Newton on (T, composition) with $dc/dT = -(dF/dc)^{-1} dF/dT$ from the factorization `hybrd` already forms, inside the sweep (`ionization_equilibrium.f90`), plus the two-pass rest mask with an acceptance class inher-**

ited by masked cells: assigned as COST2, after B3b-CO releases that file. Also fixed: the cost line divided passes over loop entries by adopted steps (could report a mean above the worst); heat_channel_state is in the checkpoint with two assertions (attempted_step 97/0, coupled_source_step 36/0; advisor rebuild and rerun green). Outside scope recorded: EXHALE_ERR EVERY read in both modes but the probe runs only in phys; mol_base_handoff in phys mode refuses 7 of its first 20 steps at the fixed point (pass cap at step 0), recovers.

- **B3b-CO, advisor verification and follow-up** (2026-09-06): private build of the live tree, physics_probe 584/0 (incl. co_destruction 27 and the scanner heating_sum_uniqueness with an EMPTY exception list after HYG-LEDGER-EPS moved the band ledger into utils_ion_eq::fuv_band_absorption_ledger), fuv_band_ledger 24/0, grid_and_gates 93/0. Accepted with the four stated departures from the design (cross section as the mean over the 912-1201 Å beam of one that ends at 1118 Å; the same flat-band cross section for a loaded spectrum, the shape-aware integral belonging to sed_read.f90; q_D1 as a table row because mion_fsp is a parameter; CO on its own ledger row, co_frac 6.39e-2 the size of decision 2's approximation). **P0 opened by the item itself and assigned at once as B3b-CO2:** the He+ + CO channel deposits its heat (65 percent of the total at the He+ front of oxygen_chemistry, $r = 1.089$) while no He+ is consumed and no C+ or O delivered anywhere; brief plan_rev2_brief_B3b-CO2.md (He+ sink in both molecular systems with Jacobian, products to the free O and carbon pools, heat rate identical to the species rate from one place, element census). Also recorded: the thermodynamic CO ceiling is still present because its deletion touches EXHALE_main.f90 34, 3641-3648, certification.f90 84, 1072-1076 (n_active_unvalidated_physics loses its only producer), write_setup_report.f90 18, 925-934 (CEILING-DEL, after B4-1 releases main); heat is deposited for carrier reactions whose products a never-covered carrier interval does not deliver (oxygen_chemistry in init mode never covers one), an operator-split defect that B4-1c and B5 address and that leaves the oxygen case without an end-to-end gate for its CO source; heat_channel_state now has 19 columns (checkpointed by B3c-COST); python/paper_data.py HEAT_FIXED stale.
- **B3b-CO2 landed: the He+ + CO channel now consumes the helium ion** (2026-09-06). Row (2) of mol_heh_rows (System_HeH_mol.f90) loses $k_{D1} \ n_{CO} \ n(He^+)$ with $k_{D1} = rk_{D1_Hep_CO}() = 1.60e-9 \text{ cm}^3 \text{ s}^{-1}$, and its turnover scale (set_mol_turnover_rates) gains $k_{D1} \ n_{CO} \ n_{He}$; the helium leaves neutral, so the free neutral He that closes the helium budget receives it and no other row moves. One row serves every path: System_HeH_mol_metals and the constrained-equilibrium residual both call it, and neither molecular system has an analytic Jacobian to extend (hybrd1, finite differences), so the test asserts the difference derivative the solver actually builds. CO is a background density of that row and not an unknown, because the one-sided model gives it no formation and therefore no local equilibrium; the reaction is split across the two operators, each reading the other's frozen value from the one rate, which makes the heat of channel 18 the heat of exactly the events the species ledger performs. The products needed nothing added: the free-atomic-oxygen closure of carrier_source and $n_{C_free} - n_{CO}$ in carrier_write_back already carry the two nuclei and element_census already counts them in CO. Tests: new physics_probe driver co_helium_ion_sink (8 assertions, RED 4 FAIL against the pre-change module relinked into the same suite, GREEN 8 PASS; the RED reads hep_row_co_sink 0.0 against -3.84e6, hep_row_derivative_co_term 0.0

against -38.4, hep_turnover_scale_co_term 0.0 against 3.84e13); physics_probe suite PASSED after; element census with EXHALE_ELEMENT_ASSERT=1 on oxygen_chemistry 150 steps exits 0 with no budget violation (a guard, not a RED/GREEN pair: the change moves no C or O nucleus). Impact, MEASURED at 1000 steps of oxygen_chemistry: the largest share of the total heating taken by channel 18 falls from 0.672 at $r = 1.081$ to 0.035 at $r = 1.117$, channel 18 at $r = 1.0894$ from $6.90e-7$ to $9.75e-9$ erg cm⁻³ s⁻¹, $n(\text{He}^+)$ there to 0.015 of its former value and to 0.0073 at $r = 1.011$, T at $r = 1.0233$ from 1156.4 K to 1088.2 K, and $\log_{10} \text{Mdot}$ stays 9.61 with exit 0. dom_cells_HeP does NOT fall (281325 -> 281382): the counter compares $k_{D1} n_{CO}$ with recombination and still says this reaction is the leading He+ loss of the layer, so what it now reports is the operator split's lag rather than a missing sink. mol_carrier 300 steps and mol_metals 400 steps byte-identical in every output file. Goldens not refreshed. Reported for owners elsewhere: the dom_cells_HeP header text in write_output.f90 558-562 still says the frozen He+ "over-states that channel"; the mirror decomposition hydrogen_helium_row_terms in constrained_chemical_equilibrium.f90 needs the same term after line 1859 or it stops summing to row 2; the n_co comment in ion_cell_state.f90 still calls CO an equilibrium closure.

- B3b-CO2 landed, advisor verification and follow-ups** (2026-09-06): the He+ row of mol_heh_rows (System_HeH_mol.f90, the one row every caller uses) loses $k_{D1} n_{CO} n(\text{He}^+)$, its turnover scale gains the same bound; CO stays a background density of the solve because the one-sided model gives it no formation channel (not a balance row). Products needed no code: the O nucleus goes to the free-oxygen closure and the C nucleus to $n_{C_free} - n_{CO}$ in the carrier write-back, element_census already counts both; the carbon CHARGE is set by the sweep's stage equilibrium, the design of record ("the step moves nuclei, the sweep moves charge"), so the reaction's C+ is not imposed (recorded as a design choice, not a defect). MEASURED, oxygen_chemistry 1000 steps: channel-18 share of the total heating 0.672 at $r = 1.081$ -> 0.035 at $r = 1.117$; $n(\text{He}^+)$ at the front 0.015 of before (0.0073 at $r = 1.011$); T at $r = 1.0233$ 1156 -> 1088 K; $\log_{10} \text{Mdot}$ 9.61 both; constrained-continuation fallbacks 932 -> 1880, all accepted; mol_carrier 300 and mol_metals 400 byte-identical. dom_cells_HeP did not fall (281325 -> 281382): it measures dominance of the channel over recombination, not over-statement; its header text now says so. Advisor follow-ups applied directly: the $D1 \text{ He}^+ + \text{CO}$ term added to the He+ decomposition of constrained_chemical_equilibrium.f90 (hydrogen_helium_row_terms, whose header requires every mol_heh_rows reaction to appear), the FUV_bands.txt/Oxygen_chemistry.txt dom_cells_HeP header rewritten, the stale n_co comment of ion_cell_state.f90 replaced. Private build: physics_probe 592/0 (incl. co_helium_ion_sink 8), fuv_band_ledger 24/0. Remaining: the operator split is now a LAG (the sweep destroys He+ at the entry CO while the CO row destroys CO at the previous sweep's He+) and oxygen_chemistry never covers a carrier interval in init mode, so this row has no end-to-end gate: B4-1c/B5 territory, recorded.
- HEATBRK closed** (2026-09-06): the mol_metals pair finished at its 12000 steps, Hydro_ioniz.txt and Ion_species.txt byte-identical, gate $1.683e-16$ with 21 columns, _adv profiles moved by at most $8.45e-3$ (the deposits the post-process copy lacked). All three impact cases of the item therefore confirm: the sweep's heat is unchanged, only the two consumers of the assembly moved.
- B4-1** (species_face_flux.f90 new; Reconstruction.f90, PLM_rec.f90, RK_rhs.f90, EXHALE_main.f90,

binary_element_diffusion.f90, Makefile, new suite src/tests/species_face_flux/): the advective half of the ELEMENT transport now rides on the hydrodynamic face mass flux, inside the Runge-Kutta stages, on the same faces, areas, volumes and time step as the mass row. The face composition is the reconstruction of the mass fraction by the same limiter as the primitive variables, taken from the side the face mass flux selects, and the closing member of the set is one minus the others, so $\sum_s F_s(j) = F_{\rho}(j)$ holds by construction; the operator asserts it at every face of every stage (largest residual MEASURED in the unit suite: $2.1\text{e-}16$ relative, against a tolerance of 64 ulp). The cell-velocity upwind advection of binary_element_diffusion.f90 is gone from the marching path: the operator advects only when it is given an advecting mass flux, which the fixed-wind relaxation supplies and the marching path does not, and the trace-metal solve lost its advective term the same way. The carriers of photochemical_transport_step are NOT in this increment (that file was held by another item): they keep their present advective term and are B4-1c. Tests, all MEASURED with a private build: a uniform composition is preserved to 2 ulp on a compressing flow, an expanding flow and one whose face mass fluxes alternate in sign, for PLM and for WENO3; the mass fraction stays in $[0,1]$ on a step profile carried across the domain in 300 steps; the closing member of the set differs from an independently reconstructed $1 - Y$ by $1.1\text{e-}16$; the face reconstruction converges at order 1.97 (PLM) and 1.92 (WENO3); the helium nucleus total of the domain changes only by its boundary fluxes to $4.9\text{e-}17$, where the cell-velocity form it replaces leaves $1.6\text{e-}9$ on the same column and the same step (RED and GREEN in one run); the injected violation of the face identity stops the operator with the face named; the new step is refusable at its injection point and the run goes on. The element-diffusion suite diffusion_tests is 35 of 35 after T11 was rewritten onto the new operator (its premise, that a face-averaged upwind decouples a cell whose face velocities straddle zero, is what the face-flux form removes). Impact, MEASURED on scratch copies, OMP_NUM_THREADS=1, mol_*/hp_* at 200 steps and mol_diffusion/lower_profile at 1000 rather than the 12000 their maxsteps pins, wasp_* at 300: 14 cases byte-identical, every one of them without element transport (wasp_full, wasp_he23off, wasp_full_newton, mol_base_handoff, mol_metals, mol_lyman_werner, mol_ir_bands, mol_sec_ion, mol_carrier, oxygen_chemistry, hp_zero_seed, hp_trace_seed, hp_front, hydrostatic_column); mol_diffusion moves by at most $7.1\text{e-}5$ in Ion_species.txt and $3.6\text{e-}3$ in Cooling_breakdown.txt, lower_profile by $1.2\text{e-}3$ and $1.1\text{e-}2$. The transported quantity, $n_{\text{He}}/n_{\text{H}}$, moves by $1.9\text{e-}10$ at the base and $3.5\text{e-}6$ at the molecular front of mol_diffusion; the base-layer mass flux $\rho v r^2$ moves by at most $9.2\text{e-}7$ over $1.00 \leq r \leq 1.16$. Those are smaller than the design's estimate because at 1000 steps neither case has separated its elements far from the reservoir ratio, and a uniform composition is what both forms carry exactly; a converged pair is what would measure the difference at the separation those cases reach, and is named as open in docs/b4_spatial_operator_design_20260906.md. No golden was refreshed.

- **B4-1 landed (element rows), advisor verification** (2026-09-06): new module species_advective_transport (species_face_flux.f90): the face mass fraction by the primitives' own limiter (PLM_rec_scalar, Reconstruct_scalar, the MC limiter and the WENO3 geometry coefficients factored, not duplicated), upwind side = sign of F_{ρ} (coincides with the HLLC contact selection wherever both are defined, ROE/LLF offer only the sign), the closing member one minus the others so $\sum_s F_s = F_{\rho}$ by construction and asserted at every face and stage, advanced inside the three SSP-RK3 stages with the mass row's dt , dV and areas;

positivity_limited_fluxes writes a replaced first-order flux back into face_flux so the species flux built on it is the repaired one. Transported set: helium and the hydrogen component (normalized), the trace metals riding on the same faces inside m_1 at the reservoir abundance (trace limit stated). binary_element_diffusion.f90 keeps the diffusive half only (element_transport_residual keeps its stationary advective term, equal where $r^2 \rho v$ is constant). Carriers NOT migrated (diffusive_photochemistry.f90 was B3b-CO's): B4-1c. Suite species_face_flux 13 + 2 whole-binary rows GREEN on the advisor's private build: uniform composition preserved to 2 ulp (PLM, WENO3, three flows incl. a sign-alternating face flux), face identity 2.1e-16, bounds exact, face reconstruction order 1.97/1.92, helium nucleus total changes only by the boundary flux to 4.9e-17 where the replaced form gave 1.58e-9 (B1 AT-4d RED/GREEN in one run), identity assertion fires, element step refusable; diffusion_tests 35/35 (T11 rewritten: its premise was the defect removed); attempted_step, grid_and_gates 93/0, carrier_retry 97/0 green. Impact (worker, control = same snapshot with the stage calls removed): 14 cases without element transport BYTE-IDENTICAL (wasp_* at 300, mol_* at 200 steps); mol_diffusion and lower_profile at 1000 steps move by at most 3.6e-3 / 1.1e-2 (cooling breakdown), n_He/n_H by $\leq 6e-5$, base $\rho v r^2$ by $\leq 1e-6$: far below the design's estimate because at 1000 steps the elements have not separated from the reservoir ratio; the converged 12000-step pair is open in the design document. Outside scope recorded: ESWENO3 measures second order on a smooth profile (1.92 against PLM 1.97, face error 8x PLM), a property of the primitive reconstruction to look at (WENO3-ORDER); hp_* cases need output/Hydro_ioniz_IC.txt in place; element_flux_profile.txt has a trailing text column.

- **CEILING-DEL** (diffusive_photochemistry.f90, write_output.f90, write_setup_report.f90, EXHALE_main.f90, certification.f90; tests carrier_retry (+ scanner no_CO_ceiling_identifier_in_src), carrier_constraint_attribution rewritten on the carbon clamp): the thermodynamic CO ceiling (clamp to co_equilibrium_density, four counters, attempted twins, three accessors, Oxygen_chemistry.txt header line, three co_ceiling_* resolved keys, the end-of-run line, validity state 4.1's only producer) is deleted per design decision 10; the domain record of B3b-CO stands in every place (seven co_domain_* resolved keys under oxygen chemistry + carrier transport, the CO items under class 4.2 beside H3+ in the certification, n_active_unvalidated_physics = -1 "not produced", kept because the five validity states are a fixed set). MEASURED before deletion on oxygen_chemistry 1000 steps: co_ceiling_cells 0, applications 0, CO removed 0.0 (with the published destruction rates in the row and the He+ sink in the sweep the carried CO never exceeds its equilibrium), so every numerical column is byte-identical before/after; mol_carrier 300, mol_metals 400, wasp_full 300 byte-identical. The design's "of order 160 cells" predates the CO rates and is retired with the clamp. Advisor verification: private build, carrier_retry, certification 29/0, carrier_constraint_attribution 11/11, fuv_band_ledger green, no co_ceiling identifier left in src/. Recorded for the docs sweep (DOCS-CEIL): the checkpoint enumerations and validity category 12 in a0_run_mode_contract, a2_certification_contract, b1a_active_equation_inventory, b3a_attempted_step_controller_design, b1_target_system (1246-1554), development_plan_20260905_rev2/rev3/review3, physics_numerics_audit_20260905, and b3b_co_destruction_design sections 2.3/4/7 still describe the ceiling as present. The domain record is not checkpointed (a statement about the state, valid whether or not the step is accepted), unlike the ceiling record it

replaces.

- **DOCS-CEIL** (documentation only, 2026-09-07): sixteen documents brought level with CEILING-DEL, the current state READ from the code (destruction row $\text{src}(\text{ic_CO}) = -(\text{k_D1} \text{ n}(\text{He}+) + \text{k_CO}) \text{ n_CO}$, conservation-only `limit_to_element_budget`, `carrier_co_domain_record`, certification class 4.2 CO block, `n_active_unvalidated_physics` = -1, seven `co_domain_*` keys); no measurement deleted, the superseded ones ("of order 160 cells"; "still cuts CO above 3000 K") kept with the measurement that replaced them beside them. Files: a0/a2 contracts, b1a inventory (4.1 row retired, 4.2 row the domain record, open question 3 closed), b3a design (checkpoint enumerations), b1 target system (5.1 split into the 2026-09-06 READ and the current one, AT-5 superseded, decision 12 moot), b3b_co design (sections 2.3, 3(iv), 4, 6, 7, 7.4 as done), development plan rev2/rev3/ review3/execution, `physics_numerics_audit` (P6 resolved), `code_status`, d0 (C26 closed, question 5 answered), `co_destruction_rates_literature` (banner), `To_be_determined_recommend`, PLAN rev2. Left as dated records: the superseded plan revisions and external reviews, and the b1 draft. Recorded: line-number citations in the 2026-09-06 design documents have drifted by up to a thousand lines (a re-anchoring sweep on routine names is a separate item, DOCS-LINES); the stale checkpoint header comment in `diffusive_photochemistry.f90` goes to B4-1c.
- **WENO3-ORDER, measurement and fix** (2026-09-07): the ESWENO3 reconstruction was SECOND order on every grid the code runs on. Probe `weno3_reconstruction_order` (`physics_probe`; exact volume averages by 8-point Gauss quadrature, face error against the point value, $N = 50$ to 800): production rate 1.997 (uniform in r), 1.988 (the `define_grid` Mixed production grid), 2.940 on a grid uniform in the volume coordinate, where the defect is invisible. Cause, isolated by variants: not the ESWENO nonlinear weights ($S = 1$ gives the same 1.997), not the floor $\text{eps} = \text{dr_j}^2$ (rate 3.000 at κ 0.1 to 10 once fixed), not the ideal ratios $D1, D2$ ($1/2 + O(h)$, harmless); TWO of the four stencil coefficients applied the NEIGHBOUR's volume share where cell j 's own belongs: $C1(j) = c/(b+c)$ on the jump to $j+1$ at the right face and $C2(j-1) = a/(a+b)$ on the jump to $j-1$ at the left face, each $1/2 + O(h)$ where $1/2 - O(h)$ belongs, an $O(h)$ coefficient error on an $O(h)$ jump. Inherited character for character from ATES v2.0 (ATES/ATES-Code-main). Advisor check of the derivation: a linear interpolant of two cell averages in the volume coordinate, evaluated at a face $dV(j)/2$ from cell j 's centre, weights the jump by $dV(j)/(dV(j) + dV(nb))$; with the corrected shares a profile quadratic in the volume coordinate is reproduced to $5.9e-16$. **Fixed (advisor, two symbols in each of Reconstruct and Reconstruct_scalar, Reconstruction.f90):** $C1(j) \rightarrow C2(j)$ on dWp in WL, $C2(j-1) \rightarrow C1(j-1)$ on dWm in WR; the probe's production mapping and header updated, the two waiting assertions enabled: `weno3_production_third_order` rate 3.000, `_mixed` 2.991 (RED before: shortfalls 0.803, 0.812); `physics_probe` 603/0. **Every golden whose case reaches the WENO3 stage moves, and not by rounding** (face error down two decades at $N = 800$); PLM-only cases are untouched. Consequence for the method: until now the two-stage PLM $\rightarrow$ WENO3 recipe was a continuation between two second-order schemes with the second having the larger error constant (1.3x uniform, 3.8x on the production grid). Mdot values are to be quoted from Newton-finished runs after the gate. Attribution of the weight form and the floor to Yamaleev and Carpenter (2009, JCP 228, 3025 and 4248) stays unverified: both papers are paywalled here; requested from the user. Also recorded: stale `.ipynb_checkpoints` copies of

Apply_BC.f90 and UW_conversions.f90 under src/modules/ define the same module names as the live files (not built; to remove, HYG-CKPT).

- **HYG-CKPT** (advisor, 2026-09-07): ten untracked, gitignored Jupyter checkpoint copies of production modules under src/modules/*/ .ipynb_checkpoints/ (Apply_BC, Reconstruction, Source, base_boundary, caloric_eos, low_mach_dissipation, cross_sec, UW_conversions, h2_photo_channels, element_census) removed: not built, but each defined the same module name as a live file and two scanners had to skip that directory (batch 2c) to stay green.
- **T15-INSTRUMENT** (attempted_step.f90 +289/-165, suite, b3a design section 9, b1 T1.5 wording; advisor verification: private build, attempted_step 43/0 with the two whole-binary rows skipping until the main-loop hook lands, coupled_source_step, certification green): the B3a instrument now states the THERMAL identity of the step, $\sum_j dV_j [u_{th}^{n+1} - u_{th}^n] = \sum_j dV_j dt (\text{heat} - \text{cool})_j + (\text{transport contribution})$, the transport contribution MEASURED between two marks on either side of the coupled source step, the residual the closure of the source step alone, Delta u_{form} reported as the reservoir's change and never added (the B3c reason stated at the routine). GATE in phys mode at the energy row's own energy_res_tol = 1e-9 (reason as_reject_source_energy = 14 at operation 9): MEASURED on a hooked scratch tree, worst lresidual/scale 7.44e-1 -> 5.40e-10 (wasp_full 300 phys), 8.63e-1 -> 1.88e-10 (mol_base_handoff), zero refusals, states byte-identical in both modes. The two hook lines in EXHALE_main.f90 (before/after the coupled_source: loop) are assigned to COST2, the file's current owner, with the rewording of the print block. **Defect found and fixed in passing:** attempted_step_checkpoint_take saved u and f_sp through a saver that reallocates from index 1, so a reader indexing the saved grid arrays by cell compared cell j with cell j - Ng; energy_identity was that reader, so every d_u_th/d_u_form printed so far (incl. the 28 percent reservoir share of the B3c report: with bounds fixed, same step, d_u_th 2.006e-5, d_u_form 1.298e-5) was formed against a state shifted by two cells; the state restore itself was unaffected (whole-array assignment). The second private copy of the eps_s table in attempted_step.f90 (no OH/H2O/CO rows) is deleted in favor of species_formation_energy. Recorded: n_projection_applied can no longer be nonzero since B3c but is still printed by the report and refused by the certification (HYG-PROJ, spans global_parameters, certification.f90, main); design section 9 decision 8 still names the projection as a live correction (dated record).
- **COST2** (measurement; the only production change is the T15 hook and print in EXHALE_main.f90, byte-identical on wasp_full 300 in both modes and mol_base_handoff 60 phys; attempted_step whole-binary rows now run, identity residual -1.8e-10, zero refusals): the brief's premise is refuted twice. (1) MEASURED with T frozen through the coupled loop on wasp_full: the composition sweep alone still takes 10-13 passes with the same contraction 0.15-0.20 and the same front walking one cell per pass; freezing the lagged coefficients as well collapses it to 2 passes; freezing only the radiation field leaves 3-4, only the rate coefficients 9-13. The pass count is the self-consistency of the photoionization field with the composition that attenuates it; the temperature is worth at most one pass. (2) A timer inside ioniz_eq (60 steps, 29.95 s): PH_heat 11.40 s (38 percent of the run), the whole cell loop 7.37 s (25 percent), pre-sweep eval_cool 2.44 s, post-sweep assembly 2.40 s; skipping

the entire cell body for resting cells saves 4.7 percent. B3c-COST's gprof attribution (78 percent in MINPACK) was inverted; the `csd_max_pass` comment still carries it (to be rewritten by COST3). (3) The two-pass rest mask was implemented, measured (6.0 percent on `wasp_full`, 0 on `mol_base_handoff`, state moved $5.5e-7$, step 0 of `mol_base_handoff` in `phys` turned into an exhausted attempt) and removed. **The lever: the field dependence is one-directional (a cell's `P_HI` is set by the attenuation of the cells outside it) and the sweep already runs outside in, so updating the column with the composition just produced turns the Jacobi iteration into a forward substitution: same fixed point, about one pass instead of ten, and `PH_heat` no longer repeated per pass.** Assigned as COST3 (`util_ion_eq.f90` `PH_heat_HHe/PH_heat_H`, `ionization_equilibrium.f90` sweep order; the OpenMP parallel cell loop becomes sequential in the field direction, so the design must weigh 10 parallel passes against 1-2 sequential ones and measure both at 1 and 16 threads). Also recorded: the pre-sweep `eval_cool` is run for its coefficients and its cooling discarded, then a second traversal forms the cooling (16 percent together); `mol_base_handoff` in `phys` refuses 7 of its first 60 steps at the fixed point on the unmodified tree.

- **B4-1c landed: the carriers ride on the species face flux** (2026-09-07): the carriers of `photochemical_transport_step` (`H2` always, `OH/H2O/CO` under Oxygen chemistry, `H+` under Ionization transport) join the transported set of B4-1 and are advected as `F_rho Y_c/m_c` inside the Runge-Kutta stages. `carrier_set_init` declares them once the keys are parsed (`advected_carrier_register` in `binary_element_diffusion.f90`, which also holds the columns and hands them back through `advected_carrier_fractions`); `EXHALE_main.f90` is untouched, the existing three calls carry both sets. The carriers ride OUTSIDE the normalized set (`n_norm = 0`): a carrier is one species inside an element, the element is already the closing member, and normalizing it beside the element would count that mass twice; what keeps carrier and element consistent is the operator's own write-back, which restores the element totals cell by cell. Their rows are advanced from cell 1 (the elements' start at 2), with the inflow composition taken from the inner ghosts: the handoff's `q_H2_base` where one is stated, the base cell's own partition where none is, one rule for both branches of decision D6 (B4-4a). `diffusive_photochemistry.f90` loses the cell-velocity advective term, `carrier_advection_correction` and the base Dirichlet branch from its MARCHING rows and their Jacobian; two paths keep the advective term, `carrier_steady_residual` (the stationary balance is the whole equation) and the fixed-wind relaxation `relax_photochemical_composition` (it takes many transport steps with no hydrodynamic stage between them, so nothing else is taking that transport), through the switch `carrier_rows_advect`. Tests: two rows added to `species_face_flux`, `uniform_carrier_partition_preserved` 0.83 ulp (bound 8) on a sign-alternating face mass flux and `h2_nucleus_total_changes_only_by_boundary_flux` $1.36e-15$ (bound $1e-12$), RED before in the strongest sense (the driver does not compile against the pre-increment module); suite 15 + 2 GREEN; `carrier_retry` 101, `carrier_returned_state_acceptance` 37, `carrier_constraint_attribution` 11, `carrier_reference_scales` 15, `acceptance_classes` 22 GREEN with no reference re-derived. Impact MEASURED against a binary built from the same tree snapshot with the two files at their pre-increment text: 12 cases BYTE-IDENTICAL, every one without Molecular carrier transport (`wasp_full`, `wasp_he23off`, `wasp_full_newton`, `mol_base_handoff`, `mol_metals`, `mol_lyman_werner`, `mol_ir_bands`, `mol_sec_ion`,

mol_diffusion, lower_profile, oxygen_chemistry, hydrostatic_column); the four that do transport carriers move. mol_carrier at 1000 steps: Ion_species 5.0e-1 (H3+ at $r = 1.100$), Cooling_breakdown 6.2e-1, Hydro_ioniz 2.6e-2 (v at $r = 1.186$); hp_zero_seed/hp_trace_seed/hp_front at 100 steps: 8e-3 to 1.1e-1, Hydro_ioniz 1.3e-3 to 1.8e-3. **The front does not move; its tail does.** The $x_{H2} = 0.5$ crossing of mol_carrier is $r = 1.0698$ at 1000 steps before and after and $x_{H2}(1.05)$ is 0.7308 against 0.7310, while $x_{H2}(1.10)$ falls from 3.19e-3 to 2.02e-3 (-37%): the molecular tail the cell-velocity form was carrying by its own numerical diffusivity, which is where the design said the difference lives, and every large entry above is a consequence of that column. Every carrier interval covered and the element limiter moved no cell, in both runs. Recorded and not fixed: oxygen_chemistry has carrier_transport on by default with the oxygen chemistry, but at 200 steps in initialization mode 201 of 201 intervals are uncovered before AND after, so the write-back never runs and the case is byte-identical for that reason, not for an absence of effect (its rows did change: worst CO row of step 1 |res| 6.03e-5 against 5.68e-5). No golden was refreshed. Also in this increment: the stale clause saying the carrier checkpoint restores "the cumulative CO record" removed from two comments in diffusive_photochemistry.f90 (the carrier_checkpoint type carries no such field since CEILING-DEL).

- B4-1c landed, advisor verification** (2026-09-07; the worker's own entry carries the numbers): every carrier of the photochemical transport operator (H2; OH/H2O/CO under oxygen chemistry; H+ under ionization transport) rides on $F_{rho} \ Y_c / m_c$ inside the RK stages through the registry of binary_element_diffusion.f90 (carriers outside the normalized set, from cell 1, base inflow composition = the inner ghosts'), species_face_flux.f90 unchanged (species-agnostic), EXHALE_main.f90 untouched; the marching rows of diffusive_photochemistry.f90 keep diffusion, drift and chemistry only; the advective term stays in carrier_steady_residual (the whole stationary equation) and in the fixed-wind relaxation relax_photochemical_composition (no hydro stage takes it there), switch carrier_rows_advect. Advisor: private build, species_face_flux 15 + 2 rows (carrier partition 0.83 ulp, H2 nucleus budget 1.4e-15), carrier_retry, carrier_returned_state_acceptance, acceptance_classes, grid_and_gates green. Impact (worker, same-snapshot control): 12 cases without carrier transport BYTE-IDENTICAL; mol_carrier 1000 steps: the H2 front does not move ($x_{H2} = 0.5$ at $r = 1.0698$ both), the tail the cell-velocity form carried by its numerical diffusivity does ($x_{H2}(1.10)$ 3.19e-3 $\rightarrow$ 2.02e-3, -37 percent), hence H3+ 0.50, cooling 0.62, heating 0.37 relative there; hp_* 100 steps $\leq 1.7e-2$; base rho $v r^2 \leq 9.7e-4$. Recorded: element_advection_* names are now too narrow (HYG-NAME, six call sites in main, after COST3); oxygen_chemistry in init mode covers 0 of 201 carrier intervals so it cannot judge carrier transport; a refused carrier interval discards a transport the mass row already took (B3a/B5 question); adv_corr still checkpointed though only the residual writes it; 52 documents under docs/ carry em-dashes (DOCS-DASH, launched).
- DOCS-DASH** (documentation only, 2026-09-07): 1354 em-dashes and 237 en-dashes removed from 60 files (docs/*.md, READMEs, test and data READMEs), each replaced by the punctuation the sentence needs (colon, comma, parentheses, or a table placeholder hyphen), en-dash ranges and joined names to the hyphen the repository already writes; 108 mechanical substitutions corrected by hand on read-back; grep over the 187-file set clean. Left: Update_EXHALE_stage1.md (append-only log, 319/59), and 1220 LaTeX --- in 27 .tex sources

that render as the same dash (DOCS-DASH2, launched: the .md/.tex twins realign there; the built PDFs are re-generated at the gate's docs pass), plus docs/lower_atmosphere_figs/README.md (11) and three READMEs under backup/ (5) outside the first sweep's paths.

- **DOCS-DASH2** (documentation only, 2026-09-07): 1283 LaTeX --- in 29 .tex sources (docs, paper, posters) and 30 spaced -- replaced by the punctuation each sentence needs; four exceptions kept and named (a direct quotation of Frelikh and Murray-Clay 2026 compared letter for letter, the AASTeX \keywords separator, a literal metals.inp listing, hyphen rules); the four READMEs of the first sweep's remainder cleared. The 24 built docs/*.pdf, literature_survey.pdf, paper/ms.pdf and the poster PDFs are now dated and are regenerated in the gate's docs pass. Left: the append-only stage logs (Update_EXHALE_stage1.tex 1614, stage0.tex 37).
- **COST3** (measurement; util_ion_eq.f90, ionization_equilibrium.f90; default output BYTE-IDENTICAL on wasp_full 300, mol_base_handoff 1500, mol_metals 1500, hydrostatic_column 300 at one thread; advisor rebuild: coupled_source_step 36, physics_probe 862/0 incl. the new photoionization_field_substitution driver (259 assertions), grid_and_gates 93/0): the star-ward columns can now be carried forward cell block by cell block (advance_starward_columns, reproducing calc_column_dens* bitwise; photoionization_field_at_cell*, the PH_heat* cell bodies extracted verbatim; xuv_field_block_cells, default 0 = whole grid). MEASURED on wasp_full 300: the substitution stops the front's walk (worst-moving cell 206 -> 177 over the passes on the lagged scheme, 202 -> 196 with the columns carried) and cuts the pass count 10.10 -> 7.13, but a pass costs the same field work either way, and the field of a whole grid is one loop sixteen threads solve at once: block width 8 buys 1.43x at one thread (136 -> 103 s) and LOSES 2.8x at sixteen (17.0 -> 42.5 s); width 64 keeps the parallel front and saves nothing. The default therefore stays the whole grid, and the width is not tied to the thread count (the widths reach states differing by 1.5e-7 at the 1e-8 tolerance, the thread count by 7e-14). **The remaining lag is the cell's OWN optical depth**, which enters its own field only on the next pass (contraction 0.17): closing it means the cell residual carries its own field, System_HeH*, i.e. B5. Folded: the pre-sweep eval_cool run for its coefficients only is now chemical_rate_coefficients (-3.6 percent wasp_full, -5.6 percent mol_metals, +3.5 percent mol_base_handoff, the last from the 27 metal slots looped with metals off: guard on thereis_metals, HYG-COL). Recorded: docs/openmp_parallelization.md (2.07x at 16 threads, bit-identical across threads) is stale: measured 8.2x and round-off 6.6e-14 (DOCS-OMP). A cp -irp of another item blocked on an interactive prompt since 2026-09-06 was killed by PID (23818, cwd checked).
- **WENO3-REF** (comments only, Reconstruction.f90 and the probe's References block; the user supplied the publisher PDFs, read with pdftotext -layout): the nonlinear weights are EXACTLY Yamaleev and Carpenter 2009, JCP 228, 3025, Eqs. (18), (20), (21), (22) (published tau already squared; $d_0 = 2/3$, $d_1 = 1/3$ entering as the ratio 1/2 that D1, D2 carry); the scheme is NOT their ESWENO: the energy-stability dissipation of Eq. (48) is absent, their analysis is a uniform-grid finite-difference flux reconstruction, and for systems only in characteristic variables (Section 5 quoted), so the code inherits the accuracy result (Eq. 24) and not the energy estimate; the geometry (C1, C2, D1, D2 on the stretched finite-volume grid) is this code's own. $\epsilon_s = dr_j^2$: a defensible stand-in, not a defect: the published h^2 scaling of Eqs. (62), (64) without the solution scale of Eq.

(65); MEASURED on the production grid the face error is flat to every digit for $\kappa \geq 0.1$ over four decades (rate 2.991) and deteriorates below $1e-2$ per Eq. (57), so in smooth regions the departure is not measurable; where it acts is the ENO biasing (b_r/ϵ spans $2.2e-6$ to $8.4e3$ over the `wasp_full` grid; the published floor would make the base LESS nonlinear), a candidate experiment for the base sound-wave work, not a change here. Case label and comments now say "third-order WENO"; identifiers unchanged. Advisor: `paper/ms.tex` 144 "the energy-stable ESWENO3" reworded to the verified attribution and the ESWENO3 name replaced by WENO3 there and in `docs/code_comparison.tex`. **Recorded for the user:** every `src/tests/*/` suite of this series (19 directories) and the 2026-09-05/06 design and plan documents are untracked in git; a fresh clone of the remote carries none of them.

- **References received** (2026-09-07, from the user, publisher PDFs in `references/`): Zhang and Shu 2010 (JCP 229, 8918), Poinot and Lele 1992 (JCP 101, 104), Thompson 1987 (JCP 68, 1) and 1990 (JCP 89, 439), Neufeld 1990 (ApJ 350, 216), Mao and Kaastra 2016 (A&A 587, A84), Verner et al. 1996 (ApJ 465, 487), Yelle 2004 (Icarus 170, 167), Larsson et al. 2008 (CPL 462, 145), Tully 1975 (JCP 62, 1893), Slinger and Black 1982 (JCP 77, 2432). Five verification items launched against them: REF-ZHANGSHU (the positivity limiter), REF-LODI (the characteristic boundary design document), REF-NEUFELD (the Ly-alpha escape-probability closure), REF-METALS (every metal photoionization and recombination fit parameter against the published tables, a transcription error being a physics fix), REF-CHEM (the H3+ network transcriptions, the H3+ dissociative recombination coefficient, O6 and the Ly-alpha H2O yields). The series gate waits for them, since REF-METALS and REF-CHEM may move coefficients.
- **Reference data fetched** (2026-09-07, at the user's direction, each with a provenance README, third-party material unchanged): `references/verner_photo/` (D. A. Verner's distribution: `phfit2.f` reference routine with its fit parameters, `table1.dat` of Verner et al. 1996, the 1995 inner-shell fits), `references/mao2016_cds/` (CDS J/A+A/587/A84 `rr_spex3.dat`, 39696 rows), `references/badnell_rr/` and `badnell_dr/` (Strathclyde TAMOC total RR and DR fit coefficients `clist_K/clist_eV`, tabulated `cfout_*`, reference routines). REF-METALS now compares every metal stage's photoionization, RR and DR coefficients against these by direct evaluation of the reference routines, and reports which source each stage uses today and any missing DR contribution.
- **HYG-MAIN** (2026-09-07; numerically inert, `wasp_full` 300, `mol_carrier` 200, `mol_diffusion` 200, `mol_base_handoff` 300 byte-identical; advisor rebuild: certification, `attempted_step`, `species_face_flux` green): `HYG-NAME` `element_advection_*` -> `species_advection_*` in `binary_element_diffusion.f90`, the main program (use list, six calls) and the two test drivers, the two design documents that still named them (advisor); `HYG-PROJ` `n_projection_applied` deleted from `global_parameters`, the `attempted_step_report` block and both `certification.f90` references (MEASURED: every reference in `src/` was a read, no assignment anywhere, so the counter was provably zero since B3c), `rep%n_unbudgeted_accepted_correction` is now `n_shapiro_applied` alone, b3a design decision 8 dated closed for row 8; `HYG-COL` the 27-slot metal column loop of `advance_starward_columns` guarded on `thereis_metals` (inert; COST3's +3.5 percent NOT reproduced: guarded 53.7/52.8 s against

unguarded 55.7/52.5 s on mol_base_handoff 300, within the run-to-run spread, kept as dead-work removal); DOCS-OMP docs/openmp_parallelization.md dated and brought to COST3's measured state (8.2x at 16 threads, 6.6e-14 between thread counts). Recorded: no reduction on a physical quantity exists in src/, so the 6.6e-14 thread-count difference comes from block boundaries set by omp_get_max_threads() in chemical_rate_coefficients/eval_cool; which routine turns a moved boundary into a moved last bit is not localized (THREAD-DET: until it is, the thread count is a silent input to every result at round-off); photoionization_field_at_cell_HHe makes an array temporary for nm_j on every call (runtime warning in physics_probe); EXHALE_main.f90:452 is 141 characters.

- **REF-ZHANGSHU** (comments, docs/p49_front_energy_mode.md 6.1, probe positivity_limiter_scaling 15 assertions; user-supplied PDF read): the citation was wrong as stated: positivity_limited_faces is not the limiter of Zhang and Shu 2010 (JCP 229, 8918) but one built on the same idea, departing on three axes (primitive, not conserved, variables; one theta per face state, not per cell, Eqs. (2.4)-(2.5), (2.10)-(2.12); a relative one-ulp floor, not their absolute 1e-13). What carries over: Lemma 2.5 (convex combination with an admissible anchor); what does not: Theorem 2.1 (the next cell average admissible under their Gauss-Lobatto set and CFL, $N = 2$ covers $k = 1$ only; Remark 2.6 names finite-volume WENO as open) and conservativity of the reconstruction (per-face theta moves the cell's end-point mean by $8.3e-2$ where a per-cell theta gives 0). Along their conserved path $p(t)$ is concave, so the code's theta is never larger than their t_{eps} (MEASURED 0.856 against 0.949 at Mach 60): the anticipated negative-internal-energy failure cannot occur. **Defect found and fixed (advisor applied the worker's proposal):** with eps one ulp of the average the bare update $q_{avg} + \theta (q_{rec} - q_{avg})$ is a cancellation of the size of the floor and lands on EXACTLY ZERO for 1013 of 200000 reconstructed values crossing zero (never negative), the value the floor exists to prevent (division in $v = m/\rho$ and in the sound speed); positivity_limited_faces now clamps the scaled density and pressure to eps after the update (same number in exact arithmetic). Moves only runs in which the limiter fired, by one ulp except where the zero would have divided; goldens at the gate. Probe: bare_update_reaches_zero (the reason), production_limiter_never_delivers_zero (the production routine over 4000 grids of zero-crossing face states) GREEN. A per-cell theta (Eq. 2.11), which would restore the reconstruction average, is recorded as a change of scheme, not applied.
- **REF-LODI** (documentation, docs/phaseC_characteristic_base_bc.md +282/-50; the three user-supplied journal PDFs read as page images since their text layers lose the displayed equations): the wave-amplitude normalization (Poinot and Lele Eqs. (15)-(23); Thompson 1990 (44)-(47) with $l_1 = (0, 1, -\rho c, 0, 0)$; Thompson 1987 (19)), the LODI relations ((24)-(36), equal to Thompson 1990 (48) term for term), the subsonic inflow two-condition closure (P&L section 3.1, Tables III/IV; Thompson 1990 Eq. (68)) and all four branch-table rows CONFIRMED; no constant wrong, the constraint count unchanged. Two statements of the note corrected: pair (D) is NOT singular at stagnation (Thompson 1990 Eqs. (69)-(70) have denominator $u_1 + c$; the note's singularity was its own closure $\rho_b = F_c/(r_b^2 v_b)$; D1 reopened with a pointer), and D6's claim that no transparent condition is expressible as a fixed point is too strong (Thompson 1990 Eq. (64) $L_5 = \rho c g_1$ and P&L Eq. (40) $L_1 = K (p - p_{inf})$ are algebraic in the face state; D6 stays the user's decision with a third option). One

gap stated: the source projection L_1 C (gravity and the spherical divergence, Thompson 1990 footnote 1 and Eq. (52); the steady form is $L_1 = -\rho c g_1$, not 0) survives only because both states of (C-) sit at the same radius; any reformulation relating cell 1's centre to the face must restore it. Carlson's "but not U and p" marked unverified against the primaries.

- **REF-CHEM** (comments and documents only; `mol_rates.f90` and `oxygen_rates.f90` function bodies byte-identical; probes `h3p_recombination_branching` 12 and `water_photolysis_lyman_alpha_yields` 6, RED with the competing published values): every `mol_rates` constant sourced to Yelle 2004 matches his Table 1 digit for digit; Yelle's R7 ($H_3^+ + H$) reference column reads "Estimated", so Frelikh's barrier-free $2.0e-9$ transcribes an estimate faithfully; the code's H_3^+ dissociative recombination R6/R7 ($2.16e-8 / 5.04e-8 \times (300/T)^{0.65}$) is already Larsson et al. 2008's 300 K total $7.2e-8$ split by their branching 0.70, with the 0.65 index inherited from the Sundstrom/Datz fits Yelle prints (Larsson gives no temperature dependence; he states the CRYRING/ASTRID values the older $2.9e-8$ rests on were too high from rotational excitation), so the code stays; the Slanger and Black 1982 Ly-alpha H_2O yields 0.78/0.10/0.12 are exact (p. 2435) with one label corrected, $\Phi_{i2} + \Phi_{i4} \rightarrow \Phi_{i1} + \Phi_{i2}$, and the paper's coupling of the three yields and the 8 percent OH(A) bound recorded; Tully 1975's $2.87e-10$ is a 300 K statistical-model value (Table II) that VULCAN flattens over 100-2100 K, and its agreeing "experiment" is a 1973 relative-rate evaluation, strengthening the audit's verdict for O6. Documentation fixed in passing: `methodology_comparison.md` 6.4 listed three-body H_2 formation as identical in value while the code's Cohen and Westberg 1983 $2.8e-31 T^{-0.6}$ differs from Ham et al. 1970 by 1.142 at every temperature; Baulch 1992 is in references/ (its $H_2 + H_2$ entry valid 2500-8000 K, above the whole layer); section 10 item on H_2 dissociative photoionization closed (the code carries it, `h2_photo_channels.f90` channel S).
- **REF-NEUFELD** (comments and documents; `lya_rt.f90` machine code identical before/after; `physics_probe` 1118/0): the Ly-alpha escape-probability closure does NOT use Neufeld 1990 or Harrington 1973. $\beta_{esc} = \pi^{(-1/4)} \sqrt{a/\tau}$ is the one-flight wing escape chance under complete redistribution; Ly-alpha scatters with partial redistribution and the published static-slab damping-wing solution is Harrington 1973 eq. (40), $\langle N \rangle = 0.9093161$ B scatterings for a mid-plane source, so $\beta = 1/\langle N \rangle$, proportional to $1/\tau$ and independent of a ; MEASURED on the stored `wasp_full` column the code exceeds it by 234x at the base and 14x at 1.20 R_p , crossing at 1.42 R_p . Neufeld contains no expression of the code's form (his escaping fractions are escape against destruction, eq. 2.25 partitions the two faces; the code's two-face PRODUCT is unpublished); the escape frequency scales as $(a \tau)^{(1/3)}$ (Harrington eqs. 37, 47; Neufeld eq. 4.32), not the code's $(a \tau)^{(1/2)}$; Neufeld's validity $(a \tau)^{(1/3)} > 10$ holds in 320 of 504 cells (inside 1.156 R_p); Neufeld is static, the Sobolev factor is Sobolev 1960 radial-ray. The header's `J_int` formula lacked the $(1 - \beta)$ factor the code applies (fixed). **Advisor decision: adopt the published solution (LYA-BETA, launched):** $\beta = 1/$ of eq. (40) joined to the thin limit, the face partition of eq. 2.25 in place of the product, the $(a \tau)^{(1/3)}$ escape frequency, a validity record, `A_2p1s` $6.3e8 \rightarrow 6.2648e8$ (spectroscopic), every case with `Jlya escape-prob`: True moves (expected at frozen state: `J_int/n_2p` up to 90x higher inside 1.02 R_p , 14x at 1.20 R_p , up to 7x lower beyond 1.47 R_p ; Mdot to be measured to the du stop). The same question for the metal lines' `line_escape_probability_one_face` (`Cool_coeff.f90`) is LYA-BETA-METALS, after REF-METALS.

- REF-METALS** (cross_sec.f90, Cool_coeff.f90; probes metal_photoionization_fits, recombination_coefficient_fits, 255 assertions; advisor rebuild green apart from LYA-BETA's in-progress probe): all 132 fit parameters of the seventeen metal photoionization cross sections match Verner et al. 1996 Table 1 (verner_photo/photo.dat) digit for digit and metal_photoion_sigma reproduces Verner's own phfit2 exactly wherever the fit is the one phfit2 uses; 220 Badnell RR and DR parameters (fourteen daughters; cfout_K witness within 0.045 dex), the H/He rows and the Mao and Kaastra n = 1 rows match. **Two fixes:** the Mg II turn-on gated at Verner's rounded 15.04 eV while the grid edge is the NIST 15.035 (e_th_MgII now; byte-identical on wasp_full, the band geometric means miss the sliver); alpha_rr_metal('CaI') used Verner rrfit.f's rrec(:,20,19), the row that FORMS Ca II (second index = the recombined ion's electron count, fixed by rnew(:,1,1) = H+ + e and rnew(:,2,2) = He+ + e), 6.05x the Ca I row (1.120e-13, 0.900) at 1e4 K; corrected. Impact (worker, scratch): wasp_full 300 Ca I 0.87 relative at its 1.196 Rp maximum, Ca II +2.8 percent at the base, T 4.3e-3, v 9.9e-3; mol_metals 400 Ca II/III x2.5 at 1.38 Rp, cool 2.9e-2; wasp_full, mol_metals, lower_profile move at the gate through calcium. **Two physics items opened:** (a) the outer-shell fit is evaluated on the whole photon grid to 1240 eV while every row carries an E_max (34-558 eV) above which phfit2 hands over to the 1995 subshell fit and adds the inner shells: the code carries 0.1-12 percent of the true metal photoabsorption there, and Fe I's row (Q = 1.54) RISES with energy to 49x the truth at 1240 eV; the handover of phfit2 itself is adopted (METALS-OUTER, launched); (b) the inner-shell absorption and its Auger cascade (an atom two or more stages up) cannot be represented by the three-stage metal ladder: To_be_determined_by_user_20260906.md item 7. Also: verner_photo/README.md corrected (photo.dat is the 1996 Table 1, table1.dat the 1995 fits); rev3 line 944 closed; e_th_MgI/e_th_MgII provenance lines in parameters.f90 to be annotated (METALS-OUTER, after LYA-BETA releases the file).
- METALS-OUTER** (cross_sec.f90; generator src/utis/metal_photoion_table.py from verner_photo/photo.dat + table1.dat; probes metal_photoionization_fits 188 and metal_photoion_table_transcription 15; docs/photoion_cross_sections.tex + PDF; advisor rebuild: physics_probe green): the seventeen sigma_XX routines are one parameter block, and metal_photoion_sigma makes phfit2's handover: the 1996 outer-shell fit below E_max, the 1995 fit of the same shell at and above it (MEASURED: einn of phfit2 equals photo.dat's E_max for all sixteen metal rows; the production routine reproduces phfit2's outer shell to 1e-10 at 17 ions x 12 energies where 110 of 204 rows disagreed before, worst Fe I at 1240 eV 24.9 Mb against 0.00172 Mb, a factor 14464). Inner shells still absent by decision item 7 (option (a) recommended; the measured sizes now in that item). Impact (scratch, metals-off mol_base_handoff byte-identical): wasp_full 300 v 0.15, T 4.7e-2, heat 7.8e-2, Fe III 0.99, C III 0.60 relative (hard photons now reach the base, the top stages fall at small radius; metal share of the heating at the base 17.4 -> 8.5 percent); mol_metals 400 v 0.56, heat 0.42; lower_profile 200 heat 8.3e-3; log10 Mdot unchanged to 0.01 dex in the bounded snapshots. Both Mg thresholds already agree with NIST to their digits (item 4 needed no move; provenance note pending parameters.f90).
- LYA-BETA-METALS** (measurement by the worker, fix applied by the advisor; Cool_coeff.f90, probe fine_structure_escape_probability, docs/resonance_line_trapping.md sections 6.1 and 10; pub-

lisher scans of de Jong, Boland and Dalgarno 1980 and Osterbrock 1962 fetched into references/): line_escape_probability_one_face is de Jong et al. 1980 eq. (B-7), both branches to round-off, applied to the eight ground-term fine-structure lines only; NO metal resonance line uses an escape probability (Mg II h&k, Ca II H&K, Na I D, Fe I/II at $\beta = 1$, justified in that memo's sections 1-5 by $n_{\text{crit,eff}} = \beta A/q \sim 1e10-1e15$ against $n_e \leq 4.3e9$), and the transit post-process uses none. The fine-structure lines have a $\sim 1e-8$ and a $\tau \leq 1e-9$, so the Ly-alpha correction does not transfer. **Defect:** (B-7) is written in the frequency-integrated depth $\sqrt{\pi} \tau_0$ (normalized profile; equal to Hollenbach and McKee 1979 eq. (5.10) once the convention is matched, 3.4 percent at worst over $\tau = 3$ to $1e6$) while the caller passed the line-centre column: β too large by 1 (thin) to 2.11 (branch point) to 1.77 (asymptotic). **Fixed:** each cell's depth is now $\sqrt{\pi}$ times the line-centre depth in fine_structure_line_transfer. MEASURED regime on the shipped columns: largest line-centre depth 0.14 ([O I] 63 μm at the hot-Uranus base), so the factor there is 1.12 and the change of the total radiative losses is below 0.1 percent (mol_metals 0.096 percent), but a factor of two in any base thick in these lines and in the Base IR field incident term the same β scales. physics_probe 1291/0 after the fix. Metals cases move at the gate at that level.

- **Decision item 7, literature check** (2026-09-07; the user supplied Locci 2022 PSJ 3, 1; Locci 2024 PSJ 5, 58; Cecchi-Pestellini 2006 A&A 458, L13 and 2009 A&A 496, 863; Fox 2008 SSRv 139, 3; all in references/): the published exoplanet X-ray photochemistry models take the inner-shell absorption as opacity summed over all ionization channels (Cecchi-Pestellini 2009 Eq. 1 with Verner 1993 cross sections), put the photoelectron and the Auger electrons into the secondary-electron cascade (Locci 2022 Eq. 3), and carry singly charged ions only, i.e. option (b) of the item with the charge under-count accepted; no escape hydrodynamics code includes inner shells. Recommendation revised from (a) to (b) in To_be_determined_by_user_20260906.md; the user decides.
- **METALS-INNER** (decision 7 option (b), user 2026-09-07; cross_sec.f90 subshell block from phfit2.f's PH1 data (its thresholds differ from table1.dat in six rows, phfit2 being the reference), set_energy_vectors.f90 (every subshell turn-on of an active metal ion a bin edge; bin count unchanged, 17 -> 60 threshold cuts on wasp_full), util_ion_eq.f90 comment only (the existing rate σ/e_v and heat photoelectron_share(E_{th}) σ integrals already implement "one stage, one electron of $h\nu - E_{\text{th}}$ (outer) to the SvS85/Dalgarno cascade" once the cross section is the total); generator metal_photoion_table.py --check; probes 392 metal assertions, physics_probe 1479/0; docs photoion_cross_sections.tex (+PDF), code_status new L row): the metal photoabsorption is the SHELL SUM phfit2 returns, reproduced to $1e-10$ at 17 ions x 12 energies (130 of 204 rows disagreed before). It also closed a cliff METALS-OUTER's handover had left at E_{max} where the 1996 fit stood for a valence complex the 1995 outer row does not cover (Fe I x0.012 and Fe II x0.0066 steps, now 1.23 and 1.13; genuine K edges C/N/O x19-21), and put E_{max} , a step of the cross section, on a bin edge (it was not). Approximation stated at the code site: ion advanced one stage, one electron of $h\nu - E_{\text{th}}$ (outer) to the cascade (MEASURED with the degradation routine: 0.95-1.09x the heat and 1.10-1.37x the secondary H I ionizations of the true two-electron event; the un-spent I_2 is 24-35 eV of a 300-550 eV photon), fluorescence neglected (with $e_{\text{top}} = 1240$ eV only C, N, O, Na have K holes; the high-Z ions

contribute L shells only). Impact (worker, same-snapshot control; metals-off mol_base_handoff byte-identical): wasp_full 300 the wind 1-3 percent hotter and faster, base heating +26 percent, T at most +9.5 percent, Fe II-I/Mg III/N III at the base x74-160 from vanishing values, **Mg II falls 84 percent at $r = 1.178$** (the h&k transit line); mol_metals base heating +26 percent, T $1.4e-3$; lower_profile heating +21 percent at the base, T $2e-5$; metals' share of the 100-1240 eV absorption 2.2 -> 3.5 percent (wasp_full), the band 0.38 percent of the whole; log10 Mdot unchanged to 0.01 dex in the snapshots. Krause 1979 (JPCRD 8, 307) K/L fluorescence yields could not be obtained (NIST 503, AIP 403): requested from the user; no yield is quoted anywhere. Advisor verification: physics_probe 1479/0, grid_and_gates 93/0; spectrum_type segfaulted in loaded_sed_threshold_edges (its threshold array was sized $n_{\text{mion}} + 16 = 43$ while the active list now reaches 126), fixed by sizing it with the module's $n_{\text{thr_max}}$; suite green after. ($n_{\text{thr_max}}$ was a local parameter of set_energy_vectors; hoisted to module scope, public, so every caller of ionization_thresholds_active sizes its array by the one bound; spectrum_type 154/0, physics_probe 1479/0, grid_and_gates 93/0 after.)

- **METALS-INNER addendum, Krause 1979 yields** (comments and documents only; user-supplied Krause_1979_JPCRD_8_307.pdf, read from page images where the OCR is noisy, alignment checked on $\omega_K(\text{Fe}) = 0.340$): Table 3 ω_K for C/N/O/Na $2.8e-3/5.2e-3/8.3e-3/0.023$, Mg/Si/S/K/Ca/Fe $0.030/0.050/0.078/0.140/0.163/0.340$; Table 5 effective L1/L2 yields (Coster-Kronig included per the text, $\nu_3 = \omega_3$) for Mg to Fe all below $6.5e-3$; Table 2 uncertainties (ω_K 40-10 percent at Z 5-10, footnote (a) on molecules and solids). With $e_{\text{top}} = 1240$ eV only C, N, O and Na I can hold a K hole, so the neglected radiative branch is at most 2.3 percent of a K-hole event (under 1 percent for the three coolants) and under 0.7 percent of any L-shell event; raising e_{top} past 1.3 keV (Mg, Si K) or 4 keV (Ca, Fe K) is the condition under which the approximation must be revisited, stated at the code site, in photoion_cross_sections.tex (new tab:kedge), code_status and decision item 7.
- **LYA-BETA landed, advisor verification** (lya_rt.f90, hydrogen_n2_rates.f90 ($A_{2p1s} 6.3e8 \rightarrow 6.2649e8$, the nonrelativistic dipole value times the reduced-mass factor, = NIST ASD's Wiese and Fuhr 2009 value), parameters.f90 comments; probe lya_escape_probability 24 assertions; advisor rebuild physics_probe 1479/0, certification 29/0): the escape probability is the published static-slab solution, $\langle N \rangle = \text{Neufeld 1990 eq. (3.27) at zero destruction} = \text{Harrington 1973 eq. (40)} (4 \sqrt{6})/\pi^2 u_2 B = 0.909316 B$ for a mid-plane source (the 0.9093161 of the REF-NEUFELD entry was a transcription slip), $\beta = 1/(1 + \langle N \rangle)$ (exact at both limits: a generated photon is emitted $1 + \langle N \rangle$ times, and the budget $n_{2p} A \beta = P$ wants the escape chance per emission), the two faces sharing one escaping population by Neufeld eq. (2.25) instead of the former product, $\Phi(\xi)$ summed in closed form; the Voigt parameter has left the escape probability. a still enters only lya_wing_penetration_coefficient (where an incident stellar photon crosses in one flight, a different question, judged at the code site). Defect found in passing: $\beta_{\text{tot}} = 1 - (1 - \beta_{\text{esc}})(1 - \beta_{\text{sob}})$ lost digits at the base ($4.8e-9$ relative), now $\beta_{\text{esc}} + \beta_{\text{sob}} (1 - \beta_{\text{esc}})$ (closure $2.2e-16$). RED before $4.79e2$ / GREEN $2.19e-16$ on the end-to-end identity. **Impact, wasp_full to its du stop (2 h 59 m): log10 Mdot 13.34 -> 13.34 (+1.6 percent, 0.0067 dex); Jint/n_2p x84 at the base, x73 at 1.05, x37 at 1.10, x10 at 1.20, x0.75 at 1.50 R_p ; T within 2 percent; the old $\min(1, .)$ cap had zeroed the internal field from**

1.15 R_p outward. H(n=2) chord column x31.5 at b = 1.00, x2.8 at 1.20, x1.005 at 1.45: **every H-alpha/H-beta transit number in this repository from a Jlya escape-prob: True run is superseded** (He 10830 not affected). Validity $(a \tau)^{1/3} > 10$ holds in 349 of 504 cells, out to 1.193 R_p, exactly where the change is an order of magnitude. Advisor follow-ups applied: the second A_{2p1s} in exhale_transit_lib.py and docs/transmission_spectrum.tex set to 6.2649e8 (one value with the wind model), the stale $E = \min(\text{boost}, 1/\beta)$ comment in parameters.f90 replaced by the implemented $1 + (\text{boost}-1)(1-\beta) T_{\text{star}}$. Pending: the B6-style print of the Ly-alpha domain record in certification.f90 (the record and accessor exist; LYA-DOM, with the gate's follow-ups).

- **LYA-DOM** (advisor; certification.f90): the Ly-alpha damping-wing domain record (lya_wing_domain_record: cell visits below the $(a \tau)^{1/3}$ limit, visits seen, the smallest value, the limit) joins the out-of-domain closure count and is printed beside the H3+ and CO records, informational like them. certification, run_mode, attempted_step green. **Every worker item of the series has landed; the series gate begins (single build, make test, the 16-case matrix, movement table, one golden refresh, fcheck, docs pass).**
- **Series gate, 2026-09-07** (PLAN rev2 steps A to C and the items above, one series): single build of the shared tree (make -q up to date, EXHALE.x md5 ba3d34e6eae); make test green apart from the two steady suites, which then passed on the wasp_full_newton run log (EXHALE_STEADY_RUNLOGS); the 16 default cases run against that binary (the harness holds a single-instance lock, so wasp_full ran through run_check.sh check and the other fifteen through a runner replicating its per-case steps, all OMP_NUM_THREADS=1); every case moves against the previous goldens, as every item of the series said it would. Movement of the physical columns over the cells where the golden value is at least 1e-3 of the column maximum (max relative; T median; the change of log10 Mdot from the mean of $r^2 \rho v$ over 1.20-1.45 R_p): | case | rho max | v max | p max | T max | T median | dlog10 Mdot | log10 Mdot new | |---|---|---|---|---|---|---| | wasp_full | 2.97e-01 | 1.28e+00 | 1.65e-01 | 9.18e-02 | 3.8e-02 | +0.0095 | 13.35 | | wasp_he23off | 2.95e-01 | 1.30e+00 | 1.62e-01 | 9.18e-02 | 3.9e-02 | +0.0084 | 13.35 | | wasp_full_newton | 6.96e-01 | 3.51e-01 | 4.30e-01 | 2.35e-01 | 8.7e-02 | +0.0414 | 13.30 | | mol_base_handoff | 1.60e-01 | 2.50e-01 | 1.52e-01 | 2.33e-01 | 2.2e-02 | -0.0440 | 10.57 | | mol_metals | 1.51e-01 | 2.25e-01 | 1.53e-01 | 2.07e-01 | 2.3e-02 | -0.0476 | 10.58 | | mol_lyman_werner | 2.12e-01 | 4.87e-01 | 1.99e-01 | 2.55e-01 | 2.6e-02 | -0.0011 | 10.54 | | mol_diffusion | 1.59e-01 | 2.50e-01 | 1.52e-01 | 2.33e-01 | 2.2e-02 | -0.0440 | 10.57 | | mol_ir_bands | 1.51e-01 | 2.26e-01 | 1.53e-01 | 2.07e-01 | 2.3e-02 | -0.0475 | 10.58 | | mol_sec_ion | 2.16e-01 | 4.03e-01 | 1.96e-01 | 2.53e-01 | 3.0e-02 | +0.0149 | 10.51 | | mol_carrier | 1.54e-01 | 6.37e-01 | 1.44e-01 | 2.24e-01 | 1.9e-02 | +0.0106 | 10.56 | | lower_profile | 9.40e-03 | 3.22e+01 | 7.50e-03 | 9.77e-01 | 8.8e-03 | -0.0335 | 7.29 | | hydrostatic_column | 2.09e-12 | 2.81e-03 | 3.54e-12 | 1.40e-03 | 2.5e-14 | -0.0000 | 8.27 | | oxygen_chemistry | 8.62e-01 | 1.29e+02 | 9.76e-01 | 7.61e-01 | 3.7e-01 | +nan | 9.61 | | hp_zero_seed | 2.25e-05 | 1.81e-04 | 1.61e-04 | 3.20e-03 | 4.4e-05 | +0.0000 | 10.56 | | hp_trace_seed | 2.24e-05 | 1.81e-04 | 1.61e-04 | 3.20e-03 | 4.4e-05 | +0.0000 | 10.56 | | hp_front | 2.07e-05 | 5.77e-04 | 3.66e-04 | 1.00e-02 | 6.7e-05 | +0.0000 | 10.56 | The maxima sit at the fronts, which move by a cell; the T medians (2-9 percent on the wind cases) and the Mdot changes (within +/-0.05 dex everywhere it is defined; oxygen_chemistry has a reversed

flow in the window) are the state of the physics after B3c, B4-1/1c, LYA-BETA, the WENO3 coefficient fix, the metals (REF-METALS, METALS-OUTER/INNER), the LW band, the CO channels, ACCEPT-1/2 and the rest. `wasp_he23off` exits 2 (its stationary claim refused by four inventory entries, as `A2tol2` measured at the default tolerance; outputs written in full); `wasp_full_newton` exits 0, certified, `log10 Mdot` 13.30. PDFs of the 25 dated documents, the paper and both posters rebuilt. **Golden refresh pending the user's hand:** the archive copy `golden` -> `golden_pre_series_20260907` and `run_check.sh golden` were refused by the session's permission classifier; the commands are in the report to the user. The final confirmation (`make check` against the refreshed goldens, serial) follows the refresh. **Golden refresh done by the user (2026-09-07 13:19):** `golden/` archived as `golden_pre_series_20260907/` (17 cases) and re-snapshotted from the case outputs of this gate (checked: `golden` equals output on `wasp_full`, `mol_metals`, `oxygen_chemistry`, `lower_profile`). The serial `make check` against the refreshed goldens runs next as the bitwise confirmation, then `run_fcheck.sh`.

- **User manual, environment-variable table** (user report, 2026-09-07): the fixed table float of the environment hooks (67 rows) overflowed page 11 into the footer; it is now a longtable with the caption at its head, breaking across pages 10-11 with a "continued" foot. PDF rebuilt. Pages 20-21 (user report): the keyword table's long file paths were unbreakable `\texttt` tokens and ran past the right margin; 18 such paths are now `\path{}` (url package, breaks at / and _), one inside a caption kept as `\texttt` (a moving argument). Rebuilt with `latexmk`, so the longtable column widths and cross-references are resolved (a single `pdfLatex` pass leaves both unresolved, which is what produced the "Table ??" and the first overflow the user saw).
- **Update log TeX twin** (user request, 2026-09-07): `docs/Update_EXHALE.tex` and its PDF now exist, GENERATED from this Markdown by `src/utls/update_log_to_tex.py` (pandoc body, the memo-class preamble of the stage logs); regenerate after appending here. 56 pages at this entry.
- **Decision item 4 closed** (user, 2026-09-07): candidate C, `wasp121b_composite_huang2023.txt` (solar XUV shape below 1700 Å rescaled to Huang et al. 2023's `F_XUV` and `Ly-alpha`, the MUSCLES WASP-17 photosphere above), is the WASP-121 b spectrum of record, as both `input.inp` files already state; A and B remain sensitivity inputs. No decision item of `To_be_determined_by_user_20260906.md` is open. **make check against the refreshed goldens: REGRESSION PASS, all 16 cases byte-identical (64 files), 13:21-16:25 KST 2026-09-07** (the harness runs the cases eight at a time, one thread each). `run_fcheck.sh` follows. **run_fcheck.sh: CLEAN** (1500 bounded steps of the HD 209458 b folder case under `-fcheck=bounds,do,mem,log10 Mdot 9.56`) after one harness fix: the scratch copy of the planet input named its spectrum by the folder's relative path, which the script now rewrites to the repository's absolute path for the `Spectrum` file and `Lower atmosphere` profile keys (the first run failed on "Cannot open file `../inputdata/sed/...`", not on a runtime check). Production `EXHALE.x` restored, `make -q` up to date, md5 `ba3d34e6eae`. **The series is closed:** goldens refreshed once, `make check` byte-identical, `fcheck` clean, docs and PDFs rebuilt. What follows is in `docs/session_handoff_20260907.md`.
- **HYG-PY** (advisor, 2026-09-07; `python/paper_data.py`, `python/make_struct_figures.py`): `HEAT_FIXED` (the headerless-file fallback) and `HEAT_CH_LABEL/HEAT_CH_ORDER` (the heating-budget panel) now carry all 19 deposit channels of `heat_channel_name` with the descriptive brackets stripped the way `_clean_name`

strips them (heat_He23S_assoc, the two Lyman-Werner halves, heat_mol_chem, heat_FUV_photolysis, heat_oxygen_collisional, the two CO channels); the panel loop already skips a channel a run's file lacks. Both scripts parse; the loader reads the file header, so the fallback list was never what a real file was read by.

- **THREAD-DET** (advisor, 2026-09-07; util_ion_eq.f90): the thread count is no longer an input to any result. Mechanism, MEASURED: Cool_coeff.o imports glibc libmvec's two-lane _ZGVbN2v_exp, _ZGVbN2v_log, _ZGVbN2vv_pow (gfortran -O3 vectorizes the rate-coefficient loops), and the vector variants agree with the scalar libm to one ulp, not bitwise; inside a block of cells the loop takes lane pairs from j_lo and an odd remainder by the scalar call, so where a block STARTS decides which cell meets which variant. chemical_rate_coefficients and eval_cool cut the grid into 2*omp_get_max_threads() blocks, so the block boundaries, and with them the last bit of every rate, moved with the thread count (the 6.6e-14 of COST3/HYG-MAIN). Fixed: both sweeps use blocks of the fixed even length xuv_rate_block = 32 (16 blocks on the 504-cell grids); the threads decide only who takes which block. MEASURED on wasp_full 300: old binary at 1 thread, new at 1 thread and new at 16 threads are byte-identical in Hydro_ioniz, Ion_species, Heating_breakdown, Cooling_breakdown (the old single-thread partition happened to start both its blocks at even offsets, so the refreshed goldens are untouched). Shared EXHALE.x rebuilt from the tree (make -q up to date).
- **DOCS-LINES** (documentation only, 2026-09-07): every file.f90:NNN citation in the nine design and status documents (d0 922 tokens on 309 lines, b1a 127, b1 64, b3b 24, a2 7, a0 5, b3a 5, b4 2) re-anchored to file + routine, declaration or labelled comment block, read against the live tree; the step tables of d0 6.1 and b1 7.1 index by the as_op_* operation tag of attempted_step.f90; the convention is stated in each document. Citations of removed code (the composition projection, the two-iteration energy update, the CO ceiling, the cell-velocity carrier advection, the element advective term, DB96 eq. (39), the H3+ clamp, the second H2 partition function) kept as dated records naming the removing item. Stale values corrected in passing: e_lw_photon_erg 1.88021e-11 erg (11.7354 eV) over 912-1201 Å, binary_element_diffusion.f90 3037 lines, a2 section 1's chemical count now passed by the marching caller. All 190 names introduced grepped present in src/. Recorded: pre-existing markdown table pipe misalignments (unescaped | in math) in a2, b4, code_status, d0; two markdown line citations of grid_and_gates/README.md left.
- **B5, carrier arm** (steady_newton.f90, diffusive_photochemistry.f90, input_read.f90; new suite steady_species_rows 41 + 8 rows; advisor rebuild: it and certification, run_mode, carrier_retry, coupled_source_step, species_face_flux green): the Newton unknown vector carries one row and one unknown per solved carrier (H2, OH, H2O, CO, H+) through a row registry in place of the H2-only nvar = 4 special case; band geometry derived, kl = ku = 3 nvar - 1, ncolor = 6 nvar - 1 (reproduces the 8/17 and 11/23 the code carried); the completion flag reads the species rows of the system it solved; EXHALE_SPECIES_JAC_TEST asserts J r inside the solve (the run-level test builds its Jacobian before the registry exists). Ionization transport + Solver: Newton supported by derivation when the proton is a Newton unknown (x_hp_fix handed to the sweep), refused with the reason otherwise; MEASURED nvar = 5, Jacobian action 7.2e-7 (all rows), 4.6e-5 (species rows), 0 unresolved colors. Two pre-existing defects fixed: frozen_residual left the species slots of an

intent(out) array unwritten (directional check relative error 1.0 from stale memory) and n_headroom_iterate was never assigned (the trial-admissibility threshold stood at 0; with the four oxygen balances as rows all 41 colors came back unresolved). wasp_full_newton stays certified (13.30, byte-identical); H2-only control byte-identical; marching path byte-identical (wasp_full, mol_base_handoff 300). **Findings:** no regression case enters the steady solver, so every golden is a marching snapshot and cert_tol_carrier/cert_tol_element still have no anchor (contract section 11 records two rows from solves that did not meet their target, 7.46e-3 and 5.82e-3, not anchors). **Open, and a physical-consistency defect:** the stationary carrier row still uses the cell-velocity advective form while the marching rows ride on $F_{\rho} Y_s^{\text{face}} / m_s$ since B4-1c (two operators, two fixed points): B5b. Also open: the element rows (element_transport_residual must return a residual per element, not the worst at each cell), the cell's own field in the cell residual (interface written: FIELD-SELF), reload consistency, three EXHALE_main.f90 sites (the registry name, EXHALE_JAC_TEST coverage, the PTC refusal text).

- **B5b** (species_face_flux.f90, steady_residual.f90, steady_newton.f90, certification.f90, binary_element_diffusion.f90, diffusive_photochemistry.f90, three sites of EXHALE_main.f90; suite steady_species_rows 67 rows; advisor rebuild: it and species_face_flux, certification, carrier_retry, coupled_source_step, run_mode, acceptance_classes green): ONE discretization of the material advective term exists in the code: the RK stages, the stationary carrier rows, the fixed-wind carrier relaxation and the stationary element rows all form it as the divergence of the same face species fluxes (species_face_fraction, species_face_flux, new species_flux_divergence) on the mass row's faces, areas and volumes; the cell-velocity upwind form, its deferred van Leer correction (carrier_advection_correction, adv_corr, carrier_slope), the base Dirichlet record it needed and the advective coefficient ntv are gone from diffusive_photochemistry.f90 (RED 76.7 relative against a stage-1 update on a sign-changing face flux, GREEN 7.5e-14). element_transport_residual returns one residual per element (the certification prints one entry per element: lower_profile C 0.977, O 0.921, N 0.968, seven elements "not applicable"); rows srow_element_he and one srow_element_trace per metal element registered before the carriers, project_element_mass_fractions the write-back. mol_diffusion reloaded with Coupled carrier solve: nvar = 4, one element row, Jacobian action 4.9e-3 / 1.3e-3, 0 unresolved colors, the solve refused at the He/H row (0.9987) with no descent direction: the element system is expressible and its Jacobian right, not yet solved from that start (the element unknown moves the whole column's opacity, which the banded model does not carry; the "all rows" bound of the suite widened 1e-4 -> 1e-2 with the measurement at the site). Reload consistency asserted on the binary. EXHALE_main.f90: set_transported_species_rows (not carrier-specific), the two EXHALE_JAC_TEST inserts, the PTC text; at that site two pre-existing defects fixed: the species slots of Yvec were read uninitialized (2.4e2 against 1.85) and the probe direction was unscaled (3.3e1 at nvar = 4); now 3.2e-6 / 4.5e-6. **By construction the stationary species rows now inherit the mass row's imbalance** ($\text{div}(F_{\rho} Y) = Y \text{ div } F_{\rho} + F_{\rho} \text{ grad } Y$; the old form was the second term alone), so on marching snapshots whose mass rows read 1.1-1.9 of their terms the rows read O(1) where they read 4e-4 to 0.4 before. Marching path byte-identical on wasp_full, mol_base_handoff, mol_carrier, mol_diffusion, lower_profile at 300 steps; wasp_full_newton reloaded certifies (13.30, 2.6e-9 of the stored state). Anchors still UNCONFIRMED

(contract 11.3). Recorded: `element_diffusion_step` / `relax_element_composition` still advect on the smoothed steady mass flux, so the elemental Picard alternation and the elemental row are two operators (B5c); `carrier_rows_advect` kept as the on/off switch of one operator; Coupled `carrier_solve` now also governs the element rows (key not renamed); `ionization_equilibrium.f90` cites `carrier_base_state`, which no longer exists (to FIELD-SELF). THREAD-DET confirmed by the official check: `make check` on the rebuilt binary, REGRESSION PASS, every case byte-identical (16:58-20:04 KST 2026-09-07; the binary of that check also carried whatever of B5 had landed by 16:58, and B5's marching path was measured byte-identical anyway). A further `make check` follows FIELD-SELF and B5c.

- B5c** (`binary_element_diffusion.f90`; suite `steady_species_rows` 60/0; advisor rebuild: it and `species_face_flux`, certification, `carrier_retry` green): the elemental relaxation and the stationary elemental row are one operator. `element_diffusion_step` takes the FACE MASS FLUX and forms its advective term as the divergence of the face element fluxes through the three shared routines (`element_advective_divergence`, one place); `relax_element_composition` reads that flux from the mass row of the state (`face_mass_flux_of_state`) and no longer smooths a steady $\dot{m}/(4\pi r^2)$; `element_transport_residual` takes both halves of the helium row from `composition_residual` and the trace rows from the new `trace_composition_residual`, the routines the step itself solves, so the fixed point is the row's zero by construction; `solve_trace_element_in_hydrogen` is a deferred-correction loop that takes one pass, term for term the old direct solve, when the advection is off (the marching path). MEASURED on a synthetic column relaxed to its own convergence: the row reads 8.65e-13 (He alone), 8.45e-13 / 5.38e-13 (He + ten trace elements) of its terms, the advection carrying 0.993 of them; the replaced form leaves 6.4e-3 on the same state. The smoothed-flux rationale does not survive: it repaired a form (the element equation minus X times the mass equation) that is gone, and its consequence is now a measured property of the equation: on a face mass flux that does not conserve mass the conservative balance has no bounded steady composition (X driven onto 0 and 1). Marching path byte-identical (`mol_diffusion`, `lower_profile` 300 steps); the direct-steady reload of `mol_diffusion` byte-identical, and its outer pass never reaches the relaxation (the solve ends `info = 2`, B5b 6.3), so the run-level RED/GREEN does not exist. **Found: `diffusion_tests.x` has not linked since B5b** (`binary_element_diffusion` now uses `steady_residual_mod`, whose chain the test's DIFT_SRC does not carry): a functions/ module depending on `time_step/steady_residual` is a layering inversion; the flux goes by argument (DIFT-LINK, launched), and `make test's diffusion_tests` row is red until it lands. Also recorded: the relaxation's convergence measure is helium-only (DIFT-LINK folds the trace drift in); the vanished-element write-back branch leaves a trace row at 1e-4 to 2e-2 where an element is driven to exactly zero (deliberate guard, measured, documented at the test site); `element_transport_residual's` dummy `v` is unread.
- FIELD-SELF** (`ionization_equilibrium.f90` self-field loop, default one pass; `util_ion_eq.f90` and `System_HeH*` comments; probe `photoionization_field_self_consistency` 12; advisor rebuild: `coupled_source_step` 36/0, `physics_probe` 1491/0, certification, `acceptance_classes` green): the cell's own optical depth CAN now be closed inside the cell solve (one pass leaves a 4.06e-2 self-consistency defect in the ionized fraction of a cell one optical depth thick, the loop closes it to 2.8e-15 in 13 passes, contrac-

tion 0.080; the field routines already formed the B4-6 cell mean from the cell's own densities, so no field code changed), **and doing so buys nothing: COST3's attribution is refuted.** MEASURED on wasp_full 300: coupled passes 10.09 -> 10.01 for 1.65x the wall time (2 self-field passes) and 1.87x (8); mol_base_handoff 1500: 7.19 -> 7.24 for 2.0x; the decisive binary with BOTH levers (carried columns at block 8 AND four self-field passes): 7.20 passes, 190 s, against COST3's 7.13 / 103 s for the columns alone. (**Superseded by COST4 below:** with both stellar lags closed at the settings that close them the count is 8.18, not seven, and the two couplings named here are worth 0.08 of it; the count is the slowest rung of a ladder of modes.) Same fixed point: state movement $\leq 1.4e-7$, at the $1e-8$ tolerance's level. Default output byte-identical on wasp_full 300, mol_base_handoff 1500, mol_metals 1500, hydrostatic_column; wasp_full_newton uncapped CERTIFIED, both files byte-identical to its golden (that binary carried B5b too). Kept in the tree at its inert default on COST3's precedent (the switch is the record of a measured negative result). Comment fixes in passing: System_HeH_metals.f90 said the sweep is serial (untrue since the parallelization; its arrays are threadprivate); the carrier_base_state citation of ionization_equilibrium.f90 rewritten to carrier_base_composition_imposed. Recorded: he_ground_singlet_density's clamp counter needs a lock when the self-field loop runs (a slot per cell would be free, composition.f90); f_vib_quench/e_vib_bound stay at the entry composition inside the loop; the COST3 attribution in that report, its log entry above and the csm_max_pass comment of EXHALE_main.f90 are superseded by this measurement (comment to be corrected when DIFT-LINK releases main).

- **DIFT-LINK** (binary_element_diffusion.f90 and its three call sites in EXHALE_main.f90, steady_newton.f90, certification.f90; suite steady_species_rows 62 rows; advisor rebuild: it and species_face_flux, certification, coupled_source_step green): the element module no longer uses steady_residual_mod; the face mass flux is a required argument of relax_element_composition and element_transport_residual, obtained by each caller with face_mass_flux_of_state. The layering inversion is gone and **make diffusion_tests links again with the Makefile and DIFT_SRC untouched: 35 passed, 0 failed** (MEASURED by the advisor on the live tree; it failed at the link since B5b). The relaxation's convergence measure now runs over every element it relaxed, each against its own reservoir (element_composition_distance): RED 5.99 / GREEN 0.0 on a column where Fe moves 1.78 while helium moves 0.25, so the helium-only measure returned a number 7x too small to bound the element that moved. Byte-identical on mol_diffusion, lower_profile 300 steps and the mol_diffusion direct-steady reload. Dead code removed (helium_mass_fraction, an unread element_mass_fractions call). **Advisor fixes on its findings:** the limiter of species_face_fraction divided by $Y_{av} - Y_f(j)$, which vanishes when the donor cell's average is itself negative (a trace species handed back at a small negative density; MEASURED as a SIGFPE under -ffpe-trap=zero): the anchor is now the average clamped into [0,1], which makes both denominators strictly positive and is what the comment already claimed; mol_diffusion 300 steps byte-identical, since at production flags the infinity was clamped to the same 0. The csm_max_pass comment of EXHALE_main.f90 and docs/steady_solver_design.md section 15 corrected to the FIELD-SELF and B5c measurements. Recorded: diffusion_tests T9 reads $1.098e-12$ against its $1e-12$ bound at -O1 -fcheck (an optimization-level rounding sensitivity of relative_settling_mass, 35/0 at -O3); the outer loop's under-relaxation damps helium only

while the drift it tests can now be set by a trace element.

- **Batch gate after B5/B5b/B5c/FIELD-SELF/DIFT-LINK/THREAD-DET** (2026-09-07/08): shared tree rebuilt (EXHALE.x md5 eba2d7f2260f, make -q up to date); make test green, the two steady suites passing on the wasp_full_newton run log and the diffusion_tests row restored; make check **REGRESSION PASS, all 16 cases byte-identical** (20:54 to 00:05 KST). Every item of this batch was designed to leave the marching path alone and the check confirms it: the goldens refreshed at the 2026-09-07 series gate still stand, and no golden was refreshed here.
- **HYG-T9** (advisor, 2026-09-08; src/tests/diffusion_tests.f90): T9's ambipolar-limit bound sat exactly on the arithmetic's noise floor, so the suite passed at -O3 and failed at -O1 with checks (DIFT-LINK's finding). MEASURED at both levels: the neutral limit is exact (no electron term), the H+ plasma reads 8.207e-13 (-O3) and 7.740e-13 (-O1), the He++ plasma 7.083e-13 and 1.098e-12. Mechanism, written at the site: the limits are exact algebra in the masses, so the deviation is arithmetic, and the ambipolar term of the plasma limits comes from a centred difference of $\ln \rho$ over dr whose logarithm is $O(b_0/r)$, so each ulp reaches dm_{eff} amplified by $1/dr$. The bound is now one decade above the largest measured ($1e-11$), the anchoring rule the certification tolerances use, and the test still catches a wrong ambipolar limit by ten decades (a wrong limit misses its target by 0.1 to 1) with the neutral limit still asserted exact. 35/0 at -O3 and at -O1 -fcheck.
- **B5d** (steady_newton.f90 only; suite steady_species_rows 77 rows; advisor rebuild: it and certification, species_face_flux, coupled_source_step, carrier_retry green): the item's own brief expected the element rows to fail because an element unknown moves the whole column's opacity, too non-local for the banded model. **MEASURED and REFUTED**: the scaled Jacobian column of an element unknown carries at most 6.3 percent of its 2-norm outside the band, the same order as a hydrodynamic column (mass 0.0018, momentum and energy 0.0000), so the diagonal correction, the rank-one column-integral update and the identity block the brief offered would each have repaired a defect that is not there; none was written. What does break the model, located by the same measurements and fixed: (1) the base element row was structurally EMPTY, cell 1 being the Dirichlet reservoir of the element operator, so the banded model was singular by construction (its LU pivot grew to 8.413e19 against a largest band entry of 5.229e3; now 1.433e6) and it now carries the boundary condition the operator states; (2) the element ROW SCALE collapses on the abandoned iterate (helium rows spanning 1.277e-29 to 2.360, so an ordinary $dF/dY = 46.7$ became a scaled band entry of 4.890e22 and $||grad\ merit|| = 1.104e44$), and the row scale is now floored at the flow-time rate of the quantity the row transports, the hydrodynamic convention with no chosen constant. Result: the same case reaches $||R|| = 1.723$ against 1.974 at outer iteration 18 instead of 8 with $||grad\ merit|| = 4.813e3$, 40 decades lower and a number the model's own entries can produce. **The element solve still does not converge and the item does not claim it does**; cert_tol_element and cert_tol_carrier are unmoved and UNCONFIRMED. New diagnostics: banded_model_conditioning (empty rows and columns stated once per registry, pivots and scales under EXHALE_BAND_REPORT), jacobian_column_reach_beyond_the_band (EXHALE_JAC_COLUMN_REACH). mol_diffusion 300 steps byte-identical; wasp_full_newton byte-identical, info = 0, CERTIFIED, 13.30, same 6 outer iterations. **Named as what is left: the globalization** (the non-

monotone search accepts steps that raise $||R||$ from 1.7 to 25; the trust-region route gives up at radius $3.8\text{e-}16$ after 12 ray-test refusals while the merit is still falling): item B5e. Also recorded: `carrier_row_term_scale` has the same collapsing structure and was not measured (no carrier row in that case); the trace arm of the floor is covered by inspection only.

- **COST4** (measurement; NOTHING in the repository changed by the item, its four candidate files byte-identical to the snapshot it took; every experiment a private binary in a scratch tree whose no-switch build is byte-identical to the control): **the coupled loop's pass count is a ladder of nested modes and equals the SLOWEST of them, not their sum.** MEASURED on `wasp_full 300`, one thread, removing one composition coupling at a time: the production loop 10.09 passes; the stellar field's dependence on the composition removed, 7.16; the temperature removed as well (the energy solve replaced by its own converged answer), 4.94; the He recombination rates too, **2.00** (which reproduces, with a different probe, the 2 passes COST2 measured with every composition-dependent input frozen). Both hypotheses of the brief are refuted: the two diffuse couplings are worth at most 0.08 of the 10.09 (Balmer off 10.09, He rec off 10.01, both off 10.01, He rec frozen after pass 1 10.01, and on `mol_base_handoff` all three read the control's 7.19 to the last digit), and **the Balmer continuum is not a lag at all**: `excited_H_update` fills its arrays once BEFORE the loop, so they are constants of it, and refreshing it every pass costs 1.5 percent of wall for +0.01 passes. The stellar lag closed IN FULL for the first time (a one-cell block and eight self-field passes): 8.18 passes for 218 s against 10.09 for 139 s, the next rung, with freezing the field entirely giving 7.16. The temperature is not the residual either: the composition sweep alone, at the temperature the step converged to, takes 10.94 passes (6.32 against 7.19 on `mol_base_handoff`). **Decision: change nothing** (no loop-internal Balmer refresh, no self-consistent He recombination coupling, no default moved); to go below seven passes every rung must come out at once. Advisor: the three statements this refutes are corrected in place, the `csm_max_pass` comment of `EXHALE_main.f90`, the FIELD-SELF entry above (marked superseded) and open item 1 of the 2026-09-08 handoff. **The one remainder, now open item 1**: on `mol_base_handoff` the ladder is flat (freezing the field RAISES the count 7.19 -> 7.41) and everything the item could reach accounts for 0.6 of its 7.19 passes; the named candidate, READ and not measured, is `n_tot`, the third-body density of the three-body molecular reactions, built by `calc_ntot` from the ENTRY composition, and with it the H2 vibrational quench fraction and bound energy. Also recorded: `xuv_field_block_cells` and `xuv_self_field_passes` are compile-time parameters, so measuring the two together needs a rebuild; `heating_of_composition` calls `he_rec_coupling` a second time whole-grid every pass for its heating alone (both calls together bounded at 1.7 percent of wall); on `mol_base_handoff` the He recombination coupling is what keeps step 0 inside the pass cap.
- **COST5** (measurement; NOTHING in the repository changed by the item, its four candidate files byte-identical to its own snapshot; probe binaries byte-identical to the control at their default): the molecular remainder COST4 left is named, quantified, and the ladder reaches its bottom rung. **The count is arithmetic, and the law is measured and then tested**: $\text{passes} = 2 + \log_{10}(\text{the step's own excursion} / \text{the stopping tolerance}) / (\text{decades per pass})$. On `mol_base_handoff` the composition contracts by 0.171 a pass and the temperature by 0.072 (0.77 decades a pass) from an excursion of $7.0\text{e-}5$ to a tolerance of $1\text{e-}8$: 7.00 predicted, 6.93 MEASURED;

on wasp_full 0.222 and 0.199 from an excursion seven times larger, 9.2 predicted, 10.09 measured (so the molecular cases are the cheap ones, not the pathological ones). Tested, not fitted: the law says the count depends on the stopping tolerance through one logarithm at 1.30 passes per decade, and moving both tolerances together gives 8.06 / 6.96 / 5.68 / 4.04 at $1e-9$ / $1e-8$ / $1e-7$ / $1e-6$, i.e. 1.34 over three decades; and it is not the cell solve, six decades of hybrid tolerance leaving the count at 6.97 for up to 1.99x the wall. Of the four candidates only the named one is an iterate: `n_tot`, the third body of R12/R13/R15 (the ONLY composition dependence of the molecular network outside the field: of seventeen hoisted coefficients exactly two see it, both linearly). Holding it: `mol_base_handoff` 7.19 -> 6.03 at 1500 steps, `mol_metals` 6.99 -> 6.01; it limits the composition mode's RATE (0.171 -> 0.073) but the count falls only 6.93 -> 6.11 because the temperature mode then binds. The other three are entry state through the operator-split field (vibrational pair 6.96, electron density 6.94, Lyman-Werner 6.96, against 6.96), and the metal element totals are invariant across a sweep by construction (6.99 against 6.99, exact). With every input AND the temperature held the sweep needs **2.01 passes**, reproducing COST2's and COST4's atomic bottom rung on the molecular path. **Decision: nothing closed.** The mode that sets the count is the temperature/composition alternation (`EXHALE_main.f90`, `energy_semi_implicit.f90`, not that item's files); closing the `n_tot` lag exactly needs two sites in `System_HeH_mol.f90` (also not its), and the in-file alternative costs a second cell solve per pass, which the cell-tolerance scan bounds at more than the 0.80 passes it returns. The production treatment is not wrong: at the fixed point the entry and returned compositions are one state, so the third body the network is solved with is the third body of the accepted state, to `esm_comp_tol`. Advisor: the law is now in the `esm_max_pass` comment; the misleading indentation of `composition.f90`'s `calc_ntot` branches fixed. Recorded: the loop's stopping test compares an ABSOLUTE mass-fraction change against `esm_comp_tol`, so it is a relative $1e-8$ on H2 and no constraint on a $1e-12$ species. **The one lever that keeps both the fixed point and the tolerance is sequence extrapolation** on the (T, c) pair after the third pass (the decay is geometric with a ratio stable from pass 2 and close across cases): ESTIMATED at three passes of seven on the molecular cases and four of ten on wasp_full, item COST6.

- **HYG-4** (advisor, direct; the four hygiene items of the 2026-09-08 handoff): (i) `he_ground_singlet_density`'s run-wide clamp counter is incremented with `!$omp atomic update` inside the function, so the named critical section around its one call inside a parallel region (`ionization_equilibrium.f90`, self-consistent field loop) is gone and the three comments that justified serial column formation by the counter are reworded (the columns are still formed before the blocks are solved, for the ordering, not for the counter); (ii) `element_transport_residual` no longer takes the cell velocity it never read (four call sites: `certification.f90`, `steady_newton.f90`, two test rows), the balance being the divergence of the FACE mass flux handed in; (iii) the outer Picard loop of `steady_wind_with_element_diffusion` halved the element under-relaxation omega whenever `comp_drift` failed to fall, but `comp_drift` is the worse of the element drift and the carrier drift while omega damps the element relaxation alone (the carrier pass is bounded by `carrier_trust`), so a carrier drift that failed to fall slowed the element alternation for nothing; the damping test now reads the element relaxation's own drift (`elem_drift`) and the exit test keeps the worst of the two. No regression case runs both He_diffusion and the carrier transport, so the matrix cannot move. (iv) `f_vib_quench/e_vib_bound` staying at the entry composition inside the self-field loop is CLOSED BY MEASUREMENT (COST5): they are

entry state through the operator-split field, and holding them at the first-pass value is worth 0.00 passes (6.96 against 6.96). Suites: steady_species_rows 74/0, certification 29/0, diffusion_tests 35/0.

- B5e** (steady solve globalization; steady_newton.f90, the steady_species_rows suite, design doc section 17, contract section 11.5): **the first stationary solve with a species row that meets its own target.** MEASURED on mol_diffusion reloaded with Coupled carrier solve: True (one element row, nvar = 4): the scaled trust region takes $||R||$ from 1.986 to **4.345e-09** against a target of 1e-08 in 66 outer iterations (54 accepted), where the pseudo-transient line search aborts at iteration 18 on "no descent direction exists for the banded model" at 1.702. Two defects were named by instrumenting the eleven exits of trust_region_step and repaired where the measurement pointed: (i) the radius collapse was the CAUCHY LEG ASCENDING the model, because its gradient is the transpose of the BANDED A and has no sign (MEASURED $r_0 \cdot A g = -2.1e13, -2.2e19$ at the collapsing iterates, pred negative and exactly linear in the radius); where $r_0 \cdot A g \leq 0$ the leg is now dropped and the dogleg reduces to the Krylov leg cut to the ball, which always predicts a reduction (GMRES gives $r_0 \cdot A s_N < -||A s_N||^2/2$); and pred is formed as $-(r_0 \cdot A s) - ||A s||^2/2$ instead of the cancelling difference of two squared norms, whose sign on a short step was rounding; (ii) the best-iterate ledger ranked on $||R||$ (three hydrodynamic rows) while the state is judged on those AND every species row, so it could hand back the iterate with the worse element balance; it now ranks on distance_from_certification, the largest of the judged row measures over their tolerances, formed once inside eval_residual under an optional argument; the three-unknown branch keeps its comparison character for character. **Refused by measurement:** bounding the STEP by the judged rows (implemented in all three step controls) stalled the element solve at $||R|| = 1.92$ on 13 refusals; and the banded-gradient Cauchy length $(r_0 \cdot A g)/||A g||^2$, more defensible on paper, left the same solve crawling at 1.8e-4 after 81 iterations against 4.3e-9 in 66 with the original length, so the original stays, recorded at the site with both numbers. **Default moved:** use_tr = (nspec_row > 0), the trust region is the step control of the coupled route (EXHALE_TRUST_REGION=0 restores the line search for measurement); no regression case sets Coupled carrier solve: True, so no golden can move. **Anchor:** the elemental transport He/H row of the returned state reads **4.633e-11 against cert_tol_element = 1e-8** (215x inside), the first measurement under that tolerance; it is NOT moved, because contract sections 9 and 10 want a ladder (the same case at Resid tol 1e-9 and 1e-10, not run) to tell a target-limited row from one at its floor. The state is not certified: its cell-1 hydrodynamic mass row reads 6.2e-11 against 3e-12, a base-cell matter and not an element-row one. **The carrier arm is not rescued:** on mol_carrier the trust region stagnates at 1.873 (line search 1.855) and 18 of its 49 iterations end on "no Krylov direction could be sampled", so cert_tol_carrier stays UNCONFIRMED, and the measurement says it is not a step-length problem. Worker MEASURED wasp_full_newton byte-identical (it 6, $||R|| = 8.131e-09$, info 0, CERTIFIED, log10 Mdot 13.30) and the mol_diffusion trust-region run identical to the pre-item text under EXHALE_TRUST_REGION=1 (every landed change inert on a path that already worked). Recorded, not fixed: EXHALE_SPECIES_JAC_TEST=1 perturbs the run it measures (iterations 1-5 agree to every printed digit, iteration 6 differs, the two runs then diverge; some module state survives a probe eval_residual; the same root writes n_eq_sweeps_model and the frozen background); and the ray test can refuse a step that reduced the merit far MORE than the model promised (ratio = 7187), which Algorithm 4.1 of Nocedal and Wright never does, the obvious amendment (gate it on

ratio $\leq$ eta_accept) left unmade for want of a control measurement. The worker was stopped by a rate limit after writing its report; advisor verified the tree: build clean; suites steady_species_rows 74/0 (six new driver rows), certification 29/0, species_face_flux 16/0, carrier_retry 101/0, coupled_source_step 30/0; one "per-iteration" phrase in run.sh reworted; wasp_full_newton re-run by the advisor against its golden with the B5e binary: all four files byte-identical, info 0, CERTIFIED, log10 Mdot 13.30.

- **Batch gate 2 (2026-09-08, 05:04 to 08:08):** make check on the tree as of B5e + HYG-4 (before COST6 landed; the harness built at 05:04, COST6's first edit is 05:17) **PASS, all 65 files of the matrix byte-identical** to the goldens refreshed by the user on 2026-09-07. make test in the same script failed 7 suites on their build-staleness precheck only (the shared build/ had not been remade after HYG-4 when the suites ran, the harness remade it afterwards); the suites themselves were run on the advisor's private build of the same tree: steady_species_rows 74/0, certification 29/0, species_face_flux 16/0, carrier_retry 101/0, coupled_source_step 30/0 then 39/0 with COST6, diffusion_tests 35/0, physics_probe 1491/0, and energy_update, grid_and_gates, spectrum_type 53/0, 93/0, 154/0. Every suite of make test is therefore green on this tree. No golden touched.
- **COST6** (extrapolation of the coupled loop's geometric decay; EXHALE_main.f90 +532 lines, coupled_source_step +3 whole-binary rows; **default OFF**, EXHALE_CSM_EXTRAP=1 turns it on): implemented, guarded, measured. The object is the GLOBAL pair (T, f_sp), which is exactly the loop's iterate (the energy solve rewrites u(3,:) and W(3,:) in full from T and the entry anchor each pass, so an undo of the pair is exact, and the guard-forced-refusal row reproduces the off run byte for byte); one Rayleigh-quotient ratio per block over all cells and species, the projection onto the dominant mode COST5 measured stable, and one damped ray, so no cell can move in a direction the global mode lacks. That is how it differs from COST2's per-cell Aitken (12.00 against 11.05 passes): a ratio taken cell by cell spreads 200 percent across the grid and cannot be tested for validity; the global one is tested ($||d_k - \theta d_{k-1}|| \leq 0.5 ||d_k||$) before use. Admissibility is by construction (fractions in [0,1], He 2³S inside its He I column, T inside the energy solve's own bracket, all linear in the step, so the largest admissible multiple is taken); MEASURED, the worst violation is identically zero on every case at 300 and 1500 steps, 1 and 16 threads, and the step was never damped below 0.95. The stopping test and its tolerances are untouched: a candidate is kept only if the pass taken FROM it meets them. Three things the measurement taught: the mode ALTERNATES (cosine of consecutive increments -0.85 to -1.00 on mol_base_handoff, +0.71 to +0.98 on wasp_full; the first implementation demanded $\theta > 0$ and refused 1188 of 1188 candidates); the one-mode model does not reach back into early passes (the ratio drifts 0.77, 0.47, 0.31, 0.22, 0.17, 0.14, 0.11 over one wasp_full step), hence a REACH guard, take the limit only where the modelled remaining error is already inside the tolerance ball (reach scan 6.61 / 6.15 / 6.36 / 7.81 / 7.74 / 7.75 at 1/2/3/10/30/100, control 6.96); and undoing a mediocre candidate costs more than it saves (7.29 strict against 6.94 lenient, control 6.99 on mol_metals). MEASURED, sweeps control -> on: mol_base_handoff 1500 10780 -> 10102 (**-6.3 percent**, matched pair one binary 285.4 -> 268.0 s), mol_carrier 300 -2.3, mol_metals 400 -0.75, wasp_full 300 -0.63, hydrostatic_column 0; state movement 8.3e-8 to 5.0e-7 relative, log10 Mdot unchanged. **The brief's estimate of three passes of seven is NOT reach-**

able, and the reason is the item's real finding: the stopping test measures the INCREMENT, and the error is $l\theta/(1-\theta)$ times it, one to two decades smaller, so the loop is one to two decades stricter than its tolerances say and spends 1.0 to 1.6 passes on that; the extrapolation's ceiling is the last factor $1/l\theta$. Redefining the test on the estimated error is a change of what the loop returns and is put to the user as decision item 8 of docs/To_be_determined_by_user_20260906.md (recommendation (b)). Tests: coupled_source_step RED 2 rows against the control, GREEN 39/0 (extrapolation_lowers_the_pass_count 6.09 against 6.96 at 300 steps; refused_extrapolation_costs_nothing identical at 6.80; mean_passes_within_bound re-measured 6.80, unchanged); physics_probe 1491/0. Worker-introduced defect fixed inside the item: a ledger write with six integer descriptors for seven items. Recorded, not fixed: write_setup_report.f90 does not echo the switch; n_csm_x_kept can only count candidates the stopping test did not meet and reads 0 everywhere (the number that matters, met on the very next pass, is printed beside it); the csm_max_pass comment block is now some 60 lines of measured tables and belongs in a docs/ memo. Advisor: the measured tables and the item history that had grown to some 230 comment lines around the coupled-step constants are moved to docs/coupled_source_loop_cost.md (sections 1-7, decision item 8 linked); the code site keeps each constant's conclusion and a section pointer. The setup report does not echo environment switches (none of them, by design), so the switch is reported by the run counter line alone. Advisor verification: coupled_source_step 39/0 and physics_probe 1491/0 on the advisor build; default-off byte-identity against the GOLDENS at matrix length (which checks B5e, HYG-4 and COST6-off together): mol_base_handoff 12000 steps and wasp_full to its own stop, both Hydro_ioniz.txt and Ion_species.txt IDENTICAL to the goldens (both runs end with status 2, "not certified", exactly as the golden runs do).

- **COST7** (decision item 8, option (b), user 2026-09-08; EXHALE_main.f90, coupled_source_step suite, docs/coupled_source_loop_cost.md sections 7-8): **the coupled source loop stops on the ESTIMATED REMAINING ERROR of the pair (T, composition), not on the last increment.** The estimate is $l\theta/(1-\theta)$ times this pass's increment in each of the two norms the test uses, θ the global Rayleigh-quotient ratio, formed in ONE routine (coupled_pair_geometric_error_estimate) that the COST6 extrapolation now reads too (its reach guard is literally the stopping test at a loosened tolerance; shared objects renamed csm_x_* -> csm_geom_*, env hooks EXHALE_CSM_GEOM_*). Where the sequence gives no estimate (passes 1-2, a guard failing) the increment test stands, and that fallback (EXHALE_CSM_GEOM_REFUSE=1) is byte-identical to the pre-COST7 binary on all 10 written files of wasp_full 300 and all 7 of mol_base_handoff 1500. **The item's finding:** the estimate is an L2 quotient against two max-norm tolerances, and MEASURED with a probe that runs the loop on to $1e-12$ and rolls back, the unfactored estimate accepts states OUTSIDE csm_comp_tol on 100 of 299 steps of mol_base_handoff (1.8x the tolerance at worst); a factor 2 leaves 1 of 299; **a measured safety factor csm_err_safety = 3 leaves none**, the accepted state's distance to its own fixed point reading $7.4e-9$ and below (composition) and $2.1e-9$ and below (temperature) on every one of the 2496 accepted states probed. The price is that almost none of the passes COST6 estimated survive: wasp_full 300 10.09 -> 10.01 passes (sweeps -0.9 percent), mol_base_handoff 1500 7.19 -> 6.96 (-3.1), mol_metals 400 6.99 -> 6.93 (-0.9), **mol_carrier 300 6.01 -> 5.01 (-16.6)**, hydrostatic_column 1.01 -> 1.01 (+1 sweep of 302: on its single three-pass step the estimate EXCEEDS the increment, so the new test is the stricter one there; it is not uniformly looser). log10 Mdot

unchanged on every case; state movement $4.3\text{e-}10$ to $1.1\text{e-}7$ relative, at the tolerance level. What the change buys is the MEANING of the two tolerances, the decision's own first reason, not passes. The COST6 extrapolation under the new test is worth nothing (6.79 against 6.82 passes) and stays default off. Tests: coupled_source_step RED 41/3 against the control, GREEN 44/0; mean_passes_within_bound re-measured 6.23 (was 6.80), both at the site; extrapolation_lowers_the_pass_count rewritten as extrapolation_does_not_cost_passes; the fallback-is-one-path row replaces the brief's byte-identity row (a suite cannot hold a golden of the previous binary; the identity is measured in the report). physics_probe 1491/0. Recorded, not fixed: the composition increment of mol_base_handoff has a floor near $5\text{e-}9$ (127 of 1500 probes cannot reach $1\text{e-}12$ in 59 passes), so csm_comp_tol = $1\text{e-}8$ is within a factor of two of what that case can reach at all; COST6's n_csm_x_kept still reads 0. **Every golden of the matrix moves at the 1e-8 level:** gate 3 below.

- **Gate 3 (2026-09-08, 09:29 to 12:00)**, tree after COST7 (with COST6 default off, HYG-4, B5e): shared build remade, **make test all suites green**; make check verdict FAIL against the $1\text{e-}3$ relative tolerance, as expected for a change of the stopping test, and every movement examined. log10 Mdot unchanged on all 14 cases that print one (13.35, 13.35, 13.30, 10.57, 10.58, 10.54, 10.57, 10.58, 10.51, 10.56, 7.29, 8.27, 9.61, 7.98). Movement table (largest relative change of any column, new against golden): wasp_full $6.6\text{e-}5$ (v at the base, cell 136), wasp_he23off $3.8\text{e-}4$ (v at cell 146), wasp_full_newton $1.1\text{e-}10$, mol_base_handoff $2.8\text{e-}6$, mol_metals $2.4\text{e-}6$, mol_lyman_werner $4.1\text{e-}7$, mol_diffusion $2.8\text{e-}6$ (adv file $2.2\text{e-}3$ at the outer boundary, cell 504), mol_ir_bands $2.4\text{e-}6$, mol_sec_ion $4.5\text{e-}3$ ($v = 0.091$ cm/s at cell 177, a near-zero base velocity, adv $1.5\text{e-}3$), mol_carrier $5.2\text{e-}7$, lower_profile $9.3\text{e-}6$, hp_zero_seed / hp_trace_seed / hp_front identical, roundtrip within. Two cases move by order one and both are pre-existing base artefacts, not COST7's: **oxygen_chemistry** (1000-step snapshot) reads 42x in n(H II) at cell 22 because the GOLDEN carries an odd-even sawtooth there (cells 21, 24, 25, 27 at $7\text{e}3\text{--}9\text{e}3$ against $2\text{e}5$ in their neighbours) and the new run is smooth ($1.6\text{e}5$ to $3.9\text{e}5$ across cells 18-26; rho, p, T within $1\text{e-}6$); **hydrostatic_column** reads 0.999 in the adv file's T at cell 350 because the adv post-process gives $T_{\text{adv}} = 0.78$ K inside 1.40 Rp in this static column (the original T is 1124 K throughout) and the edge of that region moved by one cell; the non-adv files move by $4.6\text{e-}8$ (Ion_species by 0.16 at cell 2 on values of $1\text{e-}182$, denormal noise). The adv defect is recorded as an open item. **Goldens to be refreshed by the user** (archive first), then make check re-run.

- **ADV-STATIC** (post_process_adv.f90, new suite adv_static_limit, docs/postprocess_advection_validity.md): **the brief's premise is refuted and the 0.78 K of hydrostatic_column's _adv file is named.** MEASURED: the flow where it sits is Mach 0.23 to 0.82 (the case is a 300-step relaxation snapshot, not a static column: ρv^2 spans seven decades); the IONIZATION correction is held at equilibrium in 503 of 503 cells (Ion_species_adv.txt equals Ion_species.txt to the bit) and its Damkohler condition already gives the continuous $v \rightarrow 0$ limit the brief asked for, so nothing there was touched. The number is the exact solution of the equation the _adv ENERGY solve poses: the steady internal-energy equation with the enthalpy flux of the mass-flux divergence DROPPED, $\rho v \frac{de}{dr} - p v \frac{d \ln(\rho)}{dr} = \text{heat} - \text{cool}$, which equals the exact $\dots + h \text{div}(\rho v) = \text{heat} - \text{cool}$ only where ρv^2 is stationary; with the radiation off it reduces to the discrete adiabat $T_j/T_{j-1} =$

$\rho_j / [\rho_j - (\gamma - 1)(\rho_j - \rho_{j-1})]$, and integrating that on the run's density reproduces the reported temperature to 6e-11 over the 347 cells before the first fallback (1138 K $\rightarrow$ 0.78 K at 1.396 R_p; the dropped term exceeds the kept ones in 480 of 502 cells). With heat/cool negligible the equation is homogeneous in v , so no $v \rightarrow 0$ condition can change its answer (99.96 percent departure at $|v| \leq 1.4e-3 c_s$). Landed: a validity condition of the energy solve's own, the thermal Damkohler number $Da = t_{\text{cross}} |heat - cool| / u_{\text{th}}$ (new pure thermal_damkohler_number; above one the cell keeps the run's own T), the counterpart of ionization condition (ii) that the bare $v \leq 0$ sign test lacked; the stationary-mass-flux assumption stated at the site with its exact dropped term and MEASURED at run time (largest ratio, where, how many cells exceed one; nothing acts on it yet); the first eval_cool writes a named tcool_in instead of a discard slot; the stale "is not evaluated" sentence of the validity memo rewritten with both tables. Impact: byte-identical everywhere measured (hydrostatic_column full case all four files; wasp_full, mol_base_handoff, lower_profile post-processed from the same golden state by both binaries, all four files; max thermal Da over tested cells 0.951 / 0.534 / 0.055 / 3e-19, 0 cells above one anywhere). Suite RED/GREEN (rows 1-7 cannot link before; row D measured=silent before), advisor 10/0. Also MEASURED: no matrix case has a stationary mass flux by the cell-to-cell measure ($\rho v r^2$ over outflowing cells spans 1.27x its median in wasp_full, 2.2x in mol_base_handoff, 83x in lower_profile); the du stop is taken on a restricted window and does not bound this. **Follow-up decided by the advisor under the physical-correctness rule: restore the dropped term (item ADV-ENERGY)**, since an energy equation that is exact only for a stationary mass flux is wrong on every snapshot and at the 1e-3 level on every converged wind; MEASURED worth on the static column 569 K at 1.40 R_p and 121 K at 2.96 R_p in place of 0.78 K and 6.0 K. The alternative, refusing the column where the assumption fails, would rewrite 162 of 502 _adv temperatures of wasp_full (median 5 percent) and is decision 14 (T8.1) of the target-system document, left to the user. Advisor fixed the stale provenance paragraph of the case README.

- **ADV-ENERGY** (T_{equation}.f90, ion_cell_state.f90, post_process_adv.f90, adv_static_limit +7 rows, docs/postprocess_advection_validity.md): the advection post-process's energy equation is now the EXACT steady internal-energy equation. The enthalpy flux of the mass-flux divergence, $h \operatorname{div}(\rho v)$ with $h = e + p/\rho$, enters T_{equation} through a new zero-default field of the cell record, teq_cell%div_rho_v, as a separate additive term in both branches (caloric $\mu_m \operatorname{div}_{\rho v} (E(x_{H2}, T) + T)$, monatomic $\gamma \mu_m \operatorname{div}_{\rho v} T$; μ_m, μ_p , coeff untouched; zero field an exact zero, tested to the bit). $\operatorname{div}(\rho v)$ is the mass row's own operator $(A_p F_p - A_m F_m)/dV$ over the control volume between the two points the upwind difference uses. MEASURED corrections to the briefs: T_{equation} has no caller outside post_process_adv (no marching solve to protect; the bitwise property holds anyway); and ADV-STATIC's 569 K prediction had put back $p \operatorname{div}(v)$ alone, not $e \operatorname{div}(\rho v)$; the whole term gives **1.7e7 K at 1.40 R_p and 1.6e9 K at 2.96 R_p** on the static-column snapshot, the closed form $w \sim \rho^{(\gamma-1)} F^{(-\gamma)}$ with $F = \rho v r^2$ falling 8 percent from cell to cell. Tests RED (nine "not a member" compile errors before) / GREEN (closed-form solution $T \sim r^{(-\gamma)}$ on a diverging column to 6.9e-4 at 400 cells, first-order convergence 5.59e-3 / 2.79e-3 / 1.39e-3 / 6.95e-4, the adiabat departing by 2.17); advisor adv_static_limit 17/0, physics_probe 1491/0. **Impact, MEASURED, same golden state post-processed by both binaries:** marching files bitwise in

every case; T_adv movement wasp_full_newton 1.4 percent (the only Newton-converged state, He 2³S 1.9 percent), wasp_full (du stop) 16 percent at 1.088 R_p (He 2³S 16 percent), mol_base_handoff snapshot 301 percent, lower_profile snapshot 456 percent (He 2³S 238x), hydrostatic_column 4.2e9 (reaching 2.6e10 K). Restored term against the kept ones: wasp_full 67x at 1.033 R_p in 161 of 502 cells, lower_profile 25x in 299 cells, wasp_full_newton 24x in 2 cells. **The _adv temperature of a state that violates continuity is now unbounded above where it was bounded below, and the transit tools read those columns:** on such a state NO steady post-process has a meaning, and what to output there is decision 14 of the target-system document, now put to the user as decision item 9 with a recommendation. Golden: every _adv file moves by design; refresh deferred by the user to this item's gate.

- **B5g** (steady_newton.f90 +467/-35, steady_species_rows suite, design doc sections 18, 18.5): **a residual evaluation is now a function of its arguments.** Three class-(b) caches survived a probe eval_residual and were read back by the solve: the elemental rows and scales (erow_he/escale_he/erow_tr/escale_tr), the carrier measures (carrier_relnorm_last...), and the carrier arrays of diffusive_photochemistry (col_scale_car, row_scale_car, row_terms, headroom_car, the frozen backgrounds); the Newton ROW SCALE, the acceptance gate, the best-iterate ledger and the choice of the state handed back read them. Fixed on the reader side: eval_residual gained state_is_discarded and a probe holds/puts back that set (reusing the save_/restore_carrier_module_state pair certification.f90 already uses); the loop-top row scale is taken AFTER the iterate's own evaluation; the gates read a new carrier_relnorm_state. The two self-tests also tag their sweeps as candidate sweeps (under the iterate tag they were writing the run's acceptance ledger and ieq_nonroot_streak, whose limit stops the run). PROVEN: EXHALE_SPECIES_JAC_TEST=1 and the untested run are the same run, every (JFNK) it line identical over 17 iterations, Ion_species.txt byte-identical (RED: 82 differing lines, first at iteration 6). wasp_full_newton byte-identical (info 0, 6 iterations, CERTIFIED, 13.30). **What the fix uncovered is bigger than the item:** the mol_diffusion element solve that B5e brought to 4.3e-9 in 66 iterations now STAGNATES at 1.957 after 17; the runs agree for five iterations and part on a 3 percent difference of Drow between two evaluations of the SAME state, because the residual's seed dependence (resid_sc_probe) is **0.96 in units of Resid tol** against $||R|| = 1.986$, and tightening the sweep tolerance (1e-12, 1e-14 with pass caps 200, 500) never satisfies the increment test from either seed: **the eliminated composition residual of that state has no fixed point it reaches**, and B5e's convergence was a property of one sequence of evaluations. The change is kept (a row scale must be the row's own terms of the same evaluation). Part 2, the carrier arm: all 18 "no Krylov direction" exits and all 514 refused samples were ONE thing, the H2 carrier of cell 500 at -7.4e-19 (a species unknown on its lower bound), not the element budget (the counter's label named one screen of several; corrected, census added); jv_product now takes the backward difference $[F(Y) - F(Y - \text{eps } v)]/\text{eps}$ only after every forward halving is refused: 49/0/0 accepted/rejected/ refused where 31/18/18, GMRES truncations 18 -> 0, radius 8e-3 -> 2.05; $||R||$ unchanged at 1.873 because the dogleg cuts steps by 2⁻²⁵..2⁻⁴⁹ against the same bound from the trial side, a bound-constrained unknown space (decision, not repair). cert_tol_carrier UNCONFIRMED. Suites (advisor build): steady_species_rows 74/0 (88/0 with the self-test pair logs), certification 29/0, coupled_source_step 44/0, carrier_retry 100/0, species_face_flux green; advisor re-ran wasp_full_newton in full with the B5g binary: Hydro_ioniz.txt

and Ion_species.txt IDENTICAL to the gate-3 output, CERTIFIED, log10 Mdot 13.30.

- **B5f** (docs/a2_certification_contract_20260906.md section 11.6; no constant moved): the cert_tol_element ladder on mol_diffusion at Resid tol 1e-8 / 1e-9 / 1e-10 gives ONE state: the 1e-8 solve meets its target in 66 iterations (element row 4.633e-11, reproducing B5e digit for digit, so COST7 leaves the stationary solve untouched); the 1e-9 and 1e-10 solves abort after 88 iterations with the radius at 9.3e-23 (9 of 31 refusals on "the ray difference is below its cancellation floor") and hand back the SAME state, bitwise. The row is floor-limited at 4.63e-11 (knee at ||R|| about 1e-7, 1.5 percent movement over the last factor 26); proposed value by the contract's rule 5e-10, NOT applied (two rungs are not solves that met their target; a decade above one configuration's floor is not a decade above the row's floor, which the operator's header puts between 1e-12 and 1e-5 across configurations; no second configuration exists). **Read together with B5g:** B5f ran on the pre-B5g text, and B5g shows the state it anchored on is reached by one sequence of evaluations only (with the probe caches removed the same solve stagnates at 1.957), so this rung stands as a measurement of that state's row, not as evidence that the solve converges. B5f also MEASURED that the coupled route's convergence depends on eq_sweep_reltol (1e-10 loses the 1e-9 solve at iteration 17), the same fact from the other side. Recorded: the JFNK of a reloaded 1e-8 state starts from ||R|| = 1.226 because the marching loop takes two CFL steps first (the O(dt) kick, eight decades, seen for the first time from a stationary state); the certification prints no cell-resolved record of the species rows, which blocks any floor attribution.
- **ADV-REFUSE** (decision item 9, option (a), user 2026-09-08; post_process_adv.f90 +227/-43, write_output.f90 +94/-10, examples/exhale_io.py, EXHALE_transit.py, README.md, README_HOWTO.md, user manual section 4 (PDF rebuilt, 76 pages), docs/postprocess_advection_validity.md, adv_static_limit +13 rows): **the advection post-process refuses the correction in every cell whose own energy-equation term ratio says the flow is not stationary; that cell keeps the run's temperature and the ionization equilibrium at it; a new integer column adv_status at the end of Hydro_ioniz_adv.txt records the reason row by row** (0 corrected; 1 thermal Damkohler; 2 not stationary; 3 ionization conditions (i)-(iii); 4 cell solve failed; five named module parameters in write_output, the # columns line extended, a # adv_status_counts header line counted from the array written). The criterion is enthalpy_flux_term_ratio (new pure function beside thermal_damkohler_number), the restored term over the LARGER kept term, refused on a STRICT > 1 so a stationary flux with round-off is never refused; formed ONCE on the state handed in over every cell, so the refused set cannot move with the iterate and the converged product does not depend on the iteration history; applied to temperature AND composition (a non-stationary cell has no steady advective ionization correction either). write_output gains an optional adv_status dummy (the four 'eq' calls untouched, Hydro_ioniz.txt keeps seven columns); exhale_io.load_hydro now takes names from the header (seven historical keys kept, Run.adv_status exposed, None for an equilibrium file); EXHALE_transit.py prints how many rows inside 1 <= b <= Rib are refused, by reason; EXHALE_plots.py and the four other header parsers adapt unchanged. MEASURED (same golden state, "Do only PP"; marching files bitwise everywhere): hydrostatic_column's corrected temperature falls from 2.6e10 K to **1182.32 K, exactly the run's own maximum, no row above it** (435 before); status counts 0/1/2/3/4: wasp_full_newton

501/0/2/1/0, mol_base_handoff 470/0/33/1/0, wasp_full 205/0/297/2/0 (160 of the 297 beyond conditions (i)-(iii), matching ADV-STATIC's route-2 estimate of 162; the larger total is the reference point, the state handed in over every cell including the inflow cells), lower_profile 182/0/321/1/0, hydrostatic_column 0/0/481/23/0. Every refused row carries the run's T to the BIT and the equilibrium density of every species exactly (one $1.8\text{e-}16$ reassociation in lower_profile's He I, pre-existing). Movement against the pre-ADV-ENERGY goldens, T_adv: wasp_full_newton 3.3 percent, wasp_full 8.0, mol_base_handoff 64, lower_profile 1.8x, hydrostatic_column 1525x (0.78 -> 1182 K); species: He III 5.4 percent, He 2^3S 29 percent, He 2^3S 110x (at the base of mol_base_handoff), H II 11x (lower_profile base), He III 16 percent. **Transit effect on wasp_full**: 295 of 500 rows inside the impact-parameter range are refused (printed), and the metrics move at $1\text{e-}4$ relative or less (He 10830 red 4.102 percent both, blue 2.843 -> 2.844, FWHM 0.899 Å both; largest line movement O I 1302 +0.55 percent relative), because the refused rows sit in the base under the extended wind. Tests RED (seven "not found in module" errors; measured=7 columns on the old binary) / GREEN; advisor adv_static_limit 30/0, physics_probe 1491/0. Worker also fixed a dangling \ref{sec:retiredbase} in the manual (the four retired base keys named instead). Recorded: two documents describe the _adv validity conditions (manual, HOWTO) and had both drifted; four # columns parsers exist in python; dV_cv division now appears twice unguarded (a zero is a broken grid).

- **B5h** (ionization_equilibrium.f90 +68/-2 diagnostics, steady_newton.f90 +211/-2 probe only, steady_selfconsistent_residual assertion 4; NO behaviour changed, wasp_full_newton and the element run byte-identical to a control built from the same source with the additions removed): **why one state has two residuals**. MEASURED on mol_diffusion's element configuration with a lockstep two-seed trace of the composition elimination (EXHALE_RESID_SC_TRACE), a nine-seed basin ladder (EXHALE_RESID_SC_BASIN) and a cell acceptance report (EXHALE_IEQ_REPORT_CELL): the amnesty is never used (classes 4 and 6 zero at every pass, every cell class 1 with reaction residual $1\text{e-}25$ against $1\text{e-}6$), the self-field lag is not it (xuv_self_field_passes 1 -> 20, identical to every digit), the He recombination coupling is not it (off, identical to four digits), the stopping test is not it (a fixed count of one pass per call, chained, the difference frozen from pass 2 to 60). **Two things are**. (1) Two separated ROOTS of the network at the same incident field: from a seed displaced five cells the sweep reaches a composition 8.9x apart in x(H2) at the H2 front (cell 214) and 19 percent apart in x(H I) at the ionization front (cell 192), bitwise identical from cell 216 outward, both exact roots (class 1, $1\text{e-}25$). (2) **A marginal direction: the local-equilibrium elimination does not determine the shielded layer's H2 CONTENT**. Retention of a seed perturbation of the H2 partition is 0.77 at the base rising to 0.84 at $r = 1.064$ and falling to 0 outside cell 216, the SAME curve to three digits at perturbations of $1\text{e-}6$ and $1\text{e-}2$, so the sweep's map has eigenvalue 1 in that direction and the seed dependence is a derivative, linear over four decades with gain $3.2\text{e}3$ (the base cell's energy row amplifies the ghost composition by $\sim 4\text{e}3$), reaching Resid tol at a seed change of $3\text{e-}12$. READ from the network: the fast $\text{H2} \rightarrow \text{H2}^+ \rightarrow \text{H3}^+ \rightarrow \text{H2}$ cycle returns every nucleus, so the H2 row as scaled by its turnover pins the PARTITION and leaves the total to $k15 \ n_{\text{HI}}^2$ against $k12 \ n_{\text{tot}} \ n_{\text{H2}}$, a detailed-balance pair that in the shielded layer is far below the turnover; and pinning it would pin the fully molecular limit (0.9997) where the state carries 0.376: the content is upstream and transport data, exactly why it is imposed at the ghosts. **The repair the code already owns, MEASURED**: with Molecular carrier transport: True ($n(\text{H2})$ a

Newton unknown) the same ladder falls from $3.155\text{e-}3$ to **$9.736\text{e-}10$** at $\Delta 1\text{e-}6$ and $x(\text{H}_2)$ of the front cell is bit-identical from all nine seeds; the residual is a function of its arguments below $\text{Resid}_{\text{tol}}$. What remains at large seeds (0.98) is the ionization front's two roots, a physics decision (which root a stationary residual may land on, or a statement that the front is unresolved on this grid), reported. The element solve with H_2 carried then meets B5g's carrier obstruction ($|R|$ flat at 1.986 over 19 iterations, carrier row 0.91, run stopped by PID at 4 min an iteration). Also MEASURED: eq_sweep_reltol below $\sim 1\text{e-}10$ is unreachable by construction, the increment measure $\max|df|/\max(|f|, 1\text{e-}12)$ reporting round-off of He I and H_2^+ fractions $\sim 1\text{e-}20$ in the ionized wind (plateau $6\text{e-}11$ for sixty passes), which is what B5f's and B5g's "never met at $1\text{e-}12$, $1\text{e-}14$ " was. Tests: assertion 4 $\text{residual_is_one_number_for_basin}$ RED $3.155\text{e-}3$ with H_2 eliminated, GREEN $9.736\text{e-}10$ with H_2 carried (the tolerance read from the run's own log); $\text{steady_species_rows}$ 86/0 with the self-test pair; B5g's part 1 reproduced as a by-product. A defect of the item's own fixed inside it (a cell-report default of -1 was the lower ghost). **Decision put to the user as item 10:** refuse Coupled carrier solve: True on a molecular configuration without carrier transport, or turn the transport on for it (input_read.f90 , not the item's).

- **B5i** (decision item 10, option (a), user 2026-09-08; $\text{input_read.f90} +51/-2$, grid_and_gates new row $\text{coupled_carrier_h2}$ (+5 assertions), user manual (a Coupled carrier solve entry, which the manual had NOT had; the Molecular carrier transport entry corrected from "inert without oxygen chemistry" to "needs molecular chemistry"), README.md, README_HOWTO.md (new block; the oxygen paragraph's dangling "It" corrected), docs/steady_solver_design.md section 19): **Coupled carrier solve: True with Molecular chemistry: True and Molecular carrier transport: False is refused at startup**, after every key is resolved (so the oxygen default of the transport key is seen: MEASURED, Oxygen chemistry: True plus the coupled key is accepted, the same with the transport stated False is refused), with a message naming what was asked, why there is no root (B5h's numbers) and the remedy key; $\text{error_stop } 1$, exit status 1 asserted (2 is the certification's refusal of a complete run). Scope exactly carrier_in_newton .and. thereis_mol .and. $\text{.not. carrier_transport}$, not gated on the Newton key (the direct steady route $\text{EXHALE_PTC}=1$ runs a coupled solve too, and it is the route B5b to B5h used); the atomic coupled solve with an element row is accepted (asserted). No shipped configuration is refused (MEASURED by grep: no *.inp in the tree sets the key; 62 of 81 B5-series scratch inputs would be, as the decision says). Two molecular cases byte-identical before/after at 200 steps. The parse-site comment that claimed "checked below" for a check that did not exist is corrected. A refused run leaves a zero-byte EXHALE_setup.out and no $\text{EXHALE_resolved.out}$, like every other input_read refusal. Tests RED (the eliminated- H_2 coupled run completed and printed a Mdot) / GREEN; advisor grid_and_gates 98/0, physics_probe 1491/0. Advisor corrected docs/input_schema.md K15e ("needs K15d" now qualified to the molecular configuration). Recorded: EXHALE_main.f90 opens EXHALE_setup.out (truncating) before input_read , so any refusal erases the previous run's setup report (fix: open it after the input is accepted); the other "checked below" comments of input_read.f90 deserve a sweep against the checks that exist.
- **HYG-SETUP** (advisor, direct; EXHALE_main.f90): the setup report EXHALE_setup.out is opened AFTER input_read returns instead of before it, so a configuration input_read refuses no longer truncates the re-

port a previous accepted run left in the same directory (B5i's finding). `input_read` writes nothing to that unit. `grid_and_gates` green on the advisor build; the marching path is untouched.

- HYG-CHECKED** (`input_read.f90` comments plus one new report block, `docs/input_schema.md`, `README_HOWTO.md`, user manual (PDF 77 pages), `grid_and_gates` new row `carrier_transport_inert` +4 assertions): every requirement statement of `input_read.f90` and every "Needs K.." / "refused with" clause of `docs/input_schema.md` checked against the code that enforces it. 40 enforced as stated. ONE not enforced (Molecular carrier transport: True with the molecular network off) is now a startup WARNING, not a refusal, by the code's own convention for a present-and-inert key (every consumer is guarded by `thereis_mol`, MEASURED inert; eight shipped inputs all carry the network). THREE enforced differently from the statement, one with the sense INVERTED: the schema and the HOWTO said Ionization transport is refused with Solver: Newton and with Coupled carrier solve, where the code REQUIRES the second whenever the first is set (K15e does carry a proton row; what is refused outright is the EXHALE_PTC=1 route); "Solver: PTC refuses it" named a solver value that does not exist; the base-grid resolution check lives in `write_setup_report` (warns below 10 cells per scale height). SIX stale statements corrected: `known_keys` also feeds the duplicate-key REFUSAL (MEASURED, all 99 entries used, no hole); the sub-Lyman absorbers are not mutually exclusive; `Kzz_base/He_Kzz` are NOT inert with `He_diffusion` off (the carrier transport reads `kzz_cell`), corrected in four places; Molecular reaction heat default True is guarded by `with_molecules` (a refusal would refuse every atomic run); the coronal cutoff window 0.08-0.13 superseded by `coronal_cutoff_width.md` 7.2; four K15b cross-references that pointed at the Stellar LW flux row corrected to K15. MEASURED byte-identical on `wasp_full` 100 and `mol_base_handoff` 200 (17 of 17 files). Advisor `grid_and_gates` 102/0, `physics_probe` 1491/0. Recorded: all 63 "F" line citations of the schema are stale by 300-1200 lines (marked stale at the convention's definition; recommend deleting the tokens); Coupled carrier solve: True is silently inert in an atomic run without `He_diffusion` (a report belongs where the row registry is); the user manual has no Ionization transport entry.
- Gate 4 (2026-09-08, 15:39 to 18:45)**, tree after COST7, COST6 (off), B5e, B5g, B5h, HYG-4, ADV-STATIC, ADV-ENERGY, ADV-REFUSE (B5i, HYG-SETUP and HYG-CHECKED landed during the matrix; they touch startup checks and a file-open order only, and were verified on the advisor build with the full `grid_and_gates` suite, 102/0): shared build remade, **make test all suites green**; `make check` FAIL against 1e-3 as expected. **The marching files are unchanged from gate 3 to the digit**: `Hydro_ioniz.txt` and `Ion_species.txt` carry exactly COST7's movement on every case (identical max-rel values in both logs, the three `hp_*` cases still identical), so B5g, B5h, the three ADV items, B5i and the hygiene items moved nothing on the marching path, and `log10 Mdot` is unchanged on all 14 cases. The `_adv` files move by design: `Hydro_ioniz_adv.txt` now has eight columns (`adv_status`, the harness reports SHAPE 8 vs 7), and `Ion_species_adv.txt` moves by 0.05 (`wasp_full_newton`, He I at the outer edge) to order one (the molecular cases and the `hp_*` cases, whose refused base cells now carry the equilibrium densities; the relative measure is element-wise, so a species the old correction had moved off its equilibrium value reads 1). **Goldens to be refreshed by the user, once, for COST7 + ADV-ENERGY + ADV-REFUSE together** (archive first), then `make check` re-run by the advisor.

- **Golden refresh (2026-09-08 19:05, run by the advisor on the user's instruction, the user being away from the machine):** the previous set archived to backup/regression/golden_pre_cost7_20260908/ (verified identical to golden/ before the refresh), then `run_check.sh golden` snapshotted the gate-4 outputs of all 16 cases. The refreshed goldens carry COST7 (the stopping test on the estimated error), ADV-ENERGY and ADV-REFUSE (decisions 8b, 9a). Gate 5, a make check on the current tree (B5i, HYG-SETUP, HYG-CHECKED on top, all startup-only), follows.
- **HYG-DOCS** (documents only: docs/input_schema.md, user manual (PDF 82 pages), README_HOWTO.md): the 91 stale "F" line tokens (77 lines) and the 37 GUI<n> numbers (MEASURED stale by a uniform 13 lines) are gone; a key is located by its label and the header says so; nine Notes cells that held only a citation got real notes READ from the parse sites. The manual gained the missing Ionization transport entry (in the sense HYG-CHECKED corrected) and the four other keys the schema documented and the manual did not (Reconstruction continuation, Atomic rate set, Caloric EOS, Photoelectron heating); MEASURED afterwards, all 97 schema keys have a manual entry. Three stale manual entries corrected (Resid norm is retired; Resid tol's norm is the per-row scaling of sections 143/145, not $|R|/\max|u|$; Flux spread tol default $2e-5$, not $5e-3$); the validity paragraph's "one of three" made four. The `_adv` validity conditions live in ONE place (manual section 4, `\label{sec:advvalidity}`), the HOWTO points there; the HOWTO names `exhale_io.py` as the loader every new script should use (four # columns parsers exist). Advisor: the same three stale statements in parameters.f90's comments corrected (default $1e-5$; the row scaling; the retired base valve), and the four schema rows whose code spans carried a | (a strict renderer split them) rewritten; table widths verified.
- **B5j** (steady_newton.f90 +1061/-41 only; steady_residual.f90 and diffusive_photochemistry.f90 bit-wise unchanged; steady_species_rows +14 rows, steady_selfconsistent_residual assertion 5): **the carrier arm's lower bound, MEASURED as two mechanisms in sequence.** The outermost carrier row is linear in its own unknown: at the hand-off its root is $+2.1e-11$ with the iterate 4 percent above (positivity-preserving, the Newton right to reduce it); twenty iterations later the row's constant term has changed sign and the root sits at $-5.5e-13$, below the bound, so the row has no admissible root and the unknown is pinned; and at the cell that binds the solve (215) the root is $+2.75e-6$ while the Newton has driven the unknown to exactly zero, eleven decades below it, the row reading -1.00000 of its own scale. A species unknown at zero whose row wants it positive cannot be sampled at all (a two-sided difference leaves the box on both sides). Landed: the carrier mass fraction clamped onto $[0,1]$ in `write_species_rows_into_composition` (the rule the element fractions had); the trial projected onto the faces of the species box with the predicted reduction recomputed for the step taken; Bertsekas (1982) epsilon-active set held for the step and applied to every sampled direction; the probe step cut to the box with the side chosen by the room (subsuming B5g's backward sample); the Cauchy point as the step when the Krylov leg cannot be sampled; a defect of the item's own found and fixed (the accepted trial re-evaluated as $Y + D$ s rather than as the vector evaluated, whose round trip does not close for a projected component: a carrier at $-4.8e-35$). MEASURED on `mol_carrier`, three arms, none converging: baseline 49 iterations ||R|| 1.929, 527 refused samples, cut-backs to 2^{-49} ; (a) bound-aware region 63 it, 1.415, 0 refused, 2^{-4} ; (b) carrier as `ln n` (EXHALE_CARRIER_LOG_UNKNOWN=1 (the spelling at B5j; since B5k the log unknown is the default and

=0 restores the density)) 65 it, **0.9972**, merit 17.6 -> 0.924, smallest adopted unknown 2.2e-11. (b) is better by every measure and is NOT the default (it changes the unknown space of a route a shipped key selects): decision item 11. **Part 2:** the elimination's stopping test is now the change of the three quantities the residual reads the composition FOR (particle count, H2 caloric share, the radiative pair), EXHALE_EQ_MEASURE=species restoring the old one: at eq_sweep_reltol 1e-12 wasp_full_newton reaches it in 33/49 passes where the species measure never does (plateau 2e-11); the element case's remaining plateau (1.85e-11) is the radiative channel at cells 441-459 wandering, that case's own non-convergence (B5h), not the measure's floor. Seed dependence unchanged to four digits (9.736e-10). **wasp_full_newton moves 1.1e-9 relative**, isolated to the new measure by EXHALE_EQ_MEASURE=species reproducing the control byte for byte: the solve converges FURTHER ($||R||$ 8.1e-9 -> 1.7e-9 in 8 iterations instead of 6, info 0, CERTIFIED, 13.30); below the 0.1 percent rule, the golden is not touched (the harness will read "within"). Suites (advisor build): steady_species_rows 93/0, steady_selfconsistent_residual 3/0, certification 29/0, coupled_source_step 44/0, physics_probe 1491/0; one "per-iteration" phrase in run.sh reworded. Recorded: a carrier row at an active bound is an error to the certification (row 1.00000 at cell 215) and a constrained optimum to the solver, so a bound-constrained stationary state has a multiplier there, not a zero row (decision item 12); 28 of 64 steps of arm (a) are the Cauchy point alone (Bertsekas' two-metric projection not taken); load_IC refuses a molecular restart written at the input's own He/H by comparing printed digits for equality (7.9307E-02 against 0.0793). Advisor: wasp_full_newton re-run on the matrix path (marching then Newton) with the B5j binary against the refreshed golden: movement 5.9e-9 (Hydro_ioniz.txt) and 9.6e-9 (Ion_species.txt), 13 outer iterations to $||R|| = 4.3e-9$, CERTIFIED, log10 Mdot 13.30; below the 0.1 percent rule, golden untouched.

- **Gate 5 (2026-09-08, 19:03 to 22:02):** make check on the tree as of B5i, HYG-SETUP, HYG-CHECKED (built 19:03, before B5j landed) against the refreshed goldens: **REGRESSION PASS, all 64 files of the 16 cases byte-identical.** The refreshed goldens are therefore the current tree's own outputs, and the three startup-only items moved nothing. B5j's 1.1e-9 movement of wasp_full_newton (the Newton converging further under the new increment measure) is not in this gate's binary and will read "within" at the next one; below the 0.1 percent rule it refreshes nothing.
- **B5k** (decision item 11, option (a), user 2026-09-08; steady_newton.f90 +346/-85, steady_species_rows +137/-6 (114/0), design doc section 20, schema K15e, manual entry, PDF 82 pages): **the molecular carrier densities of the coupled steady solve are carried as $\ln n$ by default** (EXHALE_CARRIER_LOG_UNKNOWN=0 restores the density unknown; the element fractions untouched; no certification change, decision 12 (c)). The floor under $\ln n$ is the transport operator's own admissibility floor, 1e-20 of the carrier's element budget in that cell (carrier_headroom), the fraction carrier_residual already puts under its row scale; MEASURED 5.6e-28 to 4.6e-21 across mol_carrier's column (B5j's arm had $\log = -\text{huge}$, unsound: exp underflows to an exact 0). Stated at the site: the column factor n of the Jacobian (both the banded model and the Krylov action probe a RELATIVE change), the region asymmetric in the density ($n \rightarrow n \exp(+\Delta)$), and $\ln 0$. **A real finding at the arming point:** carrier_headroom does not exist when the solve reads its controls (allocated inside carrier_steady_residual, first reached by the opening eval_residual), so the first implementation

silently fell back to the density arm on every coupled run; the opening state is now packed as a density and rewritten as logarithms after that evaluation, sound because the residual is a function of the state and not of its coordinates. Floor array made rank 2 (cell AND carrier row; two carriers of one cell have different elements). MEASURED, one binary, the two arms one variable apart: mol_carrier density 63 it ||R|| 1.415, 28 Cauchy-only steps, 10 unknowns on a bound, 716 adopted carriers at the floor; **ln n 66 it ||R|| 0.9957, merit 0.960, 0 Cauchy-only, 0 on a bound, 0 backward products, 0 at the floor, smallest adopted carrier 2.2e-11 = 2.5e15 times its floor**; mol_diffusion (H2 carried + element row) density arm 0 accepted steps of 12, log arm 50 accepted, ||R|| 1.986 -> 1.982. **Neither converges, and the next obstruction is named and is NOT the density arm's**: the log arm's steps are model-accurate (ratio 0.39-1.09) and SHORT (||s|| 0.09-0.17 against a radius of 2.5 sitting at its ceiling for 30 of 68 iterations), cut by 3-5 halvings before the trial is describable, and the refusing screen is the ELEMENT BUDGET (157 samples, against 0 for a negative carrier): the carrier CEILING, which species_unknowns_outside_their_bounds deliberately does not treat as a face of the box. Bertsekas' two-metric projection is not what this arm needs (it holds no bound). Gates: wasp_full_newton byte-identical twice (full case control vs new, and the reload identical to B5j's binary including every JFNK line); marching bitwise on mol_carrier 200 steps. New rows: default is ln_n, switch reaches density, the floor identical in both arms, carrier_is_above_its_own_floor (RED on the density arm: 0, 716 at the floor). Advisor: steady_species_rows 93/0, steady_selfconsistent_residual 3/0, certification 29/0, physics_probe 1491/0; the two documents that named the old =1 spelling annotated. Recorded: gm_m = 40 is a literal at the call site in EXHALE_main.f90 and the log arm sits at it in 58 of 66 iterations (unmeasured whether raising it helps, the cycle reports iterations used, not the residual reached); carrier_headroom returns huge as an availability sentinel. **Next item B5l**: the element budget as a face of the unknown box.

- **B5l** (steady_newton.f90 +392/-82; diffusive_photochemistry.f90 +51/-6, the headroom accessor only; steady_species_rows suite +161 and +108; docs/steady_solver_design.md section 21; certification.f90, EXHALE_main.f90, steady_residual.f90, ionization_equilibrium.f90, input_read.f90 untouched): **the element budget is a face of the unknown box and the projection treats it exactly**. freeze_species_unknown_box is now the one place the admissible set of a species unknown is written down (one lower and one upper bound per unknown of the flat vector, formed once per outer iteration from the iterate and from the element budget the carrier operator froze at that iterate); the three routines that had each computed their own bounds (species_unknowns_outside_their_bounds, fix_active_species_bounds, largest_step_inside_the_species_box) read it. **A drift between the three was found and fixed on the way**: the projection had read the carrier ceiling from the trial's own density slot while the other two read it from the iterate, so the set a trial was written onto and the set the Krylov cycle was sampled under were not the same set. Faces of a carrier unknown: below, zero (carrier_log_floor in ln n); above, the cell density and the element budget (carrier_headroom, for H2 half the hydrogen the carriers may take), the tighter of the two. The budget is frozen for the step, not evaluated at the trial, which is the carrier operator's own choice (headroom_car assigned under weno_mode == 1, the one evaluation that IS the iterate; refreshing it per trial was tried there and is worse), so box and screen describe one state. The budget face is raised to the iterate where the iterate stands outside it (counted, n_budget_face_at_the_iterate; 0 on both coupled cases with

the default step control); it is stored one floating-point step inside the budget (budget_face_inside = 1 - 8 eps, representability in $\ln n$, not a margin). An unknown whose box is a point (zero budget) is held without asking the gradient. EXHALE_SPECIES_BUDGET_FACE=0 restores the entry text's box (measurement switch, default on) [retired by N4b: the budget became a constraint row, EXHALE_ELEMENT_CONSTRAINT_ROWS=0 is the measurement arm now]. pgmres reports the relative residual it REACHED against the tolerance asked; EXHALE_GM_M overrides gm_m (the caller's 40 stays the default and the literal at EXHALE_main.f90 line 5343 is untouched). **MEASURED on mol_carrier (coupled, $\ln n$), face off vs on: 66 vs 52 outer iterations, both ended by the stall detector; ||R|| handed back 0.9957 vs 1.174; trials halved 3 or more times 38 of 68 vs 0 of 53; residual samples refused by the element budget 157 vs 0; trials written onto a face 0 vs 29; adopted carriers above their own budget 0 vs 0 (largest ratio 0.9886 / 0.9878); iterations ending "the model promises no reduction" 0 vs 19; worst Krylov relative residual reached 0.938 vs 0.569 where 0.1 was asked; log10 Mdot 10.63 both.** The halvings and refusals against the budget are gone exactly; ||R|| does not improve; the limit moved to a named place, the Krylov leg of the dogleg. **The Krylov budget is not the limit:** gm_m 40 / 80 / 160 give ||R|| 1.174 / 1.103 / 1.271, worst relative residual reached 0.569 / 0.630 / 0.614, ||s|| of the last accepted steps 0.04-0.24 in all three, log10 Mdot 10.63-10.64; the cycle spends every product and stops at the same 0.6 of its right-hand side, so it stagnates on this operator (preconditioner or operator, not subspace). On mol_diffusion (H2 carried) the face does the same to the step control (halved 6-11 times in 50 of 50 vs 0 of 41; budget refusals 388 vs 0) and the solve stalls where it always has (H2 front, cell 212, element row at 1.000 of its scale; ||R|| 1.982 vs 1.943; log10 Mdot 7.97 both), with the probe showing instead 419 backward-side products, 26 unknowns held on a bound, 12 Cauchy-only steps and 18 truncated cycles. A defect in the item's own first measurement was found and fixed (the adopted carrier was recorded against the PREVIOUS iterate's budget, which had reported 67 adopted carriers above budget, worst 1.058; the call now follows the iterate's own residual evaluation, arithmetic unchanged iteration for iteration). **Gates:** wasp_full_newton bitwise against the control binary (Ion_species.txt bitwise, Hydro_ioniz.txt identical but for the provenance line, every (JFNK) line identical, ||R|| = 4.299E-09, log10 Mdot 13.30, 13 iterations); marching bitwise on mol_carrier 200 steps; EXHALE_SPECIES_BUDGET_FACE=0 reproduces the entry-text binary exactly; steady_species_rows 118/0 with the new rows RED without the face and GREEN with it; physics_probe PASSED. **Advisor:** the five diffs match the tree against the worker's entry snapshots; forbidden-word scan clean; the current tree builds warning-free (build_lwv); the worker's wasp_full_newton output against the current golden moves 5.87e-9 / 9.58e-9, exactly the B5j movement already on record, so B5l itself moves nothing on the matrix path. No golden refreshed. **Recorded, not repaired (outside this item's files):** carrier_element_headroom states a box side per carrier but no joint bound for carriers sharing an element (the physical constraint $2 n(\text{H}_2) + n(\text{OH}) + 2 n(\text{H}_2\text{O}) + n(\text{H}^+) \leq n\text{H}_{\text{avail}}$ is a hyperplane the code does not state), and OH/H2O hold hydrogen nuclei their own headroom does not mention, so a state can pass every headroom and still leave the write-back's hydrogen remainder negative (the branch that zeroes H I, H II, H2+, H3+); neither reaches the {H2} carrier set both measured cases use. Also: the epsilon-active band (one finite-difference step) never engages on mol_carrier with the face (29 trials projected, 0 held; no switch, a wider band unmeasured); largest_step_inside_the_species_box returns zero for the whole direction when

one component has no room (18 truncated cycles on mol_diffusion); note_adopted_species_unknowns is not called for a converged state (pre-existing, cannot affect a stall). **Next obstruction on the carrier arm: the Krylov leg (banded preconditioner or operator), not bounds and not subspace.** Per the user's instruction of 2026-09-09 ("leave running things alone, stop code modification for now") no further item is started; B5m (the atomic element-row solve with no Krylov direction) stays planned.

- **Preparation of PLAN_20260909_rev1 (items N-1 and N17; advisor, documentation and file moves only, no source change; user's go-ahead 2026-09-09 with the instruction not to rerun regression checks after simple changes).** docs/PLAN_20260909.md was written from the review of the issues inventory (docs/ISSUES_20260909_review.md), reviewed (docs/PLAN_20260909_review.md) and revised to docs/PLAN_20260909_rev1.md, the plan of record (items N-1 to N17, stages A to E, decisions 13 to 19). Before writing rev 1 the advisor re-read the review's new source claims: the sampled integration-error estimate is printed and dropped and no production caller issues as_reject_int_error (EXHALE_main.f90 2806-2821); after a retry inside the full pass the half passes start from $0.5 * dt_step$ and the clock advances by the saved dt_step (1706-1734, 2797-2799, 2820, 2836); the audit harness asserts the OLD failure (if (ieee_is_finite(x(1))) error stop 4); use_tr = (nspec_row > 0) (steady_newton.f90 5763) so wasp_full_newton never enters the trust region, which corrects a statement of the first plan. All confirmed. **N-1:** the B5m reproducer moved from the session scratchpad into the tree as backup/regression/atomic_elem_newton/ (input, metals, setup and resolved reports, run.log, the two handed-back state files, run.sh, README; 6.8 MB, of which run.log 6.2 MB; not in DEFAULT_CASES); the common worker brief became docs/worker_rules.md with the session paths generalized and three rules added (one owner per source file per increment; affected-path validation only; named outcomes in reports); the review harness docs/audit_20260905/run_issues_review_20260909.sh was run once on the current tree, its output kept as docs/audit_20260905/run_issues_review_20260909.out (zero-radius step norm 0; Cauchy merit 0.5 to 40.5, line minimizer 0; identity GMRES 1; zero-operator GMRES non-finite with reported residual 0; enthalpy ratio 0.5 on a nonzero mass divergence; He/H pair 8.83e-5; hydrogen inventory 2 at the individual ceilings; outer-shell chord 11.83; "defects reproduced, not repaired"), the harness itself untouched. **N17:** ISSUES_20260909.md corrected per rev 1 section 3 (seed retention no longer stated as an eigenvalue of one; B5f marked a withdrawn, uncertified reference; 3.5 scoped to the present fixed-point map; the restart row now states the $heh_dev > 1e-6$ branch and the 8.83e-5 discrepancy; carrier_headroom_known noted; hydro_row_max and use_tr bullets; the _adv statement in section 5 made conditional; new section 3.8 on the physical step never being refused for its integration error). code_status_20260905.md: the Huang et al. 2023 shell-opacity note added to the photoionization row; four pre-existing spelling/forbidden-phrase slips fixed on the way (per-cent to percent twice, two per-molecule phrases reworded). Verification: forbidden-word scans of the four documents; the reproducer's run.sh resolves its default binary path; nothing built or rerun.
- **N16a** (PLAN_20260909_rev1 Stage A, the physical step can be refused and the clock is right; EXHALE_main.f90, attempted_step.f90, src/tests/attempted_step/{run.sh, attempted_step_tests.f90}; user 2026-09-09 "N16a 병행"): **the sampled step-doubling comparison is one transaction over one macro-interval H:** the full pass covers

H, the two half passes $0.5 H$ each and end at $t + H$ by construction (pass 3 no longer inherits whatever dt held); a refusal inside any pass, a finite estimate above one, or a nonfinite estimate rejects the WHOLE macrostep (as_reject_int_error now issued by the production caller), restores as_entry, shortens H by the controller and recomputes both trajectories; exhaustion of the transaction's retry budget (n_macro_retry, cap 8) is a named failure that stops with status 2, names the pass and prints the checksum of the restored state (stop_on_exhausted_retry_budget, one report for both exits). The clock, the accepted count and the physical accumulations move only at the adoption boundary below the macrostep loop. **The order of the complete split update is MEASURED as $p = 1$:** at one fixed state the estimate scales as H^2 over five decades ($p + 1 = 1.964, 1.993, 1.9985, 1.9997, 1.9999, 2.0004, 2.0000$ for H from $2e-2$ s to $2.6e-7$ s; the coarsest $1e-1$ s is outside that regime and excluded), so attempted_step_error_estimate now returns the retained full step's error $\Delta/(1 - 2^{-p}) = 2 \Delta$ (err_order_p, err_full_step_factor, convention at the declaration) [superseded by N16d/N16e: $p = 1$ was the base-face leak; the conservative and electron/T rows are $p = 3$, factor $8/7$, the species rows $p = 1$] and the reduction is $0.9 e^{(-1/(p+1))}$. The brief's first procedure (one interval by 1, 2, 4, 8 prescribed steps, ratios of the output rows) was INCONCLUSIVE, p 0.4 to 4.0, because the third difference ($2.2e-8$ relative) sits at the coupled source step's fixed-point tolerance: the inner error is not below the temporal error there (review F3's condition, not yet enforced). Test hooks, default off: EXHALE_AS_ERR_INJECT[_COUNT], EXHALE_AS_REJECT_IN_PASS, EXHALE_AS_FIXED_DT (seconds; in initialization mode with local stepping it flattens the local steps, stated at the hook). EXHALE_STEP_CLOCK=1 now also prints an err-pass: line per pass and state= (checkpoint checksum) per accepted step. **Tests:** suite RED on the entry binary 117/16 (the sixteen new rows: passes cover one interval, forced error rejects, clock unchanged, shorter retry accepted, refusal aimed at pass 1/2/3 keeps the endpoint and names the pass, exhaustion in each pass and on the error is named and restores the accepted state by checksum), GREEN on the final binary 133/0; advisor reran the suite with the worker's final binary (md5 5ea530d3251a8b8ce77e72415ddef0ab): 133/0. **Runs:** mol_carrier marching 200 steps byte-identical apart from the provenance line (initialization mode never sets n_err_passes = 3; the guard lines quoted in the report); the suite's case in physical mode, 60 steps: identical through step 19, at step 20 $e = 1.215$ (raw 0.607, the number the old code printed and dropped) rejects, H 1.118 $\rightarrow$ 0.913 (e 1.77) $\rightarrow$ 0.618 s (e 0.89) accepted, t_{phys} 66.40 $\rightarrow$ 65.92 s; two rejections in 60 steps, all by the integration error. Not run: the matrix, wasp_full_newton, run_fcheck.sh (out of the affected path; the module compiles clean under the suite's -fcheck=all). No manual entry (the manual carries no run-mode or clock text; nothing in it became false). **Advisor:** file scope as briefed; six British per cent (two new, four pre-existing) changed to percent. **Reported, not fixed (outside the item):** (1) the accepted state of a retried macrostep depends on which pass was rejected, up to $9.1e-5$ relative in the base cell's velocity ($5e-6$ elsewhere), with the checkpoint round trip exact, so some state read by the coupled source step is neither saved nor rebuilt (seeds or caches of ionization_equilibrium, constrained_chemical_equilibrium, diffusive_photochemistry): a rejected attempt is not yet without consequence; (2) one interval of the order sequence (H $6.48e-5$ s from the 0.405 controller reduction) sits 2.3x off the H^2 law the other seven follow, unexplained, for N16b; (3) err_rtol = $1e-3$, err_atol = $1e-12$ are flat, unvalidated and cover no species (N16b); the gate now shortens physical-mode steps by about two at the sampled steps on this case.

- **N0, N3, N1, N2** (PLAN_20260909_rev1 Stage A, one worker, four diffs; steady_newton.f90 +738/-143 over the four, new suite src/tests/krylov_and_dogleg/; no other file). **N0**: 14 rows calling the PRODUCTION pgmres, dogleg_step and the factored step-length and predicted-drop code (a test-operator hook in pgmres in the style of carrier_headroom_set_for_test); RED on the entry text 10/7, among them zero_operator_step_is_finite = F, approximate_gradient_length_decreases_the_model = 40.5. **N3**: seven named linear outcomes (gm_tolerance_reached, gm_subspace_exhausted, gm_no_direction_sampled, gm_jacobian_action_unusable, gm_preconditioner_failed, gm_reduced_system_singular, gm_nonfinite); a zero right-hand side returns before any factorization; hypot rotations; rank tested before the triangular division and the leading nonsingular block solved; LAPACK status and finiteness checked after each preconditioner apply and on the returned step; the TRUE scaled residual verified with one product at a breakdown; the cycle exits at a breakdown. Reorthogonalization not added: worst max $|V_i \cdot V_j|$ MEASURED $6.9\text{e-}13$ on the carrier fixture. **N1**: absent elements (reservoir abundance zero, the melem_ab ≤ 0 test set_energy_vectors and load_IC use) are no longer registered as rows; initial_trust_region_radius with named statuses, holding the active bounds before its own cycle; a positive floor tr_delta_min = $1\text{e-}6 \sqrt{\text{neq}}$ in scaled coordinates (257x and 1157x below the first radius the two fixtures take); tr_dmax follows the radius; "no feasible descent" is info = 3. **N2**: the length along the direction used, $\tau = (r \cdot A \cdot g) / \|A \cdot g\|^2$, the point named an approximate descent point, the paragraph defending the old length by B5e removed, a dropped Cauchy leg with a Krylov residual ≥ 1 exits as tr_no_direction_reduces_model. Suite after diff 4: 31/0, also under -fcheck=bounds,do,mem.

Root cause of B5m (R1's mechanism, its cause): the 1500 unknowns on a bound of the atomic element reload were the three elements metals.inp does not carry (Si, K, S x 500 cells), registered with an identity row and sitting exactly on 0, so the fraction-to-the-boundary rule zeroed ANY direction with a component there and the first cycle took 0 products; the radius became 0 and stayed so. MEASURED after (worker and advisor agree): unknowns on a bound 1500 -> 0; initial radius 0 -> $1.909\text{e-}2$ from a Krylov step that reached tolerance (3 products, $6.1\text{e-}2$ against 0.1); first step accepted, ratio 0.995; $\|R\|$ 1.888 constant -> $8.3\text{e-}2$ at iteration 50, then an oscillation between $6.9\text{e-}2$ and $1.1\text{e-}1$ with the merit creeping $7.29\text{e-}3$ -> $7.23\text{e-}3$ over 258 iterations (advisor run stopped by its 20-minute cap; 178 accepted, 80 refused; every cycle spends all 40 products). Not converged: the next obstruction is the model quality of the element rows near $\|R\|$ $1\text{e-}1$. **Carrier reload** (control reproduces B5l's run exactly): after, 54/1 accepted, $\|R\|$ handed back 1.174 -> 1.151 , worst achieved Krylov residual 0.569 -> 0.575 , radius range unchanged, $\log_{10} \text{Mdot}$ 10.63 both, outcomes now named (8 tolerance reached, 47 subspace exhausted). **The concurrent N16a edits are inert on every fixture** (paired controls bitwise). **wasp_full_newton** (once, final binary): not bitwise, largest relative movement $2.1\text{e-}10$ (Ion_species.txt, Ca I) and $8.8\text{e-}11$ (Hydro_ioniz.txt), info = 0, $\log_{10} \text{Mdot}$ 13.30; by code path the hypot rotation of N3 (use_tr is false there); below the 0.1% rule, identical, no refresh. lower_profile (the only default case with He_metal_diffusion and a partial reservoir) bitwise to its golden in both arms (it stops at its marching cap). Other suites (worker): steady_species_rows 106/0, certification 29/0, steady_selfconsistent_residual 3/0; physics_probe does not include steady_newton (source_closure.py). **Reported, not fixed**: largest_step_inside_the_species_box still

annihilates a whole direction when any unheld unknown has no room on the side the direction points (an element physically zero in one cell can trigger it; the outcome is now named and the radius survives, the direction is lost): an active-set rule decision, for N4b. Two dead locals removed from pgmres.

- **Stage A gate (advisor, 2026-09-09).** Tree after N16a, N0, N3, N1, N2. make OBJDIR=build_lwv EXE=EXHALE_lwv.x warning-free, md5 dd233d6a (identical to the worker's final binary, so the whole tree is what both workers measured). Suites on that build: krylov_and_dogleg 31/0, steady_species_rows 106/0, steady_selfconsistent_residual 3/0, certification 29/0, attempted_step 133/0 (with the N16a binary, md5 5ea530d3, same source). N7's first two iterations on the atomic reload logged above with named outcomes. Not run, by the affected-path rule: the matrix, physics_probe, run_fcheck.sh; the shared build/ and EXHALE.x remain behind the sources. No golden refreshed. Gate PASSED; Stage B (N4a, N5, N8a) and the N7 diagnostic list are next.
- **N13** (PLAN_20260909_rev1 Stage D, transit refusal census; EXHALE_transit.py, exhale_transit_lib.py, new Python suite src/tests/transit_census/; no Fortran, no output format change, so decision 17 is untouched): the census mask ($r \geq 1$) & ($r \leq \text{Rib}$) was geometrically wrong (the ray integral uses every shell with $|r| \geq b$); one function chord_shell_indices now serves the census, resonance_depth and resonance_spectrum, and the census reports, line by line at eight impact parameters and at its maximum, the share of the line-center optical depth carried by rows whose correction was refused (refused_line_center_tau_share, the trapezoid rule rewritten as a sum over rows so that shares of disjoint subsets add to one), labeled a CONTRIBUTION, not an uncertainty. MEASURED: every tpm_*.txt of wasp_full (10) and mol_base_handoff (5) bitwise before/after; the defect is latent when $\text{Rib} = r[-1]$ (both cases as they stand) and real when the domain passes the stellar radius (wasp_full read at EXHALE_TRANSIT_RSTAR_RSUN=0.30: 419 -> 500 sampled rows, 81 above the cap); on wasp_full the rays near $b = 1$ take 99 percent of their Ly-alpha line-center depth from uncorrected rows. Suite RED on the pre-change copies (11 rows, the mask present and no shared selection), GREEN 11/0 on the tree; advisor rerun 11/0; +0.3 s on the whole script. Advisor: one per-line in the library header reworded. Reported, not fixed: the wavelength-resolved share and the file metadata wait for decision 17; np.loadtxt(..., max_rows=1) at line 896 warns on every run and duplicates the column count of line 325.
- **N12** (PLAN_20260909_rev1 Stage D, the upstream caloric energy at the upstream composition; T_equation.f90, post_process_adv.f90, ion_cell_state.f90 (two fields x_h2_up, e_up), src/tests/adv_static_limit/ (new driver upstream_caloric_energy.f90, rows U0-U4)): the upwind term of the _adv energy residual now receives the upstream cell's SPECIFIC internal energy $e_{\text{up}} = E(x_{\text{H2}}, \text{up}, T_{\text{up}}) / \mu_{\text{up}}$ formed by the caller, and the caloric branch is taken when EITHER side carries H2 (an atomic cell below a molecular neighbor no longer loses the incoming rovibrational energy); the monatomic branch is verbatim when both sides are atomic. **A second, larger defect found and fixed in the same path:** x_h2_cell was assigned only inside the $k == 1$ stationarity loop (old 650) and read by the temperature loop for EVERY cell (old 1316), so the whole molecular layer was solved at the H2 fraction of the outermost ghost (1.5e-7 on mol_base_handoff, 2.3e-5 on mol_carrier, against 0.84 at the base); advisor confirmed both

lines in the pre-change source. The stationarity diagnostic's lower energy `e_lo` is likewise evaluated at the upstream composition (the ratio's definition, threshold and operator unchanged). Energy convention unchanged: the caloric EOS carries the bound rovibrational ladder only, the 4.478 eV bond lives in the formation reservoir (`molecular_reaction_heat.f90`), no term is counted twice. **Tests RED/GREEN (MEASURED):** isothermal H2 gradient residual 1.508 -> 0; atomic cell below molecular upstream 1.006 -> 0; dissociation-layer march against the analytic discrete solution, largest departure 0.155 -> 2.4e-11 (RED marched 1000, 892, 782, 615, 560 K against the discrete 1000, 975, 863, 728, 596; GREEN both 1000, 974.76, 943.62, 880.65, 853.52); controls unchanged. Suite 36/0; advisor rerun on `build_lwv` (md5 8d027f8b, the tree with N12 and N13) 36/0 (a first advisor run reported one `no_output` FAIL from a relative EXHALE_EXE path, the advisor's mistake, not the suite's). **Impact** (post-process restart Do only PP: True of each case's own state; the before binary is the Stage A binary): `wasp_full_adv` files bitwise (atomic, both `x_h2` zero, monatomic branch verbatim); `mol_base_handoff` and `mol_carrier_adv` move in 469 and 467 of 504 rows, `T_adv` by up to 94.2 percent (218 -> 424 K at `r` 1.0941) and 103.2 percent (169 -> 343 K), split about 35/45 and 41/46 percent between the stale-composition and the upstream-composition defects; both a warming of the molecular layer (5/2 kT per H2 instead of 3/2); `p_adv`, `cool_adv` follow, `heat_adv` at most 4.8 percent; `adv_status` changes in one cell of `mol_carrier` (row 26, 0 -> 2, from the `e_lo` composition in the ratio); run states (`Hydro_ioniz.txt`) bitwise in all three. **The molecular_adv goldens of the matrix will move; refresh at the Stage D gate, reported then, not now.** The writer's warning that molecular and oxygen columns are not advection-corrected stays; the product is still not a molecular stationary solution. Advisor: one per-thread phrase in `ion_cell_state.f90` reworded; the worker corrected four British spellings in comments. Reported, not fixed: a stale scalar formed in a `k == 1` block and read later is a class worth a sweep of the file; `post_process_adv.f90` is named by both N11 and N12 (one owner at a time; N11 waits for decision 16).

- **N9** (PLAN_20260909_rev1 Stage D, the restart He/H discrepancy; `load_IC.f90` +84/-12, new `grid_and_gates` row `restart_round_trip` (9 assertions); `write_output.f90` untouched, no key, no tolerance change): **the reported failure was a message defect:** the refusal printed the He/H of the TOP physical cell while the branch tests the largest departure over all physical cells; on the failing state (B5j/scan500_stag, a coupled-carrier JFNK iterate of the hot-Uranus gate) the top cell agreed with the input to 17 digits and cell 199 was off by 3.96 percent, so the message read "7.9307E-02 against 7.9307E-02 ... cannot be rescaled". Review R5's 8.83e-5 is the distance between the INPUT values of two different runs, not a property of any state. Fixed: the comparison records the cell it attained and `report_deviating_heh_cell` prints cell, radius, file and input ratios at ES16.9, the departure and the threshold (`heh_dev_tol` = 1e-6, now named, value unchanged); RED/GREEN on a state with helium raised by half at cell 199. **The round trip is clean** (MEASURED on the 12000-step `mol_base_handoff` state with EXHALE_DUMP_IC=1): `r`, `v`, `p` bitwise; every species column and every inventory (H, He, O, C nuclei, charge, He/H, H2 ratio) within 4.2e-16; `T` 5.6e-16; the `rho` column 4.9e-12 because the loader never reads it and rebuilds mass from the species (perturbing the column by 1e-15 changes nothing, bitwise), so 4.9e-12 is the writer state's own mass-closure defect (4.1e-14 on a 40-step state); a second trip 5.0e-16, no drift; no clock advanced. **The producing path moved the ratio, not the loader** (MEASURED): He/H departs from the input by 1.0e-11 and 8.9e-12 after 12000 marching steps with carrier transport off and on, but

by 3.9 to 7.9 percent on every coupled-carrier density-unknown JFNK state and 41 to 43 percent on the $\ln$ states (B5j, B5k): the carrier column is written after the element projection precisely so the Newton's carrier is not rescaled (write_species_rows_into_composition 1439-1469), the sweep reads the nuclei totals from the state (hydrogen_helium_nuclei_density), and with helium diffusion off no row re-imposes He/H. This is the element constraint N4a/N4b are about, seen from the restart side; reported, not touched. Also fixed on the way: parse_labels had no bound on 80-long label arrays (a file with more than 80 columns wrote past them); it now stops with a message (present schema at most 42). Atomic restart path MEASURED unchanged: wasp_full_newton's state reloaded before/after bitwise, no He/H line printed. Advisor: restart_round_trip 9/0 on build_lwv (conda gfortran; the worker's private build used /usr/bin gfortran 13, against the standing toolchain rule, so the advisor's run on the standard toolchain is the one on record). **Reported, not fixed:** the window integrals of the assembled residual are not Lipschitz in the state in the odd-even band above the base (one round trip at $4.9\text{e-}16$ moves the 1.03-1.10 mass window by 16 percent and flips cell residual signs at cells 183-187; one ulp on every species column reproduces it), so a round-trip residual budget must be stated on the row maxima (N5); restart_grid_guard.sh carries a stale header describing a fixed defect; nothing checks the writer's rho column against its species (decision 15's contract); EXHALE_RESIDUAL=1 runs one sweep unconditionally, so a true "no chemistry on load" evaluation has no entry point yet (N10).

- **N7** (PLAN_20260909_rev1 Stage C, the atomic element-row diagnostic after Stage A; steady_newton.f90 +477/-0, diagnostic prints under EXHALE_ELEM_DIAG=1 (iterations 1, 2, 60; =n moves the third), one extra Jacobian product per selected iteration with the refusal statistics held aside; no arithmetic of the step changed, MEASURED: every (JFNK) line identical hook off/on over 2027 lines, across four builds). **Verdict, MEASURED on the reload fixture:** the oscillation at $||R||$ $7\text{e-}2$ to $1.1\text{e-}1$ is NOT a model failure (predicted/actual ratio 0.9946, 1.0001, 1.0003 at iterations 1, 2, 60; the refused iteration's model is right to $3\text{e-}4$), NOT a bound effect (0 unknowns held, 0 samples refused in 293 iterations), and NOT limited by the linear solve although the solve fails its tolerance (40 of 40 products, achieved 0.6685 against 0.1, TRUE residual 0.6685 by one extra product, Arnoldi gap $2.3\text{e-}4$; the dogleg takes $2.1\text{e-}4$ of the Newton leg so that inaccuracy is second order). **Proximate cause: the ray test** (the finite-difference check of the model slope) refuses the step: at the refusing iteration model slope $-9.13\text{e-}9$ against ray slope $-1.48\text{e-}9$ (factor 6.2 against the allowed 4) with $|f_{2p} - f_{2m}| = 3.0\text{e-}12$ only 11x above the floor $\text{eq_sweep_reltol} * |f_2| = 2.7\text{e-}13$ (at iteration 1 the same test has a factor 1371 of room and the slopes agree to 0.5 percent); 82 of the 91 refusals in 293 iterations are this test, 3 its sign form, 6 a real over-prediction. Limit cycle read off the radius: $9.78\text{e-}4$ accepted $\rightarrow 1.96\text{e-}3$ accepted $\rightarrow 3.91\text{e-}3$ refused by the ray test $\rightarrow$ cut to $9.78\text{e-}4$, period three; the merit falls $1\text{e-}4$ of itself per accepted step, so $\text{Resid tol } 1\text{e-}8$ is seven decades away. **Underlying cause: the merit and the gate are different functionals.** At iteration 60 the eight element rows hold 0.437 of the squared merit (He 0.305) and NOTHING of $||R||$ (energy row holds 1.000 of $||R||$, 0.560 of the merit); their own certification measures sit at $1.2\text{e-}3$ to $7.4\text{e-}3$ against $\text{cert_tol_element} = 1\text{e-}8$; accepted steps swing $||R||$ by 40 percent ($7.8\text{e-}2$ to $1.09\text{e-}1$) while the merit stands at $7.291\text{e-}3$ to four digits. At iteration 20 the brief's option (c) is met outright: $||R||$ 1.400 set entirely by the mass row, the energy row 0.810 of the merit, seven of eight elements worst at cells 12 or 17. **The Cauchy leg is worth nothing here:** $||sU||$ $6.6\text{e-}9$ against a radius $3.9\text{e-}3$ ($g = B^T r_0$ from the band, $||A g||$

1e4 times r_0 . A g), predicted decrease 1.3e-9 against the dogleg's 7.6e-9; one product per iteration for 1.7e-6 of the step. **The band is not the defect** (0 zero, nonfinite or round-off columns, `dbgtrf` info 0), though it conditions worse along the solve (`dbgcon` 7.3e-8 $\rightarrow$ 9.4e-10, smallest pivot always a trace element of cell 500). **gm_m probe**: unlike the carrier arm, the larger subspace changes the achieved residual on this operator (first divergence at iteration 17: 40 of 40 at 0.1035 against 48 of 80 at 0.09999; deep in the stall 0.67 against 0.29) and the solve stalls WORSE ($m = 80$ plateau $||R||$ 0.27 to 0.34, merit 1.57e-2, against 7.7e-2 to 1.1e-1 and 7.3e-3 at $m = 40$); longer steps lift the ray probe off its floor and the refusal becomes an honest over-prediction, and the solve still goes nowhere. **Proposals, not implemented**: P1 measure the ray test's floor from two evaluations of one state instead of assuming `eq_sweep_reltol * |f2|`; P2 do not let the slope comparison overrule a reduction ratio within [0.998, 1.001] (every ray-size refusal inspected had one; the over-prediction guard stays live); P3 drop or rescale the banded-gradient leg where $||sU||$ is below a small fraction of the radius (one product saved); P4 a GMRES restart buys $m = 80$'s reduction at $m = 40$'s memory but does not fix the stall; **P5 the acceptance functional and the merit must be reconciled** (a merit that reads the element rows on their certification scales, or a separate control for them): a design decision, decision item 20 (section below). Also fixed in the same file: the `tr_exit_text/gm_outcome_text` buffers (48 and 64 characters) truncated two 67-character messages; all three are 80 now. **Reported, not fixed**: the worst j, k on the (JFNK) it line is the merit's largest scaled row while $||R||$ on the same line is a certification maximum, and the two need not be the same row (iteration 20: $||R||$ is the mass row's, worst is an energy cell); the brief's belief that Solver: Newton 100.0 caps the iterations is wrong (the third word is `newton_du_switch`; the JFNK cap is the literal 500 at EXHALE_main.f90 3509 and 3000 on the direct route, no key or hook; runs were capped by wall clock); `atomic_elem_newton/README.md` is right but incomplete (the refused trials have a correct model). Advisor: rebuild warning-free, `krylov_and_dogleg` 31/0 on the tree with N7 and N9.

- **N4a** (PLAN_20260909_rev1 Stage B, one elemental constraint map with three contexts; new `src/modules/lower_atmosphere/element_inventory.f90` (745 lines, one SRC line in the Makefile), `diffusive_photochemistry.f90` (its three element densities, hydrogen budget and five carrier box sides now derived from the map with unchanged arithmetic), `steady_species_rows` suite, `docs/element_inventory_contexts_20260909.md` corrected; `steady_newton.f90` untouched): **the map reports, nothing enforces it yet**, so the Stage B gate may not claim from this item that trial construction, chemistry and certification enforce one constraint (N4b). `nuclei_per_particle` is the one place a stoichiometric coefficient is read (`bsp_nH/nHe/nO/nC, mion_elem`); `element_inventory_of_cell` gives the full inventory grouped carried/locked/reserved/closure with the budgets and the shared-constraint breach as a MAGNITUDE (against the budget the sides spend, and against the budget the chemistry closes with, separately); `element_inventory_of_candidate` evaluates a trial's carriers against the frozen element densities; EXHALE_INVENTORY_REPORT=1 prints it at the operator's entry. **Physical judgment**: the separate box sides are a wrong feasible set by stoichiometry (two carriers of one element spend one budget); a cell at both ceilings breaches by exactly 1.0 of the available hydrogen (MEASURED). **Seven corrections to the advisor's draft**, each checked in the source: `n0_free/nC_free` include the neutral stage while `cbg_n0ion` excludes it (the F7 gap as a number: the oxygen box side spends the element total while `carrier_source` closes against the total less

the ion stages, so the box admits states whose source rows clamp free O at zero; N4b's feasible set); nH_avail is flooded; H I and He I are the closure, not rows; CO is fixed data of the local solve (its destruction row is the carrier operator's); **the shared sums ARE enforced on the marching path by limit_to_element_budget** (what has no shared statement is the coupled stationary solve, whose eval_residual compares a COUNT of cells); **max(rest, 0) in the write-back CREATES nuclei** (with rest < 0 the carriers keep what they hold and the closure cannot go negative, so the written element total exceeds the handed one by |rest|; only element_census_verify at EXHALE_ELEMENT_ASSERT 1 or 2 sees it, off by default); absent elements are no longer registered (N1). **Tests** 149/0 (advisor on build_lwv: 145/0, the four coupled-log rows unset), RED shown by EXHALE_INVENTORY_SHARED=0 (the three infeasibility rows fail in that arm); all nine molecular goldens classified: worst carrier breach exactly 0.0, tightest hydrogen occupancy 0.96 to 0.99 (oxygen_chemistry occupancy 1.0 is carbon's, 175 of 500 cells with all C in CO). **The N9 finding caught live:** inside a coupled carrier JFNK the He/H departure from the reservoir grows 1.4e-11 -> 4.9e-5 (third iterate, the first with a Newton n(H2)) -> 1.1e-1 (thirteenth) while every carrier budget stays at breach 0.0, so the fix belongs at the write-back of each iterate (the element row or a projection there), not at the end of the solve; the two saved coupled states have carrier breach 0.0 and He/H departures 4.0e-2 and 6.7e-1. **Marching:** mol_carrier 200 steps byte-identical (two independent before/after pairs); mol_diffusion 12000 steps byte-identical over the 7511 steps both arms had covered at report time (step trace character for character, the step-7001 snapshot files byte-identical apart from provenance); the pair finished later in the session scratch and the advisor compared its end: Hydro_ioniz.txt and Ion_species.txt at 12000 steps IDENTICAL apart from the provenance line, log10 Mdot 10.57 in both arms. Reported, not fixed: element_ratio_gate = 1e-9 now exists in two modules because element_census_tolerance() is private. Rule breach reported by the worker: one pkill -f on its own shell pattern (it matched only its own shell; the other workers' runs were verified alive). Advisor: five British per cent in the two files fixed; the advisor's earlier pgrep for the worker's binary name missed its renamed scratch copies, a false "finished" the worker corrected.

- **N7b** (N7 proposals P1 and P2, and the JFNK iteration hook; steady_newton.f90 +249/-40, krylov_and_dogleg +12 rows): **P2 opens the cycle; P1 is implemented and MEASURED inactive, and the measurement refutes N7's reading of the refusals.** The ray test's decision is extracted into ray_slope_verdict (public, tested), its floor is max(eq_sweep_reltol * |f2|, ray_floor_factor * measured_reproducibility) with the reproducibility measured once per outer iteration by two evaluations of the merit at one state (cost one residual evaluation per outer iteration, +1.5 percent on the carrier reload): MEASURED 1.8e-15 at a merit of 14.95 and 0 to 4e-19 at 2.7e-5, four to seven decades BELOW the assumed part (2.6e-13), which therefore governs everywhere; the refused trials' |f2p - f2m| (1e-10 to 3e-10) stand two to three decades above the floor, so they are real differences and NOT the probe near its cancellation floor as N7 and ISSUES 3.1 (d) said (corrected there). What they are: at the refusing iterate the ratio is 0.999 while the model slope (-5.5e-7) and the ray slope (+1.7e-7) have OPPOSITE signs; the model slope is set by the banded-gradient leg, which is not in the step (||sU|| 6.6e-9 against a radius 3.9e-3) but carries ||A g|| 3.1e3 into r0 . A s, so the slope test is right that the slope is wrong and the leg is what is wrong (N7 P3). ratio_sound_band = 1e-2: within it the ray verdict is formed and printed but not acted on; tr_ratio_below_acceptance untouched.

EXHALE_JFNK_MAXIT read once per solve, resolved by jfnk_outer_iteration_cap, printed with its origin. **Atomic reload, 150 iterations, before -> after:** accepted 106 -> 129, ray-size refusals 35 -> 1, ratio-below-eta refusals 6 -> 15 (the guard takes over), radius $9.8\text{e-}4 \dots 9.8 \rightarrow 3.8\text{e-}2 \dots 2.7\text{e}8$ (no longer binds; the accepted step is the interior Krylov point at $||s|| \ 1.0\text{e-}5$, ratio 0.72), $||R|| \ 7.27\text{e-}2 \rightarrow 6.92\text{e-}2$, merit $7.26\text{e-}3 \rightarrow 7.08\text{e-}3$, achieved linear residual still 0.69 against 0.1 with 40 of 40 products: **the limiter is now the linear solve and the model behind it** (N7 items 4 and 7, P3/P4), not the step length. No convergence claimed. **Carrier reload:** every numeric (JFNK) line identical; 20 of 55 iterations had carried a STALE refusal label ("the model promises no reduction" printed on ACCEPTED iterations because refuse was reset only inside .not. model_ok; pre-existing, fixed): earlier reports quoting "iterations that ended on N: the model promises no reduction" counted that defect wherever N exceeds the refused count (B5I's 19 and 20 included). wasp_full_newton BITWISE on all ten output files (three log lines differ: two wall clocks, one new cap line). Tests RED on the extraction stage (4 of 43 fail: the measured floor rows and the sound-ratio rows), GREEN 43/0; advisor rebuild warning-free, krylov_and_dogleg 43/0, binary md5 6d4bf63e identical to the worker's final. Two dead locals removed (md5 unchanged). Advisor: four British neighbour spellings in comments fixed. Advisor decision: P1's measured floor stays (correct in form, one evaluation per iteration). **Reported, not fixed:** the ray test runs before the acceptance test, so a catastrophic over-prediction (ratio -533, -1493 at iterations 47, 50) is named a slope disagreement; ray_h = $1\text{e-}3$ is not a measured interval; tr_dmax doubles without an absolute cap and with P2 runs to $2.7\text{e}8$ (harmless, the interior point is taken, but delta stops being a diagnostic); n_tr_exit has no index guard.

- **N5** (PLAN_20260909_rev1 Stage B, the residual contract; steady_newton.f90 +877, ionization_equilibrium.f90 +274/-6, EXHALE_main.f90 replay driver +134/-60, src/tests/residual_determinism/ rewritten with a suite run.sh (cases: the two in the tree plus EXHALE_RESID_RELOADS=<dir>[:<dir>], binary EXHALE_RESID_EXE)): **the old determinism suite had a hole:** it replayed the THREE-unknown residual in every configuration (0 of 1500 entries, the species-row registry is empty before a solve); the replay now registers the rows and returns 0 of 1500 / 2000 / 2500 / 5500 entries differing (3, 4, 5, 11 unknowns per cell) after three trials, a discarded trust-region trial and a jv_product probe with the active bounds held (**contract 1 GREEN on four configurations;** a rejected macrostep is not reachable from the driver, the one part unmeasured). **Contract 2, conditional closure convergence, MEASURED and it exceeds the certification tolerances:** the judged row maximum moves, per unit relative change of the eliminated partition's seed, by $4.2\text{e}2$ cert_tol_carrier on the carrier reload, $3.1\text{e}1$ cert_tol_element and $1.7\text{e}1$ Resid_tol on the atomic element reload, $6\text{e-}8$ on the element + carrier reload; a seed above $2.4\text{e-}3$ (carrier) or $3.3\text{e-}2$ (atomic) moves the judged row past its tolerance; the retention of a scaled seed at the fronts is ABOVE one (1.39, 3.69, 3.46) where B5h had 0.77 at the base. Two suite rows stay FAIL by design as the standing measurement (closure_spread_within_the_row_tolerance_{atomic_elem_newton, det_car}, 61.5 and 8.5 against < 1). **Contract 3, branch discovery** (EXHALE_RESID_BRANCH_REPORT=1, roots grouped by a branch-datum distance $1\text{e-}1$ anchored on B5h's two roots): with H2 transported B5h's two-root H2 front does NOT reproduce (x(H2) at the front bitwise identical over 10 seeds, the transported value replaces the seed in the H2 row); the H I partition shows one root on the carrier reload and TWO on the element + carrier reload (9 and 1 seeds, $1.05\text{e-}1$ apart); whether the second is a distinct branch or a slow iteration is N6's

question. **Chemical decay against transport** on the resolved gradient length: H2 $t_{adv} * \lambda$ 3.68 (base), 4.0e-3 (front), 0.82 (wind); H2+, H3+, HeH+ 3 to 12 decades faster than transport everywhere, so the transported-H2 / eliminated-ion split stands and no promotion is proposed (conservation modes are removed by the parameterization itself: the neutral stage closes each element). **Rounding floor of the species rows in the judged measure is exactly epsilon** (2.2e-16) against tolerances 1e-8, 7.65 decades apart, condition numbers 2 to 1e3: compensated summation (the Al-Refaie et al. 2024 reading) is not a lever. Validation: krylov_and_dogleg 43/0, steady_species_rows 145/0, mol_carrier 200 steps bitwise, wasp_full_newton bitwise on all ten files (one new log line, N7b's cap). Advisor: rebuild warning-free, md5 913e6579 identical to the worker's final; krylov_and_dogleg 43/0; residual_determinism 5 PASS / 1 FAIL, the FAIL being the atomic closure-spread row by design (a first advisor run with the wrong variable name EXHALE_EXE tested the stale shared EXHALE.x, the advisor's mistake); two British per cent in ionization_equilibrium.f90 fixed, the worker fixed five neighbour. The worker's session was cut once by the Opus limit (09:50 KST) and resumed; nothing was lost. **Reported, not fixed:** a discarded evaluation restores n_eq_sweeps_last, so a caller reads the previous state's pass count; report_accepted_cell_state writes inside the parallel region without a lock; carrier_diffusion_coefficient reads the marching operator's pct_Dco, which the stationary residual never fills (a pure stationary path would read zero; diffusive_photochemistry.f90); the global count shadows the intrinsic.

- **N16b** (PLAN_20260909_rev1 Stage E brought forward, species-aware integration error; attempted_step.f90 +546/-28, EXHALE_main.f90 macrostep code +125/-7, attempted_step suite +289): the step-doubling estimate reads four row classes on their own scales (conservative rows; transported species fractions, the f_sp columns carrier_state builds fc from: H+, H2, OH, H2O, CO; element mass fractions under He_diffusion; electron fraction and T through the caloric EOS), the species scale being the species' own value with an absolute floor of element_ratio_gate (1e-9) of the largest density the species can reach in the cell, read from the N4a map's stoichiometry; the flat 1e-12 no longer judges any species row (a carrier at 1e-20 changed by 1 percent: e = 20 on its own scale, 2e-10 on the flat floor). **The inner nonlinear error is collected cell by cell** (the coupled pair's safety-factored geometric factor on the cell's own increment, and the carrier substep's row residual) and the estimate carries the interval e_lower ... e_upper the three inner errors leave it in: reject when above one at the bottom, accept when below one at the top, UNRESOLVED in between (named outcome, counted in the end-of-run block); the brief's per-class fraction rule (1/10) is kept as the report, not as the verdict, because applied literally it called EVERY estimate of a healthy run unresolved (species difference 6e-17 five decades below the pair's own error) and would have made F1's rejection inert. **p = 1 confirmed on every class** (atomic case, eight levels over five decades: species p + 1 1.9987, 1.9799; electron/T 1.9943 to 1.9997; conservative 1.9809 to 2.0004; molecular case coupled ladder H = 5, 2.5, 1.25 s: species 1.949, 2.299, electron/T 1.912, 1.983), so the retained-step factor 2 holds for all rows. On the molecular case at H <= 0.4 s the species class reports itself unresolved (inner over difference 2.7e5): review F3's condition, MEASURED and detected. **The 60-step physical run is unchanged to 17 digits** (t_phys, checksum, both files): the conservative rows still decide it. mol_carrier marching bitwise. **N16a's 9.1e-5 state leak IDENTIFIED** (not fixed, files not owned): Apply_BC.f90 base_face_W and base_face_lower_W (lines 48, 50; written only by Apply_BC, read by Rec_BC

in the first Reconstruct of a retaken pass BEFORE the pass's first Apply_BC, so the base face's left state is the refused attempt's) and caloric_eos.f90 caloric_mixture_active, nk_per_mass, x_h2, molecular_cell (125, 145-147; written only inside get_species_densities, which rebuild_state_from_checkpoint calls AFTER U_to_W, so the restored p and T use the refused attempt's caloric composition); the two reinforce each other at the base cell, which is the measured signature (v $9.1e-5$, T $6.8e-8$, cooling $1.7e-5$). Row rejected_pass_leaves_no_state_behind stays RED at $9.1162e-5$, labeled with the files and lines. N16a's 2.3x outlier reproduced and bounded (only at controller-formula levels, conservative and charge classes, always the base cell; 0.665 and 0.53 of the H^2 prediction), not explained; two candidates named (the unbounded inner error of the mass/momentum rows through the second half's coupling; the base-face leak). err_rtol, err_atol unchanged (the CFL step 20 would need $1.21e-3$; the estimate RISES at the first shortened interval, a second reason not to anchor on this case). Tests 155/1 (the one FAIL by design), RED on the entry text 150/6 (an estimate that was the solvers' own error rejected ten macrosteps there; now 0, named). Advisor: suite rerun with the worker's final binary (md5 7bd041dd) 155/1; four modelled in EXHALE_main.f90 comments changed to modeled. **Reported, not fixed:** the mass and momentum rows carry no inner error; the carrier substep's number is a row residual, not a solution error (conditioning unmeasured); NCO_col, k_co_diss, theta_co_shield are outside the checkpoint by an unstated exception (read only after their write within a sweep).

- **N16d** (the checkpoint's restore is complete; EXHALE_main.f90 rebuild_state_from_checkpoint 4 code lines, comments in Apply_BC.f90, caloric_eos.f90, attempted_step.f90, the suite row extended to both files): **closed by rebuilding, not saving.** The five leaking quantities (base_face_W, base_face_lower_W; caloric_mixture_active, nk_per_mass, x_h2, molecular_cell) are functions of (u , f_{sp}), so the rebuild now runs $\rho = u(1,:)$, get_species_densities, Apply_BC, then U_to_W and comp_T_from_p (composition first because the pressure map and the boundary read it; boundary second so the face states are the restored state's). Every one of the six call sites follows a restore, so no restore-free path can move. **MEASURED:** rejected_pass_leaves_no_state_behind $9.1162e-5 \rightarrow 0.0$ bitwise on both files; suite 156/0 (advisor rerun with the worker's final binary md5 abdf17f7: 156/0). **The fix changes the measured order of the conservative rows:** the full-versus-two-half difference was 40x the real splitting difference (the leaked base face carried an $O(H)$ error into one face of one RK stage of an $H/2$ step, an $O(H^2)$ contribution) and now falls as H^4 ($p + 1$ 3.9966 to 4.0104 over the ladder), the local-error order of the three-stage Runge-Kutta; the electron/T rows follow it; the transported species stay H^2 (unchanged to all printed digits). N16a's 2.3x outlier is EXPLAINED: the two controller-formula levels now sit on the H^4 law to 0.1 percent; the leaked base face carried the difference of two intervals whose ratio was not the ladder's. **Consequence not yet fixed (N16e, launched):** err_order_p = 1 was calibrated on the leak; the conservative and electron/T estimates carry a factor 2 where $8/7$ is right (inflated 1.75x) and the reduction exponent is $-1/2$ where $-1/4$ is; the species class is genuinely $p = 1$, so the order is a property of each class. mol_carrier marching 200 steps bitwise (0 refused attempts, no restore); with one injected refusal the initialization-mode state moves $2.1e-5$ (v at r 1.0185) and $1.2e-5$ (He 2^3S), the fix of the same defect with both channels live (atomic hydrostatic_column had the base-face channel alone). wasp_full_newton not run (no restore unless an attempt is refused; whether it has zero refusals was not measured). The reload route does not share the rebuild (checked). Advisor: two Per-cell phrases in

caloric_eos.f90 reworded. **Reported, not fixed:** the same composition-before-pressure ordering stands at three other sites of EXHALE_main.f90 (1075-1078, 5162-5166, 5312-5326) and upmap_bc_roundtrip measures an identity its comment says is not one (N16e); the added Apply_BC adds one boundary evaluation per restore to the run-total diagnostic counters.

- **N20** (decision 20 a with N7's P3; steady_newton.f90 +314/-84, krylov_and_dogleg +13 rows, steady_species_rows +1 row; certification.f90 unchanged, cert_scale_floor was already public): **one functional, implemented exactly, with a measured consequence that is a decision, not hidden.** merit_row_scale_from_certification is the one denominator a species-row-route row is read through; cell_row_scales builds the hydrodynamic rows through residual_row_scale(k,j,u), the element rows through their certification scale, the carrier rows through the term sum the certification divides by; base cells of element rows keep the unknown's own scale (Dirichlet reservoir); the three-unknown route is untouched by construction ($D_r = D$ when $n_{\text{spec_row}} \leq 0$). **MEASURED:** on both reloads every element row's largest scaled merit row **EQUALS** its certification measure at the same cell and the merit's largest scaled row **IS** $||R||$'s row (before: factor 2.5 apart at iteration 1, 112 at iteration 60). **P3:** the banded-gradient leg is dropped from the step **AND** its image when shorter than $\text{cauchy_leg_min_fraction} = 1e-3$ of the radius (measured $1.4e-5$ to $8.7e-7$ on both fixtures, so everywhere), so the slope handed to ray_slope_verdict is the step's own; products saved: zero, because tau_c needs A g (a lazy leg would save one per iteration, P-N20-1, reported). A quotient verdict at $\text{ratio} \leq \text{eta_accept}$ is named tr_ratio_below_acceptance. tr_dmax capped at $1e3 \sqrt{\text{neq}}$ ($7.4e4$ on the atomic fixture; binds on neither fixture now, would have bound the $2.7e8$ of N7b). **The consequence:** the carrier reload improves 2.6x (best $||R||$ 1.151 $\rightarrow$ 0.446, both arms on the stagnation detector, +16 percent evaluations, dgbcon $5.4e-10 \rightarrow 1.9e-11$) while **the atomic element reload no longer solves:** the certification row scales span 11.4 decades ($3.7e-10$.. 92.7) against 8.3 before, dgbcon $7.3e-8 \rightarrow 8.9e-10$ at iteration 1, the Krylov point collapses to $||s||$ $2.5e-15$ inside a radius 19.6 at iteration 11 and the solve aborts on the no-descent watchdog at $||R||$ 1.777 (before: $7.38e-2$ at 150 iterations). A third arm (element/carrier rows on certification scales, hydrodynamic rows on the state scales) does not collapse but is worse than the entry text on both fixtures (0.741, 1.026): the hydrodynamic reconciliation is what helps the carrier arm and kills the atomic one. The merit's row scaling **IS** the operator's ($r_0 = F/D_{\text{row}}$, $A = D_{\text{row}}^{-1} J D$), so the remaining freedom is the column scaling D and the preconditioner: N8a's, launched with the atomic collapse as its RED. Tests RED on the entry objects (the element-row scale row 4.0 against 1.0; the ratio-naming row), GREEN krylov_and_dogleg 56/0, steady_species_rows 145/0, certification 29/0; wasp_full_newton BITWISE on all ten files (two wall-clock lines). Advisor: rebuild warning-free, steady_newton.o md5 6f3aef50 identical to the worker's, the three suites 56/0, 145/0, 29/0. Two pre-existing defects fixed (a dead local; [diag 7] printed at cell 0 for the carrier). **Advisor decision:** (a) stays in the tree; the atomic arm's collapse is a conditioning problem of the scaled operator and is the first deliverable of N8a, not a reason to give the step control a functional other than the gate's; the state of the atomic arm is reported as **WORSE** than before N20 until N8a lands. **Reported, not fixed:** EXHALE_main.f90 1182 should pass u to cell_row_scales; docs/p54_base_layer_mass_flux.md 10.4 now has a second measurement behind it.

- **N16e** (the order of each row class is its own integrator's; attempted_step.f90 +106/-50, EXHALE_main.f90 +60/-21, attempted_step suite +75/-15): err_order_p(n_err_class) = [3, 1, 1, 3] (hydro, species, element, charge/T) with the measured ladders at the declaration (conservative 3.9966 to 4.0104; electron/T 3.9962 to 3.9867; species 1.9987, 1.9799; the element class is assigned the species' order by construction, said so); err_full_step_factor is a class's own (8/7 for the RK3-carried rows, 2 for the first-order split ones), the estimate records the deciding class (the one whose lower bound reached the gate) and attempted_step_reduced_dt uses that class's $-1/(p+1)$. MEASURED: the conservative and electron/T estimates fall by exactly 4/7 (inflation 1.7500004 removed), the 60-step physical run's trajectory does not move (N16d had already removed the step-20 rejection, so the brief's expected "1.21 rejected -> 0.69 accepted" belonged to N16b's entry text, not to this item's; stated); on the forced-error ladder the controller with exponent -1/4 asks for 0.846 and 0.226 of the interval where -1/2 asked 0.601 and hit the 0.2 floor, so the floor no longer does the sizing. Stale sites: EXHALE_RESIDUAL=1 and physical_handoff_check NOT reordered, the composition cannot have moved there (init and equilibrate_loaded_composition both end with get_species_densities), the invariant is now stated at both; update_map_begin_step's install branch reordered (composition before U_to_W; EXHALE_UPDATE_MAP on mol_carrier moves 0.0); upmap_bc_roundtrip renamed upmap_bc_interior, required exactly zero (Apply_BC writes only the ghosts), MEASURED 0. Tests 163/0 (156 before; the unit rows cannot be run RED against a scalar err_order_p, the RED is the entry suite's own contradiction, $p = 1$ asserted and $p + 1 = 4$ measured in one run; the whole-binary row 0 of 2 -> 2 of 2). mol_carrier marching bitwise. Advisor: suite rerun with the worker's final binary (md5 0ae8d8d9) 163/0. **Reported, not fixed:** update_map_begin_step's install branch owes the next Reconstruct an Apply_BC (same family as N16d, a diagnostic route); err_atol = $1e-12$ still floors the conservative rows on a code-unit density; the mass/momentum inner error is still unbounded (the deciding row's interval is a point); no fixture has the species class deciding a step, so the exponent switch is covered by unit rows only.
- **N8a** (PLAN_20260909_rev1 Stage B, scales, boundary equations, derivatives and the inner budget, with N20's atomic collapse as its RED; steady_newton.f90, krylov_and_dogleg +17 rows, steady_species_rows +2 rows; certification.f90 and steady_residual.f90 unchanged, no tolerance moved): **the collapse was one exactly empty Jacobian column, not the spread of the row scales.** At iteration 11 of the atomic reload the carbon unknown of cell 500 had been driven to zero, its column scale sat on the absolute floor $1e-20$, the probe step $\sqrt{\text{eps}} * 1e-20$ underflowed the residual difference, the column was exactly 0 and the banded preconditioner singular (dgbcon 3.98e-27, smallest pivot 3.7e-19); the $2.5e-15$ Krylov point is what a triangular solve with that factor returns. Iteration 1's $8.9e-10$ was only the 7.3-decade spread of the certification row scaling (no empty column). **Two repairs, the model untouched to the last bit** ($r0 = F/Drow$ and A unchanged, MEASURED by the scaled column norms): (a) element_unknown_column_scale = $\max(|x|, 1e-4 |reservoir|, 1e-20)$, the arithmetic at the function (the column is resolvable while the probe step stands far above $\text{eps} * X_{reservoir}$); (b) equilibrate_the_preconditioner: two-sided equilibration of the band that is FACTORIZED only, powers of two, banded_preconditioner_solve applying C dgbtrs(R v). Attribution (first solve, accepted/refused; empty columns at it 11; dgbcon at it 11): entry 14/16, 1, $4e-27$; floor alone 69/16, 0, $4.9e-11$; both 114/36, 0, $1.1e-9$. **Atomic reload (EXHALE_JFNK_MAXIT=150):** entry abort at $||R|| 1.777$ (no descent, 30 it-

erations); floor alone 1.189 (stagnation, 86); both **3.800e-2 at the 150 cap, still descending** (2.5e-1 to 3.8e-2 over the last thirteen iterations), dgbcon at iteration 1 8.9e-10 -> 1.65e-6, the brief's target (the pre-N20 7.4e-2) passed by 1.94x with one functional kept; refusals by name shift from ray-floor to ratio-below-eta (35) and one slope-size. **Carrier reload** 0.446 -> **0.1513** (2.95x), dgbcon 1.9e-11 -> 8.7e-6, +30 percent evaluations. The model's own column equilibration (EXHALE_COLUMN_EQUIL=1, default off) was built, MEASURED and NOT adopted: it moves the trust-region coordinates, the radius runs to the cap and trials ascend by 4e4 times the predicted decrease. **First cell rows declared once** (species_row_base_equation: element row = Dirichlet reservoir, READ from element_transport_residual; carrier row = control-volume balance, READ from carrier_residual's $j == 1$ branch; no equation changed). **Base derivatives**: the directional-derivative sweep at the base is a clean V with its minimum 4.8e-8 at $\sqrt{\text{eps}}$, the step build_banded_jac_full takes; whole-system rows 8.2e-8. **Closure map at the base** (measure_the_closure_map_at_the_base, four evaluations once per solve): the sweep exits after ONE pass at 9e-12, four decades inside the 1e-8 asked, so the reference is a pass count; quotients 2.1e-4 (atomic, attenuation) and 5.34 (carrier); outer_residual_closure_target (what the outer residual may keep, what the ray floor reads) is now separate from eq_sweep_reltol (what the sweep is asked = target / measured amplification; 1.87e-9 on the carrier reload). **B5h's ~4e3 base amplification is not reproduced on either fixture** and is not claimed. Manufactured states in krylov_and_dogleg (columns spanning eight decades, pivot ratio 5e-18 -> 0.54 after equilibration; exactness rows: the map and the unknown-space step unchanged to the last bit). Tests krylov_and_dogleg 73/0, steady_species_rows 147/0, certification 29/0; the new binary with both changes off reproduces the entry arm to every digit; wasp_full_newton BITWISE (ten files; nspec_row = 0 short-circuits every new path, the hydrodynamic column scales untouched). **Advisor**: rebuild warning-free, steady_newton.o md5 c7621824 identical to the worker's, the three suites 73/0, 147/0, 29/0; the atomic reload rerun on the advisor binary reproduces the first solve to the digit ($||R||$ 3.800e-2 at 150, info = 1), and the run's SECOND stationary solve (the code re-enters the Newton from the handed-back state, another 150 iterations) reaches $||R||$ **5.273e-4, flux spread 2.3e-6**, still not certified ($\log_{10} \text{Mdot}$ 10.10; exit 2), 32 min at 16 threads: the arm now descends four decades in 300 iterations where it stood still since B5m. Not converged. **Reported, not fixed**: n_eq_sweeps_model = 1 so the pass count, not eq_sweep_reltol, controls the closure on both fixtures; EXHALE_JFNK_MAXIT caps each solve, not the run; fix_active_species_bounds lost its D argument (the epsilon-active width now on state_column_scale); the [diag 4] true-residual row is not valid for long steps (its finite-difference product is out of range). The fixture README updated by the advisor.

- **Advisor measurement after N8a (atomic reload, EXHALE_JFNK_MAXIT=600, 16 threads, 18 min).** One solve with the cap at 600 stops on the stagnation detector at iteration 224 with $||R||$ 2.671e-2 (info = 2; 170 accepted, 55 refused, all but three by "the reduction ratio is below eta"; $\log_{10} \text{Mdot}$ 10.06), WORSE than the same binary's 150 + 150 route, where the run's re-entry into a second solve from the handed-back state reached 5.273e-4. [The advisor's reading here was WRONG and N21 corrected it: the run does not re-enter the Newton from the handed-back state; info = 1 sends it back to MARCHING for 2000 steps (the log's Newton finish at step 2002), and the second solve starts from the marched state at $||R||$ 6.06e-2. The two decades are gained by the marching, not by a fresh trust region; N21 measured every reset and none helps.] Not converged either way.

- **N4b** (decision 14 route (i), a step that moves along the element-budget surface; steady_newton.f90, element_inventory.f90 +94 (the derivatives of the map, element_constraint_derivatives), steady_species_rows +35 rows; diffusive_photochemistry.f90 untouched, the frozen budgets recovered exactly through the public carrier_headroom): **the acceptance row is met by ten decades**. On the carrier reload the He/H nucleus ratio against its reservoir stays at 1.39e-11 at every one of 68 iterates where the entry text drifted from 1.39e-11 to 2.016e-1 (20 percent of the ratio the run was given) over 79 iterates: the write-back of each iterate now returns the element totals it was handed (the N9 / N4a finding closed where those reports said it belonged). Best ||R|| 0.1513 -> **0.0899** at 14 percent fewer residual evaluations (5690 -> 4891). The shared constraint $2 n(\text{H}_2) + n(\text{H}^+) + n(\text{OH})$

- $2 n(\text{H}_2\text{O}) \leq n\text{H}_{\text{avail}}$ (and the O and C rows) replaces the five separate corners as the step limit (each corner is implied by the row, so nothing is lost and the direction along the surface is gained); a nonlinear feasibility screen by magnitude at every candidate; a restoration phase for an infeasible start (on the element-and-carrier reload 52 trials restored over 1165 cell-steps, worst violation 1.630 -> 2e-16; 6 candidates refused by magnitude; 167 write-back cells whose closure could not return an element total, counted). ****Stated, not claimed****: on both reload fixtures the shared hydrogen row is never within a probe step of being tight (occupancy 0.985 to 0.989, active band 1e-8), so the active-set PROJECTION onto the constraint hyperplane, the part route (i) is named for, fired on ZERO iterations of either run; its correctness is stated by the suite rows, and the measured improvement comes from the write-back, the budget leaving the coordinate box and the shared step limit. The atomic element reload (element rows, no carrier row) is identical line for line over its 150-iteration first solve (||R||3.800e-2 both). Tests steady_species_rows 182/0 (RED arm EXHALE_ELEMENT_CONSTRAINT_ROWS=0 fails 20 rows), krylov_and_dogleg 73/0, certification 29/0; mol_carrier marching 200 steps bitwise; wasp_full_newton bitwise (||R||4.213e-9, info = 0, log10 Mdot 13.30). ****Advisor****: rebuild warning-free, steady_newton. omd542fcf223 and binary md59c9ba732 identical to the worker's; the three suites 73/0, 182/0, 29/0; wasp_full_newton's two matrix files compared by the advisor against the Stage A reference: IDENTICAL. Two earlier builds of the item were discarded and their arms rerun (report section 8). Certification unchanged (decision 12 c). ****Stage B gate****: N4a map, N4b constrained step and element-conserving write-back, N5's three contracts measured (contract 2 fails by design on two reloads), N8a scales and closure map: the gate's rows are in; what is NOT met is convergence of any species-row solve, which is Stage C's criterion, not Stage B's. N4b's user-visible switches: EXHALE_SPECIES_BUDGET_FACE is retired and replaced by EXHALE_ELEMENT_CONSTRAINT_ROWS(=0 restores the five corners), EXHALE_ELEMENT_WRITE_BACK added (the element-conserving write-back);

no input key, output column or unknown layout changed. Also found by N4b:dc/d(nd) = c/nd exactly (the constraint is homogeneous of degree one in the density, so an active row cannot push the mass unknown); two earlier builds of the item were discarded on the atomic fixture's line-for-line check (a blocked probe component zeroed unconditionally made the sampled operator nonlinear, factor 33 regression; the probe sides compared before capping). The advisor annotated the retired switch in the B5l entry and indocs/steady_solver_design.md'.

- N21** (what limits the species-row solves now; steady_newton.f90 +417, krylov_and_dogleg +31 rows): **no reset of the trust-region state helps, and the advisor's premise was wrong.** The run's second stationary solve does not start from the handed-back state: info = 1 returns the run to MARCHING for about 2000 steps and the second solve enters at ||R|| 6.06e-2 (advisor's own log: Newton finish at step 2002, then step 4002, the third solve reaching 4.26e-4 on the stagnation detector), so the two decades the 150 + 150 route gains are gained by the marching between the solves. MEASURED on the atomic reload from the SAME 150-iteration prefix, EXHALE_JFNK_MAXIT=300: continue (control) 227 iterations, stagnation, ||R|| 2.671e-2 (170/55, reproducing the 600-cap run); resetting the pseudo-transient shift, the acceptance memory or the closure map: bitwise identical logs (three exact no-ops: dtau already sits on its floor dtau0, the Grippo window is reported and never acted on on the coupled route, the closure map re-measures to amplification 1.000); resetting the radius and ceiling 2.851e-2; returning to the best iterate 2.921e-2; all five together (what a re-entry does) 3.161e-2; a second Krylov cycle of 40 2.927e-2; a restart every 10 iterations without improvement runs to the 300 cap at the same 2.671e-2. Carrier reload: control reproduces N4b (8.993e-2), periodic restart 1.153e-1, second Krylov cycle 1.728e-1, both worse. **The periodic restart trigger is delivered default OFF** (tr_restart_stall_default = 0) by the brief's own condition. **The linear solve is not the limiter:** after N8a the atomic cycle reaches a median 0.157 of its right-hand side over iterations 1 to 150 (N7: 0.57 to 0.69) but 0.74 to 0.82 over 151 to 225; the carrier cycle 0.963; doubling the products halves the linear residual and makes the outer solve WORSE on both fixtures; in the second atomic solve a cycle reaching 0.98 coexists with a two-decade fall of ||R||. The radius reset is worse for a stated reason: the carried radius 9.85e-9 accepts steps at ratio 1.000 while a re-initialized 1.15e2 over-predicts by 39x then 3392x and costs four refused iterations. READ: the band is refactored every outer iteration (a re-entry changes the preconditioner only through the shift) and the Krylov cycle starts from zero every iteration. Tests RED (13 symbols absent from the entry module), GREEN krylov_and_dogleg 104/0, steady_species_rows 182/0; with everything disarmed the delivered binary reproduces the entry text's 20-iteration solve line for line; wasp_full_newton BITWISE (ten files). Advisor: rebuild warning-free, steady_newton.o md5 cdbdf535 and binary e140284a identical to the worker's, the two suites 104/0, 182/0; the advisor's own n8a_reload log confirms the marching between the solves. **The finding that names the next item (N22):** the predicted decrease stops scaling with the step below about 1e-7 (from iteration 140: the step falls 4x and pred stays 7.099e-4; every longer step of the resulting period-two cycle is an ascent of 3e-3 against a promise of 3e-4, ratios -4, -8, -13, refused, quartered, then accepted at ratio exactly 1.000); the production dogleg geometry is cleared by two new rows (quartered radius quarters the step and the predicted decrease to 2.5e-1 and 2.50009e-1), so what remains

is `trust_region_step`'s bookkeeping at those lengths or a precision floor in $r_0 \cdot A_s$ and $||A_s||^2$; the separating measurement is one diagnostic line ($||A_s||$, $r_0 \cdot A_s$ beside $||s||$ at two radii of one iterate).
 README of the fixture updated by the advisor.

- N13b** (decision 17 option a, transit spectra carry their validity; `EXHALE_transit.py`, `exhale_transit_lib.py`, `transit_census` +8 rows; Python only): one reader `read_adv_validity` locates the status columns by name in the `# columns` line and distinguishes three cases: schema 2 (N11's two fields: a row is refused for the census when `adv_T_status` is not 0, the composition field reported beside it), schema 1 (today's single column), legacy (no schema: validity UNKNOWN, and both the console and the saved metadata say UNKNOWN rather than 0 refused). Every `tpm_*.txt` gains a comment metadata block (transit_schema 1, the input profile and its schema, its `adv_input_certified` / stationarity operator / model restrictions lines copied, its provenance lines, the tool's own identity by git hash and file mtime, the census for that line with the eight-point refused-share ladder and its maximum, the CONTRIBUTION-not-uncertainty sentence, the `EXHALE_TRANSIT_*` overrides in effect); the two hand-written products get the same block. MEASURED: numerical columns bitwise on all 10 `wasp_full` and 5 `mol_base_handoff` products; the schema-2 read tested on synthetic files and on a relabeled `wasp_full` profile (the relabeling is the worker's, not N11's mapping; the production file comes with N11); every consumer in the tree (`python/make_transit_figures.py`, the WASP-52b and LHS 1140 b notebooks, two LHS scripts) reads through `np.loadtxt` or skips `#`, none edited; the planet folders' `tpm_*.txt` untouched (mtime 2026-08-19). Suite RED 8 rows (no reader existed), GREEN 19/0; advisor rerun 19/0. Fixed on the way: the second `np.loadtxt(..., max_rows=1)` column count (NumPy warning on every run) replaced by one rule. Reported: the refused share is taken over the temperature field only (a companion share over the composition field would be the honest partner); still line center only; README_HOWTO.md should mention the metadata block (advisor added the sentence).
- N11** (decision 16 option a; `post_process_adv.f90`, the `_adv` writer of `write_output.f90`, `examples/exhale_io.py`, `EXHALE_plots.py`, `adv_static_limit` +driver M1-M5 and rows R8-R9): **stationarity of the input state is judged by the certification's own mass operator** (`assemble_residual`, `residual_row_scale`, the mass row $|R_1|/s_1$ against `adv_mass_row_tol = cert_tol_mass`), the enthalpy term ratio kept as a printed sensitivity diagnostic that refuses nothing; **two status fields** `adv_T_status`, `adv_comp_status` (0 corrected, 1 retained, 2 failed, 3 unsupported, 4 not_evaluated) in BOTH `_adv` files, columns 8 and 9, with `write_adv_validity_header` writing `# adv_schema 2`, the two legends, `# adv_status_counts T <5> comp <5>`, `# adv_input_certified <T|F> <reason>` (the certification's verdict on this very state), `# adv_stationarity_operator face_flux_mass`, `# adv_product` (a ONE-WAY correction on the input's density and velocity field; a converged scalar temperature solve does not make a row self-consistent), `# adv_model_restrictions`; `adv_unsupported` where the omitted molecular and oxygen species carry more of the cell's particle count than the carried ones (structural, no threshold); `exhale_io` exposes both fields and the header (`load_adv_header`, `load_adv_status`), a legacy file reads schema 1 with both fields ADV_UNKNOWN, never a translation of the old column; `EXHALE_plots.py` shades the refused rows. **MEASURED, the physics of the item**: the ratio screen and the mass row are not interchangeable and the ratio is the weaker:

on the analytic outflow fixture the ratio is $1.1\text{e-}13$ everywhere while the mass row stands at $2.1\text{e-}7$ (RED of the old criterion); on `wasp_full` the ratio refused 297 of 503 rows, the mass row refuses 503 of 503 (largest 1.04 at $r\ 1.032$); `mol_base_handoff` 33 vs 503 (0.245 at $r\ 4.21$); `wasp_full_newton` (the one certified state) 2 vs 3. So on the two relaxation snapshots every row is retained and the `_adv` product IS the run's own state (T_{adv} equal to T in 504 of 504 rows; the old product differed by up to 6.4 and 207 percent); on the certified state the change is confined to the base (p , $T\ 7\text{e-}5$; species $7.7\text{e-}2$ at $r\ 1.0002$). Run states bitwise in all three; ρ , v in the `_adv` files moved by exactly zero. Tests 48/0 (advisor rerun on `build_lwv`, md5 4320f5b7 identical to the worker's: 48/0). No golden refreshed; every `_adv` golden will move. **Advisor decision (N11b, launched):** the criterion is the right OPERATOR but the wrong TOLERANCE for a conditional correction. `cert_tol_mass = 3\text{e-}12` is the tolerance of a certified stationary state; the advective correction is a first-order local correction whose own error is of the order of the mass row's measure (the fractional change of the face mass flux across the cell), so a du-stopped or snapshot state with a mass row of $1\text{e-}3$ is exactly what the tool was built for and is what review R6 calls the "approximate local use" with "a tolerance tied to the required error". N11b writes the mass row measure itself as a column of `Hydro_ioniz_adv.txt`, sets `adv_conditional_tol = 1\text{e-}2` (the correction is described as accurate to that fraction, stated in the header beside `adv_input_certified`), and keeps the certification verdict as the separate statement it is; the product is then a CONDITIONAL correction with its condition written in every row. Advisor: two per-field and two per cent (one in a printed message) fixed. **Reported, not fixed:** `assemble_residual` leaves `R(:,1-Ng)` undefined (`RK_rhs` assigns $2\text{-}Ng..N+Ng$ only; no caller reads it today; `RK_rhs.f90/steady_residual.f90`); README.md 154, README_HOWTO.md 959 and the manual section 4 still describe the single `adv_status` column (N11b updates them with the final schema); a suite row for corrected needs a converged fixture.

- **N11b** (the `_adv` product is a CONDITIONAL correction with its condition in every row; `post_process_adv.f90`, the `_adv` writer of `write_output.f90`, `exhale_io.py`, `EXHALE_plots.py`, `EXHALE_transit.py`, `exhale_transit_lib.py`, `adv_static_limit` (53 rows), `transit_census` (21 rows), README.md, README_HOWTO.md, manual section 4): the row decision is taken against `adv_conditional_tol = 1\text{e-}2` (the fraction a corrected row is accurate to in the mass flux, the first-order argument at its declaration), the measure $m_j = |R_1|/s_1$ is written as column 10 `adv_mass_row` of `Hydro_ioniz_adv.txt` and named in # columns, the header carries # `adv_conditional_tol` and # `adv_mass_row`, and # `adv_input_certified` stays the certification's verdict with `cert_tol_mass`; schema stays 2 (readers are header-driven; the one place that counted columns from the end of the row was a test row, fixed to locate by name). MEASURED against the N11 binary: corrected rows (T) `wasp_full` 0 -> 215 of 504, `mol_base_handoff` 0 -> 341, `wasp_full_newton` 500 -> 502 (the two others are ghost rows below the energy loop's first cell); the column equals the certification's own printed measure on the same state to every digit (1.942392 at cell 432); run states and ρ , v of the `_adv` files bitwise. `exhale_io` returns `adv_mass_row` and the tolerance; the transit metadata block copies the tolerance line and the largest measure; the census still refuses on `adv_T_status`. Tests 53/0 and 21/0 (advisor rerun on `build_lwv`, md5 3131cab0 identical to the worker's: 53/0, 21/0). **Every `_adv` golden moves, far on the two snapshot cases: refreshed at the Stage D gate.** For the record: the first-order argument bounds the dropped term; in some corrected cells the KEPT enthalpy-flux term of the mass divergence still dominates the

energy equation (452 of 503 cells of hydrostatic_column against 390 retained), and whether one percent of the mass flux bounds the temperature error of such a cell is not measured; a corrected row is not close to the run's own row and is not expected to be (the certified state moves T by $6.7\text{e-}2$ at m_j $3\text{e-}16$). Advisor: one percent in README.md fixed; docs/postprocess_advection_validity.md marked stale with a pointer.

- **N10** (decision 15 option a, the restart contract; input_read.f90 +137, load_IC.f90 +551, write_output.f90 +13, EXHALE_main.f90 +166, grid_and_gates +1 row script of 20 assertions, docs/input_schema.md K43 and appendix D.2, manual entry; PDF not rebuilt): the key Restart intent: trajectory | relaxation | stationary [evaluate|equilibrate] with the three consistency refusals (default trajectory in phys, relaxation otherwise, so no existing input changes behavior); both state files carry the eight-line block (restart_schema 1, reservoir, species_columns, grid, constants, options with 20 switches, t_phys[s], source), built and parsed in load_IC.f90 (one definition), every number ES23.16; no block = legacy, loaded as before and marked provenance unknown in the log and in every file the run writes (the mark is inherited); a disagreeing reservoir (within heh_dev_tol), grid, constants or options is a refusal naming the field, species_columns and source are reported; a pair whose halves carry different blocks is refused. stationary_state_of_the_loaded_restart: WENO3 fixed, the N16d rebuild order, one equilibrium sweep (part of the residual's definition), the loaded composition's departure from that sweep's root printed, the residual and the certification held to the FILE's own certified=, then evaluate writes back and stops or (default) enters steady_wind_with_element_diffusion at once; no CFL step. **MEASURED:** the atomic element fixture AS LOADED stands at ||R|| 1.8924 (the marched hand-off read 1.888: the two CFL steps); the certified wasp_full_newton state is NOT the fixed point of its own sweep (particle count moves $2.1\text{e-}11$, species $5.0\text{e-}10$, cooling $3.6\text{e-}10$; the cell solve's $x_{\text{tol}} = \sqrt{\text{eps}}$ and the file's digits), reported at every evaluation rather than equilibrated away; stationary evaluate takes 0 steps and returns the certification unchanged; stationary on that state enters the JFNK at $7.133\text{e-}9$ and exits info = 0 with no step. Tests 20 rows GREEN (RED on the entry binary: the key warned and ignored, one step marched, no block, no refusal); grid_and_gates 130/0 (advisor rerun 131/0 on build_lwv). mol_carrier 200 steps numeric-line bitwise (eight new # lines); run_check.sh line 297 excludes # lines, so the block cannot change a matrix verdict; wasp_full_newton cold numeric-line bitwise against a control binary of the same tree without N10 (its $5.8\text{e-}9$ / $9.4\text{e-}9$ movement against the golden is B5j's recorded movement, not N22's as the worker supposed). **Fixed on the way:** the header loop matched columns as a SUBSTRING (index(line, 'columns') > 0), so the _adv files' NOTE prose or the new species_columns field would have replaced the species labels silently; now the first token (comment_field_is); the same hazard in restart_round_trip.sh's Python readers fixed. Deviations from the design, stated: no EXHALE_STATIONARY_EVALUATE_ONLY env hook (the second word is the one answer); the provenance unknown mark is not yet in EXHALE_setup.out (write_setup_report.f90 not owned; one line); species_columns reported not refused (the by-label species map is a documented path); source is the git revision, not an executable md5. **Consequence for the user (decision item 21):** the exact options comparison forbids the arm-ladder workflow (converge without an option, restart with it on); legacy pairs are unaffected, every new rung would be refused; the disciplined fix is a key naming the tokens allowed to differ. Also corrected: docs/input_schema.md appendix D listed the retired valve/fluxconst fields and the wrong base_bc

values. Reported: Run mode is absent from input_schema.md; the EXHALE_PTC=1 route writes no setup report; a re-evaluated state writes sec_ion_step=0.

- N22** (the predicted decrease stops scaling below $||s|| \cdot 1e-7$; steady_newton.f90 +195/-10, krylov_and_dogleg +16 rows): **all three candidates of the brief refuted by measurement, and the cause named:** the trials that break the scaling are exactly the trials whose species unknown was written onto a face of its box (nout = 1), and that write is a move of the state no model of the step contains (11 of 11 such trials refused with a fixed ascent of $3e-3$ over five decades of step; 10 of 10 others accepted at ratio 1.000). When the radius is cut the SCALED step quarters but the step in the unknowns the residual reads, $||D \cdot s||$, moves 0.4 percent, so the model image and pred are unchanged to seven digits; $r0 \cdot A \cdot s = -||A \cdot s||^2$ to every digit ($pred = 0.5 \cdot (r0 \cdot A \cdot g)^2 / ||A \cdot g||^2$, radius-free, the interior minimizer along the leg: not a defect); a fresh Jacobian action at the same step reproduces pred to every digit (no stale cache); the probe displacement is $\sqrt{\epsilon}(1 + ||Y||) = 1.8e-7$ whatever the direction, nine decades above the residual's reproducibility (no floor; the reverse: the stalling step is $2.7e-3$ of the probe arc). **The fix:** hold_the_step_where_it_leaves_the_species_box zeroes the step at an unknown it would carry out of its box (the iterate is feasible, so the trial is, and the step is one the operator can be sampled along); probe_length_of_the_jacobian_action is the one definition of the probe arc; a per-trial diagnostic line under EXHALE_ELEM_DIAG. **Atomic reload, cap 300, before -> after:** 227 -> 113 iterations (both stagnation), accepted/refused 170/55 -> 91/19, negative ratios 49 -> 14, best $||R|| \cdot 2.353e-2$ -> **1.333e-2**, handed back $2.671e-2$ -> **1.356e-2**; the period-two cycle is gone (iterations 95-98: pred doubles with the step, all accepted at 0.998 to 1.001, steps $1e-2$ to $1e-1$ instead of $1e-9$); the closing refusals are ray/floor verdicts at POSITIVE ratios. Carrier reload BITWISE (the branch never fires); wasp_full_newton BITWISE (ten files; the probe-arc refactor is shared through jv_product and this settles it). Tests: the twelve-rung ladder rows are GREEN at entry (the geometry was never the cause), the new rows RED by four absent symbols, GREEN 120/0; steady_species_rows 182/0. Advisor: rebuild warning-free, steady_newton.o md5 d70a078b identical to the worker's delivered object, the two suites 120/0, 182/0. Not converged. **Reported, the next item (N23):** the approximate-gradient leg is 0.7 percent of the step and supplies ALL of its model image ($||sU|| / \Delta = 6.9e-3$ passes `cauchy_leg_min_fraction = 1e-3` while $||A \cdot s|| = \tau_c \cdot ||A \cdot g||$ to four digits): the criterion must be the share of the IMAGE, not the length ratio (N7b's finding at a length the threshold admits).
- N10b** (decision 21 option a; load_IC.f90 +319/-20, input_read.f90 +145/-1, write_setup_report.f90 +62/-6, grid_and_gates +1 script of 11 rows, docs/input_schema.md K44 and D.2, manual entry): the key Restart option change: `<token>[, ...]`; the options field is compared TOKEN BY TOKEN from one table (`opt_name(20)`, `opt_value`, the emitted text bitwise unchanged), only the named tokens may differ, every other difference refuses the load naming the token with its from -> to; six layout tokens (metals, mol, oxychem, carrier, carrier_newton, iontrans) may never be named (they decide the rows the state carries: a cold start, not a restart); an unknown token, a grid/reservoir/constant word, the key with no token or without Load_IC? True are input errors; the change is written into both halves of the new state as `# option_change` from -> to at restart of `<source>`, inherited (history of 32 lines); EXHALE_setup.out gains Restart

provenance: UNKNOWN | UNKNOWN by descent | states its configuration and the permitted changes (N10's deviation 2 closed; the "by descent" wording was added because the first wording was measured false on a ladder's second rung). **Correction of the advisor's brief:** LW is a FLUX (Stellar LW flux), not an option token, so the armA_noLW -> armA_LW rung never needed this key; and every state written before the Jupiter-radius unification (those two included) is refused by the grid guard at $2.25e-2$ in the outermost cell, so the arm directories are configurations, not restart sources. The ladder step was measured on the armA_noLW configuration with base_ir, mol_ir flipped: refused by token without the key, accepted with it, the provenance line in both halves. Tests RED 11/11 at entry (the key was an unrecognized line), GREEN 11/11; grid_and_gates 141/1 with the one FAIL attributed (below); mol_carrier 200 steps numeric-line bitwise (the rebuilt # options line bitwise). Also fixed: N10's test stage A read the LIVE wasp_full_newton/output/, which the matrix rewrites (a false failure of the legacy-mark row as soon as the pair carried a block); it now strips the block from its own copy. **Advisor:** rebuild warning-free, md5 b10641e5 identical to the worker's; grid_and_gates 141/1, the FAIL being stationary_evaluate_rebuilt_within_sweep_budget reading the live output/ of wasp_full_newton, which the running Stage D matrix had left as a mid-run relaxation snapshot ($3.2e-3$ against $1e-8$; the same pair evaluated by the entry and the N10b binary is bitwise, so not N10b's); the advisor pinned the certified state as backup/regression/wasp_full_newton/IC/ (copied from the golden, certified=T, README there) and pointed the row at it. Reported: restart_grid_guard.sh's stale header (third time); sec_ion_step=0 on a re-evaluated state; whether molbase, he_diff, he_metal_diff belong with the layout tokens is a reviewer's question (the worker's reading: they add operators, not rows).

- **N23** (the approximate-gradient leg admitted by its share of the model image; steady_newton.f90 +220/-5, krylov_and_dogleg +16 rows): two constants cauchy_leg_image_share_max = 0.5, cauchy_leg_step_share_max = 0.1, two pure routines (cauchy_leg_shares_of_step_and_image, cauchy_leg_is_image_without_step: image share above the first AND step share below the second, so a leg that IS the step keeps its image), and a branch in the dogleg's trial loop that DROPS such a leg from the step and its image and rebuilds the dogleg at no extra product (drop, not rescale: a rescale would either shorten A sU without sU, which is not a step of the operator, or move the step off the dogleg path); the model slope handed to ray_slope_verdict is the slope of the step taken in both branches. **MEASURED, the reading ([diag 9]):** at the reload's second solve quartering the radius moves the promise by 0.9959 with the leg admitted and by 0.2501 with it dropped (six iterates: 0.9959 to 0.8938 against 0.2501 to 0.2517), so the promise follows the radius again where it did not. **Honest impact: the new rule fires ZERO times on both fixtures and all three runs are BITWISE** (atomic 113 iterations, $1.356e-2$; carrier $8.993e-2$; wasp_full_newton 13.30); on the post-N22 trajectories the length test reaches every image-dominant leg first (117 length drops, 0 image drops); the shares printed at every trial say the image test is the right criterion nonetheless (0.9986 of the image at $3.4e-5$ of the step; eleven of seventeen read trials satisfy the predicate on their own), and the leg N22 saw at iteration 145 sat on a trajectory the N22 hold removed. Tests RED (four symbols absent), GREEN 136/0; steady_species_rows 182/0; -Wall warnings 6 before and after. **Advisor:** rebuild warning-free, the two suites 136/0 and 182/0 on build_lvw (the object md5 differs from the worker's delivered one by a post-measurement comment edit that moved line numbers, which the I/O statements carry; the worker's line-for-line reproduction of the solve covers

it). Not converged; the rule is not claimed to improve either fixture. Reported: `cauchy_leg_min_fraction` is now a proxy for a test the code can make directly (letting the image test alone decide is the next increment, not taken because no fixture exercises the difference); the trial diagnostic's evaluations enter the cost counters.

- **N24** (what refuses the remaining trials of the atomic element arm; `steady_newton.f90` +51/-2, `krylov_and_dogleg` +13 rows): **the obstruction is the linear model, and the row is named**. Atomic reload, cap 300: 110 trust-region iterations (the "113" of earlier reports counted three summary lines), 91 accepted / 19 refused. The 19 are two populations, neither a step-control defect: 14 ascents at LONG steps ($||s||$ 1.6 to 51, five with zero species unknowns held, so N22's mechanism is not what is left) and five closing refusals at positive ratios ($||s||$ $2.3e-4$ to $8.9e-7$ with the promise quartering exactly while the true decrease keeps a step-independent floor of $3e-11$ to $1.6e-10$). One accepted step (outer 59, ratio 0.948, $||R||$ $7.37e-2 \rightarrow 2.12e-2$) moves the binding row from the energy equation to **the sodium element row of the outermost cell (cell 500, r 4.06 Rp)** and the solve never leaves it: from outer 60 to 110 that row holds 0.435 of the squared merit and the whole judged distance ($5.3e-2$ of its own scale, $5.36e6$ at `cert_tol_element` $1e-8$) while $||R||$ stands at $1.333e-2$. At outer 105 everything else is sound (0 held, 0 refused samples, 0 empty columns, `dgbcon` $1.25e-6$, step ratio 0.9998); what fails is the linear model: 40 of 40 products, true relative residual 0.58 against 0.1, **the Arnoldi image 21 percent away from the operator** ($1.6e-6$ at iteration 1, $2.3e-4$ at N7's iteration 60), $||sN||$ $8.6e2$ against a radius $5.9e-2$, so the region uses $7e-5$ of an inaccurate direction; the stuck row is at once the smallest scaled column and the smallest scaled row of the band. The stagnation detector read the judged distance, which IS that row; $||R||$ was not falling ($1.337e-2 \rightarrow 1.333e-2$ over 32 iterations). **The image-only leg test** (`EXHALE_CAUCHY_LEG_BY_IMAGE=1`, default off) measured and NOT adopted: worse on both fixtures (atomic $2.30e-2$ against $1.36e-2$; carrier 0.324 against 0.090, to the cap), the arms parting at outer 7, so N23's "the rules agree" was that trajectory's property. Deliverable 3 STOPPED as briefed (linear solve / model / one row's physics, not the step control). Tests RED (two symbols absent), GREEN 149/0; `steady_species_rows` 182/0; the delivered binary reproduces the entry solve line for line over 925 lines; `wasp_full_newton` BITWISE. Advisor: rebuild warning-free, 149/0, 182/0 on `build_lvw` (the advisor's binary of N23 had been built from an earlier revision of `steady_newton.f90`, as the worker showed; the suites and fixtures were the check). **Proposal (N25)**: (1) explain the 21 percent Arnoldi gap (`EXHALE_GM_ORTHO`; Jv reproducibility by two products); (2) ask the cycle for the direction ON THE BALL (Steihaug-Toint truncation) instead of solving to 0.1 a leg four decades beyond the radius; (3) the sodium row's own terms at cells 490 to 500: the element operator gives cell 500 a zero-gradient outer ghost, so nothing outward constrains it (a boundary question of the element operator, `binary_element_diffusion.f90`); (4) `cert_tol_element` $1e-8$ is what makes this row the whole judged distance (UNCONFIRMED, ISSUES 3.1 e; not moved). Reported: the (JFNK) it line prints `k=*` for a species row without naming the element (on this arm the identity was the whole diagnosis); the stagnation messages quote $||R||$ where the detector read the judged distance.
- **Stage D gate and golden refresh (advisor, 2026-09-09 21:14 to 2026-09-10 00:27, make check on the shared build/; refresh 00:40)**. Tree after every item to N24 (N25 was running on a private build). Verdict against the goldens of 2026-09-08 19:05: on all 16 default cases the two marching files are **data identical** (15 cases

bitwise; wasp_ full_newton WITHIN at 5.78e-9 / 9.41e-9, B5j's recorded movement, which the 0.1 percent rule counts as identical) and all 32 _adv files FAIL by SHAPE (10 against 8 and 40/36/43 against 38/34/41 columns: the two status fields and the measure column of N11/N11b), so make check exits 2 as expected. The _adv movement is the physics of N12 (stale composition, upstream energy), N11 (mass-operator stationarity) and N11b (conditional tolerance): T_adv changes in 477 of 504 rows on mol_base_handoff (max 61.6 percent, 364 -> 588 K at r 1.093), 479 of 504 on mol_carrier (max 138 percent, 379 -> 901 K at r 1.098), 215 of 504 on wasp_full (max 2.3 percent, 5637 -> 5505 K at r 1.131). **Goldens refreshed:** the previous set archived as backup/regression/golden_pre_stageD_20260910/, run_check.sh golden snapshotted the outputs the gate had just produced (17 cases, including the three hp_* and oxygen_chemistry extras), and the advisor compared the 68 snapshot files against the case outputs with the comparator's own #-filter: 68 identical, 0 differ (this comparison, not a three-hour rerun, is the "check again" of the standing rule, since a snapshot is a copy; the user asked that regression checks not be repeated). The shared build/ and EXHALE.x are now current with the tree (md5 f6ccb498 at the gate's build; N23/N24 built after it touched steady_newton.f90 only, so the marching path of the matrix is the gate's). State files now carry the N10 metadata block and the _adv files the N11/N11b header; both are # lines the comparator ignores.

- **N25** (the Arnoldi gap, a direction on the ball, the sodium row at the outer boundary; steady_newton.f90, krylov_and_dogleg +25 rows, binary_element_diffusion.f90 a default-off print only, 33 lines, no arithmetic): **(1) the 21 percent Arnoldi gap is the NONLINEARITY of the finite-difference action across directions**, shown positively: every one of the 40 Arnoldi columns matches its own product to 1e-16 to 3e-16, two products of one direction agree exactly, the basis's orthogonality loss is 4.4e-13, yet the action of $V_1 + V_2$ differs from the sum of the two actions by 1.1e-4 and the assembled gap tracks the step length (8.5e-5 and 2.5e-4 per unit of $||sN||$ at outer 1 and 105; $||sN||$ 12 -> 865), the signature of a curvature term over the fixed probe arc $\sqrt{\epsilon}(1 + ||Y||) = 1.8e-7$. Reorthogonalization NOT adopted (loss 1e-13 to 3e-12 against the 1e-8 level; as a measured arm EXHALE_GM_REORTH0=1 it is worse on both fixtures: 1.487 and 0.43). **(2) Steihaug-Toint truncation** is in as EXHALE_KRYLOV_ON_THE_BALL=1, default off: products per iteration 33 -> 7 (atomic), 40 -> 1.5 (carrier); atomic better on the judged distance (1.69e6 against 5.31e6, lowest $||R||$ 9.4e-3 against 1.33e-2) but worse on $||R||$ handed back (1.69e-2 against 1.36e-2); carrier worse (0.486 against 0.090); the brief's condition fails, default stays off. **(3) The sodium row of cell 500 is not a well-posed outflow condition, and the reason is a DEFECT of the stationary residual evaluation, not of the element operator:** the outer ghost of the element mixing ratio is FROZEN at the pre-solve composition (1.73111e-6, the metals.inp reservoir the handed-in state carried) throughout the solve while cell 500 falls to 1.0e-11, so at the stagnating iterate the ghost is 1.7e5 times the cell it "continues"; the marching path refreshes that ghost (the second solve, entered at step 2002, has ghost = cell 500 again); the accepted step of outer 59 did not move fX(500) at all (it moved the outer face mass flux); the scaled Jacobian row is dominated by the momentum of cells 499/500 and the sodium of 499 while its own diagonal is 1.3e-4 of the largest entry (the row is nearly independent of its own unknown). The physically stated condition (zero gradient at the top, no diffusive flux) requires the ghost to follow the ITERATE; the repair is in the composition write-back / ghost policy of eval_residual, which is N26 (launched). **(4) the (JFNK) it line names the element (worst row: element Na of cell 500) and**

the stagnation messages quote the judged distance and dj_best. Tests RED (eight symbols absent), GREEN 174/0; steady_species_rows 182/0; default path unchanged (atomic 109 iteration lines identical to the digit; carrier identical); wasp_full_newton BITWISE. Advisor: rebuild warning-free, 174/0, 182/0 on build_lwv; three British neighbour in the element operator's comments fixed. Not converged; the ball arm's best iterate is chosen by the judged distance while the control is ||R|| (the two rank iterates differently, which limits arm comparisons).

- **N26** (the upper ghost of a transported element or carrier follows the iterate; steady_newton.f90 +135/-20, steady_species_rows +7 rows): in write_species_rows_into_composition (the ONE place a trial composition is assembled for the residual: eval_residual and the three best-iterate restores) the transported columns of the upper ghosts take the outermost cell's value by the rule the marching path uses (READ: Yetr(N+1:N+Ng,q) = Yetr(N,q) in species_advection_stage; ftry/fc(N+1:N+Ng,ic) = ... (N,ic) in the carrier operator; Apply_BC never touches f_sp), the element totals of a ghost restored after the carrier copy, a ghost deficit clamped as the operator clamps its own; lower ghosts untouched (Dirichlet reservoir); the three-unknown route returns before reading anything. The mixing ratio follows the mass fraction exactly (equal He mass fraction at N and its ghost gives equal $n_{\text{He}}/n_{\text{H}}$, hence equal $n_{\text{X}}/n_{\text{H}}$; MEASURED 1.4e-16). **MEASURED, atomic element reload (EXHALE_JFNK_MAXIT=300):** ||R|| handed back 1.356e-2 -> **3.719e-4** (36x), judged distance 5.36e6 -> 6.21e4 (86x), flux spread 6.6e-4 -> 5.5e-7, 91/19 -> 235/18 accepted/refused, the sodium row of cell 500 binds on **0 of 252** iterations (was 51 of 109) and no row of cell 500 is ever the worst; the solve now stagnates on the INTERIOR helium row of cell 246 (r 1.155), info = 2, uncertified. **Carrier reload:** binding cell (H2 of cell 205, interior) and judged distance (4.807e7) unchanged to four digits, ||R|| handed back 8.993e-2 -> **2.519e-1** (worse by 2.8x, a path difference of a stagnating solve, reported not judged), flux spread down 12 percent. mol_carrier marching bitwise (no caller outside solve_steady_jfnk); wasp_full_newton BITWISE. Tests RED on the entry text with only the export added (4 rows: ghost mass fraction 6.7e4 off, mixing ratio 1.0e5 off, the outermost element row of a flat column 1.19e-2 instead of 0, the carrier ghost 0.80 off), GREEN 189/0 (a non-vacuous row keeps the stale ghost and reads 1.19e-2 in both arms); krylov_and_dogleg 174/0. Advisor: rebuild warning-free, steady_newton.o md5 4bd3138a and binary e56609ff identical to the worker's, 174/0 and 189/0 on build_lwv; one neighbours fixed. **Reported, the next small item (N26b):** the relaxation solvers solve_trace_element_in_hydrogen and element_diffusion_step refill their working arrays' upper ghosts before every pass while element_transport_residual, documented as the same operator, reads the ghost from f_sp with no refill, so the operator's boundary depends on its caller (binary_element_diffusion.f90, two lines). Not claimed: convergence; an explanation of the collapse of the outermost cells to 1e-11 of the reservoir.
- **N26b** (the element operator refills its own upper ghosts; binary_element_diffusion.f90 +26/-0, new suite src/tests/element_operator/ with 5 rows): element_transport_residual now writes $X_{\text{he}}(N+1:N+Ng) = X_{\text{he}}(N)$, $f_X(N+1:N+Ng) = f_X(N)$ and $wY_{\text{tr}}(N+1:N+Ng) = wY_{\text{tr}}(N)$ (the third line the relaxation twin gets for free through project_elements) so the operator poses ONE outer boundary (zero gradient, no diffusive flux at faces 0 and N) whatever the caller left in the ghosts; the lower ghosts stay the Dirichlet reservoir; the header

states the boundary. Both outermost rows (N-1 and N, WENO3 reaches cell N+1 at face N-1) are covered. MEASURED: the relaxation's own fixed point handed to the residual with stale ghosts read $1.19\text{e-}2$ of its scale in the outermost row before and $1.5\text{e-}13$ after (the interior level is $1.3\text{e-}13$); the same column with stale and with zero-gradient ghosts now gives the two outermost rows bitwise. Suite RED 3 of 5 on the entry text, GREEN 5/0; certification 29/0; the atomic reload's first two JFNK iterations identical but for the merit's reproducibility going $2.8\text{e-}14 \rightarrow 0$ exactly (the residual no longer reads a ghost the sweep leaves at a different last bit), the outputs moving at $7.9\text{e-}12$ (T) and $1.6\text{e-}10$ (an `_adv` status/measure column) because N26's write-back hands over the ghost to $2\text{e-}16$ and the refill to the bit; `mol_diffusion` marching 200 steps BITWISE (the marching path reaches the relaxation twin, never the residual). Built and measured in an isolated source tree with `steady_newton.f90` pinned, because N27 was editing it. Advisor: rebuild warning-free, `element_operator` 5/0, certification 29/0 on `build_lwv`; one per cent fixed. Reported: the write-back's ghost is the cell's to round-off, not exactly (any other consumer of those ghosts reads a $2\text{e-}16$ gradient).

- **N27** (the helium row of cell 246 and the H2 row of cell 205 read side by side; `steady_newton.f90` +191/-3 diagnostic only, `steady_species_rows` one row restated): **the helium row of cell 246 is held by the LINEAR SOLVE, not by its physics, its boundary, a bound, the closure or the tolerance.** At outer 245 the row is $+2.43\text{e-}22 = \text{diffusive (settling)} + 2.40\text{e-}22$ plus advective $+3.3\text{e-}24$, same sign (they add); the Newton removed the advective term (a factor 7200 since outer 1) and barely touched the diffusive one ($\times 1.28$); the measure $6.17\text{e-}4$ is the maximum of a flat plateau ($5.9\text{e-}4$ at 240 to $5.4\text{e-}4$ at 253), not a bad cell; 0 held, 0 refused samples in 246 iterations, band healthy (`dgbcon` $7.5\text{e-}7$), merit reproducibility exactly 0, the trust region inert (ratio 0.933, 0 cuts, `aN` 1.0) and the Krylov cycle spends 40 of 40 products on a direction whose true relative residual is **0.9917** against 0.1 with the Arnoldi image 14.7 percent off the operator; the step buys $4.7\text{e-}5$ of the merit per iteration. The helium Jacobian triple is a proper tridiagonal stencil with the diagonal largest (the unknown controls its row; the wind dominates the row's entries). The element rows hold 0.98 of the merit (He 0.63, Fe 0.36); $||R||$ is the energy row at $1.37\text{e-}4$; the stagnation detector is honest (merit and $||R||$ fall while the judged distance rises $6.12\text{e4} \rightarrow 6.21\text{e4}$). **A number that does not fit N25's reading:** the Arnoldi gap is 0.147 at $||sN||$ $1.8\text{e-}2$, 8 per unit against N25's $2.5\text{e-}4$ per unit, so the gap is not a function of the assembled length alone. **The H2 row of cell 205** is the same shape on a coarser front: measure 0.480 (not a cancellation, half its own terms), `f(H2)` falling $1.4\text{e-}1 \rightarrow 1.1\text{e-}3 \rightarrow 4.9\text{e-}4$ over cells 195-207 (logarithmic slope -757), `f(HI)` flat (1.7 percent), so cell 205 is NOT a two-root HI front (consistent with N5); linear 40/40 at 0.97, gap $3.1\text{e-}2$. Deliverable 3 STOPPED (nothing in the step control to change at the binding iterate). Deliverable 4: `cert_tol_element` would have to be $6.2\text{e-}4$ (4.7 decades; far above the $1\text{e-}5$ the operator's header gives as its loosest floor); the closure's reproducibility ($3.1\text{e-}7$ in the row's measure per unit seed, N5) is not what holds it. Proposals: re-measure the Arnoldi gap at three iterates (it no longer tracks $||sN||$); the untried freedom is the COLUMN preconditioner (row scaling is fixed by decision 20 a); whether the detector should stop a solve whose merit still descends while the judged row does not is an acceptance decision; the carrier front is a grid statement. Tests 189/0, 174/0; advisor rerun 189/0, 174/0. **Incident, found by N27 and CONFIRMED by the advisor:** with N26b's ghost refill in `element_transport_residual` the atomic element arm parts from N26's arm at outer 10 and stands at $||R||$ 1.199 at iteration 40 where N26's arm was at $3.21\text{e-}2$ (advisor run, cap 40, current tree: 1.888, 1.891, 1.677, 1.484, 1.518, 1.582, 1.199 at iterations

1, 5, 10, 15, 20, 30, 40 against N26's 1.888, 1.891, 1.678, 1.955, 1.758, 0.912, 0.0321); N27's own diagnostic is inert (hook on = hook off). N26b's validation (the first two iterations, the certification suite, mol_diffusion marching) did not reach it, and the advisor's verification of N26b did not rerun the arm: the advisor's mistake. N26b's refill is therefore NOT accepted as it stands; N26c (launched) finds why an operator that poses its own boundary breaks a solve whose caller already hands it that boundary to 2e-16, or reverts the refill. N27 restated the row a_stale_upper_ghost_is_read_ by_the_outmost_row to its negation because N26b made the original false; N26c reconciles the two rows with whichever text stands.

- N26c** (why N26b's ghost refill broke the atomic solve; binary_element_diffusion.f90 refill REVERTED with the header restated, element_operator rows restated (6), N26's steady_species_rows row restored): **the refill changes the residual by one element's outermost row in its twelfth digit and nothing else, and the atomic element arm does not survive a last-bit change of the upper ghosts, whatever makes it.** MEASURED: the caller's ghosts already equal the cell to 1 to 4 units in the last place (Xhe 9e-16, fX 1.9e-15, wYtr 6e-16 over the whole solve); the two texts' outermost rows agree to 17 digits on every element but nitrogen's, which differs by 9.8e-13 relative (6.8e-17 in its own measure); no other ghost of the routine is read (face coefficients stop at N-1); the band's stencil already carries the dependence of rows N-1, N on cell N through the write-back. **A control that multiplies the same three ghost quantities by $1 + 2e-16$ and poses no boundary at all breaks the arm as badly** (||R|| at iteration 40: caller's ghosts $3.212e-2 = N26$ to the digit; refill 1.199; one-ulp control 1.125; the three part at outer 10-11 and from each other). Where it amplifies: the Krylov leg at its one-digit tolerance (relative residual reached $8.86e-2 / 8.89e-2 / 8.93e-2$ at outer 10, pred 0.4 percent apart, then 2.7x apart at outer 11, then different branches of the acceptance at outer 12: accepted 0.874 / REFUSED -1.98 / accepted 85); the solve's own merit reproducibility prints $1.4e-14$, a hundred times the ghost movement. **The refill is physically right and is reverted anyway:** no version of it can be bit-inert on the production caller, and the arm's acceptance is a trajectory no last-bit change preserves. Delivered text reproduces N26's atomic arm line for line over 252 records (||R|| 3.719e-4, judged 6.22e4, He of cell 246, 27543 samples, exit 2) and N26's carrier arm over 63 (0.2519, 4.806e7, H2 of cell 205); mol_diffusion marching bitwise; suites certification 29/0, element_operator 6/0 (restated to ASSERT the measured caller dependence, $1.194e-2$, so the suite exits 0; the advisor accepts that choice), steady_species_rows 189/0, krylov_and_dogleg 174/0; advisor rerun of the four on build_lwv (md5 3201310a, the worker's delivered binary): 6/0, 29/0, 189/0, 174/0. **The finding that matters beyond this item:** the atomic element arm is NOT a reproducible quantity; its ||R|| at a fixed iteration count is set by which Arnoldi vector crosses a one-digit tolerance at the last bits of the residual, so "reproduces the previous arm" rejects correct changes as readily as wrong ones, and what the arm can carry is named outcomes (binding rows, refusal counts, flux spread), not ||R|| at iteration N. Recorded in ISSUES 3.1 (d) and in the fixture's README. The asymmetry of the element operator (the relaxation twins write their ghosts, the residual takes the caller's) stays in the tree, documented and measured ($1.19e-2$ on a stale caller). Advisor: a cp -irp of the N4b worker hung on its interactive prompt since 2026-09-09 (PID 61163, cwd the repository root, copying from the session scratch) was killed by PID. CORRECTION within the hour: killing the cp let the worker's shell run its NEXT statement, which re-applied N4b's first (discarded) patch onto the delivered element_inventory.f90 (832 -> 926 lines, a duplicate routine, would not compile); the N4b worker,

woken by the stale task, rebuilt the file from its saved entry copy plus the delivered patch and it reproduces `diff_element_inventory.diff` line for line; the advisor confirmed 832 lines, one routine, a warning-free build (md5 3201310a, N26c's delivered binary) and `element_operator` 6/0, `steady_species_rows` 189/0. Lesson written into `docs/worker_rules.md`: `\cp -f / \mv -f` always, and a stale command is cancelled as a task, not killed as a process.

- **Direction after N26c (user, 2026-09-10: "권고안대로 진행").** The advisor's recommendation adopted: (b) N8b, anchoring the species-row certification tolerances on the two candidate states by the review's five measurements (reproducibility floor, derivative convergence, a manufactured steady solution of the element operator, grid refinement of the candidate state, conserved-flux error), reported per row class and regime with a PROPOSAL for the user (decision item 22) and no value moved; and in parallel (c) N28, the code items of ISSUES 3.7/3.8 (run-wide JFNK cap, one `element_ratio_gate`, the update map's install branch owing an `Apply_BC`, the undefined innermost ghost column of `assemble_residual`, the PTC route's setup report, `sec_ion_step` on a re-evaluated state, the stale test header, `Run` mode in the input schema). (a), the column preconditioner, waits for N8b's numbers. The user manual PDF was rebuilt by the advisor (85 pages; it was two keys behind its source: `Restart` intent, `Restart` option change).
- **N8b** (anchoring the species-row certification tolerances; `certification.f90` +173 additive, no value moved; `steady_newton.f90` one diagnostic; `certification` suite 29 -> 42 rows; memo `docs/certification_tolerance_anchoring_20260910.md`): **1e-8 is refused by no floor of the arithmetic, the closure or the representation (five to eleven decades below it); what refuses it is the DISCRETIZATION and the ELEMENT FLUX CONSERVATION.** MEASURED on the two candidates: reproducibility exactly 0; closure-seed floor $5.3\text{e-}15$ (the state's own sweep seed $1.7\text{e-}10$); state-perturbation amplification 0.23 to 0.30 in the wind but $1.1\text{e}3$ to $2.0\text{e}3$ at $r < 1.01$ (floors $5\text{e-}17$ and $4.4\text{e-}13$); derivative: a clean V at the base ($4.5\text{e-}7$), a PLATEAU at $6.4\text{e-}5$ over four decades of probe length at the binding cell 245 (the part of the derivative outside the band) so the smallest row a Newton step resolves there is $3.8\text{e-}8$; manufactured relaxed column at 801/401/201 cells $5.4\text{e-}13$ / $2.47\text{e-}6$ / $7.42\text{e-}6$, order 1.59 (the flat column $2.7\text{e-}14$); grid refinement of the candidate at r 1.1515: the He/H row $5.97\text{e-}4$ ($N = 500$) -> $2.48\text{e-}4$ ($N = 1000$), remap control $1.6\text{e-}5$, so **about 78 percent of the $6.0\text{e-}4$ is discretization** (a C2 interpolant is needed; below r 1.03 the remap dominates); element flux spread 39.3 in the layer, $2.6\text{e-}2$ in the wind, $2.2\text{e-}3$ over the escape window against the $5.5\text{e-}7$ mass-flux spread the arm is quoted with. Memo section 7 is the proposal (decision item 22, added by the advisor to `To_be_determined`): (a) $1\text{e-}5$ in the wind with the layer reported, (b) keep $1\text{e-}8$ and state the fixtures cannot certify, (c) two regimes gating; no proposal certifies either candidate. **Two defects found outside the item's files, reported:** (1) **the candidate's species do not carry the run's own density:** mass closure $6.68\text{e-}3$ at the top against $1.9\text{e-}9$ on the three-unknown control; reproduced in the suite on a column closed to $6\text{e-}16$ that reads $3.37\text{e-}3$ after `ONE relax_element_composition`, so `project_elements` does not return to hydrogen the mass it takes from helium (N29, launched); (2) the writer-to-loader round trip does not return the state: the atomic candidate reloads at `||R||` 0.486 instead of $3.7\text{e-}4$ because `load_IC` rebuilds `rho` from the species (a consequence of 1). Tests 42/0 (RED at the build level: the suite does not compile against the entry

objects); the default certification report byte-identical (anchoring under EXHALE_CERT_ANCHOR=1). Advisor: rebuild warning-free, certification 42/0; one neighbourhood and two per cent fixed.

- **N28** (the code items of ISSUES 3.7; EXHALE_main.f90, RK_rhs.f90, steady_residual.f90, element_census.f90, element_inventory.f90 and attempted_step.f90 (the duplicate constant), grid_and_gates +2 rows and headers, docs/input_schema.md K45): (1) EXHALE_JFNK_MAXIT caps the RUN: the three call sites of steady_wind_with_element_diffusion pass one named default jfnk_outer_iterations_default = 3000 (the marching hand-off's literal was 500, the one default that moved; wasp_full_newton converges in 13 and cannot reach either), and under the variable the run takes its FIRST stationary solve and refuses later entries by name (MEASURED: three solves of 2 iterations -> one); (2) element_ratio_gate has one owner, element_census.f90 (public, element_census_tolerance() public with its override), read by element_inventory.f90 and attempted_step.f90; (3) update_map_begin_step's install branch calls Apply_BC in the N16d order (the EXHALE_UPDATE_MAP diagnostic's second block on mol_carrier moves 5 percent in the mass and energy rows: the fix of a defect, a boundary of one state under an interior of another); (4) assemble_residual defines the innermost ghost column (dF, S, R zero at 1-Ng; two callers read it in whole-array expressions); -fcheck clean on the certification driver and on a 20-step mol_carrier (the run that reaches it); (5) the EXHALE_PTC=1 route writes its setup report (was 0 lines; new row RED/GREEN); (6) a Restart intent: stationary evaluate keeps the file's sec_ion_step (2402 -> 2402, was 0; new row RED/GREEN); (7) restart_grid_guard.sh's stale header rewritten and the script made to honor EXHALE_TEST_OUT; (8) Run mode documented as K45. Item 9 (n_eq_sweeps_last on a discarded evaluation) is steady_newton.f90's, reported. Validation: mol_carrier 200 steps and wasp_full_newton cold BITWISE on every output file (the one log difference: cap 500 -> 3000, from the caller); grid_and_gates 146/0; attempted_step 162/1 with the FAIL proven pre-existing (rebuilt from entry sources: same row); advisor rerun on build_lvw: grid_and_gates 146/0, attempted_step **163/0** (the N28b fix below had reached the tree by then). **Found, reported, N28b launched:** bg_cell (type(ion_rates)) is allocated without initialization and the type has no default values, so before the first sweep the checkpoint checksum reads 8.7e244 and the row a_perturbed_state_is_not_the_checkpoint could never fire (a perturbation of 1e2 lost in the round-off of 1e244). Advisor: three forbidden phrases fixed (per-element in a printed line, neighbouring, per-region). One deviation stated by the worker: an extra private -fcheck build (the shared run_fcheck.sh runs make distclean).
- **N28b** (bg_cell allocated and never initialized; ion_cell_state.f90 the type only, ionization_equilibrium.f90 the allocation site, attempted_step suite +1 row and a deterministic dirty-heap setup): **the defect is real, wider than bg_cell, and includes a LOGICAL.** Three grid-sized objects of ionization_equilibrium came out of allocate holding the recycled block: bg_cell and ieq_rate_cell (both type(ion_rates), no default values) and the real array heat_chem (the one array left out of the routine's zero list). MEASURED with a probe that frees a 16 kB block before the allocation: T_K, P_HI 8.7e244 and **x_h2_fixed TRUE** (an imposed H2 partition the caller never asked for, read from garbage, standing in the array the carrier transport reads), on a fresh heap all zero (which is why N28 saw the dead

row fail and the advisor's rerun saw it pass: the defect appears only when a freed block is recycled). Fix: ion_rates carries its zero in every one of its 37 fields (34 reals, 3 logicals) as default initializers, stated as the invariant "a cell not yet swept has no gas, no rates and no imposed partition", which reaches bg_cell, ieq_rate_cell, bg_cell_adopted, bg_cell_best, the closure scratch and the four checkpoint copies at once; heat_chem = 0 at the allocation. Two comments that claimed a default initializer would not reach threadprivate copies were corrected (MEASURED with a four-thread program: it does, as OpenMP specifies). Tests: RED on the entry text with the new driver 68/2 (the_unswept_background_carries_no_rates -3.4e250; a_perturbed_state_is_not_the_checkpoint h1 == h0), GREEN 164/0 (the intermediate arm located heat_chem at 8.7e244 by a field-by-field peek). mol_carrier 200 steps bitwise (every reader of the three objects is gated on bg_ready/ieq_rates_ready, set after the loops that write them; heat_chem has no physics reader, only the checkpoint); wasp_full_newton not run (atomic: the arrays never enter a physics path). Advisor: rebuild warning-free, attempted_step 164/0. Reported: three test fixtures enumerate all 37 fields of bg_cell by hand (now redundant) and element_census_tests.f90 set two fields and left the rest to the allocator (now covered by the type).

- **N29** (the element projection does not conserve mass; binary_element_diffusion.f90 +258/-113, element_operator suite 6 -> 14 rows): **the leak is the mass the write-back is told to hand back, not the write-back.** project_elements took the mixture mass as msum = m_1 n_H + m_He n_He with m_1 = mass_per_H_nucleus_without_He(), the RESERVOIR mass per hydrogen nucleus (the trace metals at melem_ab), while the cell carries the metals at whatever the trace transport left; the projection then wrote the difference into hydrogen at every call. MEASURED at one cell of the certification column (metal/H 0.024401 against the reservoir's 0.028227): sum m_i f_i 1.0000000 -> 1.0000636 after one step, helium's gain exceeding the hydrogen and metal loss by 6.4e-5: mass created. A second leak of the same kind: the trace-metal arm wrote metal densities with no counter-flux in the hydrogen it diffuses through (3.0e-3 on the suite column with the metals frozen). **Fix:** mixture_mass_split reads the mass the cell's species actually carry (bsp_mass and melem_A, no species weight restated), project_elements(f_sp, Xhe, msum, scale_metals) scales component 1 as a whole, every helium-free species included (CO too), the metals written first and the hydrogen group closing the mixture; one projection after the metal loop puts the metals' mass change back on hydrogen; settling_coefficient takes its component-1 mass from the same split. Metals off is bitwise by construction (m_1 = m_H = 1 exactly). **MEASURED:** certification column closure after the relaxation 3.37e-3 -> 2.3e-15; the handed-back atomic candidate 6.44e-3 -> 1.94e-9 (the file's own precision, the three-unknown control's 1.93e-9); the writer-to-loader round trip of that state now returns it (writer ||R|| 1.112 and reload 1.1123 on the SAME row, against 3.2e-2 vs 0.4385 on different rows before: memo section 6 closed); mol_diffusion and wasp_full_newton BITWISE (metals off / no element row); **lower_profile MOVES and should:** its stored golden is 3.6e-4 out of closure (O/H 4.61e-4 against the reservoir 4.90e-4), a 1500-step cold start goes 6.77e-3 -> 3.14e-9 in closure with T at 1.6 Rp moving 1.3 percent and H I 6.3 percent: **a golden that moves for a physics fix, refreshed at the next gate.** Atomic reload cap 40 by named outcomes: closure 6.44e-3 -> 1.94e-9, 32/8 -> 31/9, refusals 0 in both, binding row energy of cell 2 -> cell 4, flux spread 2.3e-3 -> 1.4e-2, same stop. Tests RED 4 rows on the entry text (3.44e-3,

4.78e-4, 3.44e-3, 3.82e-4), GREEN 14/0; certification 42/0, diffusion_tests 35/0, krylov_and_dogleg 174/0. **The worker's control discipline:** the advisor's EXHALE_lwv.x is rebuilt at every verification and is a build of the live tree, so the worker built its own control (the delivered objects with only the entry-text binary_element_diffusion.o) and discarded three comparisons that had used the advisor's binary as "before"; workers are to build their own controls (added to the rules below). **Advisor:** steady_species_rows_tests.f90 referenced element_ratio_gate through element_inventory, which since N28 imports it with use, only and does not re-export it, so the suite would not build (N28's validation did not include this suite; the advisor's did not either): one use element_census, only: element_ratio_gate line added by the advisor; rebuild warning-free (md5 a9627a14), element_operator 14/0, certification 42/0, steady_species_rows 189/0, krylov_and_dogleg 174/0; one per cent fixed. Reported: the metal arm's counter-flux is closed on hydrogen but not on momentum or internal energy (a trace approximation, named); element_census_verify gates the element ratios and only reports the mass closure (the 6.4e-3 state passed every gate it has): whether it should gate is a decision.

- **N8c** (the five anchors re-measured on the states after N29; the memo docs/certification_tolerance_anchoring_20260910.md only, 344 -> 524 lines, no source file changed): **the reading is unchanged and is now taken on states worth measuring.** Post-N29 candidates (advisor-verified: no src file newer than the brief): atomic reload 275 iterations, 254/22, ||R|| 2.271e-4, binding row He/H of cell 245 (r 1.153), flux spread 9.8e-7, **mass closure 1.95e-9** (was 6.68e-3), 11 of 13 entries refuse; carrier reload byte-identical to N26's (no metals, no element row; the corrected projection is never reached), closure 1.93e-9. The anchors: two evaluations of one state 0 exactly; the state's own sweep seed 1.7e-10 -> 1.6e-14 (the loader no longer repairs a composition whose mass missed the density), closure floor 5e-19; state-perturbation amplification 1.2e3 to 2.0e3 at $r < 1.01$ and 0.4 to 0.7 elsewhere; derivative: a clean V at the base (7.6e-8), the PLATEAU at 6.6e-5 at the binding cell stays (smallest resolvable row 3.9e-8 there, 6.5e-9 in the wind; both taken at the first JFNK iterate of a relaxation restart, two CFL steps from the loaded state, stated); manufactured column order 1.589, closure 2.3e-15 (was 3.4e-3), extrapolated 1.52e-6 at the production spacing; grid refinement of the candidate 5.93e-4 (N = 500) -> 2.47e-4 (N = 1000), ratio 2.40, remap control 1.5e-5, Richardson limit 1.31e-4: **78 percent of the row is discretization**; element flux spread in the layer **39.3** -> **9.79** (He), 37.3 -> 9.09 (H), the wind 2.6e-2 -> 2.7e-2, the escape window 2.0e-3. Memo: sections 3 (mass closure) and 6 (the reload) marked CLOSED by N29 with the readings; section 5 rewritten; section 7's proposal re-stated with the new floors and **option (a) supported as written** (1e-5 gating at $r \geq \text{cert_regime_wind_r}$, the layer reported and not gating); two of its three prerequisites discharged, the layer's flux conservation (9.8) standing. No proposal certifies either candidate (wind element row 3.05e-4 and 7.2e-2 against a proposed 1e-5). certification 42/0. Reported: the iron wind row of the atomic candidate rose 18x between the two chaotic states (a size, not a trend); the derivative anchor is not measured at the loaded state itself (needs the probe under stationary evaluate, steady_newton.f90).
- **Decision 22 (user, 2026-09-10): option (a).** cert_tol_element and cert_tol_carrier become 1e-5, gating, for cells at $r \geq \text{cert_regime_wind_r}$ (1.20); the layer's element and carrier rows ($r < \text{cert_regime_layer_r}$,

1.10) are reported and do not gate until the layer's element flux conservation (9.8 after N29) is repaired; the hydrodynamic tolerances do not change; a wind-certified state says so in one line. Anchor: the operator's stated floor and a decade above the manufactured column's discretization at the production spacing (1.52e-6); no existing state is admitted by it (3.0e-4, 7.2e-2). Implemented as N30 (launched).

- **Gate after N29 and the lower_profile refresh (advisor, make check 2026-09-10 07:58 to 11:03; refresh 11:10).** Tree after N28, N28b, N29 (N30 running on a private build). Against the 2026-09-10 00:40 goldens: **15 of 16 default cases data identical on all four files** (wasp_full_newton now identical too, its golden being the 00:40 snapshot); **lower_profile FAILS on all four, by design** (N29's mass fix: its golden was 3.6e-4 out of closure; v moves up to 4.8 percent at r 3.18 in 85 of 504 rows, rho 1.0 percent in 4 rows, T 0.5 percent in 2 rows; the _adv measure column 57 percent at one row; log10 Mdot 7.29). The failing case was identified by its own section of the log (the advisor's first parse attributed the summary line to hp_front, which PASSES). Refreshed: golden/lower_profile archived as golden_pre_n29_20260910/lower_profile, run_check.sh golden_lower_profile snapshotted the gate's output, the four files compared identical to the case output with the comparator's filter. No other golden touched.
- **N30** (decision 22 a; certification.f90 +296/-54, steady_newton.f90 +50/-30, certification suite 42 -> 57 rows, steady_species_rows 189 -> 194, docs/input_schema.md the cert_reason row): cert_tol_element and cert_tol_carrier are gone as single numbers; cert_tol_element_wind = cert_tol_carrier_wind = 1e-5 with one accessor each (cert_tol_*_at(r), a step at cert_regime_wind_r = 1.20; cert_tol_reported_only = huge below), the anchoring written at the declaration with the memo cited; **one reduction** certification_species_row_gate decides a species row over the GATED cells (no gated cell = refused, never a satisfied row); the entries carry gated and reported maxima with their cells; the report prints gated at r >= 1.2 against 1e-5: <measure> at cell <j> and, for a certified state with species rows, the one line certified IN THE WIND ... the rows below it are reported and do not gate: worst <measure> at cell <j>; the state file carries cert_reason=certified_in_wind (the field already existed). **The band 1.10 <= r < 1.20 is reported and does not gate** (the decision was silent; the band holds the candidates' binding cell at r 1.153 where the discretization is 4.6e-4, so a 1e-5 gate would ask the residual to fall 1.5 decades below the error of its own discrete equation; no anchor for the band exists). distance_from_certification and certified_row_measures read the gated measures and the accessors, so **the trust region's ranking changes and both arms move** (that is the decision's intended reach): atomic reload 275 -> 166 iterations at the same log10 Mdot 10.10, binding row He/H 2.896e-4 at cell 288 (a WIND cell) against 1e-5, NOT certified; carrier reload 63 -> 76 iterations, H2 7.284e-2 at cell 285 against 1e-5, NOT certified; the hydrodynamic tolerances, the H(n=2) level balance (4.8e-10 against 1e-6) and steady_gates_met's own carrier gate (carrier_resid_th, a volume ratio, not this tolerance) untouched. wasp_full_newton bitwise, still certified with no reason. Tests RED (the superseded rule refused a layer-1e-2 / wind-5e-6 row and a wind-5e-5 row alike, on a layer cell; four judged-distance rows moved onto the new divisor), GREEN 57/0, 194/0, krylov_and_dogleg 174/0. No key added (the radii are properties of the flow, constants of certification.f90). Advisor: rebuild warning-free, 57/0, 194/0, 174/0 on build_lwv; the stale comment in EXHALE_main.f90 (580) rewritten; write_output.f90's

`adv_input_certified` T branch now carries the reason (a wind-certified state keeps its qualification in the `_adv` header); superseded notes appended to `a2_certification_contract`, `steady_solver_design` and `element_inventory_contexts`; one neighbourhood fixed. Reported: whether `steady_gates_met` should read the anchored tolerance is a separate decision on a shared interface.

- **DOC1 to DOC3** (the documentation brought to the stage-2 state, 2026-09-10, three concurrent documentation workers, no source touched, no binary run): **DOC1** `EXHALE_BC_and_IC.tex` (29 pages): the whole lower-boundary chapter described the retired component-wise ghost closure (Dirichlet density, one-way valve, isothermal ghost pressure, four keys that `input_read.f90` now refuses with error `stop`); rewritten as the characteristic (p, s) face condition of `base_boundary.f90` (branches by face Mach number, the smoothstep of width `base_face_mach_blend`, the Gauss-Legendre ghost averages, the three surviving controls), the upper boundary as `free_outflow_ghost`'s isothermal continuation, a new section on the transported columns' boundaries (N26, N26c, N8a, N29), the checkpoint rebuild order (N16d), and the old test matrix retitled as the retired closure's, not reproducible by today's binary. `composition_restart_and_base_handoff.tex` (18): thirteen base species in the nuclei count (OH, H₂O, CO added), every `load_IC.f90` line citation re-derived, a new section for the restart contract (intent, the eight-line metadata block field by field with refuse/report on each, provenance unknown, the grid guard, `Restart option change`), the round-trip and projection evidence of N9 and N29 labeled as read from this log. `transmission_spectrum.tex` (11): a new section for `read_adv_validity` (schema 2, the five statuses, `adv_conditional_tol`), the `chord_shell_indices` census with the CONTRIBUTION statement, and the `transit_schema 1` block. **DOC2** `steady_solver_memo.tex` (16): a new section 7 for the species-row solve as it stands (merit on certification scales, the trust region with named outcomes and the leg rules, the constraint rows, the empty-column repair and the equilibrated preconditioner, the three arms measured and disarmed, the linear solve as the limiter with the additivity finding, the ulp-level chaos, the residual contract, the certification by regime), eleven hooks added to its table. `molecular_hydrogen_treatment.tex` (23): the `_adv` row corrected (the energy residual reads the upstream caloric energy since N12), the layer table RE-MEASURED on the current `mol_lyman_werner` output (144/184/476 cells, H₂ peak 5.15e12, 1 percent edge 4.48e9 against the stale 227/251, 4.63e13, 4.42e9), thirteen of sixteen cases pinned, a new section on the element budget, the H₃⁺ departure factors marked STALE in place, the `mol_ir_bands` peak re-measured (2.148e-8 at r 1.0059 against the stale 2.934e-7). `lower_atmosphere_coupling.tex` (28): a new section 7.4 on the element operator (N29's conservation, N26b/N26c's boundary, N8c's flux spread table as the open item); two pre-existing overfull boxes fixed. `EXHALE_user_manual.tex` (88): certification by regime in section 2.7, the `Run mode` key entry, eleven hooks, the coupling header example replaced by the line the code writes (`cert_reason`), `Cooling_breakdown.txt` columns 12 to 14 (H₂, H₂O, CO infrared) restored to the column list. **DOC3** `README.md` (sixteen-case matrix, 25 suites, the restart contract, the transit metadata, certification stated, nineteen examples), `README_HOWTO.md` (`Resid tol` default 1e-5 not 1e-3, `Flux spread tol` default 2e-5 on the face flux not 5e-3, a certification table, a restart section, an eleven-row hook table, the sixteen cases and 25 suites), the `NEW docs/code_status_20260910.md` (418 lines, replaces the 2026-09-05 file as the read-first document; section 3.3 states what is and is not certified and what limits the rest), the project `CLAUDE.md` pointer line, `TO_BE_DONE.md` (an index of what is open, three headings corrected, 35 em-dashes removed),

steady_solver_design.md section 22. All eleven PDFs rebuilt, `grep -c '^!' 0` on every log. Advisor: every default and constant quoted checked against the source (`flux_spread_th_default 2.0e-5`, `resid_max 1e-5`, `cert_tol*_wind 1e-5`, `cert_regime_wind_r 1.20`, `ratio_sound_band 1e-2`, `tr_delta_min_dimensionless 1e-6`, the `Cooling_breakdown` header); one constant name corrected in the memo; pre-existing British spellings fixed in the workers' files, in `ISSUES_20260909.md` and in the handoff; the 51 em-dashes of the workspace `CLAUDE.md` replaced. **Reported by the workers and CONFIRMED by the advisor** (ISSUES 3.7): the base reservoir is stated once at startup and the refreshes of `dp_bc` and `ntot_bc` after every sweep are dead writes (the boundary itself is right: the reservoir equals the user's base pressure by construction); `golden/mol_ir_bands/` carries no `Cooling_breakdown.txt`; `examples/README.md` covers part of nineteen folders under the wrong planet; the `.md` twins of two `.tex` documents have parted from them; the `max(rest, 0)` write-back clamp (N4b) restated. ISSUES section 5 rewritten to the state after N30; the handoff opens with a dated 2026-09-10 state block. The carrier reload is still a scratch recipe (`B5k/mkrun.sh carrier`), not a pinned fixture: to pin with N31.